

Differential Geometry Homework

*Solutions by Ari Feiglin (ari.feiglin@gmail.com)
Hebrew University of Jerusalem, Spring 2026*

Contents

1	Homework 1	1
2	Homework 2	6
3	Homework 3	12
4	Homework 4	18
5	Homework 5	24
6	Homework 6	33
7	Homework 7	43
8	Homework 8	50
9	Homework 9	60
10	Homework 10	70
11	Homework 11	79
12	Homework 12	87
13	Homework 13	97

1 Homework 1

1.0.1 Problem (Do Carmo, Riemannian Geometry, page 33 problem 11)

Consider the two differentiable structures on $\mathbb{R}$, given by the smooth global charts

$$(\mathbb{R}, \varphi_1), (\mathbb{R}, \varphi_2), \quad \varphi_1(x) = x, \quad \varphi_2(x) = x^3$$

- (1) Demonstrate that the identity $\text{id}: (\mathbb{R}, \varphi_1) \rightarrow (\mathbb{R}, \varphi_2)$ is not a diffeomorphism, and as such the two charts are not smoothly compatible (i.e. the smooth structures induced by these charts are distinct).
 - (2) Show that $f: (\mathbb{R}, \varphi_1) \rightarrow (\mathbb{R}, \varphi_2)$ given by $f(x) = x^3$ is a diffeomorphism. Thus despite the smooth structures being distinct, the manifolds are diffeomorphic.
- (1) For id to be a diffeomorphism, it and its inverse (itself) must be smooth. That is to say, $\varphi_2^{-1} \circ \varphi_1$ must be a diffeomorphism. Note that this map is $x \mapsto x^{1/3}$, which is not differentiable at $x = 0$ and thus cannot be smooth, in particular it is not a diffeomorphism.
 - (2) We must show that $\varphi_2^{-1} \circ f \circ \varphi_1$ and $\varphi_1^{-1} \circ f^{-1} \circ \varphi_2$ are smooth. Notice that $f = \varphi_2$, so both of these maps are just the identity, which is obviously smooth.

1.0.2 Problem

In this exercise we demonstrate the existence of a smooth structure $\mathcal{S}$ on the real line $\mathbb{R}$ so that $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3$ is a diffeomorphism.

- (1) Show that it is sufficient to find a homeomorphism $\psi: \mathbb{R} \rightarrow \mathbb{R}$ such that $\psi \circ f \circ \psi^{-1}$ is a diffeomorphism with respect to the standard smooth structure on $\mathbb{R}$.
 - (2) Let $g, h: (0, 1) \rightarrow (0, 1)$ be homeomorphisms such that $g(x), h(x) > x$ for all $x \in (0, 1)$. Show that there exists a homeomorphism $\varphi: (0, 1) \rightarrow (0, 1)$ such that $\varphi \circ g = h \circ \varphi$.
 - (3) Construct a diffeomorphism (with respect to the standard structure) $k: \mathbb{R} \rightarrow \mathbb{R}$ such that the following conditions hold:
 - (i) for $x \in (0, 1)$ we have $k(x) < x$,
 - (ii) for $x \in (1, \infty)$ we have $k(x) > x$,
 - (iii) for $x \in (-1, 0)$ we have $k(x) > x$,
 - (iv) for $x \in (-\infty, -1)$ we have $k(x) < x$,
 - (v) $k(0) = 0, k(1) = 1, k(-1) = -1$.
 - (4) Note that the above conditions are satisfied by $k = f$ (other than it being a diffeomorphism). Apply point (2) repeatedly to construct the homeomorphism from point (1).
- (1) Given such a homeomorphism, the chart $(\mathbb{R}, \psi^{-1})$ is clearly smoothly compatible with itself and thus induced a smooth structure. We want $f: (\mathbb{R}, \psi^{-1}) \rightarrow (\mathbb{R}, \psi^{-1})$ to be a diffeomorphism, which requires that $\psi \circ f \circ \psi^{-1}$ and $\psi \circ f^{-1} \circ \psi^{-1}$ be smooth (wrt the standard smooth structure), as these are inverses this is the same as requiring $\psi \circ f \circ \psi^{-1}$ be a diffeomorphism, as desired.
 - (2) Note that g, h must be monotonic as they are homeomorphisms. In particular they must be increasing, as if say g were decreasing let $x \in (0, 1)$. Then for every $z > x$ we have $x < z < g(z)$ and so g maps $(x, 1)$ into $(x, 1)$. But for $z < x$ we have $g(z) > g(x) > x$ and so g maps $(0, x)$ into $(x, 1)$ as well, meaning the image of g is contained in $(x, 1)$, a contradiction. And since g, h are increasing so too must their inverses.

Furthermore we must have that their limits at 0 and 1 are 0 and 1 respectively. For example, let $x_n \rightarrow 0$ be decreasing then $g(x_n)$ is decreasing as well, and must have infimum 0 and thus the limit of $g(x_n)$ is zero as well. Thus we can extend g, h to homeomorphisms $[0, 1] \rightarrow [0, 1]$ by defining $g(0) = h(0) = 0$ and $g(1) = h(1) = 1$.

Now for $x \in (0, 1)$ consider the sequence $x_n = g^n(x)$ for $n \in \mathbb{Z}$. This is an increasing sequence since $g(x) > x$, and its supremum is 1. Suppose otherwise and let $s = \sup x_n < 1$, so that $g^n(x) \leq s$ for all n . But then $g^n(x) \leq g^{-1}(s) < s$ for all n , a contradiction. Similarly its infimum is 0. Thus the sets $[x_n, x_{n+1}]$ cover $(0, 1)$.

Now choose another $y \in (0, 1)$ and form the same construction via h to give $[y_n, y_{n+1}]$ covering $(0, 1)$. Take an increasing homeomorphism $\varphi_0: [x_0, x_1] \rightarrow [y_0, y_1]$. Now we define $\varphi_n: [x_n, x_{n+1}] \rightarrow [y_n, y_{n+1}]$ as

$$\varphi_n = h^n \circ \varphi_0 \circ g^{-n}.$$

This is an increasing homeomorphism as a composition of increasing homeomorphisms. Thus in addition we can extend φ_n to a homeomorphism $[x_n, x_{n+1}] \rightarrow [y_n, y_{n+1}]$ by $\varphi_n(x_{n+1}) = y_{n+1}$ (as demonstrated before, this is true for increasing homeomorphisms).

So we have a collection of homeomorphisms φ_n defined on $[x_n, x_{n+1}]$ which all agree on the endpoints (meaning $\varphi_n(x_{n+1}) = \varphi_{n+1}(x_{n+1})$). Thus by the pasting lemma, since the collection of these closed intervals is locally finite, we can paste these together to form a homeomorphism $\varphi: (0, 1) \rightarrow (0, 1)$ restricting to φ_n on the relevant interval.

In particular for $x \in [x_n, x_{n+1}]$ we have

$$\varphi \circ g(x) = \varphi_{n+1}g(x) = h^{n+1} \circ \varphi_0 \circ g^{-n}(x)$$

and

$$h \circ \varphi(x) = h \circ \varphi_n(x) = h^{n+1} \circ \varphi_0 \circ g^{-n}(x)$$

so that $h \circ \varphi = \varphi \circ g$ as desired. Notice that defining $\varphi(0) = 0, \varphi(1) = 1$ extends this result to $[0, 1] \rightarrow [0, 1]$.

- (3) Consider the scalar multiple of $\sinh$,

$$k(x) = \frac{e^x - e^{-x}}{e - e^{-1}}.$$

This is a smooth function as a linear combination of smooth real valued functions. Its inverse is smooth as well; it is sufficient to show this regarding $\sinh$. We can write the inverse of $\sinh$ as

$$\sinh^{-1}(x) = \log(x + \sqrt{x^2 + 1})$$

and thus compute its first two derivatives

$$\frac{1}{\sqrt{x^2 + 1}}, \quad -\frac{2x}{x^2 + 1}$$

The second derivative is smooth, as the quotient of two smooth functions ($x^2 + 1$) has no zeroes. Thus $\sinh^{-1}$ is, as desired.

$k(x)$ has the desired properties: these can be deduced by a routine analysis of $k(x) - x$.

- (4) We can dualize our conjugacy result above with the condition that $g(x), h(x) < x$. The result holds also for intervals of the form $[a, \infty)$ by the same construction. Thus we can define homeomorphisms

$$\psi_{-\infty}: (-\infty, -1] \rightarrow (-\infty, -1], \quad \psi_-: [-1, 0] \rightarrow [-1, 0], \quad \psi_+: [0, 1] \rightarrow [0, 1], \quad \psi_{\infty}: [1, \infty) \rightarrow [1, \infty)$$

such that $\psi_{\bullet} \circ f = k \circ \psi_{\bullet}$ for each $\psi_{\bullet}$, and that $\psi_{\bullet}$ agree on the endpoints of the intervals (e.g. $\psi_-(-1) = -1, \psi_-(0) = 0$). Thus we can paste them together to get a homeomorphism $\psi: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$\psi \circ f \circ \psi^{-1} = k$$

a diffeomorphism, as desired.

1.0.3 Problem

Let K be a field, and define $G_K(k, n)$ to be the set of all k -dimensional subspaces of K^n . We will define a smooth structure on $G_K(k, n)$. Let $V \subseteq K^n$ be a linear subspace of dimension k and let $W \subseteq K^n$ be any complementary subspace, necessarily of dimension $n - k$. Define

$$U_{V,W} = \text{hom}(V, W) \cong K^{k(n-k)}$$

Also define

$$\mathbf{x}_{V,W}: U_{V,W} \rightarrow G_K(k, n)$$

which maps a linear map $\ell \in \text{hom}(V, W)$ to its graph: $\mathbf{x}_{V,W}(\ell) = \{v + \ell(v) \mid v \in V\} \subseteq K^n$.

- (1) Show that $\mathbf{x}_{V,W}$ is injective.
- (2) Show that $\bigcup_{V,W} \mathbf{x}_{V,W}(U_{V,W}) = G_K(k, n)$.
- (3) Show that for another pair V', W' , $\mathbf{x}_{V',W'}^{-1} \circ \mathbf{x}_{V,W}(U_{V,W})$ is open in $U_{V',W'}$.
- (4) Show that $\mathbf{x}_{V',W'}^{-1} \circ \mathbf{x}_{V,W}$ is smooth.

- (1) Let $\ell, \ell': V \rightarrow W$ such that $\mathbf{x}(\ell) = \mathbf{x}(\ell')$ ($\mathbf{x} = \mathbf{x}_{V,W}$). This means that for every $v \in V$ there is a $u \in V$ such that $v + \ell(v) = u + \ell'(u)$. Since V and W are complementary, any representation of a vector as a sum of one from V and one from W is unique; thus $v = u$ and $\ell(v) = \ell'(u) = \ell'(v)$. So we have shown that $\ell = \ell'$, meaning that $\mathbf{x}$ is injective as desired.
- (2) Let $V \in G_K(k, n)$ be a k -dimensional subspace of K^n with complement W . Consider then the zero morphism $0 \in \text{hom}(V, W)$, and we note that $\mathbf{x}_{V,W}(0) = V$. Thus we have the desired result.
- (3) First we make the following claim:

$$\mathbf{x}_{V,W}(U_{V,W}) = \{V' \in G_K(n, k) \mid V' \cap W = 0\}.$$

Indeed, one direction is clear: if $v + \ell(v) \in W$ then by uniqueness of $V \oplus W$, we must have that $v = 0$ and so $v + \ell(v) = 0$, thus $\mathbf{x}_{V,W}(\ell) \cap W = 0$. Conversely consider the projection $\pi_V: K^n \rightarrow V$ parallel to W . This map defines an isomorphism $\pi_V|_{V'}: V' \rightarrow V$, since the kernel of the projection is W and thus the restriction is injective and thus an isomorphism by considering dimension. We also can consider the projection $\pi_W: K^n \rightarrow W$ parallel to V and define the composition

$$\ell = V \xrightarrow{\pi_V|_{V'}^{-1}} V' \xrightarrow{\pi_W|_{V'}} W$$

Let us denote by $I_V: V \rightarrow K^n$ the inclusion of V in K^n . So for $\ell \in U_{V,W}$, $\mathbf{x}(\ell)$ is simply the image of $I_V + I_W \ell: V \rightarrow K^n$. Since $\pi_V|_{V'}$ is an isomorphism $V' \rightarrow V$, the image of $I_V + I_W \ell$ and $I_V \pi_V|_{V'} + I_W \ell \pi_V|_{V'}$ are the same, and

$$I_V \pi_V|_{V'} + I_W \ell \pi_V|_{V'} = I_V \pi_V|_{V'} + I_W \pi_W|_{V'} = \text{id}_{V'}$$

since this is the restriction of $I_V \pi_V + I_W \pi_W = \text{id}_{K^n}$. In particular $\mathbf{x}(\ell) = \mathbf{x}(\text{id}_{V'}) = V'$ as desired.

Thus we have

$$\mathbf{x}_{V',W'}^{-1} \circ \mathbf{x}_{V,W}(U_{V,W}) = \{\ell' \in U_{V',W'} \mid \mathbf{x}(\ell') \cap W = 0\},$$

and we wish to show this is open. The image of a linear map $T: V' \rightarrow K^n$ has trivial intersection with W iff $\pi_V \circ T: V' \rightarrow V$ is injective, equivalently invertible. Indeed, it is injective iff T 's image intersects non-trivially with π_V 's kernel, W . Thus the set in question is

$$\{\ell' \in U_{V',W'} \mid \pi_V \circ (I_{V'} + I_{W'} \ell') \text{ is invertible}\}$$

Identifying the spaces of linear maps in question with spaces of matrices, that is choosing bases for V, W, V', W' , this set corresponds to the subset of Euclidean space

$$\{L \in M_{(n-k) \times k}(K) \mid P(I_1 + I_2 L) \text{ is invertible}\}$$

where I_1, I_2 are the matrix representations of $I_{V'}, I_{W'}$ and P the matrix representation of π_V . This is the preimage of the open set $K^\times$ under the function $L \mapsto \det(P(I_1 + I_2 \cdot L))$, which is a composition of continuous functions, so the set is open, as desired.

- (4) Let $\ell' \in \mathbf{x}_{V',W'}^{-1} \circ \mathbf{x}_{V,W}(U_{V,W})$, we want to compute its image under $\mathbf{x}_{V,W}^{-1} \circ \mathbf{x}_{V',W'}$. Recall that we showed $\mathbf{x}_{V,W}^{-1}(V') = \pi_V|_{V'}^{-1} \circ \pi_W|_{V'}$. Thus

$$\mathbf{x}^{-1} \circ \mathbf{x}(\ell') = \pi_W|_{\mathbf{x}(\ell')} \circ \pi_V|_{\mathbf{x}(\ell')}^{-1}.$$

Let us define $I_{\ell'}$ to be the inclusion $\mathbf{x}(\ell') = \text{im}(I_V + I_W \ell') \rightarrow K^n$ so that $\pi|_{\mathbf{x}(\ell')} = \pi \circ I_{\ell'}$. Then we have

$$\mathbf{x}^{-1} \circ \mathbf{x}(\ell') = (\pi_W \circ I_{\ell'}) \circ (\pi_V \circ I_{\ell'})^{-1}.$$

Let us choose bases for V, W, V', W' , then these give a basis $\{v'_i + \ell' v'_i\}_i$ for $\mathbf{x}(\ell')$. Then relative to these bases, $\mathbf{x}^{-1} \circ \mathbf{x}(\ell')$ is represented by

$$(\pi_W \circ I_{\ell'})_{W'}^{\mathbf{x}(\ell')} \cdot (\pi_V \circ I_{\ell'})_V^{\mathbf{x}(\ell')^{-1}}.$$

The i th column in these matrices is $(\bullet \text{ standing in place for } V \text{ or } W)$,

$$(\pi_{\bullet} v'_i + \pi_{\bullet} \ell' v'_i)_{\bullet} = (\pi_{\bullet} v'_i)_{\bullet} + (\pi_{\bullet} \ell' v'_i)_{\bullet}.$$

Thus the representation matrix of $\pi_{\bullet} \circ I_{\ell'}$ is simply $P_{\bullet}^1 + P_{\bullet}^2 L$, where $P_{\bullet}^1$ is the representation $(\pi_{\bullet})_{\bullet}^{V'}$ and $P_{\bullet}^2$ is the representation $(\pi_{\bullet})_{\bullet}^{W'}$, and L is the representation $(\ell')_{W'}^{V'}$.

All in all, we see that $\mathbf{x}^{-1} \circ \mathbf{x}$ corresponds to the map of matrices

$$\mathbf{x}^{-1} \circ \mathbf{x}(L) = (P_W^1 + P_W^2 \cdot L) \cdot (P_V^1 + P_V^2 \cdot L)^{-1}$$

Both of these maps $L \mapsto P_{\bullet}^1 + P_{\bullet}^2 \cdot L$ are smooth, as affine Euclidean maps. And taking the inverse of $P_V^1 + P_V^2 \cdot L$ is smooth too: if $L \mapsto f(L)$ is a smooth function where $f(L)$ is always invertible, then $L \mapsto f(L)^{-1}$ is smooth as well. This is because $f(L)^{-1} = 1/\det(f(L)) \cdot \text{adj}(f(L))$ with adj being the adjugate matrix. $\det$ is a smooth function, and thus $1/\det$ is smooth where defined (the multiplicative inverse of a real/complex-valued smooth function is smooth, proven in a later answer). adj is smooth as well: in each coordinate it is the composition of a projection onto a minor and $\det$, both of which are smooth. Thus $L \mapsto f(L)^{-1}$ is smooth as claimed.

So we have shown $L \mapsto \mathbf{x}^{-1} \circ \mathbf{x}(L)$ is a smooth function, as desired.

1.0.4 Problem

An **exhaustion function** for a topological space X is a continuous function $f: X \rightarrow \mathbb{R}$ such that $f^{-1}(-\infty, c]$ is compact for every $c \in \mathbb{R}$.

- (1) Show that $x \mapsto |x|^2$ is a smooth exhaustion function $\mathbb{R}^n \rightarrow \mathbb{R}$.
- (2) Show that $f: B_1(0) \rightarrow \mathbb{R}$ defined by $f(x) = 1/(1 - |x|^2)$ is a smooth exhaustion function.
- (3) Prove that every manifold admits a smooth positive exhaustion function.

- (1) First of all, $f(x) = |x|^2$ is smooth. And $f(x) \leq c$ iff x is in the closed ball centered at 0 of radius $\sqrt{c}$, i.e. the preimage $f^{-1}(-\infty, c]$ is $\overline{B}_{\sqrt{c}}(0)$ which is compact (for negative c it is empty, still compact).
- (2) f is smooth here as well (it is the quotient of two smooth functions, which is smooth as I show in the next exercise). And

$$f(x) \leq c \iff |x|^2 \leq 1 - 1/c,$$

meaning that $f^{-1}(-\infty, c]$ is $\overline{B}_{\sqrt{1-1/c}}(0)$ when defined (otherwise empty), which is again compact.

- (3) Let $\{U_n\}_{n=1}^\infty$ be a countable cover of M by precompact open sets (open sets whose closure is compact). Such a cover exists: take a chart (U, φ) , then $\varphi(U)$ has a countable open cover of proper precompact sets (as a subspace of Euclidean space, e.g. open balls are open precompact), and lifting these back to

U via the homeomorphism φ covers U in open precompact sets. M is covered by the union of countable precompact open covers of a countable cover of coordinate charts.

In recitation we showed there exists a partition of unity subordinate to it, let it be $\{\psi_n\}_{n=1}^\infty$. Define the function

$$f(p) = \sum_{n=1}^{\infty} n\psi_n(p).$$

Since $\{\psi_n\}_n$ is a partition of unity, at each point this is only a finite sum and thus f is a well-defined function $M \rightarrow \mathbb{R}$. Moreover, for every U_n , $f|_{U_n}$ is smooth as a finite combination of smooth maps, and thus f itself is smooth (U_n covers f).

Now, notice that if $p \notin \bigcup_{i=1}^N \overline{U_i}$ then in particular p cannot be in the support of ψ_i for $1 \leq i \leq N$ and so

$$f(p) = \sum_{n=1}^{\infty} n\psi_n(p) = \sum_{n=N+1}^{\infty} n\psi_n(p) \geq (N+1) \sum_n \psi_n(p) = N+1$$

Thus if $p \notin \bigcup_{i=1}^N \overline{U_i}$, $f(p) > N$.

Thus $f^{-1}(-\infty, N]$ is a closed subset of $\bigcup_{i=1}^N \overline{U_i}$, a compact set as the finite union of compact sets. M is Hausdorff and so $f^{-1}(-\infty, N]$ is compact as a closed subset of a compact space. For a real $c \in \mathbb{R}$ we can take an integer $N > c$ and then we have that $f^{-1}(-\infty, c]$ is a closed subset of the compact $f^{-1}(-\infty, N]$ and thus is compact.

f is clearly positive, so we have constructed a positive smooth exhaustion function.

1.0.5 Problem

Let $C^\infty(M)$ denote the $\mathbb{R}$ -algebra of smooth functions $M \rightarrow \mathbb{R}$ for a smooth manifold M . Prove that every $\mathbb{R}$ -algebra morphism $\varphi: C^\infty(M) \rightarrow \mathbb{R}$ is given by evaluation at some $p \in M$; i.e. $\varphi(f) = f(p)$ for all $f \in C^\infty(M)$.

Since ring morphisms preserve the multiplicative unit, $\ker \varphi$ is a proper ideal. We will show this first for the case that M is compact.

Let $f \in \ker \varphi$, then we claim that f has a zero. Indeed, if f has no zeroes then it has a multiplicative inverse $g(p) = 1/f(p)$. To show that g is smooth we must prove that for every chart $(U, \mathbf{x}: U \rightarrow \mathbb{R}^n)$, $g \circ \mathbf{x}^{-1}$ is smooth. Indeed:

$$g \circ \mathbf{x}^{-1} = \frac{1}{f \circ \mathbf{x}^{-1}},$$

and 1 over a never-zero smooth function is always smooth. Since f has a multiplicative inverse, it cannot be contained in a non-trivial kernel and thus $f \notin \ker \varphi$ when f has no zeroes.

For $f \in \ker \varphi$ define $Z(f) = \{p \in M \mid f(p) = 0\}$ to be its set of zeroes, i.e. $Z(f) = f^{-1}\{0\}$ (in particular $Z(f)$ is closed in M as the preimage of a closed set). If $f, g \in \ker \varphi$ are two functions in the kernel, then we claim $Z(f) \cap Z(g)$ is nonempty. Indeed, if the intersection were empty then $f^2 + g^2$ would be a smooth function having no zeroes, but also in the kernel, a contradiction.

Thus the finite intersection of zeroes of functions in the kernel is non-empty. By compactness, the intersection of the zeroes of all the functions in the kernel is non-empty. Indeed, if $\bigcap_{f \in \ker \varphi} Z(f) = \emptyset$ then $\{Z(f)^c\}_{f \in \ker \varphi}$ would be an open cover of M and by compactness have a finite subcover. That is, $\bigcup_i Z(f_i)^c = M$ for finitely many $f_i \in \ker \varphi$, meaning $\bigcap Z(f_i) = \emptyset$, which is a contradiction.

So let $p \in \bigcap_{f \in \ker \varphi} Z(f)$; we must have that $\varphi(f) = f(p)$ for all $f \in C^\infty(M)$. Indeed, for $f \in C^\infty(M)$ consider the constant function whose value is $\varphi(f)$. As an $\mathbb{R}$ -algebra morphism, $\varphi(f - \varphi(f)) = 0$ so that $f - \varphi(f) \in \ker \varphi$. Thus p is a zero of $f - \varphi(f)$, i.e. $f(p) = \varphi(f)$ as desired.

In the general case when M is not necessarily compact, let $u: M \rightarrow \mathbb{R}$ be a smooth exhaustion function, and define K to be the compact subset given by $u^{-1}(-\infty, \varphi(u)]$. As before, if we can show that the intersection $Z(f)$ over all $f \in \ker \varphi$ is nonempty, we are finished. So assume that the intersection is empty, then $\{Z(f)^c\}_{f \in \ker \varphi}$ form an open cover of M , in particular K . By compactness there must be a finite subcover; let $f_1, \dots, f_n$ be these functions inducing one. That is to say,

$$K \subseteq \bigcup_{i=1}^n Z(f_i)^c,$$

meaning every $x \in K$ is not the zero of some f_i .

Now notice that if $u(x) = \varphi(u)$, then x must be in K by definition. Thus $u - \varphi(u)$ is a map in φ 's kernel, all of whose zeroes lie within K . So we can define

$$g = (u - \varphi(u))^2 + \sum_{i=1}^n f_i^2.$$

This map has no zeroes: if $g(x) = 0$ then x must be a zero of $u - \varphi(u)$ and also of each f_i . But this cannot be: if $u(x) = \varphi(u)$ then x must be in K , which means it is not the zero of some f_i , a contradiction.

So g is non-zero and thus has a multiplicative inverse, but also $g \in \ker \varphi$ since it is the sum and product of functions in the kernel, a contradiction. Thus $\bigcap_{f \in \ker \varphi} Z(f)$ is non-empty, and any point p in this intersection satisfies $\varphi(f) = f(p)$ as before.

2 Homework 2

2.0.1 Problem

Let $M = G_K(k, n)$ be the Grassmanian, and for $V \in G_K(k, n)$ and a chart $(U_{V,W}, \mathbf{x}_{V,W})$ consider the induced bijection $\bar{\mathbf{x}}_{V,W}: T_V M \rightarrow \mathbb{R}^{k(n-k)} \cong \text{hom}(V, W)$. Consider the isomorphism $f_{V,W}: \text{hom}(V, W) \rightarrow \text{hom}(V, K^n/V)$, and the composition

$$T_V M \xrightarrow{\bar{\mathbf{x}}_{V,W}} \text{hom}(V, W) \xrightarrow{f_{V,W}} \text{hom}(V, K^n/V).$$

Show that this composition is independent of W , giving a canonical isomorphism $T_V M \cong \text{hom}(V, K^n/V)$.

Let W, W' be two complementary subspaces of V . Then the projection $\pi_W: K^n \rightarrow W$ parallel to V restricts to an isomorphism $\pi_W|_{W'}: W' \cong W$. Choose a basis $\{w'_i\}_i$ for W' , and then $w_i = \pi_W(w'_i)$ is a basis for W , and also choose a basis $\{v_j\}$ for V . Then we have bases E_{ij}, E'_{ij} for $\text{hom}(V, W), \text{hom}(V, W')$ respectively mapping v_j to w_i, w'_i respectively.

Now let $\gamma_{ij}, \gamma'_{ij}$ be the bases for $T_V M$ corresponding to E_{ij}, E'_{ij} ; i.e. the preimages of these under $\bar{\mathbf{x}}_{V,W}$ and $\bar{\mathbf{x}}_{V,W'}$. If we let $T = \mathbf{x}_{V,W'}^{-1} \circ \mathbf{x}_{V,W}$ be the transition function, we know that $\gamma_{ij}, \gamma'_{ij}$ are related via the Jacobian of T :

$$\gamma'_{ij} = \sum_{t,\ell} \frac{\partial T^{t,\ell}}{\partial x^{i,j}}(0) \cdot \gamma_{t,\ell}.$$

From our previous problem set, we know that relative to the bases chosen,

$$TX = (P_W^1 + P_W^2 X) \cdot (P_V^1 + P_V^2 X)^{-1}$$

where

$$P_W^1 = (\pi_W)_W^V = 0, \quad P_W^2 = (\pi_W)_{W'}^{W'} = I, \quad P_V^1 = (\pi_V)_V^V = I, \quad P_V^2 = (\pi_V)_{V'}^{V'} = 0$$

and so we have that

$$TX = X,$$

meaning its Jacobian is the identity, so that

$$\gamma'_{ij} = \gamma_{ij}.$$

Since the composition is linear, we need only check how it acts on the basis vectors γ_{ij} . By definition, we have that

$$f_{V,W} \circ \bar{\mathbf{x}}_{V,W}(\gamma_{ij}) = \iota_W \circ E_{ij}, \quad f_{V,W'} \circ \bar{\mathbf{x}}_{V,W'}(\gamma_{ij}) = \iota_{W'} \circ E'_{ij},$$

where $\iota_W: W \rightarrow K^n/V$ is the isomorphism $\iota_W(w) = w + V$. The lefthand side maps v_i to $w_j + V$, and the righthand side maps v_i to $w'_j + V$. These cosets are equal since $\pi_W(w_j - w'_j) = w_j - \pi_W(w'_j) = w_j - w_j = 0$, and the kernel of π_W is V . Thus the composition is indeed independent of choice of W .

2.0.2 Problem

Let $f: N \rightarrow M$ be smooth, and define the **global differential**

$$df: TN \rightarrow TM,$$

whose component on $T_p N$ is $df_p: T_p N \rightarrow T_{f(p)} M \subseteq TM$. Show that the global differential is smooth.

Let $n = \dim N, m = \dim M$. We have smooth atlases of TN by charts $(U \times \mathbb{R}^n, \bar{\varphi})$ where (U, φ) is a chart for N and $\bar{\varphi}$ is defined via

$$\bar{\varphi}(\bar{x}, \bar{v}) = (\varphi(\bar{x}), \hat{\varphi}_{\bar{x}}(\bar{v}))$$

where $\hat{\varphi}_{\bar{x}}(\bar{v})$ is the curve $t \mapsto \varphi(\bar{x} + t\bar{v})$. Its inverse is the map

$$\bar{\varphi}^{-1}(p, \gamma) = (\varphi^{-1}(p), \tilde{\varphi}_p(\gamma)),$$

where $\tilde{\varphi}_p(\gamma) = (\varphi^{-1} \circ \gamma)'(0)$.

So for a chart (U, φ) of N and (V, ψ) of M , we have that the coordinate representation of df is

$$\begin{aligned} \bar{\psi}^{-1} \circ df \circ \bar{\varphi}(\bar{x}, \bar{v}) &= \bar{\psi}^{-1} \circ df(\varphi(\bar{x}), \hat{\varphi}_{\bar{x}}(\bar{v})) \\ &= \bar{\psi}^{-1}(f \circ \varphi(\bar{x}), f \circ \hat{\varphi}_{\bar{x}}(\bar{v})) \\ &= (\psi^{-1} \circ f \circ \varphi(\bar{x}), \tilde{\psi}(f \circ \hat{\varphi}_{\bar{x}}(\bar{v}))) \end{aligned}$$

The first coordinate is smooth, since f is. The second coordinate is the map

$$(\bar{x}, \bar{v}) \mapsto (\psi^{-1} \circ f \circ \varphi(\bar{x} + t\bar{v}))'(0).$$

If we let $\hat{f} = \psi^{-1} \circ f \circ \varphi$ be f 's coordinate representation, then we can rewrite it as

$$(\bar{x}, \bar{v}) \mapsto (\hat{f}(\bar{x} + t\bar{v}))'(0) = J\hat{f}(\bar{x})\bar{v},$$

which is smooth as well (since $\hat{f}$ is smooth, its Jacobian is).

2.0.3 Problem

Let $\mathbb{RP}^n$ be the real projective space.

- (1) Define $f: \mathbb{RP}^n \rightarrow \text{Mat}_{n+1}(\mathbb{R})$ by $f_{ij}(x) = x_i x_j / \sum_k x_k^2$. Show that f defines a smooth embedding, whose image is all symmetric matrices A of trace 1 where $A^2 = A$.
- (2) Show that $\mathbb{RP}^n$ is compact.

- (1) We can write f as $f(x) = xx^\top / x^\top x$; this is clearly well-defined. We first demonstrate that f is smooth: we consider the atlas of $\mathbb{RP}^n$ whose charts are $(\mathbb{R}^n - 0, \varphi_i: \mathbb{R}^n \rightarrow U_i)$ where $U_i \subseteq \mathbb{RP}^n$ is the set of $[x]$ where $x_i \neq 0$, and

$$\varphi_i(u) = [u_1, \dots, 1, \dots, u_n], \quad 1 \text{ is inserted in the } i\text{th index,}$$

with inverse

$$\varphi_i^{-1}(x) = 1/x_i(x_1, \dots, \hat{x}_i, \dots, x_n).$$

Then f 's coordinate representation is $\hat{f} = f \circ \varphi_i$:

$$\hat{f}(u) = \frac{\varphi_i(u) \cdot \varphi_i(u)^\top}{u^\top u + 1},$$

viewing φ_i as the map $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$ which inserts 1 at the i th index. This is smooth: φ_i is smooth, the inner product is smooth, and $u^\top u + 1 \neq 0$. Thus f is smooth.

To show that f is a smooth embedding, we need to show that f is a smooth immersion, i.e. that its differential has full rank. By reordering indices, we can focus on f 's coordinate representation for $i = n + 1$. Then $\hat{f}^{ij}$ is the map $\hat{f}^{ij}: \mathbb{R}^n \rightarrow \mathbb{R}$ mapping u to $u_i u_j / (u^\top u + 1)$ where $u_{n+1} = 1$. Let us focus on when $j = n + 1$, letting $\hat{f}^i = \hat{f}^{i, n+1}$, which maps

$$\hat{f}^i(u) = \frac{u_i}{u^\top u + 1}.$$

Its partial derivatives are

$$\frac{\partial \hat{f}^i}{\partial x_i}(u) = \frac{u^\top u + 1 - 2u_i^2}{(u^\top u + 1)^2}, \quad \frac{\partial \hat{f}^i}{\partial x_j}(u) = \frac{-2u_i u_j}{(u^\top u + 1)^2}.$$

So the submatrix of the Jacobian corresponding to $\hat{f}^1, \dots, \hat{f}^n$ is

$$\frac{1}{u^\top u + 1} \left(I - \frac{2}{u^\top u + 1} U \right),$$

where $U_{ij} = u_i u_j$. This means that v is in its nullspace iff $Uv = \frac{u^\top u + 1}{2}v$. Now, U is symmetric and $U^2 = \text{tr}(U)U$, since $\text{tr}(U) = u^\top u$, this means U is unitarily diagonalizable with eigenvalues 0, $\text{tr}(U)$. This means that

$$\text{tr}(U) = \frac{\text{tr}(U) + 1}{2} \implies \text{tr}(U) = 1,$$

so when $u^\top u \neq 1$, this matrix is invertible and thus f has full rank.

When $i = n + 1$, we have the map $\hat{f}^{n+1}(u) = 1/(u^\top u + 1)$ whose derivatives are

$$\frac{\partial \hat{f}^{n+1}}{\partial x_i}(u) = \frac{-2u_i}{u^\top u + 1}.$$

In the problematic region of norm 1, this is equal to $-u_i$. Let j be an index where $u_j \neq 0$, and consider the minor of the Jacobian where we replace the j th column of the matrix above $(1/2(I - U/2))$ with the gradient computed above, $-u$. If this were singular then u could be written as a linear span of the columns of $I - U/2$, without the j th, so

$$u = \sum_{i \neq j} \alpha_i e_i - \frac{\alpha_j u_j}{2} u \implies \left(1 + \sum_{i \neq j} \frac{\alpha_i u_i}{2} \right) u = \sum_{i \neq j} \alpha_i e_i.$$

But the j th component of the right-hand side is zero, and $u_j \neq 0$, which means that both sides must be equal to zero. Thus all $\alpha_i = 0$ since the e_i s are linearly independent. Thus the Jacobian is invertible, and f is a smooth immersion.

We now show that f 's image consists of all symmetric involutions of trace 1. Firstly clearly f 's image is contained in this: $f(x)$ is clearly symmetric, and has trace 1, and

$$f(x)_{ij}^2 = \sum_t f_{it}(x) f_{tj}(x) = \frac{1}{(x^\top x)^2} \sum_t x_i x_t^2 x_j = f_{ij}(x),$$

as desired. Let A be such a matrix, then since A is symmetric, it is unitarily diagonalizable. Furthermore since $A^2 = A$, its eigenvalues are either 1 or 0; since its trace is 1, the multiplicity of 1 must be one. Thus its diagonal form is $\text{diag}(1, 0, \dots, 0)$. Thus there exists a unitary matrix P such that

$$P^\top A P = \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix} = f(e),$$

where $e = (1, 0, \dots, 0)$ is the first standard basis vector. Thus

$$A = P f(e) P^\top = \frac{P e e^\top P^\top}{e^\top e} = \frac{(P e)(P e)^\top}{(P e)^\top (P e)} = f(P e),$$

so that this is indeed f 's image.

f is also injective. We note that every class in $\mathbb{RP}^n$ has a representative of the form Pe where P is unitary and e is the first standard basis vector: this is simply because every nonzero vector of norm 1 can be extended to an orthonormal basis, and the matrix which represents this basis has the vector as its first column. We showed above that when $f(Pe) = A$,

$$P^\top AP = \text{diag}(1, 0, \dots, 0),$$

which means that P is a matrix of eigenvectors of A , in particular P 's first column, Pe , is A 's eigenvector with eigenvalue 1. So if $f(Pe) = f(Qe) = A$, then Pe and Qe are both eigenvectors with eigenvalue 1 of A , and since this eigenspace has dimension 1, Pe and Qe span the same subspace, i.e. are equal in $\mathbb{RP}^n$. Now for two matrices A and B in $\text{im} f$, their distance squared is

$$d(A, B)^2 = |A - B|^2 = \text{tr}((A - B)^\top (A - B)) = \text{tr}((A - B)(A - B)) = \text{tr}(A - AB - BA + B) = 2 - 2\text{tr}(AB).$$

If $A = f[x]$ and $B = f[y]$ where x, y are representatives of norm 1, then

$$|A - B|^2 = 2 - 2\text{tr}(xx^\top yy^\top) = 2 - 2(x^\top y)\text{tr}(xy^\top) = 2 - 2(x^\top y)^2,$$

and the distance squared of x and y is

$$d(x, y)^2 = |x - y|^2 = (x - y)^\top (x - y) = 2 - 2x^\top y.$$

We want to show that f is an open map onto its image: for $U \subseteq \mathbb{RP}^n$ open, $f(U) \subseteq \text{im} f$ is open. U is open when $\pi^{-1}U \subseteq \mathbb{R}^{n+1} - 0$ is open where $\pi: \mathbb{R}^{n+1} - 0 \rightarrow \mathbb{RP}^n$ is the projection map. For $[x] \in U$ with x of norm 1, let $\varepsilon > 0$ be a radius such that $B_\varepsilon(x) \subseteq \pi^{-1}U$. Now for another $[y] \in \mathbb{RP}^n$, we have

$$|f[x] - f[y]|^2 = 2 - 2(x^\top y)^2.$$

Potentially by changing signs, we can assume $x^\top y > 0$ and so,

$$d(f[x], f[y])^2 = 2 - 2(x^\top y)^2 = 2(1 - x^\top y)(1 + x^\top y) \geq 2(1 - x^\top y) = d(x, y)^2.$$

Thus if $d(f[x], f[y]) < \varepsilon$ then $d(x, y) < \varepsilon$ as well, meaning $y \in B_\varepsilon(x)$ so that $[y] \in U$. Thus $B_\varepsilon(f[x]) \subseteq fU$, so that fU is open.

Thus f is an injective continuous open map onto its image, meaning it is a topological embedding. Since it is also a smooth immersion, it is by definition a smooth embedding.

(2) f forms a homeomorphism $\mathbb{RP}^n \cong \text{im} f$, which we showed is

$$\text{im} f = \{A \in \text{Mat}_{n+1}(\mathbb{R}) \mid A^2 = A, A^\top = A, \text{tr}(A) = 1\}.$$

This is a bounded closed subset of Euclidean space, and thus compact. It is bounded since the norm of A is precisely its trace, which is 1. And it is closed since we can write it as the intersection of closed sets

$$\text{im} f = g_1^{-1}\{0\} \cap g_2^{-1}\{0\} \cap \text{tr}^{-1}\{1\},$$

where $g_1(A) = A^2 - A$ is continuous, and $g_2(A) = A^\top - A$ is as well, as is the trace. Thus their preimages of closed sets are closed, so $\text{im} f$ is. So $\mathbb{RP}^n$ is homeomorphic to a compact space, and is thus compact.

2.0.4 Problem

Let M be a smooth manifold and $p \in M$. Let $T_p M$ denote the tangent space of M at p ; $D_p M$ the set of **derivations** at p on $C^\infty(M)$, i.e. linear maps $\xi: C^\infty(M) \rightarrow \mathbb{R}$ satisfying $\xi(fg) = f(p)\xi(g) + g(p)\xi(f)$; and finally $\mathfrak{m}_p \subseteq C^\infty(M)$ the ideal of functions vanishing at p .

- (1) Show that $D_p M$ is a vector space, with pointwise addition and multiplication.
- (2) Let $\mathfrak{m}_p^2$ be the product of $\mathfrak{m}_p$ with itself; show that $\mathfrak{m}_p/\mathfrak{m}_p^2$ inherits the structure of a vector space from $C^\infty(M)$.

- (3) Let $\psi: M \rightarrow N$ be smooth such that $\psi(p) = q$. Let $(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$ be the dual space of $\mathfrak{m}_p/\mathfrak{m}_p^2$. Define

$$T_p\psi: (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee \rightarrow (\mathfrak{n}_q/\mathfrak{n}_q^2)^\vee$$

by

$$T_p\psi(\ell)([f]) = \ell([f \circ \psi]), \quad \ell \in (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee, \quad f \in \mathfrak{n}_p.$$

Show that $T_p\psi$ is well-defined and linear.

- (4) Show that if ψ maps a neighborhood of p diffeomorphically to a neighborhood of q , then $T_p\psi$ is an isomorphism.
- (5) Define the map $I: T_pM \rightarrow D_p$ by

$$I([\gamma])(f) = (f \circ \alpha)'(0).$$

Show that I is well-defined, linear, and injective.

- (6) Let $\xi \in D_pM$; show that $\xi|_{\mathfrak{m}_p}$ defines a linear map $\widehat{\xi}: \mathfrak{m}_p/\mathfrak{m}_p^2 \rightarrow \mathbb{R}$.
- (7) Define $J: D_pM \rightarrow (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$ by $J(\xi) = \widehat{\xi}$. Show that J is linear and injective.
- (8) Show that $\dim \mathfrak{m}_p/\mathfrak{m}_p^2 = \dim M$.
- (9) Show that I and J are isomorphisms.

- (1) If ξ, ξ' are derivations so are $\xi + \xi'$ and $c\xi$ for $c \in \mathbb{R}$. They are linear as the linear combinations of linear maps, and are derivations since

$$(\xi + \xi')(fg) = \xi(fg) + \xi'(fg) = f(p)\xi(g) + g(p)\xi(f) + f(p)\xi'(g) + g(p)\xi'(f) = f(p)(\xi + \xi')(g) + g(p)(\xi + \xi')(f),$$

and

$$(c\xi)(fg) = c\xi(fg) = c(f(p)\xi(g) + g(p)\xi(f)) = f(p)(c\xi)(g) + g(p)(c\xi)(f).$$

- (2) If A is an R -algebra, then its ideals are R -modules. Indeed, if $I \leq A$ is an ideal, it is closed under addition and for $r \in R$ and $x \in I$, $rx = (r1)x \in I$. In particular if $J \leq I \leq A$ are ideals of A , then they are R -modules and so I/J is an R -module as a quotient of R -modules. Here $R = \mathbb{R}$, $A = C^\infty(M)$, $I = \mathfrak{m}_p$, and $J = \mathfrak{m}_p^2$.
- (3) To show that $T_p\psi$ is well-defined, we show that $T_p\psi(\ell): \mathfrak{n}_p \rightarrow \mathbb{R}$ given by $T_p\psi(\ell)(f) = \ell([f \circ \psi])$ is linear and $\mathfrak{n}_p^2$ is contained in its kernel. This map is well-defined: $f \circ \psi$ is smooth $M \rightarrow N \rightarrow \mathbb{R}$ and $f \circ \psi(p) = f(q) = 0$ so that $\ell[f \circ \psi]$ is defined. It is clearly linear, viewed as the composition of linear operators,

$$T_p\psi(\ell): \mathfrak{n}_q \xrightarrow{\bullet \circ \psi} \mathfrak{m}_p \xrightarrow{[\bullet]} \mathfrak{m}_p/\mathfrak{m}_p^2 \xrightarrow{\ell} \mathbb{R}.$$

If $f, g \in C^\infty(N)$ vanish at q then $fg \circ \psi = (f \circ \psi)(g \circ \psi) \in \mathfrak{m}_p^2$ so that $\ell[fg \circ \psi] = 0$, so that $T_p\psi(\ell)(fg) = 0$. Such products generate $\mathfrak{n}_q^2$ so that its contained in the kernel, so we can quotient $T_p\psi(\ell)$ by $\mathfrak{n}_q^2$, so the given function is well-defined.

- (4) We make two claims: if $f: U \rightarrow \mathbb{R}$ is a smooth map defined on an open subset of M , then for $A \subseteq U$ closed there is a smooth map $\tilde{f}: M \rightarrow \mathbb{R}$ whose restriction to A is equal to f , and support is contained in U . Indeed, $U, M - A$ is an open cover and thus has a partition of unity subordinate to it, φ_0, φ_1 . So define $\tilde{f}(x) = f(x)\varphi_0(x)$, taken to be zero when $x \notin U$. This is smooth on M , since $\text{supp } \varphi_0 \subseteq U$, and for $x \in A$ we have that $\varphi_1(x) = 0$ and so $\varphi_0(x) = 1$, meaning $\tilde{f}(x) = f(x)$. Finally, $\tilde{f}$'s support is contained in φ_0 's which is contained in U .

Secondly, if $f, g \in \mathfrak{m}_p$ agree in a neighborhood of p , then $[f] = [g]$. Equivalently if f is zero in a neighborhood of p , then $f \in \mathfrak{m}_p^2$. Let this neighborhood be U , and consider the open cover $U, M - p$ and a partition of unity subordinate to it φ_0, φ_1 . Then $f\varphi_1$ is equal to f : for $x \in U$ we have that they

are both zero, and for $x \notin U$ we have that $\psi_1(x) = 1$ since $\psi_0 + \psi_1 = 1$. Since $\psi_1(p) = 0$, we have that $f = f\psi_1 \in \mathfrak{m}_p^2$.

If ψ maps $p \in U$ diffeomorphically to $q \in V$, then suppose $T_p\psi(\ell) = 0$, we want to show that $\ell = 0$. Let $\tilde{\psi} = \psi|_U: U \rightarrow V$ be the diffeomorphism, and let $g \in \mathfrak{m}_p$. Then $g \circ \tilde{\psi}^{-1}: V \rightarrow \mathbb{R}$ is zero at q , and for $B \subseteq V$ open with $\bar{B} \subseteq V$, let $f: N \rightarrow \mathbb{R}$ be a smooth map where $f(x) = g \circ \tilde{\psi}^{-1}(x)$ for $x \in \bar{B}$ and whose support is contained in V , i.e. $f = g \circ \psi^{-1}$ in a neighborhood of q . Then $f \circ \psi = g$ in a neighborhood of p , and as we showed this means that $[f \circ \psi] = [g]$. So if $T_p\psi(\ell) = 0$ then $\ell[g] = 0$ for all $[g] \in \mathfrak{m}_p/\mathfrak{m}_p^2$, meaning $\ell = 0$. Thus $T_p\psi$ is injective.

Similarly let $\psi^{-1}: V \rightarrow U$ be its inverse diffeomorphism. Then for $B \subseteq V$ open with $\bar{B} \subseteq V$, we can extend ψ^{-1} to $\tilde{\psi}^{-1}: N \rightarrow M$ restricting to ψ^{-1} on B . Now,

$$T_p\psi \circ T_q\tilde{\psi}^{-1}(\ell')[f] = T_q\tilde{\psi}^{-1}(\ell')[f \circ \psi] = \ell'[f \circ \psi \circ \tilde{\psi}^{-1}],$$

and since $f \circ \psi \circ \tilde{\psi}^{-1} = f$ in a neighborhood of q , this is equal to $\ell'[f]$. Thus $T_p\psi \circ T_q\tilde{\psi}^{-1} = \text{id}$ and $T_p\psi$ is injective, thus $T_p\psi$ is an isomorphism, as desired.

- (5) By definition $[\gamma] = [\gamma']$ if $(f \circ \gamma)'(0) = (f \circ \gamma')'(0)$ for all $f \in C^\infty(M)$, so that I is well-defined as a map. And its image is in D_pM ; i.e. $I[\gamma]$ is a derivation. It is clearly linear by linearity of the derivative. And by the product rule,

$$I[\gamma](fg) = ((fg) \circ \gamma)'(0) = ((f \circ \gamma)(g \circ \gamma))'(0) = f(p)(g \circ \gamma)'(0) + g(p)(f \circ \gamma)'(0) = f(p)I[\gamma](g) + g(p)I[\gamma](f),$$

so that $I[\gamma]$ maps tangent vectors to derivations.

It is also clearly injective: if $I[\gamma] = I[\gamma']$ then $(f \circ \gamma)'(0) = (f \circ \gamma')'(0)$ for all $f \in C^\infty(M)$, which by definition means $[\gamma] = [\gamma']$.

Let (U, φ) be a chart centered at p , so that we have an isomorphism $\mathbb{R}^n \rightarrow T_pM$ mapping $v \mapsto \varphi_v$ which maps t to $\varphi(tv)$. To show that I is linear, it is sufficient to show that precomposing with this isomorphism is linear. This maps $v \in \mathbb{R}^n$ to

$$I[\varphi_v](f) = (f \circ \varphi_v)'(0) = (f \circ \varphi \circ (tv))'(0) = \nabla(f \circ \gamma)|_0 \cdot v,$$

where ∇ is the gradient. This is clearly linear by linearity of inner products.

- (6) For $\xi \in D_pM$, it is linear and so its restriction to $\mathfrak{m}_p$ is as well. If $f, g \in \mathfrak{m}_p$ then since $f(p) = g(p) = 0$,

$$\xi(fg) = f(p)\xi(g) + g(p)\xi(f) = 0.$$

So $\ker \xi \subseteq \mathfrak{m}_p^2$ (as elements of the form fg generate it), and thus $\hat{\xi} \in (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$ defined by $\hat{\xi}[f] = \xi(f)$ is well-defined.

- (7) This is clearly linear as restriction of a function is linear. If $J(\xi) = 0$ then $\xi|_{\mathfrak{m}_p} = 0$, which in turn means $\xi = 0$. Indeed, for $f \in C^\infty(M)$ we have $f - f(p) \in \mathfrak{m}_p$ and so $0 = \xi(f - f(p)) = \xi(f) - \xi(f(p))$. But derivations of constants are zero: by linearity it is sufficient to prove this for 1, and

$$\xi(1) = \xi(1^2) = \xi(1) + \xi(1) \implies \xi(1) = 0.$$

Thus $\xi(f) = 0$, proving that J is linear.

- (8) Let (U, φ) be a chart centered at p , so that φ^{-1} is a diffeomorphism from a neighborhood of p to $U \subseteq \mathbb{R}^n$. As we showed, this means that for $\mathbf{n}_0 \leq C^\infty(U)$, we have that

$$(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee \cong (\mathbf{n}_0/\mathbf{n}_0^2)^\vee.$$

So if we let $n = \dim M$, it is sufficient to show that $\dim(\mathbf{n}_0/\mathbf{n}_0^2) = n$. Let us define $D: \mathbb{R}^n \rightarrow (\mathbf{n}_0/\mathbf{n}_0^2)^\vee$ by mapping $v \in \mathbb{R}^n$ to the directional derivative evaluated at 0: $D_v|_0$. $D_v|_0$ is a derivation (by the product rule), and as above we showed it descends to an element of $(\mathbf{n}_0/\mathbf{n}_0^2)^\vee$.

This is linear, since we can write $D_v|_0 = v \cdot \nabla \bullet$ (where ∇ is the gradient), which is clearly linear in v by linearity of the dot product. This map is injective, since if we denote by x_i the i th projection we have that $x_i \in \mathbf{n}_0$ and $D_v|_0(x_i) = v_i$. So if $D_v|_0 = D_u|_0$ then $v_i = u_i$ for all i .

Conversely it is surjective: if $\xi \in (\mathbf{n}_0/\mathbf{n}_0^2)^\vee$ then for $f: U \rightarrow \mathbb{R}$ we have by Taylor,

$$f(x) = f(0) + \sum_i x_i \frac{\partial f}{\partial x_i}(0) + \sum_{i,j} x_i x_j \int_0^1 (1-t) \frac{\partial^2 f}{\partial x_i \partial x_j}(tz) dt,$$

and since the right-most sum is a sum of elements in $\mathfrak{n}_0^2$ (the integral is smooth, so these are x_i and x_j times the integral), we have that if $f \in \mathfrak{n}_0$, $f(0) = 0$ and so

$$[f(x)] = \sum_i \frac{\partial f}{\partial x_i}(0)[x_i].$$

Thus

$$\xi[f] = \sum_i \frac{\partial f}{\partial x_i}(0) \cdot \xi[x_i].$$

So define $v_i = \xi[x_i]$ and we have that $D_v|_0[x_i] = v_i = \xi[x_i]$ and thus

$$D_v|_0[f] = \sum_i \frac{\partial f}{\partial x_i}(0) D_v|_0[x_i] = \sum_i \frac{\partial f}{\partial x_i}(0) \xi[x_i] = \xi[f],$$

so that $D_v|_0 = \xi$, meaning this map is surjective.

So we have an isomorphism $\mathbb{R}^n \cong (\mathfrak{n}_0/\mathfrak{n}_0^2)^\vee \cong (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$.

- (9) Putting this all together, we have a sequence of injective morphisms

$$\mathbb{R}^n \cong T_p M \xrightarrow{I} D_p M \xrightarrow{J} (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee \cong \mathbb{R}^n.$$

So $J \circ I$ is an injective map between two linear spaces of the same dimension, so it must be an isomorphism. Since J is injective, this means it must be an isomorphism, and thus so too must I be.

3 Homework 3

3.0.1 Problem

Let M be a set and $\mathcal{A} = \{(\mathbf{x}_\alpha, U_\alpha)\}_\alpha$ an atlas on M . For α, β define the map

$$\mathbf{x}_{\alpha\beta}: \mathbf{x}_\alpha^{-1}(\mathbf{x}_\beta(U_\beta)) \rightarrow U_\alpha \times U_\beta, \quad x_{\alpha\beta}(u) = (u, \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(u)).$$

- (1) Show that if $\mathbf{x}_{\alpha\beta}$ is always proper, then the induced topology on M is Hausdorff.
- (2) Show that for $\mathcal{BK} = \mathbb{R}, \mathbb{C}, \mathbb{K}\mathbb{P}^n$ is Hausdorff.

- (1) We can view M as the quotient manifold of $\coprod_\alpha U_\alpha$ under the equivalence that $p \in U_\alpha$ and $q \in U_\beta$ are identified when $\mathbf{x}_\alpha(p) = \mathbf{x}_\beta(q)$. If we are given two distinct points $p \in U_\alpha, q \in U_\beta$ then (p, q) cannot be in the image of $\mathbf{x}_{\alpha\beta}$ as that would imply that $\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(p) = q$, i.e. that $\mathbf{x}_\beta(q) = \mathbf{x}_\alpha(p)$, a contradiction.

$\mathbf{x}_{\alpha\beta}$ is proper and thus closed (since all spaces in question are locally compact Hausdorff), and so its image is closed in $U_\alpha \times U_\beta$. Thus there exist open sets $p \in V_\alpha \subseteq U_\alpha$ and $q \in V_\beta \subseteq U_\beta$ which don't intersect $\mathbf{x}_{\alpha\beta}$'s image. These are open in M , and they must be disjoint: if $r \in V_\alpha, r' \in V_\beta$ are such that $\mathbf{x}_\alpha(r) = \mathbf{x}_\beta(r')$ then r is in $\mathbf{x}_{\alpha\beta}$'s domain and

$$\mathbf{x}_{\alpha\beta}(r) = (r, \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha(r)) = (r, r') \in V_\alpha \times V_\beta,$$

a contradiction to $V_\alpha \times V_\beta$ not intersecting $\mathbf{x}_{\alpha\beta}$'s image.

- (2) The charts of $\mathbb{K}\mathbb{P}^n$ are given by $\mathbf{x}_i: \mathbb{K}^n \rightarrow \mathbb{K}\mathbb{P}^n$ where $\mathbf{x}_i(v)$ inserts 1 at the i th index of v (and takes the equivalence). Its image is the set of classes of the form $[v]$ where $v_i \neq 0$, let this be U_i . Its inverse is $\mathbf{x}_i^{-1}: U_i \rightarrow \mathbb{K}^n$ given by $\mathbf{x}_i^{-1}([v]) = 1/v^i \hat{v}^i$ where $\hat{v}^i$ denotes the removal of the i th index.

The maps $\mathbf{x}_{ij}$ then are

$$\mathbf{x}_{ij}: \mathbf{x}_i^{-1}(U_j) \rightarrow U_i \times U_j, \quad v \mapsto (v, \mathbf{x}_j^{-1} \circ \mathbf{x}_i(v)).$$

$\mathbf{x}_j^{-1} \circ \mathbf{x}_i(v)$ is equal to $1/v^j(v_1, \dots, \widehat{v}_j, \dots, v_{i-1}, 1, v_i, \dots, v_n)$ when $j < i$ (for $j > i$ the indices are ordered a bit differently; $\widehat{v}_j$ denotes the removal of v_j). Now, $\mathbf{x}_i^{-1}(U_j)$ is the set of all $v \in \mathbb{K}^n$ such that $\mathbf{x}_i(v)$'s j th index is nonzero, this is just the set of vectors in $\mathbb{K}^n$ whose j th index is nonzero. So $\mathbf{x}_{ij}$'s image is all (v, u) such that $\mathbf{x}_i(v) = \mathbf{x}_j(u)$. This occurs iff $(v_1, \dots, 1, \dots, v_n)$ and $(u_1, \dots, 1, \dots, u_n)$ are linearly dependent: $\lambda(v_1, \dots, 1, \dots, v_n) = (u_1, \dots, 1, \dots, u_n)$ for some λ .

Let us denote insertion of 1 at index i of y_i , so $\mathbf{x}_{ij}$'s image is

$$\text{im} \mathbf{x}_{ij} = \{(v, u) \in U_i \times U_j \mid \lambda y_i(v) = y_j(u) \text{ for some } \lambda \in \mathbb{K}\}.$$

If $i < j$ then λ must be u_i so that the image is

$$\{(v, u) \mid u_i y_i(v) = y_j(u)\}$$

which is the preimage of zero of the continuous map $(v, u) \mapsto u_i y_i(v) - y_j(u)$. Thus the image is closed. Similar for $i > j$, and when $i = j$ the image is just the diagonal which is again closed.

We showed above that $\mathbf{x}_{\alpha\beta}$'s image being closed is enough to ensure the manifold being Hausdorff.

3.0.2 Problem

Let $\text{Mat}_n(\mathbb{C})$ denote the space of $n \times n$ complex matrices. Denote by $M^\dagger$ the conjugate transpose of $M \in \text{Mat}_n(\mathbb{C})$. A matrix M is **unitary** if $M^\dagger M = I$, and the group of unitary matrices is denoted U_n . A matrix M is **skew-Hermitian** if $M^\dagger = -M$, and denote by $\mathfrak{u}_n$ the space of skew-Hermitian matrices. We will show that U_n is a Lie group and we can identify its Lie algebra with $\mathfrak{u}_n$.

- (1) Let $\text{Herm}_n \subseteq \text{Mat}_n(\mathbb{C})$ be the space of Hermitian matrices $M = M^\dagger$. Use the obvious maps $\mathbb{C}^{n^2} \rightarrow \text{Mat}_n(\mathbb{C})$ and $\mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n \rightarrow \text{Herm}_n$ to define smooth structures. Define $F: \text{Mat}_n(\mathbb{C}) \rightarrow \text{Herm}_n$ by $F(M) = M^\dagger M$. Show that the identity I is a regular point of F .
- (2) Use the invariance of F under left multiplication by unitary matrices to show that all $U_n \subseteq \text{Mat}_n(\mathbb{C})$ are regular points of F .
- (3) Conclude that $I \in \text{Herm}_n$ is a regular value of F , and thus U_n is an embedded submanifold of $\text{Mat}_n(\mathbb{C})$ of dimension n^2 .
- (4) Let $P \subseteq M$ and $Q \subseteq N$ be embedded submanifolds, and $F: M \rightarrow N$ smooth with $F(P) \subseteq Q$. Show that $F|_P: P \rightarrow Q$ is smooth.
- (5) Show that multiplication $U_n \times U_n \rightarrow U_n$ and inversion $U_n \rightarrow U_n$ are smooth, and thus U_n is a Lie group.
- (6) Let $\iota: U_n \rightarrow \mathbb{C}^{n^2}$ be the inclusion map. Show that the composition

$$T_e U_n \xrightarrow{d\iota_e} T_e \mathbb{C}^{n^2} \xrightarrow{\zeta} \mathbb{C}^{n^2}$$

induces an isomorphism of vector spaces between $T_e U_n$ and $\mathfrak{u}_n$.

- (1) The map $\mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n \rightarrow \text{Herm}_n$ is given by mapping $(z_i) \oplus (x_i)$ to the Hermitian matrix whose diagonal is given by the x_i s, above the diagonal by z_i s, and below by $\bar{z}_i$ s. This is a real linear isomorphism, in particular a bijection, and thus induces a smooth structure. Let $\Phi: \mathbb{C}^{n^2} \rightarrow \text{Mat}_n(\mathbb{C})$ be the obvious isomorphism and $\Psi: \text{Herm}_n \rightarrow \mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n$ the inverse of the above bijection.

Then the coordinate representation of F maps $M \in \mathbb{C}^{n^2}$ to the upper triangle of $M^\dagger M$ and its diagonal. We have that

$$(M^\dagger M)_{ij} = \sum_t \bar{M}_{ti} M_{tj},$$

and each of these coordinates is smooth over $\mathbb{R}$ (complex multiplication is smooth, and so is conjugation as it just the map $(x, y) \mapsto (x, -y)$). Thus all of F 's coordinates are smooth, and so F is.

- (2) For a unitary matrix U , we have that

$$F(UM) = M^\dagger U^\dagger U M = M^\dagger M = F(M),$$

and since the action of matrix multiplication in $\text{Mat}_n(\mathbb{C})$ is a smooth action by $\text{GL}_n(\mathbb{C})$, we have that left multiplication by $U \in U_n \subseteq \text{GL}_n(\mathbb{C})$ is a diffeomorphism of $\text{Mat}_n(\mathbb{C})$. Thus (denoting left multiplication under U by U), we have that for all $M \in \text{Mat}_n(\mathbb{C})$,

$$d(F \circ U)_M = dF_{UM} \circ dU_M = dF_M.$$

Since U is a diffeomorphism and thus of full rank, dF_{UM} and dF_M must have the same rank for all $U \in U_n, M \in \text{Mat}_n(\mathbb{C})$. In particular if U, Q are unitary then so is UQ^{-1} , and thus $dF_{UQ^{-1}Q} = dF_U$ and dF_Q have the same rank.

In particular, I is unitary and so for $U \in U_n$, dF_U and dF_I have the same rank, which we showed is full. Thus every $U \in U_n$ is a regular point of F .

- (3) $I \in \text{Herm}_n$'s level set consists of all unitary matrices, which are all regular points. Thus I is a regular value, and its level set U_n is an embedded submanifold of $\text{Mat}_n(\mathbb{C})$. F 's rank at $U \in U_n$ is full, thus equal to $\dim \mathbb{C}^{n(n-1)/2} \oplus \mathbb{R}^n = n(n-1) + n = n^2$. This means U_n 's codimension is n^2 , and thus its dimension is $2n^2 - n^2 = n^2$.
- (4) Since $P \subseteq M$ is a submanifold, the inclusion $\iota_P: P \hookrightarrow M$ is smooth. $F|_P: P \rightarrow M$ is just $F \circ \iota_P$ and thus is smooth.

Now we want to show that we can restrict the codomain. Let F' denote the restriction on the codomain $F' = F|_P: P \rightarrow Q$; we note that it is still continuous. Since $\iota_Q: Q \hookrightarrow N$ is a smooth embedding, and $\iota_Q \circ F' = F$. Since ι_Q is in particular a smooth immersion and thus of constant rank, at every point of Q there is a chart (U, ψ) and a chart $(V, \tilde{\psi})$ of N such that $U \subseteq V$, and the coordinate representation is of the form

$$\tilde{\psi}^{-1} \circ \iota_Q \circ \psi: \bar{x} \mapsto (\bar{x}, 0).$$

If we denote by π the projection $(\bar{x}, \bar{y}) \mapsto \bar{x}$ we see that

$$\pi \circ \tilde{\psi}^{-1} \circ \iota_Q \circ \psi = \text{id} \implies \psi^{-1} = \pi \circ \tilde{\psi}^{-1} \circ \iota_Q.$$

Then for $p \in P$ we can choose coordinate charts $\psi \subseteq Q$ and $\tilde{\psi} \subseteq N$ such that $F(p)$ is in ψ (and thus $\tilde{\psi}$'s domain). We then choose a coordinate chart $\varphi \subseteq P$ around p which is mapped by F' into ψ 's domain (since F' is continuous, its preimage of ψ 's domain is open). Then F' 's coordinate representation is

$$\psi^{-1} \circ F' \circ \varphi = \pi \circ \tilde{\psi}^{-1} \circ \iota_Q \circ F' \circ \varphi = \pi \circ (\tilde{\psi}^{-1} \circ F \circ \varphi).$$

Since F is smooth, $\tilde{\psi}^{-1} \circ F \circ \varphi$ is smooth, and the projection π is smooth as well. Thus $\psi^{-1} \circ F' \circ \varphi$ is smooth, meaning F' has a smooth coordinate representation in some neighborhood of any point $p \in P$, so F' is smooth as desired.

- (5) $\text{GL}_n(\mathbb{C}) \subseteq \text{Mat}_n(\mathbb{C})$ and $U_n \subseteq \text{Mat}_n(\mathbb{C})$ are embedded submanifolds. This means that U_n is an embedded submanifold of $\text{GL}_n(\mathbb{C})$: its inclusion in $\text{Mat}_n(\mathbb{C})$ is the composition of inclusions $U_n \hookrightarrow \text{GL}_n(\mathbb{C}) \hookrightarrow \text{Mat}_n(\mathbb{C})$. This composition is a smooth immersion, and thus the first inclusion $U_n \hookrightarrow \text{GL}_n(\mathbb{C})$ must be a smooth immersion as well; thus $U_n \subseteq \text{GL}_n(\mathbb{C})$ is embedded. Thus we can restrict multiplication and inversion of $\text{GL}_n(\mathbb{C})$ to give the multiplication and inversion of U_n , and these remain smooth.
- (6) The dimension of $\mathfrak{u}_n$ is n^2 , as a skew-Hermitian matrix is determined by the imaginary values on its diagonal, and its complex upper triangle. That is, there is an obvious isomorphism between it and $\mathbb{C}^{n(n-1)/2} \oplus (i\mathbb{R})^n$, which has dimension n^2 . Thus $T_e U_n$ and $\mathfrak{u}_n \subseteq \mathbb{C}^{n^2}$ have the same dimension. Since $\iota: U_n \rightarrow \text{Mat}_n(\mathbb{C})$ is a smooth immersion, $d\iota_e$ is injective. Thus we need only show that the composition lands in $\mathfrak{u}_n$, as then we have an injective map $T_e U_n \rightarrow \mathfrak{u}_n$ and since they are of the same dimension, it must be an isomorphism.

Let $\gamma: (-\varepsilon, \varepsilon) \rightarrow U_n$ be smooth and centered at I , then $\iota \circ \gamma$ is a smooth curve which lands in $U_n \subseteq \text{Mat}_n(\mathbb{C})$. Identify γ with $\iota \circ \gamma$; we want to show that if $\gamma(t) \in U_n$ for all t then $\gamma'(0) \in \mathfrak{u}_n$. Notice that the Hermitian transpose is smooth (over $\mathbb{R}$) whose differential is itself. Since $\gamma(t)\gamma(t)^\dagger = I$, differentiating both sides at zero gives

$$0 = \gamma(0)\gamma'(0)^\dagger + \gamma'(0)\gamma(0)^\dagger = \gamma'(0)^\dagger + \gamma'(0),$$

meaning $\gamma'(0)$ is indeed skew-Hermitian, as desired.

3.0.3 Problem

For $M = \mathbb{R}^n$, let $r: M \rightarrow M$ be a smooth retraction, meaning $r \circ r = r$.

- (1) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be smooth with a differential of constant rank k . Show that there is a coordinate frame $0 \in (U, \mathbf{x})$ on $\mathbb{R}^n$ such that

$$f \circ \mathbf{x}(x_1, \dots, x_n) = f \circ \mathbf{x}(x_1, \dots, x_k, 0, \dots, 0).$$

- (2) Let $T \subseteq \mathbb{R}^n$ be connected and for $t \in T$ define $A(t): \mathbb{R}^n \rightarrow \mathbb{R}^n$ so that $t \mapsto A(t)$ is continuous. Further suppose that $A(t)$ is a projection for all t , $A(t)^2 = A(t)$. Show that the rank of the matrix $A(t)$ must be constant.

- (3) Show that there is a neighborhood Q of $\text{im} r$ such that dr_q has constant rank for $q \in Q$.

- (4) Show that for each $q \in \text{im} r$ there is a coordinate system $(U, \mathbf{x})$ on M centered at q such that $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$ and

$$r \circ \mathbf{x}(x_1, \dots, x_n) = r \circ \mathbf{x}(x_1, \dots, x_k, 0, \dots, 0).$$

- (5) Show that $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U)) \cap \mathbf{x}(U)$

- (6) Show that $r(M)$ is an embedded submanifold.

- (1) This proof is essentially the same as that of the first part of the rank theorem. Let us write $F(x, y) = (Q(x, y), R(x, y))$ for $Q: \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $R: \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$. Then F 's Jacobian is

$$JF = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j} & \frac{\partial Q^i}{\partial y^j} \\ \frac{\partial R^i}{\partial x^j} & \frac{\partial R^i}{\partial y^j} \end{pmatrix}.$$

Since it has rank k , it has a $k \times k$ invertible submatrix, which we can assume without loss of generality is the upper left. We can assume this via a change of coordinates, and noting that by continuity if a submatrix has nonzero determinant, it has nonzero determinant in an open neighborhood of the point.

Let us then define $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\varphi(x, y) = (Q(x, y), y)$. Its Jacobian is

$$J\varphi = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j} & \frac{\partial Q^i}{\partial y^j} \\ 0 & \delta_{ij} \end{pmatrix},$$

which is invertible (in the appropriate neighborhood). Thus by the inverse function theorem has an inverse $\varphi^{-1}(x, y) = (\tilde{Q}(x, y), \tilde{R}(x, y))$, and we see then that comparing $\varphi \circ \varphi^{-1}$, $\varphi^{-1} \circ \varphi$ with the identity, we have $\tilde{R}(x, y) = y$ and that $Q(\tilde{Q}(x, y), y) = x$ and $\tilde{Q}(Q(x, y), y) = x$.

Now consider $F \circ \varphi^{-1}$, which is equal to

$$F \circ \varphi^{-1}(x, y) = (x, R(\tilde{Q}(x, y), y)) = (x, A(x, y))$$

and has Jacobian

$$J(F \circ \varphi^{-1})(x, y) = \begin{pmatrix} \delta_{ij} & 0 \\ \frac{\partial A^i}{\partial x^j} & \frac{\partial A^i}{\partial y^j} \end{pmatrix}.$$

φ^{-1} is a diffeomorphism and thus the rank of this Jacobian is still k . Since the first k columns are linearly independent, this means that the remaining $n - k$ must be 0. That is, each A^i is independent of y , since its partial derivatives with respect to each y^j is zero (in other words, $A(x, \bullet)$ is constant). So $A(x, y) = A(x, 0)$ and thus

$$F \circ \varphi^{-1}(x, y) = (x, A(x, y)) = (x, A(x, 0)) = F \circ \varphi^{-1}(x, 0),$$

so φ^{-1} defines the desired chart.

- (2) Notice that for a projection P , its rank is just its trace. This is because if it projects onto V parallel to W , taking bases of V and W , P 's representation relative to the union of this base is $\text{diag}(1, \dots, 1, 0, \dots, 0)$, whose trace is its rank. Trace is a continuous function $\text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R}$, and thus the rank of $A(t)$ depends continuously on t . The set of ranks of $A(t)$ is a set of integers, which is disconnected when not a singleton. The set of ranks is the continuous image of $T \rightarrow A(t) \rightarrow \text{tr}A(t)$ and since T is connected, so must the set of traces be, thus it is a singleton, as desired.

- (3) For $q \in \text{im}r$, we have that since $r \circ r = r$,

$$dr_q = d(r \circ r)_q = dr_{rq} \circ dr_q = dr_q \circ dr_q.$$

Thus dr_q is a projection, and the map $q \rightarrow dr_q$ is continuous so that by above dr_q 's rank is constant (since $\text{im}r$ is the continuous image of a connected space). Let this constant rank be k , then note it is also r 's maximum rank: for all $q \in M$ we have that $dr_q = dr_{rq} \circ dr_q$, and since composition does not increase rank, dr_q 's rank is bound by $dr_{rq} = k$.

Let Q be the set of all points $q \in M$ where dr_q 's rank is equal to k . This contains $\text{im}r$ as already shown, but it is also open. Indeed, for $q \in Q$, dr_q has rank k and in a suitable neighborhood of q the rank of r remains k .

This is because we can look at dr 's matrix representation relative to open charts around q and rq (i.e. the Jacobian of r 's coordinate representation). Since dr_q has rank k , the matrix has an invertible $k \times k$ submatrix. Consider then the function which maps a point p (which lies in the chart around q) to the determinant of this $k \times k$ submatrix of dr_p . If a point p is mapped to a nonzero value, then this $k \times k$ submatrix of dr_p is invertible thus dr has rank $\geq k$ at p ; since k is the maximal rank, dr_p 's rank is equal to k . So the preimage of the open set $\mathbb{R} - 0$ is a neighborhood of p contained in Q , so Q is open.

- (4) Our first point only required that F have constant rank in a neighborhood, which is true here (the neighborhood is Q). Thus there is $(\mathbf{x}, U)$ with the desired property, but not necessarily $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$. We want $r(\mathbf{x}(u)) \in \mathbf{x}(U)$ for all $u \in U$, i.e. that $u \in (r \circ \mathbf{x})^{-1}(\mathbf{x}(U))$, which is open. Thus intersecting $(r \circ \mathbf{x})^{-1}(\mathbf{x}(U))$ with U and restricting $\mathbf{x}$ gives us the property. Indeed,

$$r(\mathbf{x}((r \circ \mathbf{x})^{-1}(\mathbf{x}(U)) \cap U)) \subseteq r(\mathbf{x}(U)) \cap \mathbf{x}(U) = \mathbf{x}((r \circ \mathbf{x})^{-1}(\mathbf{x}(U)) \cap U).$$

- (5) Clearly we have that $r(M) \cap \mathbf{x}(U) \subseteq r(\mathbf{x}(U)) \cap \mathbf{x}(U)$. Conversely if $r(p) \in \mathbf{x}(U)$ then $r(p) = r(r(p)) \in r(\mathbf{x}(U))$, as desired. Since $r(\mathbf{x}(U)) \subseteq \mathbf{x}(U)$, we have that $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U))$ even.
- (6) Consider a chart $(U, \mathbf{x})$ as above, then we can make $r(M) \cap \mathbf{x}(U) = r(\mathbf{x}(U))$ an embedded k -manifold. Indeed, consider the projection $\pi: \mathbb{R}^n \rightarrow \mathbb{R}^k$ and inclusion by concatenating zeroes, $\iota: \mathbb{R}^k \rightarrow \mathbb{R}^n$. So we have $r \circ \mathbf{x} = r \circ \mathbf{x} \circ \iota \circ \pi$ by assumption. Define the chart (U', y) with $U' = \pi(U) \subseteq \mathbb{R}^k$ by $y = r \circ \mathbf{x} \circ \iota$, i.e. $y(x) = r \circ \mathbf{x}(x, 0)$. This is surjective by construction. Now, we know that from our first point, $r \circ \mathbf{x}(x, y) = (x, R(x))$ for some function $R: \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$. Thus y has the form $x \mapsto (x, R(x))$ and is thus clearly injective. Its inverse is $(x, y) \mapsto x$, which is clearly smooth thus y is a diffeomorphism.

It remains to show that $y(U')$ is open in $r(M)$. Notice that since $r \circ \mathbf{x}(x, 0) = r \circ \mathbf{x}(x, y)$,

$$y(U') = \{r \circ \mathbf{x}(x, 0) \mid x \in U'\} = \{r \circ \mathbf{x}(x, y) \mid (x, y) \in U\} = r(\mathbf{x}(U)) = r(M) \cap \mathbf{x}(U),$$

which is open in the subspace topology. And for y_1 and y_2 corresponding to $\mathbf{x}_1, \mathbf{x}_2$, we note that $y_1^{-1} \circ y_2$ is just the identity. As we said, y has the form $x \mapsto (x, R(x))$ with inverse $(x, y) \mapsto x$, so $y_1^{-1} \circ y_2$ maps x to x , so this does define a smooth atlas on $r(M)$.

To show that $r(M) \subseteq M$ is a submanifold, we need to show that the inclusion is an immersion. Let $\iota: r(M) \hookrightarrow M$ be the inclusion, then relative to $(U, \mathbf{x})$ and (U', y) (since $y(U') \subseteq \mathbf{x}(U)$) it is equal to

$$\mathbf{x}^{-1} \circ \iota \circ y(x) = \mathbf{x}^{-1} \circ r \circ \mathbf{x}(x, 0),$$

which is smooth and has constant rank k since r is and does, and U' 's dimension is k so this is indeed a smooth immersion. Since $r(M)$ has the subspace topology, the inclusion is a smooth embedding, as desired.

3.0.4 Problem

Consider the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$. Let $A: S^n \rightarrow S^n$ be the antipodal map $p \mapsto -p$. Using any oriented atlas of S^n , show that A is orientation preserving iff n is odd.

Consider the stereographic projections at e_1 and $-e_1$, denoted by φ_+ and φ_- , $S^n - \{\pm e_1\} \rightarrow \mathbb{R}^n$. These map

$$\varphi_+(x_0, \dots, x_n) = \frac{1}{1-x_0}(x_1, \dots, x_n), \quad \varphi_-(x_0, \dots, x_n) = \frac{1}{1+x_0}(x_1, \dots, x_n).$$

These are restrictions of the obviously smooth maps $\{x \in \mathbb{R}^{n+1} \mid x_0 \neq \pm 1\} \rightarrow \mathbb{R}^n$, and since S^n is a submanifold, the restrictions are smooth as well. For $(X_1, \dots, X_n) \in \mathbb{R}^n$, let $N^2 = |\bar{X}|^2 = \sum_i X_i^2$, then their inverses are given by

$$\varphi_+^{-1}(X_1, \dots, X_n) = \frac{1}{N^2+1}(N^2-1, 2X_1, \dots, 2X_n), \quad \varphi_-^{-1}(X_1, \dots, X_n) = \frac{1}{N^2+1}(1-N^2, 2X_1, \dots, 2X_n).$$

These again are smooth, as the restrictions of the obviously smooth maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$, and again we can restrict the codomain to S^n since it is a submanifold.

The transition map is

$$\varphi_- \circ \varphi_+^{-1}(X) = \frac{X}{|X|^2},$$

and its Jacobian can be computed:

$$J_{ij} = \begin{cases} -\frac{2X_i X_j}{|X|^4} & i \neq j \\ \frac{|X|^2 - 2X_i^2}{|X|^4} & i = j \end{cases}.$$

This has negative determinant: for $X = (1, 0, \dots, 0)$ it is $\text{diag}(-1, 1, \dots, 1)$.

In lecture we showed that this means we can replace φ_- with $r \circ \varphi_-$ where r is the map which flips the sign of the first coordinate, and obtain an oriented atlas. For the antipodal map A , notice that $\varphi_- = A \circ \varphi_+ \circ A$, and thus A 's coordinate representation is

$$r \circ \varphi_- \circ A \circ \varphi_+^{-1} = r \circ A \circ \varphi_+ \circ \varphi_+^{-1} = r \circ A,$$

since $A^2 = \text{id}$. This is the map $X \mapsto (X_1, -X_2, \dots, -X_n)$, whose Jacobian is $\text{diag}(1, -1, \dots, -1)$ which has determinant $(-1)^{n-1}$. This is positive only when n is odd, as desired. For $\varphi_+ \circ A \circ \varphi_-^{-1}$ we get the same result by symmetry.

3.0.5 Problem

Let S^{2n+1} be the unit sphere in $\mathbb{C}^{n+1}$. Define an action of S^1 on S^{2n+1} by

$$z(w^1, \dots, w^{n+1}) = (zw^1, \dots, zw^{n+1}).$$

- (1) Show that this action is smooth.
- (2) Show that this action is free and proper, and that the orbit space S^{2n+1}/S^1 is diffeomorphic to $\mathbb{CP}^n$.

- (1) This action is just the restriction to $S^1 \times S^{2n+1} \rightarrow S^{2n+1}$ of the map $\mathbb{C} \times \mathbb{C}^{n+1} \rightarrow \mathbb{C}^{n+1}$ given by scalar multiplication:

$$z(w^1, \dots, w^{n+1}) = (zw^1, \dots, zw^{n+1}).$$

This is a smooth map, and $S^1 \subseteq \mathbb{C}$ and $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ are embedded submanifolds, thus the restriction of scalar multiplication to $S^1 \times S^{2n+1} \rightarrow S^{2n+1}$ remains smooth, by our above proof.

- (2) The action is free since if $z(w^1, \dots, w^{n+1}) = (w^1, \dots, w^{n+1})$ then since w is a point in S^{2n+1} it must have some nonzero coordinate, say w^i . Then $zw^i = w^i$ implies that $z = 1$.

The action is proper since S^1 is compact. Since if (z_i) is any sequence in S^1 , it has a convergent subsequence by virtue of S^1 being compact (without even considering a sequence (w_i) of S^{2n+1}).

Consider the quotient map $\pi: \mathbb{C}^{n+1} - 0 \rightarrow \mathbb{CP}^n$ mapping w to the linear subspace spanned by it, $\pi(w) = [w]$. Since $S^{2n+1} \subseteq \mathbb{C}^{n+1} - 0$ is an embedded submanifold, we can restrict this to $\pi: S^{2n+1} \rightarrow \mathbb{CP}^n$. Define S^1 to act on $\mathbb{CP}^n$ trivially: $z[v] = [v]$ for all $[v] \in \mathbb{CP}^n$. Then π is S^1 -equivariant, since $\pi(zw) = [zw] = [w] = z[w]$, and thus has constant rank. Furthermore, π is surjective (any vector can be normalized), and thus it must be a smooth submersion. This is because locally π has the coordinate representation $(x_1, \dots, x_n) \rightarrow (x_1, \dots, x_k, 0, \dots, 0)$ which is a surjection only when the rank k is full.

And also because π is S^1 -equivariant, it quotients to $\pi/S^1: S^{2n+1}/S^1 \rightarrow \mathbb{CP}^n$ given by $S^1 \cdot w \mapsto \pi(w)$. Notice that the following diagram commutes:

$$\begin{array}{ccc} & S^{2n+1}/S^1 & \\ \pi_1 \nearrow & & \searrow \pi/S^1 \\ S^{2n+1} & \xrightarrow{\pi} & \mathbb{CP}^n \end{array}$$

where π_1 is the covering map of the quotient manifold. Since $\pi = (\pi/S^1) \circ \pi_1$ is a submersion, we have that π/S^1 must be a submersion as well: for any $w \in S^{2n+1}$ we have that

$$d\pi_w = d(\pi/S^1)_{\pi_1(w)} \circ d(\pi_1)_w$$

is surjective, and thus so too must $d(\pi/S^1)_{\pi_1(w)}$. Since π_1 is surjective, this means π/S^1 is a submersion.

But π/S^1 is also bijective: as the quotient of the surjective π , it remains surjective. And if $\pi(w) = \pi(u)$ then $w = zu$ for some $z \in \mathbb{C}$, and since both w, u have norm 1 this means $z \in S^1$. Thus $S^1 \cdot w = S^1 \cdot u$ so that π/S^1 is injective. A bijective submersion is a diffeomorphism (it has the local representation $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_k)$ and since it is bijective we must have that $n = k$, so that it is a local diffeomorphism, and bijective local diffeomorphisms are global diffeomorphisms).

So we have shown that $\pi/S^1: S^{2n+1}/S^1 \rightarrow \mathbb{CP}^n$ is a diffeomorphism, as desired.

4 Homework 4

4.0.1 Problem

Let G be a Lie group, and H a closed embedded Lie subgroup. Define actions of G on itself by left and right multiplication:

$$L, R: G \times G \rightarrow G, \quad L(x, y) = xy, \quad R(x, y) = yx^{-1}.$$

- (1) Show that the action of H on G by R is smooth, proper, and free.
- (2) By the above point, we can form the quotient space G/H with respect to the action R ; let the quotient map be $\pi: G \rightarrow G/H$. Show that there is a unique G -action $\bar{L}$ on G/H such that the following diagram commutes

$$\begin{array}{ccc} G \times G & \xrightarrow{\quad L \quad} & G \\ \text{id} \times \pi \downarrow & & \downarrow \pi \\ G \times G/H & \xrightarrow{\quad \bar{L} \quad} & G/H \end{array}$$

- (1) First of all since H is embedded, we can restrict R to $H \times G \rightarrow G$, as shown in the previous homework. So the action is smooth. Clearly the action is free: if $R(x, y) = y$ then $yx^{-1} = y$ so $y = e$. To show that the action is proper, let $h_i \in H$ and $g_i \in G$ be sequences such that g_i and $R(h_i, g_i)$ both converge; we need to show that h_i has a convergent subsequence. Since g_i and $R(h_i, g_i) = g_i h_i^{-1}$ converge, so does

$R(h_i, g_i)^{-1} = h_i g_i^{-1}$ and thus $R(h_i, g_i)^{-1} g_i = h_i$ converges in G . Since H is closed, it contains all of its limit points, thus h_i converges in H , as desired.

- (2) The orbit of $x \in G$ by H under R is

$$H \cdot x = R(h, x)[h \in H] = \{xh^{-1} \mid h \in H\} = xH.$$

If $\bar{L}$ exists it is unique, as it is given by

$$\bar{L}(g, xH) = \pi \circ L(g, x) = gxH.$$

Note that this is well-defined: if $xH = yH$ then $gxH = gyH$. It is also a group action, since $\bar{L}(e, xH) = xH$ and $\bar{L}(g_1 g_2, xH) = g_1 g_2 xH = \bar{L}(g_1, \bar{L}(g_2, xH))$. We need only show that it is smooth.

Note that since id, π are surjective submersions, so is $\text{id} \times \pi$ (the rank of the product of two maps is the sum of their ranks). Thus $\bar{L}$ is smooth if and only if $\bar{L} \circ (\text{id} \times \pi)$ is. This is because if $M \xrightarrow{\pi} N$ is a smooth submersion, then $N \xrightarrow{E} P$ is smooth iff $F \circ \pi$ is smooth: one direction is clear, and the other follows by existence of local sections. If $F \circ \pi$ is smooth, let $p \in N$ and $q \in M$ such that $\pi(q) = p$. Then there is a local section $\sigma: U \rightarrow M$ with $q \in U$ and $\sigma(q) = p$, so

$$F|_U = F|_U \circ \text{id}_U = F|_U \circ (\pi \circ \sigma) = (F|_U \circ \pi) \circ \sigma,$$

which is smooth as the composition of smooth maps; so F is smooth around every point in N , and thus smooth.

Since $\bar{L} \circ (\text{id} \times \pi) = \pi \circ L$ is smooth, $\bar{L}$ is smooth as desired.

4.0.2 Problem

Let G be a Lie group acting transitively on M .

- (1) Let $f: N \rightarrow Q$ be a smooth surjective map of constant rank. Show that f is a submersion.
 - (2) For $p \in M$ show that $G_p \subseteq G$ is a closed embedded Lie subgroup. Conclude that G/G_p is a smooth manifold.
 - (3) Let $\pi: G \rightarrow G/G_p$ be the quotient map, by the first question G acts on G/G_p by left multiplication. Show there is a unique G -equivariant diffeomorphism $F: G/G_p \rightarrow M$ such that $F(\pi(e)) = p$.
 - (4) The unitary group $U(n)$ acts on $\mathbb{C}^n$ by matrix multiplication, and thus it acts on the Grassmanian $G_{\mathbb{C}}(k, n)$. Show that this action is smooth and transitive. Further show that the stabilizer of a point is the subgroup of block-diagonal matrices $U(k) \times U(n-k)$. Conclude that $G_{\mathbb{C}}(k, n)$ is diffeomorphic $U(n)/(U(k) \times U(n-k))$.
- (1) Suppose f has rank k , and let $n = \dim N, q = \dim Q$; suppose that f is not a submersion so $k < q$. Then locally f has the representation $\varphi^{-1} \circ f \circ \psi: (x_1, \dots, x_n) \mapsto (x_1, \dots, x_k, 0, \dots, 0)$, where φ, ψ are coordinate charts of Q, N respectively. This is a map between open subsets of Euclidean space, and since f is surjective and φ, ψ are diffeomorphisms, it must be surjective. But its image consists of vectors whose final $q-k > 0$ coordinates are zero, and this cannot be open in $\mathbb{R}^q$ (any open ball centered on one of these points has points whose last coordinates are nonzero), a contradiction. Thus f must have rank $k = q$, and is thus a submersion.
 - (2) Let $\theta: G \times M \rightarrow M$ be the group action, and define $\theta^p: G \rightarrow M$ by $\theta^p(g) = \theta(g, p) = gp$. Since G acts on M transitively, it is surjective. Furthermore it has constant rank since it is G -equivariant relative to G 's smooth action on itself by left multiplication:

$$\theta^p(g'g) = g'gp = g'\theta^p(g).$$

So by the previous point, it is a smooth submersion.

Thus its level sets are embedded submanifolds, in particular $G_p = (\theta^p)^{-1}(p)$ is. It is also a subgroup of G , and since it is an embedded submanifold we can restrict the domains and codomains of multiplication and inversion to show that multiplication and inversion on it are smooth (as per the previous homework

again). It is the preimage of a singleton under a continuous function, so it is closed as well, so is a closed embedded Lie subgroup. By the previous question, G/G_p exists.

- (3) If $F: G/G_p \rightarrow M$ is G -equivariant, it must satisfy

$$F(gG_p) = gF(G_p) = gF(\pi(e)) = gp$$

for all $gG_p \in G/G_p$; so F is unique if it exists. It is surjective since G acts transitively on M , and injective since if $gp = g'p$ then $g^{-1}g'p = p$ so $g^{-1}g' \in G_p$ and thus $gG_p = g'G_p$. Now we need to show that it is smooth.

As we showed in the previous question, F is smooth iff $F \circ \pi$ is, and $F \circ \pi$ is the map $g \mapsto gp$, which is smooth as G acts smoothly on M . Thus F is a smooth G -equivariant bijection. Since it is G -equivariant, it has constant rank, and thus is a smooth submersion by our first point. Similarly it is an immersion: injective maps of constant rank are immersions by a similar argument to our first point (but even simpler, since $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_k, 0, \dots, 0)$ is injective only when $k = n$, we don't need to even consider topology). Also we could just note that $\dim G/G_p = \dim M$ so a smooth submersion is a smooth immersion. Then F is a local diffeomorphism, again by the constant rank theorem, and since it is bijective it must be a global diffeomorphism.

The first point that F is a local diffeomorphism is because by the constant rank theorem it locally has the representation $\bar{x} \mapsto \bar{x}$, which is a diffeomorphism. Since it is a bijection, F^{-1} exists, and for every point in its domain there is a neighborhood in which F is a diffeomorphism, so F^{-1} is smooth; thus F^{-1} is globally smooth.

- (4) The action of $U(n)$ on $G_{\mathbb{C}}(k, n)$ is given by $(U, V) \mapsto UV = \{Uv \mid v \in V\}$. We will assume smoothness at first, and show it at the end. This action is transitive since if $V, V' \in G_{\mathbb{C}}(k, n)$ are both subspaces of dimension k then we can take orthonormal bases of them and define U to map between them. Explicitly, let B_0, B'_0 be orthonormal bases of V, V' respectively. Extend these to B, B' orthonormal bases of $\mathbb{C}^n$ and define U to map B to B' . This is a unitary matrix as it maps an orthonormal basis to an orthonormal basis, and since $UB_0 = B'_0$, we have that $UV = V'$ as desired.

For $V \in G_{\mathbb{C}}(k, n)$, its stabilizer under $U(n)$ are all unitary matrices which map V to itself. Such a unitary matrix must map $V^\perp$ to itself as well. Thus if U is in the stabilizer, since $V \oplus V^\perp = \mathbb{C}^n$, $U = U|_V \oplus U|_{V^\perp}$, so that U is a block-diagonal matrix (up to equivalence). Since the restriction of U to a U -invariant subspace remains unitary (it still maps orthonormal bases to orthonormal bases), $U|_V \in U(k)$ and $U|_{V^\perp} \in U(n-k)$ so that $U \in U(k) \times U(n-k)$ (where $U(k)$ are the unitary transformations of V , and $U(n-k)$ the unitary transformations of $V^\perp$). Conversely any $(U_0, U_1) \in U(k) \times U(n-k)$ defines $U = U_0 \oplus U_1$, a unitary matrix in V 's stabilizer.

Thus by the previous point,

$$G_{\mathbb{C}}(k, n) \cong U(n)/U(n)_V = U(n)/(U(k) \times U(n-k)),$$

as desired.

Now to show that $U(n)$'s action on $G_{\mathbb{C}}(k, n)$ is smooth, consider its charts $\Gamma_{V,W}: \text{hom}(V, W) \rightarrow G_{\mathbb{C}}(k, n)$ mapping ℓ to its graph. Recall that $\Gamma(\text{hom}(V, W))$ is the collection of k -dimensional subspaces with trivial intersection with W . For $U \in U(n)$ and $\ell \in \text{hom}(V, W)$, then

$$U \cdot \Gamma_{V,W}(\ell) = \{Uv + U\ell v \mid v \in V\},$$

which has trivial intersection with UW , since $UV \oplus UW = \mathbb{C}^n$, so it is in $\Gamma(\text{hom}(UV, UW))$, and then

$$\Gamma_{UV, UW}^{-1} \circ U \circ \Gamma_{V,W}(\ell) = U \circ \ell \circ U^{-1},$$

since its graph is $\{Uv + U\ell U^{-1}Uv \mid v \in V\} = \{Uv + U\ell v \mid v \in V\} = U\Gamma(\ell)$. This is a smooth map, since conjugation is smooth ($\text{GL}_n(\mathbb{C})$), and thus $U(n)$ since it is embedded, acts smoothly on $\text{Mat}_n(\mathbb{C})$.

So each coordinate representation of the action is smooth, but we need to show that for each (U, V) , they lie in a neighborhood which is mapped into $\Gamma(\text{hom}(UV, UW))$. Consider the map $U(n) \times \text{hom}(V, W) \rightarrow \text{hom}(V, \mathbb{C}^n)$ which maps $(A, \ell) \mapsto A\ell$. This is smooth, as it is just matrix multiplication. Since V is just represented by $0 \in \text{hom}(V, W)$ (meaning $\Gamma_{V,W}(0) = V$), we want to show that the preimage of the maps whose graphs don't intersect UW is open. Now $\ell \in \text{hom}(V, \mathbb{C}^n)$'s graph doesn't intersect UW if for a basis $v_1, \dots, v_k \in V$ and a basis $w_1, \dots, w_{n-k} \in W$, the collection $v_i + \ell v_i$ with Uw_j remains linearly independent. That is, the determinant of $(\dots, v_i + \ell v_i, \dots, Uw_j, \dots)$ is nonzero. This is an open set, since mapping ℓ to this matrix is linear and thus continuous, and the determinant is continuous.

If we let $O \subseteq U(n) \times \text{hom}(V, W)$ be this open set, so consisting of pairs (A, ℓ) such that $\Gamma(A\ell) \cap W = 0$, then through $\text{id} \times \Gamma_{V,W}$ this gives an open set O in $G_{\mathbb{C}}(k, n)$. And since $0 \in \text{hom}(V, W)$ represents V , and $UV \cap UW = 0$, this is an open neighborhood of (U, V) . Thus for every $(U, V) \in U(n) \times G_{\mathbb{C}}(k, n)$ there is an open neighborhood O which is mapped into an open chart $UV \in \Gamma(\text{hom}(UV, UW))$, and as we showed the coordinate representation is smooth. Thus $U(n)$'s action on $G_{\mathbb{C}}(k, n)$ is smooth

4.0.3 Problem

Let M be a smooth manifold, and H a Lie group which acts on it smoothly, freely, and properly by $\varphi: H \times M \rightarrow M$. Let g be an H -invariant metric on M , and denote by $\pi: M \rightarrow M/H$ the projection.

- (1) Prove there exists a unique metric $\bar{g}$ on M/H such that π is a Riemannian submersion.
- (2) Let $M = \mathbb{R}^n$ and $H = \mathbb{Z}^n$, and φ given by translation, so $M/H = T^n$. Let $g = g_0$ be the Euclidean metric. Construct an isometric embedding $(T^n, \bar{g}) \rightarrow (\mathbb{R}^{2n}, g_0)$.

- (1) For $p \in M$, we can write $d\varphi_p$ by

$$d\pi_p: T_p(Hp) \oplus T_p(Hp)^\perp = T_p M \rightarrow T_{Hp}(M/H),$$

and since $T_p Hp = \ker d\pi_p$ and π is a submersion, we have that $d\pi_p$ forms an isomorphism $T_p(Hp)^\perp \rightarrow T_{Hp}(M/H)$, let us denote this by $d\hat{\pi}_p$. In order for π to be a Riemannian submersion, we must then define

$$\bar{g}(v, u) = g(d\hat{\pi}_p^{-1}v, d\hat{\pi}_p^{-1}u), \quad v, u \in T_{Hp}(M/H),$$

for $v, u \in T_{Hp}(M/H)$. Thus $\bar{g}$ is unique if it exists. Since g is a Riemannian metric, this is a metric at each point (in general $\langle T\bullet, T\bullet \rangle$ is an inner product when T is invertible).

Note that this is not necessarily well-defined: if $Hp = Hq$ then this definition may give different values if $d\hat{\pi}_p$ or $d\hat{\pi}_q$ is used. But since g is H -invariant, if $Hp = Hq$ suppose $q = \varphi_h(p)$ then since $\pi \circ \varphi_h = \pi$ we have that $d\pi_p \circ d(\varphi_h)_p = d\pi_q$ so by H -invariance,

$$g(d\hat{\pi}_p^{-1}v, d\hat{\pi}_p^{-1}u) = g(d(\varphi_{h^{-1}})_p d(\hat{\pi}_p)^{-1}v, d(\varphi_{h^{-1}})_p d(\hat{\pi}_p)^{-1}u) = g(d\hat{\pi}_q^{-1}v, d\hat{\pi}_q^{-1}u).$$

Now we want to show that $\bar{g}$ is smooth. Notice that we can apply Gram-Schmidt to a smooth local coordinate frame to obtain a smooth orthonormal local coordinate frame (smoothness is due to g 's smoothness). Then for $p \in M$ we can define $\Pi_p: T_p M \rightarrow T_p M$ which is the orthonormal projection onto $T_p(Hp)^\perp$. Considering a smooth orthonormal coordinate frame, we see that this maps smooth vector fields to smooth vector fields, and furthermore it is clearly C^∞ -linear, and thus $\Pi: \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ forms a $(1, 1)$ -tensor. Since π is a submersion, it has local sections: there is a neighborhood $p \in U$ and a smooth local section $\sigma: \pi(U) \rightarrow M$ with $\sigma(Hp) = p$. Now we claim that locally, $d\hat{\pi}^{-1} = \Pi \circ d\sigma$. Since for $q \in U$, $\Pi \circ d\sigma$ maps into $T_q(Hq)^\perp$, we need only show that $d\hat{\pi} \circ \Pi \circ d\sigma = \text{id}$. Since $d\pi_q$ is linear and $T_q(Hq)$ is its kernel as well as the projection Π_q 's, we see that $d\hat{\pi}_q \circ \Pi = d\pi_q$, and so

$$d\hat{\pi}_q \circ \Pi_q \circ d\sigma_{Hq} = d\pi_q \circ d\sigma_{Hq} = \text{id}.$$

Thus $d\hat{\pi}^{-1} = \Pi \circ d\sigma$ on $\pi(U)$, and thus locally

$$\bar{g}(v, u) = g(\Pi \circ d\sigma(v), \Pi \circ d\sigma(u)),$$

and since both $\Pi, d\sigma$ are smooth, so is $\bar{g}$ (locally around every point, and thus globally).

- (2) For $n = 1$, the obvious embedding $\mathbb{R}/\mathbb{Z} = T^1 \rightarrow \mathbb{R}^2$ is given by $\theta \mapsto (\cos(2\pi\theta), \sin(2\pi\theta))$, or under the natural identification $\mathbb{R}^2 \cong \mathbb{C}$, $\theta \mapsto \exp(2\pi i\theta)$. This is smooth since precomposing it with the projection $\mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$ maps $x \in \mathbb{R}$ to $\exp(2\pi ix)$ (since $\exp(2\pi ix)$ has a periodicity of 1), which is smooth. This extends to an obvious embedding $T^n \rightarrow \mathbb{R}^{2n} \cong \mathbb{C}^n$ by

$$F: (\theta_1, \dots, \theta_n) \mapsto (\exp(2\pi i\theta_1), \dots, \exp(2\pi i\theta_n)),$$

which is again smooth.

Note that $\pi: \mathbb{R}^n \rightarrow T^n$ is a local diffeomorphism, as it is a submersion and $\dim T^n = \dim \mathbb{R}^n = n$. Thus let $\partial/\partial\theta^i$ be the pushforward of the coordinate vector fields $\partial/\partial x^i$, and since x^i are orthonormal and π is a local Riemannian diffeomorphism (by construction it is a Riemannian submersion), θ^i are orthonormal as well. Then for $\bar{\theta} = (\theta_1, \dots, \theta_n) \in T^n$, (using the Jacobian of F by x^i),

$$dF_{\bar{\theta}} \frac{\partial}{\partial \theta^i} = -2\pi \sin(2\pi\theta_i) \frac{\partial}{\partial x_1^i} + 2\pi \cos(2\pi\theta_i) \frac{\partial}{\partial x_2^i},$$

where the global coordinates of $\mathbb{R}^{2n}$ are labeled $x_1^1, x_2^1, \dots, x_1^n, x_2^n$. Then $dF_{\bar{\theta}} \partial/\partial\theta^i$ are orthogonal (since $\frac{\partial}{\partial x_1^i}, \frac{\partial}{\partial x_2^i}$ are), and

$$g_0 \left(dF_{\bar{\theta}} \frac{\partial}{\partial \theta^i}, dF_{\bar{\theta}} \frac{\partial}{\partial \theta^i} \right) = 4\pi^2 \sin^2(2\pi\theta_i) + 4\pi^2 \cos^2(2\pi\theta_i) = 4\pi^2.$$

So if we define $F' = F/2\pi$, the above computations normalize to give 1. So dF' maps an orthonormal basis to an orthonormal set, and is thus a Riemannian embedding.

4.0.4 Problem

On $M = \mathbb{R}^2$ consider the map

$$\varphi: (0, \infty) \times (0, 2\pi) \rightarrow M, \quad \varphi(r, \theta) = (r \cos \theta, r \sin \theta).$$

- (1) Show that φ is a coordinate chart for M .
- (2) Let $\{\partial_r, \partial_\theta\}$ denote the coordinate vector fields. For $a, b, c, d \in \mathbb{R}$ define the vector field X on φ 's coordinate domain:

$$X = r(a \cos^2 \theta + (b+c) \sin \theta \cos \theta + d \sin^2 \theta) \partial_r + (c \cos^2 \theta + (d-a) \sin \theta \cos \theta - b \sin^2 \theta) \partial_\theta.$$

Show that X extends smoothly to a vector field on all of M .

- (3) For any $p \in M$, calculate the integral curve of X starting at p . Show that X admits global flow.
- (4) Draw the integral curves of X for the following values of a, b, c, d .

Case	a	b	c	d
Sink	-1	0	0	-2
Source	1	0	0	2
Saddle	1	0	0	-1
Spiral sink	-1	-1	1	-1
Spiral source	1	-1	1	1
Center	0	-1	1	0
Improper sink	-1	1	0	-1
Improper source	1	1	0	1

- (1) φ 's image is the set of all points (x, y) where either $y \neq 0$ or $x < 0$ ($\mathbb{R}^2$ without $[0, \infty) \times \mathbb{R}$), which is open. It is smooth, as its components are smooth, and it is injective. Furthermore, its Jacobian is

$$J\varphi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

which has determinant $r \neq 0$, so it is invertible everywhere. Thus φ is a diffeomorphism onto its image, meaning it is a chart.

- (2) Let us write $X = X^r \partial_r + X^\theta \partial_\theta$. We know that

$$\begin{aligned} \partial_r &= \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y} \\ \partial_\theta &= \frac{\partial}{\partial x} x \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y} = -r \sin \theta \frac{\partial}{\partial x} + r \cos \theta \frac{\partial}{\partial y}. \end{aligned}$$

Thus we see that $X = X^1 \partial/\partial x + X^2 \partial/\partial y$, where

$$X^1 = X^r \cos \theta - r X^\theta \sin \theta, \quad X^2 = X^r \sin \theta + r X^\theta \cos \theta.$$

We have that

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \cos^2 \theta = \frac{x^2}{r^2}, \quad \cos \theta \sin \theta = \frac{xy}{r^2}, \quad \sin^2 \theta = \frac{y^2}{r^2}.$$

Thus some tedious calculations yield

$$X = (ax + by) \frac{\partial}{\partial x} + (cx + dy) \frac{\partial}{\partial y}.$$

This can be clearly extended smoothly to all of M .

(3) Since X has the form

$$X(x, y) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix},$$

solving $\gamma'(t) = X(\gamma(t))$ boils down to the linear system of ODEs:

$$\begin{pmatrix} \gamma_1'(t) \\ \gamma_2'(t) \end{pmatrix} = A \begin{pmatrix} \gamma_1(t) \\ \gamma_2(t) \end{pmatrix}.$$

Since A is two-dimensional, it has a Jordan normal form which is diagonal, or is a 2×2 Jordan block. Suppose that $P^{-1}AP = J$ is A 's Jordan form, then a solution to $\alpha' = J\alpha$ gives a solution $\gamma = P\alpha$ to $\gamma' = A\gamma$, since

$$\gamma' = P\alpha' = PJ\alpha = AP\alpha = A\gamma,$$

and vice versa. Note that $\alpha(0) = P^{-1}\gamma(0) = P^{-1}p$.

If A is diagonalizable then say $J = \text{diag}(\lambda_1, \lambda_2)$. Solving for α is easy, as the ODE is $\alpha'_i = \lambda_i \alpha_i$ for $i = 1, 2$, and solving this gives $\alpha_i = c_i \exp(\lambda_i t)$, where $c = P^{-1}p$.

Now if A 's Jordan form is a Jordan block $J = J_2(\lambda)$, then the ODE becomes mildly more difficult:

$$\alpha'_1 = \lambda \alpha_1 + \alpha'_2, \quad \alpha'_2 = \lambda \alpha_2.$$

This gives us $\alpha_2 = c_2 \exp(\lambda t)$ as before, and then $\alpha_1 = c_1 \exp(\lambda t) + c_2 t \exp(\lambda t)$ (using the usual method for solving a non-homogeneous linear ODE).

Note that $\lambda_1, \lambda_2, \lambda$ may be complex numbers. In this case, the real and imaginary parts of the solution give valid solutions to the real linear system of ODEs. For example, if $\lambda_1 = \lambda_2 = 1 + i$ then our solutions become

$$\gamma_{1,2} = \exp((1 + i)t) = p_{1,2} e^t (\cos t + i \sin t),$$

and this gives us two solutions:

$$\gamma_{1,2} = e^t \cos t, \quad \text{and} \quad \gamma_{1,2} = e^t \sin t.$$

Then the general solution is of the form

$$\gamma_{1,2} = A e^t \cos t + B e^t \sin t.$$

In either case α , and thus γ , are defined on all of $\mathbb{R}$ and thus X has global flow.

(4)

(i) Sink: the matrix is already diagonal and the solution becomes

$$\gamma(t) = \begin{pmatrix} A \exp(-t) \\ B \exp(-2t) \end{pmatrix}.$$

Thus the flow in general is given by curves (parabolas) of the form (s, as^2) (pointing toward the origin, as when $t \rightarrow \infty$, $\exp(-t) \rightarrow 0$).

- (ii) Source: the solution becomes $\gamma(t) = (A \exp(t), B \exp(t))$, and so the flow is given by curves of the form (s, as^2) pointing away from the origin.
- (iii) Saddle: the solution becomes $\gamma(t) = (A \exp(t), B \exp(-t))$, so the flow is given by curves of the form (s, as^{-1}) pointing away from the origin.
- (iv) Spiral sink: the matrix is diagonalizable to $\text{diag}(-1+i, -1-i)$ with eigenvectors $(1, -i)$ and $(1, i)$. The solution to the diagonalized problem is $\alpha_1 = A \exp((-1+i)t)$ and $\alpha_2 = B \exp((-1-i)t)$. But notice that since our ODE is a homogeneous linear system of order 2, we need only find 2 linearly independent solutions. Furthermore, given a complex solution its real and imaginary parts are both solutions to the real problem. Thus we can consider only $\alpha = (\exp(-1+i)t), 0)$ (i.e. $A = 1, B = 0$), which gives us the complex solution

$$\gamma(t) = P\alpha = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} \exp((-1+i)t) \\ 0 \end{pmatrix} = \begin{pmatrix} \exp((-1+i)t) \\ -i \exp((-1+i)t) \end{pmatrix}.$$

The real and complex parts of this are

$$\gamma_1(t) = e^{-t} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, \quad \gamma_2(t) = e^{-t} \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}.$$

So the general flow lines are given by

$$\gamma(t) = A\gamma_1(t) + B\gamma_2(t).$$

This forms a swirl with arrows pointing toward the origin (since as $t \rightarrow \infty$, the curve goes to 0).

- (v) Spiral source: the matrix is diagonalizable to $\text{diag}(1+i, 1-i)$ with eigenvectors $(1, -i)$ and $(1, i)$. As above, we have a complex solution

$$\begin{pmatrix} \exp((1+i)t) \\ -i \exp((1+i)t) \end{pmatrix},$$

whose real and complex parts are

$$\gamma_1(t) = e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, \quad \gamma_2(t) = e^t \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}.$$

Thus the flow lines are just linear combinations of these, which look similar to the spiral sink but with arrows pointing out away from the origin.

- (vi) Center: the matrix is diagonalizable to $\text{diag}(i, -i)$ with eigenvectors $(1, -i)$ and $(1, i)$. We have a complex solution

$$\begin{pmatrix} \exp(it) \\ -i \exp(it) \end{pmatrix},$$

which has real and complex parts

$$\gamma_1(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, \quad \gamma_2(t) = \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}.$$

The general flow lines are of the form

$$\gamma(t) = \begin{pmatrix} A \cos t + B \sin t \\ A \sin t - B \cos t \end{pmatrix},$$

which are circles (with radius squared $A^2 + B^2$, going counterclockwise).

- (vii) Improper sink: this matrix is in Jordan form, and so the solution is (as proven before):

$$\gamma(t) = e^{-t} \begin{pmatrix} A + tB \\ B \end{pmatrix}.$$

These form lopsided spirals into the origin.

- (viii) Improper source: this matrix is in Jordan form, and so the solution is:

$$\gamma(t) = e^t \begin{pmatrix} A + tB \\ B \end{pmatrix}.$$

This is similar to the improper sink, but the spirals go to infinity (and are mirrored along the y -axis).

5 Homework 5

5.0.1 Problem

Solve problem 6, page 78, of “Foundations of Lie Groups and Differentiable Manifolds”, Warner. Let X, Y be vector fields on M , with $p \in M$. Let θ, ψ be the flows of X, Y , and define

$$\beta(t) = \psi_{-\sqrt{t}}\theta_{-\sqrt{t}}\psi_{\sqrt{t}}\theta_{\sqrt{t}}(p)$$

in a neighborhood of 0, forming a curve. Show that for $f \in C^\infty(M)$,

$$[X, Y]_p f = \lim_{t \rightarrow 0} \frac{f(\beta(t)) - f(\beta(0))}{t}.$$

Since our limit is over $t > 0$ (considering the domain of square roots), it is equal to

$$\lim_{t \rightarrow 0} \frac{f(\beta(t^2)) - f(p)}{t^2}.$$

Our goal is to use L'Hôpital's rule to compute this limit (by continuity the numerator tends to zero). Let us define

$$\gamma(x, y, z, w) = f(\psi_{-x}\theta_{-y}\psi_z\theta_w(p)),$$

so that $f(\beta(t^2)) = \gamma(t, t, t, t)$. So to compute the derivative of $f(\beta(t^2))$ we compute the partial derivatives of γ . We have the following situation: we want to differentiate a function of the form

$$h(t) = f \circ \theta_t \circ g(p)$$

where θ_t is the flow of some vector field X and f is real-valued. Then

$$h'(t) = df_{\theta_t(gp)}(\theta^{gp})'(t),$$

and since θ is the flow of X , this is equal to

$$h'(t) = df_{\theta_t(gp)}X_{\theta_t(gp)} = X_{\theta_t(gp)}f.$$

In our case, this gives us

$$\begin{aligned} \frac{\partial \gamma}{\partial x} &= -Y_{\psi_{-x}\theta_{-y}\psi_z\theta_w(p)}f \\ \frac{\partial \gamma}{\partial y} &= -X_{\theta_{-y}\psi_z\theta_w(p)}(f \circ \psi_{-x}) \\ \frac{\partial \gamma}{\partial z} &= Y_{\psi_z\theta_w(p)}(f\psi_{-x}\theta_{-y}) \\ \frac{\partial \gamma}{\partial w} &= X_{\theta_w(p)}(f\psi_{-x}\theta_{-y}\psi_z). \end{aligned}$$

Thus

$$f(\beta(t^2))'(0) = \sum_i \frac{\partial \gamma}{\partial x_i}(0) = -Y_p f - X_p f + Y_p f + X_p f = 0,$$

and so we can apply L'Hôpital again (the denominator is $2t$). But we need to differentiate these partial derivatives. These are functions of the form

$$k(t) = X_{h \circ \theta_t \circ f} g = (Xg)(h \circ \theta_t \circ f(p)).$$

where X is a vector field and θ is the flow of some other vector field Y , g some real-valued smooth function, and f, h some smooth functions. Then

$$k'(t) = d(Xg)_{h\theta_t f(p)} dh_{\theta_t f(p)}(\theta^{fp})'(t) = d(Xg)_{h\theta_t f(p)} dh_{\theta_t f(p)} Y_{\theta_t f(p)} = Y((Xg) \circ h)_{\theta_t f(p)}.$$

So we have (by symmetry of second-order partial derivatives):

$$\begin{aligned}
\frac{\partial \gamma}{\partial xx} &= Y(Yf)_{\psi_{-x}\theta_{-y}\psi_z\theta_w(p)} \\
\frac{\partial \gamma}{\partial xy} &= X((Yf) \circ \psi_{-x})_{\theta_{-y}\psi_z\theta_w} \\
\frac{\partial \gamma}{\partial xz} &= -Y((Yf)\psi_{-x}\theta_{-y})_{\psi_z\theta_w} \\
\frac{\partial \gamma}{\partial xw} &= -X((Yf)\psi_{-x}\theta_{-y}\psi_z)_{\theta_w} \\
\frac{\partial \gamma}{\partial yy} &= X(X(f\psi_{-x}))_{\theta_{-y}\psi_z\theta_w} \\
\frac{\partial \gamma}{\partial yz} &= -Y(X(f\psi_{-x})\theta_{-y})_{\psi_z\theta_w} \\
\frac{\partial \gamma}{\partial yw} &= -X(X(f\psi_{-x})\theta_{-y}\psi_z)_{\theta_w} \\
\frac{\partial \gamma}{\partial zz} &= Y(Y(f\psi_{-x}\theta_{-y}))_{\psi_z\theta_w} \\
\frac{\partial \gamma}{\partial zw} &= X(Y(f\psi_{-x}\theta_{-y})\psi_z)_{\theta_w} \\
\frac{\partial \gamma}{\partial ww} &= X(X(f\psi_{-x}\theta_{-y}\psi_z))_{\theta_w}.
\end{aligned}$$

So

$$f(\beta(t^2))''(0) = \sum_{i,j} \frac{\partial \gamma}{\partial x_i x_j}(0),$$

and evaluating the second order partial derivatives at 0 gives results like $Y_p Y f$, $X_p Y f$, etc. Then dividing by 2 (the denominator of the limit is 2, the second derivative of t^2) gives that the limit is equal to

$$X_p Y f - Y_p X f = [X, Y]_p f$$

as desired.

5.0.2 Problem

Prove that there is a unique smooth structure on $\mathbb{R}$ with its standard topology, up to diffeomorphism.

Let M be a smooth manifold over $\mathbb{R}$ with the same topology. Suppose that there is a nowhere vanishing vector field on M , call it X , and let $\gamma: I \rightarrow M$ be a maximal integral curve starting at 0. Then

$$\gamma'(t) = X_{\gamma(t)} \neq 0$$

means that $\gamma'(t)$ has full rank (one) always, and is thus a submersion and immersion. Furthermore integral curves are either periodic or injective; since γ has nonzero derivative it cannot be periodic, so it is injective.

Since γ is a submersion, it is an open map, so $\gamma(I) \subseteq M$ is open and connected. If γ is not surjective then $\gamma(I)$ is an open interval (a, b) with one side finite, assume b is. But then there is another integral curve of X starting at b , call it δ , and similarly its image in M is an open interval and thus intersects (a, b) . Let $\delta(s) \in (a, b)$ so that $p = \gamma(t) = \delta(s)$ for some t , and then $\gamma(x+t)$ and $\delta(x+s)$ are both integral curves starting at p , so they agree in their common domain. But then we can extend γ to part of δ 's domain, including b . This contradicts γ 's maximality, so γ 's image must be all of M .

Thus γ is bijective, an immersion, and a submersion. Thus it is a diffeomorphism, and since $I \subseteq \mathbb{R}$ is an open interval and thus diffeomorphic to $\mathbb{R}$, M and $\mathbb{R}$ are diffeomorphic.

Now we show that a nowhere vanishing X exists. Let us cover M by smooth charts on open intervals (a_i, b_i) , and by reordering and shrinking we can assume $a_i < b_{i-1} < a_{i+1} < b_i$ for all i (so every point is contained in at most two open sets). Furthermore, we can assume that this atlas is oriented: we start at $i = 0$ and work forward swapping orientation of the charts if they aren't compatible with the orientation of (a_0, b_0) 's chart. Similarly we work backward.

Now define the vector field X^i on (a_i, b_i) to be given by d/dt (relative to the chart). If we let τ^i be the transition function between (a_i, b_i) and (a_{i+1}, b_{i+1}) , then X_p^i relative to (a_{i+1}, b_{i+1}) is $(\tau_i)'d/dt$. Since the atlas is oriented, $(\tau_i)' > 0$.

Now let λ_i be a partition of unity subordinate to the open cover, and let

$$X = \sum_i \lambda_i X^i.$$

This is smooth by local finiteness, and thus forms a vector field. Furthermore for $p \in M$, it is contained in a single (a_i, b_i) or two. If it is contained in a single one, then $X_p = \lambda_i(p)X_p^i = X_p^i$ (since the partition must sum to one), which is nonzero by definition. Otherwise,

$$X_p = \lambda_i(p)X_p^i + \lambda_{i+1}(p)X_p^{i+1} = \lambda_i(p) \left. \frac{d}{dt} \right|_p + (1 - \lambda_i(p))\tau_i'(p) \left. \frac{d}{dt} \right|_p.$$

This cannot be zero, if it were then

$$\lambda_i(p)(\tau_i'(p) - 1) = \tau_i'(p),$$

But $\lambda_i(p)(\tau_i'(p) - 1) \leq \tau_i'(p) - 1 < \tau_i'(p)$ a contradiction.

5.0.3 Problem

Let $G = \text{GL}_n(\mathbb{C})$ be the Lie group of complex invertible $n \times n$ matrices. Its Lie group $\mathfrak{g} \cong T_e G$ can be identified with $\text{Mat}_n(\mathbb{C})$.

- (1) Prove an explicit formula for left-invariant vector fields on G .
- (2) Prove an explicit formula for the exponential map $\exp: \mathfrak{g} \rightarrow G$.
- (3) Let $X, Y \in \mathfrak{g}$, show in two ways that $[X, Y] = XY - YX$.
 - (i) Use part (1) and the definition of Lie brackets.
 - (ii) Prove an explicit formula for $\text{Ad}: G \rightarrow \text{Aut}(\mathfrak{g})$, use the fact that $\text{ad} = \text{Ad}_*$ and $\text{ad}_X(Y) = [X, Y]$, as well as part (2).

- (1) We know in general for $v \in T_e G$, it defines the left-invariant vector field $v^L \in \text{Lie}(G)$ given by

$$v_g^L = d(L_g)_e v.$$

So we need to compute the differentials of left multiplication. Recall the identification $\text{Mat}_n(\mathbb{C}) \leftrightarrow T_l G$ by mapping a matrix $(A_j^i) \in \text{Mat}_n(\mathbb{C})$ to

$$(A_j^i) \leftrightarrow A_j^{i,b} \frac{\partial}{\partial x_j^{i,b}}$$

where $A_j^{i,0} = \text{Re}(A_j^i)$ and $A_j^{i,1} = \text{Im}(A_j^i)$.

For a matrix A recall that

$$d(L_A)_I \left(\frac{\partial}{\partial x_j^{i,b}} \right)_I = \frac{\partial (L_A)_k^{\ell,c}}{\partial x_j^{i,b}}(A) \left. \frac{\partial}{\partial x_k^{\ell,c}} \right|_A,$$

so we compute the partial derivatives of L_A . Let $(X_j^{i,b})$ be a complex-valued matrix,

$$(L_A X)_k^\ell = (AX)_k^\ell = A_t^\ell X_k^t,$$

And thus

$$(L_A X)_k^{\ell,0} = \text{Re}((AX)_k^\ell) = (A_t^{\ell,0} X_k^{t,0}) - (A_t^{\ell,1} X_k^{t,1})$$

so that

$$\frac{\partial(L_A)_k^{\ell,0}}{\partial x_k^{t,0}} = A_t^{\ell,0}, \quad \frac{\partial(L_A)_k^{\ell,0}}{\partial x_k^{t,1}} = -A_t^{\ell,1}.$$

Similarly

$$(L_A X)_k^{\ell,1} = A_t^{\ell,0} X_k^{t,1} + A_t^{\ell,1} X_k^{t,0}$$

so that

$$\frac{\partial(L_A)_k^{\ell,1}}{\partial x_k^{t,0}} = A_t^{\ell,1}, \quad \frac{\partial(L_A)_k^{\ell,1}}{\partial x_k^{t,1}} = A_t^{\ell,0}.$$

All other partial derivatives are zero.

So we have that

$$d(L_A)_I \left(\frac{\partial}{\partial x_j^{i,0}} \Big|_I \right) = A_i^{\ell,0} \frac{\partial}{\partial x_j^{\ell,0}} \Big|_A + A_i^{\ell,1} \frac{\partial}{\partial x_j^{\ell,1}} \Big|_A,$$

and

$$d(L_A)_I \left(\frac{\partial}{\partial x_j^{i,1}} \Big|_I \right) = -A_i^{\ell,0} \frac{\partial}{\partial x_j^{\ell,0}} \Big|_A + A_i^{\ell,0} \frac{\partial}{\partial x_j^{\ell,1}} \Big|_A.$$

Under the correspondence $\partial/\partial x_j^{i,0} \Big|_I$ is the basis matrix E_{ij} , and $\partial/\partial x_j^{i,1} \Big|_I = iE_{ij}$ (multiplication by the imaginary unit). We see then that

$$d(L_A)_I(E_{ij}) = AE_{ij}, \quad d(L_A)_I(iE_{ij}) = AiE_{ij}$$

under the natural correspondence.

By linearity we see that

$$d(L_A)_I(B) = AB$$

for all matrices B . Thus the left-invariant vector field corresponding to $B \in T_I G$ is

$$B^L|_A = d(L_A)_I(B) = AB.$$

(2) For a matrix A define

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

By submultiplicity of the Frobenius norm, this converges absolutely for all A (as we can compare the series with $|A|^k/k!$ which converges to $e^{|A|}$). We claim that $\exp(A) = e^A$ for all matrices.

To do this, we need to show that for $A \in \mathfrak{g}$, its one-parameter subgroup is given by $\gamma(t) = e^{tA}$. We need to show that $\gamma(t)$ is invertible for all t (so that it lands in G), $\gamma'(t) = A^L|_{\gamma(t)}$ for all t (so it defines the one-parameter subgroup), and $\gamma(0) = I$. The last point is the easiest as it follows from the definition that $e^0 = I$.

For the second point, we need to show

$$\gamma'(t) = A^L|_{\gamma(t)} = \gamma(t)A$$

by our formula for left-invariant vector fields. Note that

$$\gamma(t) = \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$$

is a power series which converges uniformly in any closed interval, and thus we can differentiate it term by term:

$$\gamma'(t) = \sum_{k=1}^{\infty} \frac{t^{k-1}}{(k-1)!} A^k = \left(\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k \right) A = \gamma(t)A$$

as desired.

Finally to show that $\gamma(t)$ is invertible, we will show that its inverse is $\gamma(-t)$ (not surprising by exponential laws). So define $\sigma(t) = \gamma(t)\gamma(-t)$, we will show that it has zero derivative. By the product rule and what we just showed,

$$\sigma'(t) = \gamma'(t)\gamma(-t) + \gamma(t)\gamma'(-t) = \gamma(t)A\gamma(-t) - \gamma(t)A\gamma(-t) = 0$$

so $\sigma(t)$ is constant. Since $\sigma(0) = \gamma(0)\gamma(0) = I$, we see that $\sigma(t) = I$ for all t and thus $\gamma(t)$ is invertible for all t . Thus $\gamma(t)$ is A^L 's integral flow starting at I , thus the one-parameter subgroup generated by A . Then by definition

$$\exp(A) = \gamma(1) = e^A$$

as desired.

(3)

(i) We recall that

$$[A, B] = [A^L, B^L]_I$$

and for vector fields,

$$[A, B] = \left(A^i \frac{\partial B^j}{\partial x^i} - B^i \frac{\partial A^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}.$$

In our case as we showed,

$$A_X^L = XA = X_k^i A_j^k \frac{\partial}{\partial x_j^i},$$

where we split the real and imaginary parts. Then

$$(A^L)_j^i(X) = X_k^i A_j^k \implies \frac{\partial (A^L)_j^i}{\partial x_k^i} = A_j^k$$

thus

$$[A^L, B^L]_I = \left((A^L)_j^i(I) B_k^j - (B^L)_j^i(I) A_k^j \right) \frac{\partial}{\partial x_k^i} = (A_j^i B_k^j - B_j^i A_k^j) \frac{\partial}{\partial x_k^i}$$

which is $AB - BA$ under the correspondence.

(ii) By naturality of $\exp$, the following diagram commutes

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\quad \text{Ad}_* \quad} & \text{end}(\mathfrak{g}) \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\quad \text{Ad} \quad} & \text{Aut}(\mathfrak{g}) \end{array}$$

Since $\text{Aut}(\mathfrak{g})$ is $\text{GL}(\mathfrak{g})$, we already know that $\exp'$ is given by the exponential map defined above:

$$\exp'(\Phi) = \sum_k \frac{1}{k!} \Phi^k, \quad \Phi \in \text{end}(\mathfrak{g}),$$

so we write $\exp$ for $\exp'$ as well. Since $\text{ad} = \text{Ad}_*$ we have

$$\text{Ad}(\exp(A)) = \exp(\text{ad}(A)).$$

By linearity, $\text{ad}(A)$'s one-parameter subgroup is given by $\gamma(t) = \exp(t\text{ad}(A)) = \exp(\text{ad}(tA)) = \text{Ad}(\exp(tA))$. This means that $\gamma'(0) = \text{ad}(A) = [A, \bullet]$, so we want to differentiate $\text{Ad}(\exp(tA))$.

Now we claim that $\text{Ad}(A)B = ABA^{-1}$ for all $A \in G, B \in \mathfrak{g}$. Indeed,

$$\text{Ad}(A) = (R_{A^{-1}})_* \circ (L_A)_*$$

and we showed that $(L_A)_* = d(L_A)_I$ is left multiplication by A , dually $(R_A)_*$ is right multiplication by A . Thus $\text{Ad}(A)B = ABA^{-1}$ as desired.

We know that

$$[A, B] = \text{ad}(A)B = \left(\frac{d}{dt} \Big|_{t=0} \text{Ad}(\exp(tA)) \right) B = \frac{d}{dt} \Big|_{t=0} (\text{Ad}(\exp(tA))B),$$

and since $\exp(A)^{-1} = \exp(-A)$,

$$\text{Ad}(\exp(tA))B = \exp(tA)B \exp(-tA)$$

so that

$$\begin{aligned} \frac{d}{dt} \Big|_{t=0} \text{Ad}(\exp(tA))B &= \left(\frac{d}{dt} \Big|_{t=0} \exp(tA) \right) B \exp(0) + \exp(0)B \left(\frac{d}{dt} \Big|_{t=0} \exp(-tA) \right) \\ &= AB - BA, \end{aligned}$$

where the last equality is because $\exp(tA)$ is the one-parameter subgroup generated by A . So we have shown

$$[A, B] = AB - BA$$

as desired.

5.0.4 Problem

Consider the Lie group $G = U_n$ of complex unitary $n \times n$ matrices and its Lie algebra $\mathfrak{u}_n$. In a previous homework we showed a natural identification $\mathfrak{u}_n \cong T_e U_n$ with the space of skew-Hermitian matrices ($M^\dagger = -M$).

- (1) Repeat the previous problem for G .
- (2) Define an inner product on $T_e U_n$ by

$$g_e(X, Y) = \text{Re tr}(X^\dagger Y).$$

Show that g_e extends to a bi-invariant metric g on U_n .

- (3) Verify that $g([X, Y], Z) = -g(Y, [X, Z])$ for $X, Y, Z \in \mathfrak{u}_n$.

- (1) U_n is an embedded Lie subgroup of $\text{GL}_n(\mathbb{C})$ as we showed in the previous homework. Thus for $U \in U_n$, $L_U: U_n \rightarrow U_n$ is the restriction of $L_U: \text{GL}_n(\mathbb{C}) \rightarrow \text{GL}_n(\mathbb{C})$ and thus dL_U is given by the same formula as for $\text{GL}_n(\mathbb{C})$. That is, left-multiplication by U . So left-invariant vector fields are of the form

$$M_U^L = UM, \quad U \in U_n, M \in \mathfrak{u}_n.$$

Since U_n is a Lie subgroup, its exponential is the restriction of that of $\text{GL}_n(\mathbb{C})$ so it is given by the same formula (matrix exponentiation), and the Lie bracket is as well.

- (2) If we have an inner product $g_e \bullet, \bullet$ on $\mathfrak{g}$ then we showed it extends to a metric on G by

$$g(X, Y) = g_e(X_e, Y_e).$$

This is right-invariant when it is invariant under Ad . This is because for $X \in \mathfrak{u}_n$ and $U \in U_n$ we have $\text{Ad}(U)X = (R_{U^{-1}})_*(L_U)_*X = (R_{U^{-1}})X$ by left-invariance. Thus for $X, Y \in \mathfrak{u}_n$,

$$\begin{aligned} ((R_{U^{-1}})^*g)_V(X_V, Y_V) &= g_{VU^{-1}}(((R_{U^{-1}})_*X)_{VU^{-1}}, ((R_{U^{-1}})_*Y)_{VU^{-1}}) \\ &= g_{VU^{-1}}((\text{Ad}(U)X)_{VU^{-1}}, (\text{Ad}(U)Y)_{VU^{-1}}) = g_I(\text{Ad}(U)X, \text{Ad}(U)Y). \end{aligned}$$

The last equality is due to how we extended g_e : $g_U(X_U, Y_U) = g_e(X_e, Y_e)$. So if g_e is Ad -invariant then

$$((R_{U^{-1}})^*g)_V(X_V, Y_V) = g_I(X, Y) = g_V(X_V, Y_V),$$

so that g is right-invariant.

Again Ad on U_n is the restriction of Ad on $\text{GL}_n(\mathbb{C})$ so $\text{Ad}(U)X = UXU^{-1} = UXU^\dagger$ since U is unitary. Thus

$$g_I(\text{Ad}(U)X, \text{Ad}(U)Y) = g_I(UXU^\dagger, UYU^\dagger) = \text{Re tr}((UXU^\dagger)^\dagger(UYU^\dagger)) = \text{Re tr}(UX^\dagger YU^\dagger).$$

Permuting the matrices in the trace, this is equal to

$$= \text{Re tr}(U^\dagger U X^\dagger Y) = \text{Re tr}(X^\dagger Y) = g_I(X, Y).$$

So g_I is Ad -invariant as desired.

For completeness, g_I is an inner product because $(X, Y) \mapsto \text{tr}(X^\dagger Y)$ is a complex inner product (the Frobenius inner product), and taking the real part of a complex inner product is a real inner product.

(3) We compute directly:

$$g_I([X, Y], Z) = \text{Re tr}((XY - YX)^\dagger Z) = \text{Re tr}(Y^\dagger X^\dagger Z) - \text{Re tr}(X^\dagger Y^\dagger Z),$$

since X, Y are skew-Hermitian this is equal to

$$g_I([X, Y], Z) = \text{Re tr}(Y X Z) - \text{Re tr}(X Y Z).$$

And

$$g_I(Y, [X, Z]) = \text{Re tr}(Y^\dagger (XZ - ZX)) = -\text{Re tr}(Y X Z - Y Z X) = -(\text{Re tr}(Y X Z) - \text{Re tr}(Y Z X))$$

permuting cyclicly the product in the last trace gives us

$$g_I(Y, [X, Z]) = -(\text{Re tr}(Y X Z) - \text{Re tr}(X Y Z)) = -g_I([X, Y], Z)$$

as desired.

5.0.5 Problem

Let $\text{SL}_n(\mathbb{R})$ denote the group of $n \times n$ real matrices with unit determinant. We show that it is a Lie group of dimension $n^2 - 1$ with Lie algebra $\mathfrak{sl}_n$, $n \times n$ real matrices with zero trace.

We also define SU_n to be the group of $n \times n$ unitary matrices with unit determinant. We show that it is a Lie group with Lie algebra $\mathfrak{su}_n$, $n \times n$ skew-Hermitian matrices of trace 0.

- (1) Let $F: \text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R}$ be given by taking the determinant. Thus for $A \in \text{Mat}_n(\mathbb{R})$, its differential $dF_A: \text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R}$ is a linear map. Show that

$$dF_A(B) = \text{tr}(\text{adj}(A)B)$$

where $\text{adj}(A)$ is the adjugate of A .

- (2) Use the previous point to show that 1 is a regular value of $\det$. Conclude that $\text{SL}_n(\mathbb{R})$ is a smooth manifold of dimension $n^2 - 1$ and $T_2 \text{SL}_n(\mathbb{R}) = \mathfrak{sl}_n(\mathbb{R})$.
- (3) Prove that group multiplication and inversion on $\text{SL}_n(\mathbb{R})$ are smooth, so that $\text{SL}_n(\mathbb{R})$ is a Lie group.
- (4) Identify S^1 with the unit circle in $\mathbb{C}$, so a Lie group. Define the group morphism $h: U_n \rightarrow S^1$ by $h(U) = \det U$. Show that h is smooth and $\text{id} \in U_n$ is a regular point of h .
- (5) Use the fact that h is a homomorphism to show that it is a submersion.
- (6) Conclude that SU_n is a Lie group.
- (7) Let $\iota: U_n \hookrightarrow \text{Mat}_n(\mathbb{C})$ be the inclusion. Show that the composition

$$T_\iota U_n \xrightarrow{d\iota} T_\iota \text{Mat}_n(\mathbb{C}) \xrightarrow{\zeta} \text{Mat}_n(\mathbb{C})$$

carries $T_\iota \text{SU}_n \subseteq T_\iota U_n$ to $\mathfrak{su}_n \subseteq \mathfrak{u}_n$, where ζ maps $[\gamma]$ to $\gamma'(0)$.

- (1) We know that

$$dF_A \left(\frac{\partial}{\partial x_j^i} \Big|_A \right) = \frac{\partial F}{\partial x_j^i}(A) \frac{d}{dt}.$$

$\partial/\partial x_j^i|_A$ corresponds to the standard basis matrix E_j^i (zero except at the (i, j) coordinate), so we want to show that this is equal to

$$dF_A(E_j^i) = \frac{\partial F}{\partial x_j^i}(A) \frac{d}{dt} = \text{tr}(\text{adj}(A)E_j^i) \frac{d}{dt},$$

and then by linearity we get the result for all B , not just E_j^i . So we want to show that

$$\frac{\partial F}{\partial x_j^i}(A) = \text{tr}(\text{adj}(A)E_j^i) = (\text{adj}(A)E_j^i)_k^k = \text{adj}(A)_i^j = (-1)^{i+j} \det(A_{ij})$$

where A_{ij} is the minor obtained by removing the i th row and j th column of A .

For $X = (x_j^i)$,

$$F(X) = \sum_{\sigma \in S_n} \text{sgn} \sigma \prod_{i=1}^n x_{\sigma(i)}^i,$$

so

$$\frac{\partial F}{\partial x_j^i} = \sum_{\sigma(i)=j} \text{sgn} \sigma \prod_{k \neq i} x_{\sigma(k)}^k.$$

Notice that

$$\frac{\partial F}{\partial x_j^i}(A) = \sum_{\sigma(i)=j} \text{sgn} \sigma \prod_{k \neq i} a_{\sigma(k)}^k = (-1)^{i+j} \det(A_{ij})$$

as desired, since the permutations run over $\sigma \in S_{n-1}$ (identified with the subset of S_n where $\sigma(i) = j$, and this identification changes the sign by $(-1)^{i+j}$).

- (2) If
- $F(A) = 1$
- , meaning
- $\det(A) = 1$
- , then
- A
- is invertible and
- $\text{adj}(A) = A^{-1}$
- . Thus

$$dF_A(A) = \text{tr}(\text{adj}(A)A) = \text{tr}(I) = n \neq 0.$$

This means that dF_A is a nonzero linear functional, and thus surjective as its codomain is one-dimensional. In other words, A is a regular point of F , so 1 is a regular value.

Since $\text{SL}_n(\mathbb{R}) = F^{-1}(1)$, by the regular level set theorem this means $\text{SL}_n(\mathbb{R})$ is an embedded smooth submanifold of $\text{Mat}_n(\mathbb{R})$, whose codimension is $\dim \mathbb{R} = 1$. Thus its dimension is $\dim \text{SL}_n(\mathbb{R}) = \dim \text{Mat}_n(\mathbb{R}) - 1 = n^2 - 1$.

Finally we also know that $T_I \text{SL}_n(\mathbb{R}) = \ker dF_I$. This is because for a regular level set $F^{-1}q$, its dimension equals $\ker dF_p$ (since dF_p is surjective, the kernel has codimension equal to the dimension of its codomain), and F is constant on $F^{-1}q$ so $dF_p = 0$ on the level set, so $T_p F^{-1}q$ is contained in the kernel. Their dimensions are equal, so they are equal.

Since $dF_I(B) = \text{tr}(\text{adj}(I)^{-1}B) = \text{tr}(B)$, its kernel is all the matrices B of zero trace. Thus $T_I \text{SL}_n(\mathbb{R}) = \mathfrak{sl}_n(\mathbb{R})$ as desired.

- (3) $\text{SL}_n(\mathbb{R})$ is a subgroup of $\text{GL}_n(\mathbb{R})$ and an embedded submanifold of $\text{Mat}_n(\mathbb{R})$. Thus it is embedded in $\text{GL}_n(\mathbb{R})$ as well, and multiplication and inversion in $\text{GL}_n(\mathbb{R})$ restrict to $\text{SL}_n(\mathbb{R})$. We showed in a previous homework that we can restrict both the domain and codomain of a smooth function to embedded submanifolds, so doing this for multiplication and inversion (restricting the smooth maps of $\text{GL}_n(\mathbb{R})$ to $\text{SL}_n(\mathbb{R})$) shows that multiplication and inversion of $\text{SL}_n(\mathbb{R})$ is smooth.
- (4) S^1 is embedded in $\mathbb{C}$, and U_n in $\text{Mat}_n(\mathbb{C})$. The map $\det: \text{Mat}_n(\mathbb{C}) \rightarrow \mathbb{C}$ is smooth, and we can restrict it to the embedded submanifolds (on both the domain and codomain) $U_n \subseteq \text{Mat}_n(\mathbb{C})$ and $S^1 \subseteq \mathbb{C}$. Thus $\det: U_n \rightarrow S^1$ is smooth.

A similar computation as in part (1) shows that

$$dh_I(A) = \text{tr}(\text{adj}(I)^{-1}A) = \text{tr}(A).$$

Since $T_I U_n \cong \mathfrak{u}_n$, the space of matrices skew-Hermitian matrices, taking $A = iI$ we see that $dh_I(A)$ is nonzero. Since S^1 is one-dimensional, this means that dh_I has full-rank and thus I is a regular point.

- (5) h is a group homomorphism between Lie groups (by multiplicity of the determinant), and thus has constant rank. Since h has rank 1 at I , it has rank 1 everywhere. Since $\dim S^1 = 1$, this means that h is a submersion.
- (6) $SU_n = h^{-1}(1)$ is a regular level set (since h is a submersion, every level set is regular), and thus an embedded submanifold of U_n . It is also a subgroup of U_n , and since we can restrict smooth functions to embedded submanifolds, restricting multiplication and inversion of U_n to SU_n shows that they are smooth, so it is a Lie group.
- (7) SU_n 's codimension is equal to the rank of h , so 1. Thus

$$\dim SU_n = \dim U_n - 1 = n^2 - 1.$$

The dimension of $\mathfrak{su}_n$ is also $n^2 - 1$: a skew-Hermitian matrix is determined by its values on its upper triangle, with imaginary values on its diagonal. Requiring that its trace be zero gives us $n - 1$ choices of the diagonal, so

$$\dim \mathfrak{su}_n = \frac{n^2 - n}{2} \cdot 2 + (n - 1) = n^2 - 1,$$

where we multiply the first term by 2 since each element of the upper triangle is complex (real + imaginary parts). Explicitly, a basis for $\mathfrak{su}_n$ is given by

$$\left\{ E_j^i - E_i^j, iE_j^i + iE_j^i \mid i > j \right\} \cup \left\{ iE_i^i - iE_n^n \mid i < n \right\}.$$

Since $\zeta \circ d\iota_I$ is a composition of injections, it is an injection ($d\iota_I$ is an injection since U_n is embedded and thus ι is an immersion). So we need only show that it carries $T_I SU_n$ into $\mathfrak{su}_n$, and then considering dimensions this shows that the image of $T_I SU_n$ is equal to $\mathfrak{su}_n$.

Let $\gamma: (-\varepsilon, \varepsilon) \rightarrow SU_n$ be a curve starting at I . Then

$$\zeta \circ d\iota_I[\gamma] = \zeta[\iota \circ \gamma] = (\iota \circ \gamma)'(0).$$

So we identify γ with $\iota \circ \gamma$ and view it as a curve into $\text{Mat}_n(\mathbb{C})$ with values in SU_n (meaning $\gamma(t)$ is unitary with unit determinant for all t). We already showed that $\gamma'(0) \in \mathfrak{u}_n$ in a previous homework, so we need only show that $\text{tr} \gamma'(0) = 0$.

We know that $\det \gamma(t) = 1$, and thus

$$0 = \left. \frac{d}{dt} \right|_{t=0} \det \gamma(t) = d \det_{\gamma(0)} \gamma'(0) = d \det_I(\gamma'(0)).$$

We showed above that this is equal to

$$= \text{tr}(\text{adj}(I)\gamma'(0)) = \text{tr}(\gamma'(0)),$$

so $\gamma'(0)$ has trace zero, as desired.

6 Homework 6

6.0.1 Problem

Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$(y_1, y_2) = f(x_1, x_2) = (x_1^3 + x_1 x_2, x_2).$$

- (1) Graph y_1 as a function of x_1 and x_2 .
- (2) What are the critical points of f ?
- (3) What are the critical values of f ?

- (4) A line in $\mathbb{R}^2$ is a submanifold $\ell \subseteq \mathbb{R}^2$ determined by an equation of the form

$$ay_1 + by_2 = c$$

for constants $a, b, c \in \mathbb{R}$. Which lines are transverse to f ?

- (5) Draw $f^{-1}\ell$ and determine whether it is a submanifold of $\mathbb{R}^2$ for the values

$$(a, b, c) = (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, -1, 1).$$

- (1) If we keep x_2 constant, the map $y_1 = x_1^3 + x_1x_2$ has derivative $3x_1^2 + x_2$, which can be zero only when $x_2 < 0$. So the graph of f at x_2 is $x_1^3 + x_1x_2$, which is just x_1^3 when $x_2 = 0$, and then as x_2 gets more negative it becomes a more and more extreme N shape, and as x_2 gets more positive it becomes closer to a line. I have no artistic skills, and was not able to draw the graph in any comprehensible way.
- (2) We compute f 's Jacobian: let $p = (x_1, x_2)$ then

$$df_p = \begin{pmatrix} 3x_1^2 + x_2 & x_1 \\ 0 & 1 \end{pmatrix}.$$

Since the domain and codomain of f have the same dimension, df_p is surjective iff it is injective iff it is bijective, which is equivalent to df_p having nonzero determinant. The determinant of df_p is easy to compute:

$$\det(df_p) = 3x_1^2 + x_2.$$

Now p is a critical point iff df_p is not surjective, so iff this determinant is zero. That is,

$$p \text{ is critical} \iff 3x_1^2 + x_2 = 0.$$

So f 's critical points are of the form $(t, -3t^2)$ for $t \in \mathbb{R}$.

- (3) f 's critical values are just the values it takes on its critical points. For a critical point $p = (t, -3t^2)$,

$$f(p) = (t^3 - 3t^3, -3t^2) = (-2t^3, -3t^2).$$

So this is the set of f 's critical values: all points in $\mathbb{R}^2$ of the form $(2t^3, 3t^2)$.

- (4) Recall that $f: N \rightarrow M$ is transverse to $S \subseteq M$ iff $df_p(T_p N) + T_p S = T_p M$ for all $p \in f^{-1}S$. If p is a regular point of f , this obviously holds. So we are interested in lines which contain a critical value of f . First let us compute $T_q \ell$. Note that ℓ is the preimage of c under the smooth map

$$g: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad g(y_1, y_2) = ay_1 + by_2.$$

Thus $T_q \ell \subseteq T_q \mathbb{R}^2 = \mathbb{R}^2$ is just the kernel of dg_q . g 's differential is

$$dg = (a, b),$$

whose kernel is then

$$\ker dg = \ker(a, b) = \text{span}(b, -a)^\top.$$

Thus for $q \in \ell$,

$$T_q \ell = \text{span}(b, -a) \subseteq \mathbb{R}^2.$$

Now recall that we are interested only in points of the form $q = fp$ where $p = (x_1, x_2)$ is a critical point of f . Let $p = (t, -3t^2)$ be a critical point of f , then

$$df_p(T_p \mathbb{R}^2) = \text{im} \begin{pmatrix} 0 & t \\ 0 & 1 \end{pmatrix} = \text{span}(t, 1)^\top.$$

So we want to now find for which lines ℓ we have that for all p critical and $fp \in \ell$, $df_p(T_p\mathbb{R}^2) + T_{fp}\ell = T_{fp}\mathbb{R}^2 = \mathbb{R}^2$. Let $p = (t, -3t^2)$, then $fp = (-2t^3, -3t^2)$ is in ℓ when

$$-2at^3 - 3bt^2 = c.$$

And in such a case, $df_p(\mathbb{R}^2) + T_{fp}\ell = \mathbb{R}^2$ iff $(b, -a)$ and $(t, 1)$ together span $\mathbb{R}^2$, which is iff they form a basis iff

$$\det \begin{pmatrix} t & b \\ 1 & -a \end{pmatrix} = -at - b \neq 0.$$

So f is not transverse to ℓ if there is a $t \in \mathbb{R}$ such that

$$-2at^3 - 3bt^2 = c, \quad at + b = 0.$$

If $a \neq 0$ then we must have $t = -b/a$ and

$$2at^3 + 3bt^2 = 2a\left(-\frac{b}{a}\right)^3 + 3b\left(-\frac{b}{a}\right)^2 = \frac{b^3}{a^2},$$

and thus $c = -b^3/a^2$.

If $a = 0$ then we must have $b = 0$, in which case either $c = 0$ so $\ell = \mathbb{R}^2$ to which f is clearly transverse, or $c \neq 0$ so $\ell = \emptyset$ which again f is also transverse (vacuously; its preimage is empty).

So f is not transverse to lines of the form

$$ay_1 + by_2 = -\frac{b^3}{a^2}.$$

- (5) If f is transverse to S , then $f^{-1}S$ is a submanifold. For $(a, b, c) = (1, 0, -1)$, f is transverse to ℓ so $f^{-1}\ell$ is a submanifold. We also compute what $f^{-1}\ell$ is in both of these cases.

For $(a, b, c) = (1, 0, -1)$, $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = -1$. Since $x_1 = 0$ gives no solutions, this is iff

$$x_2 = -\frac{x_1^3 + 1}{x_1},$$

so $f^{-1}\ell$ is the set of all points of the form $\left(t, -\frac{t^3+1}{t}\right)$, i.e. it is the graph of $-\frac{t^3+1}{t}$.

Similarly for $(a, b, c) = (1, 0, 1)$, $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = 1$ iff $x_2 = \frac{1-x_1^3}{x_1}$. So $f^{-1}\ell$ is the graph of $\frac{1-t^3}{t}$.

For $(a, b, c) = (1, 0, 0)$ we showed that f is not transverse to ℓ . Note that $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = 0$, and the set of such points does not form a submanifold. Indeed,

$$x_1^3 + x_1x_2 = 0 \iff x_1 = 0 \text{ or } x_2 = -x_1^2,$$

so $f^{-1}\ell$ is just the union of the parabola $y = -x^2$ with the y -axis $x = 0$. This is not even a topological manifold, since it is not locally Euclidean around the origin: removing it creates 4 disconnected components, but removing a single point from one-dimensional Euclidean space creates only two. So $f^{-1}\ell$ is not a submanifold.

For $(a, b, c) = (1, -1, 1)$ we have that $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 - x_2 = 1$, and since (by polynomial division)

$$x_1^3 + x_1x_2 - x_2 - 1 = (x_1 - 1)(x_1^2 + x_1 + x_2 + 1),$$

$f^{-1}\ell$ is the union of the parabola $y = -x^2 - x - 1$ with the line $x = 1$; this is not a manifold for the same reason as before.

6.0.2 Problem

Show that the Levi-Civita connection on a pseudo-Riemannian manifold exists and is unique. Provide a full proof.

The derivation of Koszul's formula done in class does not change: the Levi-Civita connection must satisfy

$$\langle \nabla_X Y, Z \rangle = \frac{1}{2} (X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle Y, [X, Z] \rangle - \langle Z, [Y, X] \rangle + \langle X, [Z, Y] \rangle).$$

A pseudo-Riemannian metric is nondegenerate, so this uniquely defined $\nabla_X Y$. To show that this defines a symmetric metric connection, we focus on a coordinate frame given by a chart $(U, (x^i))$. The Gram matrix of a nondegenerate bilinear form is invertible, so the formula for the connection coefficients derived in class remains valid:

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

where $\partial_i = \partial/\partial x^i$. Note then that this defines a connection on $U \subseteq M$: the connection coefficients on a local frame define a connection. That is,

$$\nabla_X Y = (X(Y^k) + X^i Y^j \Gamma_{ij}^k) \partial_k$$

defines a connection on U .

We will show that this connection is symmetric and metric on U , and thus by uniqueness these definitions must coincide on the intersections of any charts, and thus gluing them together gives the desired Levi-Civita connection.

Notice that by g 's symmetry, $\Gamma_{ij}^k = \Gamma_{ji}^k$. Thus ∇ is symmetric, since

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k = \Gamma_{ji}^k \partial_k = \nabla_{\partial_j} \partial_i.$$

Then since the torsion tensor

$$\tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

is indeed a tensor, by linearity we see that since $\tau(\partial_i, \partial_j) = 0$ for all i, j , τ is identically zero. Thus ∇ is symmetric.

Now,

$$\nabla g(Y, Z, X) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z)$$

is a tensor. Additivity in each component is easily seen, as well as multiplicativity in X . Multiplicativity in e.g. Y is easily shown,

$$\begin{aligned} \nabla g(fY, Z, X) &= X(g(fY, Z)) - g(\nabla_X fY, Z) - g(fY, \nabla_X Z) \\ &= (Xf)g(Y, Z) + fX(g(Y, Z)) - g((Xf)Y, Z) - fg(\nabla_X Y, Z) - fg(Y, \nabla_X Z) \\ &= fX(g(Y, Z)) - fg(\nabla_X Y, Z) - fg(Y, \nabla_X Z) \\ &= f\nabla g(Y, Z, X). \end{aligned}$$

Showing ∇ is compatible with g is the same as showing $\nabla g = 0$, and since this is a tensor it is sufficient to show this in our coordinate frame.

We have

$$g(\nabla_{\partial_i} \partial_j, \partial_k) = g(\Gamma_{ij}^\ell \partial_\ell, \partial_k) = \Gamma_{ij}^\ell g_{\ell k}.$$

By our formula for Γ_{ij}^k in this coordinate frame, this is equal to

$$= \frac{1}{2} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}).$$

Thus

$$\begin{aligned} g(\nabla_{\partial_i} \partial_j, \partial_k) + g(\partial_j, \nabla_{\partial_i} \partial_k) &= \frac{1}{2} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij} + \partial_i g_{kj} + \partial_k g_{ij} - \partial_j g_{ik}) \\ &= \partial_i g_{jk} = \partial_i g(\partial_j, \partial_k). \end{aligned}$$

Showing that ∇ is indeed compatible with g .

As explained, our formula for ∇ defines a unique symmetric metric connection on each coordinate frame. Because it is unique, the formulas must agree where the coordinate frames intersect, thus allowing us to glue them together to give a global symmetric metric connection.

6.0.3 Problem

Let $(M, g), (N, h)$ be Riemannian manifolds, and $\alpha: M \rightarrow N$ an isometric immersion. Let $\bar{\nabla}$ be the Levi-Civita connection on N , and $\tilde{\nabla}$ be its pullback along α . Consider the orthogonal projection $\Pi_q: T_{\alpha q}N \rightarrow d\alpha_q(T_qM)$ for $q \in M$, and let $P_q: T_{\alpha q}N \rightarrow T_qM$ denote its composition with the isomorphism $d\alpha_q(T_qM) \cong T_qM$.

For $X \in \mathfrak{X}(M)$ define $\hat{X} \in \mathfrak{X}(\alpha)$ by $\hat{X}_q = d\alpha_q(X_q)$.

- (1) Show there exists a unique $C^\infty(M)$ -linear map $P: \mathfrak{X}(\alpha) \rightarrow \mathfrak{X}(M)$ such that

$$P(X)_q = P_q(X_q)$$

for all $q \in M, X \in \mathfrak{X}(\alpha)$.

- (2) Define a connection ∇ on M by $\nabla X = P \circ \tilde{\nabla} \hat{X}$. Prove that ∇ is the Levi-Civita connection on M .

- (1) Uniqueness is clear as the formula given defines P . For $f^1, f^2 \in C^\infty(M)$ and $X_1, X_2 \in \mathfrak{X}(\alpha)$ we have, by linearity of P_q ,

$$P(f^i X_i)_q = P_q(f^i(q) X_i(q)) = f^i(q) P_q(X_i(q)) = (f^i P(X_i))_q,$$

so $P(f^i X_i) = f^i P(X_i)$ as desired.

Now we show that P maps into $\mathfrak{X}(M)$. By linearity it is sufficient to show this for local frames. Since α is an immersion, locally it has a coordinate representation $\bar{x} \mapsto (\bar{x}, \bar{0})$. Let $x^1, \dots, x^m$ be local coordinates for M around q and $y^1, \dots, y^n$ be local coordinates for N around $\alpha(q)$ giving this representation.

Then

$$d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) = \frac{\partial \alpha^j}{\partial x^i}(q) \frac{\partial}{\partial y^j} \Big|_{\alpha(q)},$$

and since under these coordinates α is the map $\bar{x} \mapsto (\bar{x}, \bar{0})$, its Jacobian is just $\partial \alpha^j / \partial x^i = \delta_{ij}$. Thus

$$d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) = \frac{\partial}{\partial y^i} \Big|_{\alpha q}.$$

And so $d\alpha_q(T_qM)$ is spanned by $\partial/\partial y^i|_{\alpha q}$ for $1 \leq i \leq m$. Let Π_p^N denote the orthogonal projection of T_pN onto the subspace spanned by $\partial/\partial y^i|_p$ for $1 \leq i \leq m$. This is a smooth map, as projecting is just the application of Gram-Schmidt which is smooth, and $\Pi_q = \Pi_{\alpha q}^N$. Thus

$$P(X)_q = P_q(X_q) = (d\alpha_q)^{-1}(\Pi_{\alpha q}^N X_q),$$

which is smooth in q as a composition of smooth maps. Thus $P(X)$ is a smooth vector field, as desired.

- (2) Recall that $\tilde{\nabla}$ is given by

$$\tilde{\nabla}_X(Y \circ \alpha)|_q = \bar{\nabla}_{\hat{X}_q} Y|_{\alpha q}$$

for $X \in \mathfrak{X}(M), Y \in \mathfrak{X}(N), q \in M$.

We first show that ∇ is indeed a connection; we utilize the fact that $\tilde{\nabla}$ is a connection itself. For $C^\infty(M)$ -linearity in the first argument we have

$$\nabla_{f^i X_i} Y = P(\tilde{\nabla}_{f^i X_i} \hat{Y}) = P(f^i \tilde{\nabla}_{X_i} \hat{Y}) = f^i P(\tilde{\nabla}_{X_i} \hat{Y}) = f^i \nabla_{X_i} Y,$$

where the second-to-last equality is due to P being $C^\infty(M)$ -linear. Now notice that $\hat{\bullet}$ is also $C^\infty(M)$ -linear:

$$\widehat{f^i X_i}_q = d\alpha_q(f^i(q) X_i|_q) = f^i(q) d\alpha_q(X_i|_q) = (f^i \hat{X}_i)_q.$$

This allows us to show the rest of the properties of connections. For $\mathbb{R}$ -linearity in the second argument we have

$$\nabla_X c^i Y_i = P(\tilde{\nabla}_X c^i \hat{Y}_i) = c^i P(\tilde{\nabla}_X \hat{Y}_i) = c^i \nabla_X Y_i.$$

And for Leibniz,

$$\nabla_X fY = P(\tilde{\nabla}_X f\hat{Y}) = P((Xf)\hat{Y} + f\tilde{\nabla}_X \hat{Y}) = (Xf)P(\hat{Y}) + f\nabla_X Y.$$

Now $P(\hat{Y}) = Y$, since $P(\hat{Y})_q = P_q(\hat{Y}_q) = P_q(d\alpha_q(Y_q))$, since $d\alpha_q(Y_q)$ is already in $d\alpha_q(T_q M)$ the orthogonal projection does not affect it, so this is just mapped to Y_q . Thus

$$\nabla_X fY = (Xf)Y + f\nabla_X Y$$

as desired.

Now we show that ∇ is symmetric. It is sufficient to show this for local frames of our choice since the torsion tensor ($\nabla_X Y - \nabla_Y X - [X, Y]$) is a tensor, and thus linear. In particular we will use the coordinate frame $\partial/\partial x_i|_q$ from the first point, so

$$\widehat{\frac{\partial}{\partial x^i}}_q = d\alpha_q\left(\frac{\partial}{\partial x^i}\Big|_q\right) = \frac{\partial}{\partial y^i}\Big|_{\alpha(q)},$$

meaning

$$\widehat{\frac{\partial}{\partial x^i}} = \frac{\partial}{\partial y^i} \circ \alpha.$$

Then

$$\nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j}\Big|_q = P_q\left(\tilde{\nabla}_{\partial/\partial x^i} \widehat{\frac{\partial}{\partial x^j}}\Big|_q\right).$$

Recall that $\widehat{\partial/\partial x^i} = \partial/\partial y^i \circ \alpha$, we have that

$$\tilde{\nabla}_{\partial/\partial x^i} \frac{\partial}{\partial y^j} \circ \alpha\Big|_q = \bar{\nabla}_{\partial/\partial y^i|_{\alpha q}} \frac{\partial}{\partial y^j}\Big|_{\alpha q} = \bar{\nabla}_{\partial/\partial y^i} \frac{\partial}{\partial y^j}\Big|_{\alpha q}.$$

If we let Γ_{ij}^k be the Christoffel symbols of (N, h) , then

$$\nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j}\Big|_q = P_q\left(\Gamma_{ij}^k(\alpha q) \frac{\partial}{\partial y^k}\Big|_{\alpha q}\right) = \Gamma_{ij}^k(\alpha q) P_q\left(\frac{\partial}{\partial y^k}\Big|_{\alpha q}\right).$$

All in all this means that

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = (\Gamma_{ij}^k \circ \alpha) P\left(\frac{\partial}{\partial y^k} \circ \alpha\right).$$

Since $\bar{\nabla}$ is symmetric, $\Gamma_{ij}^k = \Gamma_{ji}^k$, and thus this shows

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i}.$$

Since the Lie bracket of coordinate vector fields is zero, this gives the desired result: ∇ is symmetric.

Finally we must show that ∇ is compatible with M 's metric. Again this can be checked locally on a local frame of our choice (since ∇g is a tensor, as shown in our previous solution). We remain with our local coordinate frame as before, so we need to show

$$\frac{\partial}{\partial x^i} \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g + \left\langle \frac{\partial}{\partial x^j}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k} \right\rangle_g,$$

where $\langle \bullet, \bullet \rangle_g$ is just $g(\bullet, \bullet)$.

We first focus on the left-hand side. Notice that for $i, j, k \leq m$,

$$\frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \Big|_{\alpha q} = \frac{\partial}{\partial y^i} \Big|_{\alpha q} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h = d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h.$$

By definition this is just

$$= \frac{\partial}{\partial x^i} \Big|_q \left(\left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha \right).$$

Recall that $\partial/\partial y^i \circ \alpha = d\alpha \partial/\partial x^i$, so this is equal to

$$= \frac{\partial}{\partial x^i} \Big|_q \left\langle d\alpha \frac{\partial}{\partial x^j}, d\alpha \frac{\partial}{\partial x^k} \right\rangle_h,$$

and since α is an isometry, this is equal to

$$= \frac{\partial}{\partial x^i} \Big|_q \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g.$$

Thus we have shown

$$\frac{\partial}{\partial x^i} \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle = \frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle \circ \alpha.$$

Now we consider the right-hand side. Recall that we already computed $\nabla_{\partial/\partial x^i} \partial/\partial x^j$, and we apply that here:

$$\left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle (\Gamma_{ij}^\ell \circ \alpha) P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g = (\Gamma_{ij}^\ell \circ \alpha) \left\langle P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g.$$

Since α is an isometry, we have

$$\left\langle P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle d\alpha P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h.$$

By definition of P this is equal to

$$= \left\langle \Pi \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h.$$

Now Π is the orthogonal projection onto the subspace spanned by $\partial/\partial y^\ell$ for $1 \leq \ell \leq m$, and thus does not affect this inner product:

$$= \left\langle \frac{\partial}{\partial y^\ell} \circ \alpha, \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h = \left\langle \frac{\partial}{\partial y^\ell}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha.$$

All in all we have

$$\left\langle \nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left(\Gamma_{ij}^\ell \left\langle \frac{\partial}{\partial y^\ell}, \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha = \left\langle \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha.$$

Putting all this together, ∇ 's compatibility with M 's metric follows from $\bar{\nabla}$'s:

$$\begin{aligned} \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g + \left\langle \frac{\partial}{\partial x^j}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k} \right\rangle_g &= \left(\left\langle \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h + \left\langle \frac{\partial}{\partial y^j}, \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha \\ &= \left(\frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha \\ &= \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right). \end{aligned}$$

Since the Levi-Civita connection is unique, ∇ must be M 's.

6.0.4 Problem

- (1) Do Carmo Riemannian Geometry, page 180 problem 3a, 3b. Consider $\mathbb{R}^{n+1}$ with the symmetric

nondegenerate quadratic form

$$Q(x_0, \dots, x_n) = -x_0^2 + \sum_{i=1}^n x_i^2.$$

This induces a pseudo-Riemannian metric on $\mathbb{R}^{n+1}$, call the resulting pseudo-Riemannian manifold L^{n+1} . For $r > 0$, let $k = -1/r^2$, and define by H_k^n the embedded submanifold of L^{n+1} given by

$$H_k^n = \{\bar{x} \in L^{n+1} \mid x_0 > 0, Q(\bar{x}) = -r^2\}.$$

Show that for $x \in H_k^n$, the vector $\eta = x/r$ is normal to the tangent space $T_x H_k^n$. Further show that $(\eta, \eta) = -1$ and it is possible to extend it to a basis $\eta, b_1, \dots, b_{n+1}$ of L^{n+1} such that $(b_i, b_j) = \delta_{ij}$ and $(b_i, b_0) = 0$. Deduce that the restriction of L^{n+1} 's metric to H_k^n is positive definite, and thus forms a Riemannian manifold.

- (2) Do Carmo Riemannian Geometry, page 184 problem 7a, 7b. Define a map $f: H_{-1}^n \rightarrow D^n$, where

$$D^n = \left\{ \bar{x} \in \mathbb{R}^n \mid x_0 = 0, \sum_{i=1}^n x_i^2 < 1 \right\},$$

as follows. For $p \in H_{-1}^n \subseteq L^{n+1}$, form a line from p to $p_0 = (-1, 0, \dots, 0)$, and let $f(p)$ be the intersection of this line with D^n . Let $f(\bar{x}) = (0, \bar{u})$, and show

$$\begin{aligned} x_i &= \frac{2u_i}{1 - \sum_j u_j^2}, & i &= 1, \dots, n, \\ x_0 &= \frac{2}{1 - \sum_j u_j^2} - 1. \end{aligned}$$

Now show that

$$-(dx_0)^2 + \sum_{i=1}^n (dx_i)^2 = \frac{4 \sum_{i=1}^n (du_i)^2}{\left(1 - \sum_j u_j^2\right)^2}.$$

Conclude that f^{-1} induces a Riemannian metric on D^n by

$$g_{ij} = \frac{4\delta_{ij}}{\left(1 - \sum_j u_j^2\right)^2}.$$

- (3) Let

$$\mathbb{R}_+^n = \{\bar{x} \in \mathbb{R}^n \mid x_n > 0\},$$

and define a metric on it by

$$g_{ij}(\bar{x}) = \frac{\delta_{ij}}{x_n^2}.$$

Let $p_0 = (0, \dots, -1) \in \mathbb{R}^n$ and define $f: D^n \rightarrow \mathbb{R}_+^n$ by

$$f(p) = 2 \frac{p - p_0}{|p - p_0|^2} + p_0.$$

Show that if we equip D with the metric from the previous point, f forms an isometry.

- (4) Equip D^n with a metric given by

$$g_{ij}(\bar{y}) = \frac{\delta_{ij}}{1 - |y|^2} + \frac{y_i y_j}{(1 - |y|^2)^2}.$$

Define $h: H_{-1}^n \rightarrow D^n$ by $y_i = x_i/x_0$. Show that h is an isometry.

- (1) So H_k^n is the level set of $-r^2$ under Q (in the open subset $x_0 > 0$), which is embedded. Then for $x \in Q$, we know $T_x H_k^n \subseteq T_x L^{n+1} = L^{n+1}$ is the kernel of dQ_x . We compute Q 's differential:

$$dQ_x = (-2x_0, 2x_1, \dots, 2x_n),$$

so

$$T_x H_k^n = \ker dQ_x = (-x_0, x_1, \dots, x_n)^\perp,$$

where $\bullet^\perp$ is the orthogonal complement relative to the usual Euclidean metric. Notice though that

$$(-x_0, x_1, \dots, x_n) \cdot (y_0, \dots, y_n) = -x_0 y_0 + \sum_{i>0} x_i y_i = Q(x, y).$$

Thus

$$\ker dQ_x = \{y \in L^{n+1} \mid (x, y) = 0\},$$

and since $\eta = x/r$, $(x, y) = 0$ iff $(\eta, y) = 0$. Thus η indeed is normal to $T_x H_k^n$.

Further notice that

$$(\eta, \eta) = \frac{1}{r^2} Q(x) = -1.$$

If we extend η to an orthogonal basis of $(\bullet, \bullet)$ we obtain new vectors $b_1, \dots, b_n$ such that $(\eta, b_i) = 0$, and (b_i, b_j) is diagonal. Since the signature of a bilinear form is invariant, we must have that $(b_i, b_j) = \delta_{ij}$ as desired. Then $b_1, \dots, b_n$ forms a basis for $T_x H_k^n$ as they are orthogonal to η , and then $(\bullet, \bullet)$ on $T_x H_k^n$ must be positive definite, by linearity (b_i form a positive orthogonal basis). Thus Q forms a Riemannian metric on H_k^n as desired.

- (2) The line between $p \in H_{-1}^n$ and p_0 is given by $tp + (1-t)p_0$, which intersects D^n when

$$tx_0 + (t-1) = 0 \iff t = \frac{1}{1+x_0}.$$

Thus

$$u_k = \frac{x_k}{1+x_0}.$$

For $(0, u_1, \dots, u_n) \in D^n$, let $U = \sum_i u_i^2$. In our case

$$U = \sum_i \frac{x_i^2}{(1+x_0)^2} = \frac{1}{(1+x_0)^2} \sum_i x_i^2,$$

and since $x \in H_{-1}^n$, $\sum_i x_i^2 = x_0^2 - 1$ so

$$U = \frac{x_0^2 - 1}{(1+x_0)^2} = \frac{x_0 - 1}{x_0 + 1}, \quad 1 - U = \frac{2}{1+x_0}.$$

And then

$$\frac{2u_k}{1-U} = \frac{\frac{2x_k}{1+x_0}}{\frac{2}{1+x_0}} = x_k$$

and

$$\frac{2}{1-U} - 1 = \frac{2}{\frac{2}{1+x_0}} - 1 = x_0$$

as desired.

Now for $X^i \partial / \partial x^i \in TH_k^n$, recall from the previous point that we have $Q(X, x) = 0$, so

$$X^i x_i = X^0 x_0.$$

Now we show that for $X^i \partial / \partial x^i, Y^j \frac{\partial}{\partial x^j} \in TH_k^n \subseteq TL^{n+1}$,

$$g\left(df\left(X^i \frac{\partial}{\partial x^i}\right), df\left(Y^j \frac{\partial}{\partial x^j}\right)\right) = \left(X^i \frac{\partial}{\partial x^i}, Y^j \frac{\partial}{\partial x^j}\right),$$

i.e. that f is an isometry. Since f can be extended to an open subset of L^{n+1} to map to L^{n+1} , we use linearity to compute $df(X^i \partial/\partial x^i)$:

$$df\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial u^j}{\partial x^i} \frac{\partial}{\partial u^j}.$$

Now,

$$\begin{aligned} \frac{\partial u^j}{\partial x^j} &= \frac{1}{x_0 + 1}, \\ \frac{\partial u^j}{\partial x^i} &= 0, \quad \text{for } i \neq j, 0, \\ \frac{\partial u^j}{\partial x^0} &= -\frac{x_i}{(x_0 + 1)^2}. \end{aligned}$$

Thus

$$\begin{aligned} df \frac{\partial}{\partial x^i} &= \frac{1}{x_0 + 1} \frac{\partial}{\partial u^i}, \quad \text{for } i \neq 0, \\ df \frac{\partial}{\partial x^0} &= -\frac{x_i}{(x_0 + 1)^2} \frac{\partial}{\partial u^i}. \end{aligned}$$

So

$$\begin{aligned} g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) &= \frac{1}{(1 + x_0)^2} g_{ij} = \delta_{ij} \frac{1}{(1 + x_0)^2} \frac{4}{(1 - U)^2} = \delta_{ij}, \\ g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^0}\right) &= -\frac{x_i}{(1 + x_0)^3} g_{ii} = -\frac{x_i}{(1 + x_0)^3} \frac{4}{(1 - U)^2} = -\frac{x_i}{1 + x_0}, \\ g\left(df \frac{\partial}{\partial x^0}, df \frac{\partial}{\partial x^0}\right) &= \frac{x_i^2}{(1 + x_0)^4} g_{ii} = \frac{x_i^2}{(1 + x_0)^4} \frac{4}{(1 - U)^2} = \frac{x_0 - 1}{x_0 + 1}. \end{aligned}$$

And finally we get

$$\begin{aligned} g\left(X^i df \frac{\partial}{\partial x^i}, Y^j df \frac{\partial}{\partial x^j}\right) &= X^i Y^j g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) = \\ &= \sum_{i \geq 1} X^i Y^i - \sum_{j \geq 1} \frac{x_j}{1 + x_0} X^0 Y^j - \sum_{i \geq 1} \frac{x_i}{1 + x_0} X^i Y^0 + X^0 Y^0 \frac{x_0 - 1}{x_0 + 1} \end{aligned}$$

Since

$$\left(X^i \frac{\partial}{\partial x^i}, Y^j \frac{\partial}{\partial x^j}\right) = -X^0 Y^0 + \sum_i X^i Y^i,$$

we want to show that

$$X^0 Y^0 = X^0 \sum_j \frac{x_j}{1 + x_0} Y^j + Y^0 \sum_i \frac{x_i}{1 + x_0} X^i - X^0 Y^0 \frac{x_0 - 1}{x_0 + 1}.$$

Recall that $X^i x_i = X^0 x_0$, and thus the right-hand side is equal to

$$2X^0 Y^0 \frac{x_0}{x_0 + 1} - X^0 Y^0 \frac{x_0 - 1}{x_0 + 1} = X^0 Y^0$$

as desired.

- (3) Let $\bar{g}$ denote the metric on D^n , so $\bar{g}_{ij} = 4 \frac{\delta_{ij}}{(1-U)^2}$. The idea here is the same as the other points: compute the Jacobian of f , and show that it maps isometrically. Here we can focus only on the coordinate frame $\partial/\partial u^i$ since D^n is open, and thus the coordinate frame spans its tangent space. Let $f(u_1, \dots, u_n) = (x_1, \dots, x_n)$, then we have

$$df \frac{\partial}{\partial u^i} = \frac{\partial x^j}{\partial u^i} \frac{\partial}{\partial x^j},$$

and we want to show that

$$\bar{g}_{ij} = g\left(df \frac{\partial}{\partial u^i}, df \frac{\partial}{\partial u^j}\right) = \frac{\partial x^k}{\partial u^i} \frac{\partial x^\ell}{\partial u^j} g\left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell}\right) = \sum_k \frac{\partial x^k}{\partial u^i} \frac{\partial x^k}{\partial u^j} \frac{1}{x_n^2}.$$

The computations here became too messy for me to do, but that's the idea.

- (4) Let $\bar{g}$ denote the metric on H_{-1}^n , so we want to show that for $X = X^i \partial / \partial x^i, Y = Y^j \partial / \partial x^j \in T_x H_k^n$,

$$-X^0 Y^0 + \sum_i X^i Y^i = \bar{g}(X, Y) = g(dh(X), dh(Y)).$$

Of course we need to compute $dh \frac{\partial}{\partial x^i}$,

$$dh \frac{\partial}{\partial x^i} = \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j},$$

so

$$\begin{aligned} dh \frac{\partial}{\partial x^i} &= \frac{1}{x_0} \frac{\partial}{\partial y^i} \\ dh \frac{\partial}{\partial x^0} &= - \sum_i \frac{x_i}{x_0^2} \frac{\partial}{\partial y^i}. \end{aligned}$$

And so

$$\begin{aligned} g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^j}\right) &= \frac{1}{x_0^2} g_{ij}, \\ g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^0}\right) &= - \sum_j \frac{x_j}{x_0^3} g_{ij}, \\ g\left(dh \frac{\partial}{\partial x^0}, dh \frac{\partial}{\partial x^0}\right) &= \sum_{i,j} \frac{x_i x_j}{x_0^4} g_{ij}. \end{aligned}$$

And thus

$$g(dhX, dhY) = X^i Y^j g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^j}\right) = X^i Y^j \frac{1}{x_0^2} g_{ij} - X^0 Y^i \frac{x_j}{x_0^3} g_{ij} - Y^0 X^i \frac{x_i}{x_0^3} g_{ij} + X^0 Y^0 \frac{x_i x_j}{x_0^4} g_{ij}.$$

Note that since $x \in H_k^n$,

$$|y|^2 = \sum_i y_i^2 = \sum_i \frac{x_i^2}{x_0^2} = \frac{x_0^2 - 1}{x_0^2} \implies 1 - |y|^2 = \frac{1}{x_0^2}.$$

Thus

$$g_{ij} = \delta_{ij} x_0^2 + x_i x_j x_0^2.$$

And so

$$g(dhX, dhY) = (\delta_{ij} + x_i x_j) \left(X^i Y^j - X^0 Y^i \frac{x_j}{x_0} - Y^0 X^i \frac{x_j}{x_0} + X^0 Y^0 \frac{x_i x_j}{x_0^2} \right).$$

So we want to show that

$$\begin{aligned} (*) \quad -X^0 Y^0 &= -X^0 \sum_i Y^i \frac{x_i}{x_0} - Y^0 \sum_i X^i \frac{x_i}{x_0} + X^0 Y^0 \sum_i \frac{x_i^2}{x_0^2} \\ &\quad + \sum_{i,j} x_i x_j \left(X^i Y^j - X^0 Y^i \frac{x_j}{x_0} - Y^0 X^i \frac{x_j}{x_0} + X^0 Y^0 \frac{x_i x_j}{x_0^2} \right). \end{aligned}$$

Note that

$$\sum_{i,j} Y^i x_i x_j^2 = Y^0 x_0 \sum_j x_j^2 = Y^0 x_0 (x_0^2 - 1).$$

Thus the right-hand side of (*) is equal to

$$-X^0 Y^0 - X^0 Y^0 + X^0 Y^0 \frac{x_0^2 - 1}{x_0^2} + x_0^2 X^0 Y^0 - (x_0^2 - 1) X^0 Y^0 - (x_0^2 - 1) X^0 Y^0 + X^0 Y^0 \frac{(x_0^2 - 1)^2}{x_0^2}.$$

All this just simplifies to give $-X^0 Y^0$, as desired.

7 Homework 7

7.0.1 Problem (DoCarmo, Riemannian Geometry page 56 question 2)

Let X and Y be vector fields on a manifold M equipped with a connection ∇ . Let $p \in M$ and $\gamma: I \rightarrow M$ an integral curve of X through p , so $\gamma(t_0) = p$ and $\dot{\gamma}(t) = X_{\gamma(t)}$. Denote the parallel transport map by $P_{t_0 t_1}^\gamma$, show that

$$\nabla_X Y|_p = \left. \frac{d}{dt} \right|_{t=t_0} (P_{t_0 t}^\gamma)^{-1}(Y|_{\gamma(t)}).$$

First we prove a few basic claims regarding parallel transport (which I'm not sure were proven in class, I apologize).

Firstly, $P_{t_0 t_1}^\gamma: T_{\gamma(t_0)}M \rightarrow T_{\gamma(t_1)}M$ is linear. Indeed, if $v \in T_{\gamma(t_0)}M$ is a vector and $X \in \mathfrak{X}(\gamma)$ is the vector field parallel along γ such that $X(t_0) = v$, then for $c \in \mathbb{R}$, cX is parallel along γ and $cX(t_0) = cv$. Indeed, since D_t is $\mathbb{R}$ -linear we have $D_t(cX) = cD_tX = 0$. Thus $P_{t_0 t_1}^\gamma(cv) = (cX)(t_1) = cX(t_1) = cP_{t_0 t_1}^\gamma(v)$.

And similarly it is additive. If $v, u \in T_{\gamma(t_0)}M$ are vectors with respective parallel vector fields $X, Y \in \mathfrak{X}(\gamma)$ starting at v, u respectively, then $X + Y \in \mathfrak{X}(\gamma)$ is parallel along γ since $D_t(X + Y) = D_tX + D_tY = 0$, and $(X + Y)(t_0) = v + u$. Thus

$$P_{t_0 t_1}^\gamma(v + u) = (X + Y)(t_1) = X(t_1) + Y(t_1) = P_{t_0 t_1}^\gamma(v) + P_{t_0 t_1}^\gamma(u).$$

Also $P_{t_0 t_1}^\gamma$ is invertible with inverse $P_{t_1 t_0}^\gamma$. Indeed if $u \in T_{\gamma(t_1)}M$ has vector field $X \in \mathfrak{X}(\gamma)$ parallel such that $X(t_1) = u$, then $P_{t_1 t_0}^\gamma(u) = X(t_0) = v$. Since X is parallel along γ and $X(t_0) = v$, we have $P_{t_0 t_1}^\gamma(v) = X(t_1) = u$. Thus

$$P_{t_0 t_1}^\gamma \circ P_{t_1 t_0}^\gamma = \text{id},$$

and by symmetry the swapped composition is the identity as well, so they are inverses.

Now if $v \in T_{\gamma(t_0)}M$, let $X_v \in \mathfrak{X}(\gamma)$ be the unique vector field parallel along γ such that $X_v(t_0) = v$. That is, $X_v(t) = P_{t_0 t}^\gamma(v)$ by definition of the parallel transport map. Now if $b_1, \dots, b_n \in T_{\gamma(t_0)}M$ form a basis, then $X_{b_1}, \dots, X_{b_n}$ form a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields. They are a local frame since $X_{b_i}|_t = P_{t_0 t}^\gamma(b_i)$, and since $P_{t_0 t}^\gamma$ is a linear isomorphism it maps bases to bases, so $X_{b_i}|_t$ is a basis for each t . In particular there exists a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields.

Now getting to the task at hand, notice that by definition of the pullback connection,

$$D_t(Y \circ \gamma)|_t = \nabla_{\dot{\gamma}(t)} Y|_{\gamma(t)} = \nabla_{X \circ \gamma(t)} Y|_{\gamma(t)} = \nabla_X Y|_{\gamma(t)}.$$

Thus since $p = \gamma(t_0)$,

$$\nabla_X Y|_p = D_t(Y \circ \gamma)|_{t_0}.$$

Let $E_1, \dots, E_n$ be a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields (which exists as shown above). Since $Y \circ \gamma \in \mathfrak{X}(\gamma)$, there are smooth $Y^1, \dots, Y^n: I \rightarrow \mathbb{R}$ such that $Y \circ \gamma = Y^i E_i$. Then since $D_t(E_i) = 0$ we have

$$D_t(Y \circ \gamma) = D_t(Y^i E_i) = \dot{Y}^i E_i + Y^i D_t(E_i) = \dot{Y}^i E_i.$$

And since E_i are parallel,

$$P_{t t_0}^\gamma(Y \circ \gamma(t)) = P_{t t_0}^\gamma(Y^i(t) E_i(t)) = Y^i(t) P_{t t_0}^\gamma(E_i(t)) = Y^i(t) E_i(t_0)$$

Putting everything together we have

$$\left. \frac{d}{dt} \right|_{t=t_0} P_{t t_0}^\gamma(Y \circ \gamma(t)) = \left. \frac{d}{dt} \right|_{t=t_0} (Y^i(t) E_i(t_0)) = \dot{Y}^i(t_0) E_i(t_0) = D_t(Y \circ \gamma)|_{t_0} = \nabla_X Y|_p.$$

Keeping in mind that $(P_{t_0 t}^\gamma)^{-1} = P_{t t_0}^\gamma$, we have the desired result.

7.0.2 Problem

Let (M, g) be a Riemannian manifold and $S \subseteq M$ a compact submanifold.

- (1) Define the **normal subbundle** ν_S by

$$\nu_S = \bigsqcup_{p \in S} T_p S^\perp.$$

And denote $\pi: \nu_S \rightarrow S$ to be the obvious projection operator. Show that ν_S is a submanifold of TM and π is smooth.

- (2) Define $\exp: \nu_S \rightarrow M$ to be the restriction of the exponential operator to ν_S . Show that there is an open subset $U \subseteq \nu_S$ with $S \cong \{(p, 0) \in \nu_S \mid p \in S\} \subseteq U$ such that $\exp(U)$ is open and $\exp$ restricts to a diffeomorphism $U \rightarrow \exp(U)$. Thus $T = \exp(U) \subseteq M$ is an open neighborhood of S , and we have a retract $r: T \rightarrow S$ given by $r = \pi \circ (\exp|_U)^{-1}$. T is called a **tubular neighborhood**.
- (3) Let $f: M \rightarrow N$ be a proper submersion. Let $p \in N$ and $S = f^{-1}p$, so that $S \subseteq M$ is a compact submanifold since f is a proper submersion. Let T, r be as in the previous point. Show that there exists an open neighborhood $p \in V \subseteq N$ and $f^{-1}V \subseteq T$. Define $g: f^{-1}V \rightarrow V \times S$ by $q \mapsto (f(q), r(q))$. Show that after possibly shrinking V , g is a diffeomorphism, and that f is a locally trivial fibration.

- (1) The restriction of the tangent bundle $TM|_S = \bigsqcup_{p \in S} T_p M$ is a vector bundle over S , and $TS \subseteq TM|_S$ is a subbundle. Let $p \in S$ and let U be a neighborhood of p in S over which there is a local frame $E_1, \dots, E_k$ of TS . By potentially shrinking U we can extend this to a local frame $E_1, \dots, E_k, E_{k+1}, \dots, E_n$ of $TM|_S$. If we then apply the Gram-Schmidt process to the E_i s, they remain smooth (since the inner product of two vector fields is a smooth function), and we obtain an orthonormal local frame $E'_1, \dots, E'_n$. The Gram-Schmidt process doesn't change the linear span of $E'_1, \dots, E'_i$ for all i , so we see that $E'_1, \dots, E'_k$ form an orthonormal local frame for TS and $E'_{k+1}, \dots, E'_n$ span ν_S (over U).

Thus in some neighborhood around every $p \in S$ there are local sections $E'_{k+1}, \dots, E'_n$ which span ν_S at the points where they are defined. And thus ν_S is a subbundle of $TM|_S$, and since $TM|_S$ is embedded in TM , we see that ν_S is embedded in TM as well.

And the projection map $\pi: TM \rightarrow M$ restricts along its domain and codomain to $\pi: \nu_S \rightarrow S$. Since these are both embedded submanifolds, π remains smooth.

- (2) Let us denote by $\exp^M$ the exponential map (from a subset of) $TM \rightarrow M$, so $\exp$ is the restriction of $\exp^M$ to ν_S . And let $S_0 = \{(p, 0) \in \nu_S \mid p \in S\} \subseteq \nu_S$. Consider the zero section $\zeta: S \rightarrow \nu_S$ given by $p \mapsto 0 \in T_p S^\perp$, whose image is S_0 . This is smooth as the restriction of the zero vector field $M \rightarrow TM$ along embedded submanifolds. We have that $\pi \circ \zeta = \text{id}$, so ζ is a topological embedding as $\pi|_{S_0}$ is its continuous inverse from its image. And $d\pi_{(p,0)} \circ d\zeta_p = \text{id}$ for all $p \in S$ so $d\zeta_p$ is injective, thus ζ is an immersion. As an immersive topological embedding, it is a smooth embedding, and so its image, S_0 , is an embedded submanifold of S_0 . Since S is compact, so is S_0 and thus it is closed in ν_S .

Let $p \in S$, we want to show that $d(\exp)_{p,0}: T_{(p,0)}\nu_S \rightarrow T_p M$ is surjective. We consider $\exp$'s action on two embedded submanifolds of ν_S : S_0 and $T_p S^\perp$ (the latter is embedded as it is the level set of p under the submersion π). Notice that on S_0 , $\exp$ maps $(p, 0)$ to p and thus is the composition of the diffeomorphism $S_0 \rightarrow S$ with the embedding $S \hookrightarrow M$. In particular, $d\exp_{(p,0)}$ maps $T_{(p,0)}S_0$ isomorphically to $T_p S \subseteq T_p M$. And on $T_p S^\perp$, $\exp$ agrees with $\exp_p^M$ which is a diffeomorphism around 0. Thus $d\exp_{(p,0)}$ is equal to $d(\exp_p^M)_0$ and thus maps $T_{(p,0)}T_p S^\perp$ (viewed as a subspace of $T_0 T_p M \cong T_p M$) isomorphically to $T_p S^\perp$ (since $d(\exp_p^M)_0$ is identified with the identity). Since $T_p M = T_p S \oplus T_p S^\perp$, we see that $d\exp_{(p,0)}$ is indeed surjective.

Now, μ_S is a vector bundle over S of rank $\dim M - \dim S$ (since this is the dimension of $T_p S^\perp$). Thus its dimension is $\dim \mu_S = \dim M - \dim S + \dim S = \dim M$. And so each $d\exp_{(p,0)}$ is an isomorphism as its domain and codomain have the same dimensions and it is surjective. So by the inverse function theorem, $\exp$ is a diffeomorphism from a neighborhood of $(p, 0) \in U_p \subseteq \nu_S$ to a neighborhood of $p \in V_p \subseteq M$.

By potentially shrinking them, we can assume each U_p is a precompact coordinate ball. And since S is compact and the U_p cover it, we take a finite subcover $U_{p_1}, \dots, U_{p_n}$. Let $U_0 = \bigcup_i U_{p_i}$, which remains precompact as the finite union of precompact sets. And $\exp$ is a local diffeomorphism $U_0 \rightarrow \bigcup_p V_p = V_0$ which is open.

Now we make the following claim: if $A \subseteq X$ is closed in a compact Hausdorff space X , and $f: X \rightarrow Y$ is locally injective and injective on A , then there is a neighborhood U of A such that f is injective on U . Suppose not, then for every neighborhood U of A there are $x_U \neq y_U$ such that $f(x_U) = f(y_U)$. This forms two nets $x_\bullet, y_\bullet$, and since X is compact they have convergent subnets, $x_{U_i} \rightarrow x$ and $y_{U_i} \rightarrow y$ (we can assume they are over the same directed set). This means that for any neighborhood $x \in V$, eventually x_{U_i} is in V and in particular $V \cap U_i$ is non-empty for some i . Since the U_i are cofinal, this means that every neighborhood of x intersects every neighborhood of A . Since X is compact Hausdorff and A is closed, A is compact, and in Hausdorff spaces disjoint compact sets have disjoint neighborhoods. Thus x must be in A , and similarly y is in A .

Since f is continuous, $f(x_{U_i}) \rightarrow f(x)$ and $f(y_{U_i}) \rightarrow f(y)$ and since $f(x_{U_i}) = f(y_{U_i})$, we see that $f(x) = f(y)$ (by uniqueness of limits in a Hausdorff space). By assumption f is injective on A so $x = y$. And by local injectivity there is a neighborhood U of $x = y$ on which f is injective. Eventually x_{U_i}, y_{U_i} are in this neighborhood and since $f(x_{U_i}) = f(y_{U_i})$ we must have $x_{U_i} = y_{U_i}$, a contradiction. (This argument can be done using second-countability or metrizability and sequences instead of nets.)

Going back to our problem, we invoke the above result with $X = \overline{U}_0$, $A = S_0$, and $\exp$. So there is a neighborhood U of S_0 on which $\exp$ is injective. Thus $\exp: U \rightarrow \exp(U)$ is a bijective local diffeomorphism, and thus a diffeomorphism. Since $\exp: U_0 \rightarrow V_0$ is a local diffeomorphism to an open subset $V_0 \subseteq M$, it is in particular an open map and thus $\exp(U)$ is open.

And $r = \pi \circ (\exp|_U)^{-1}$ is indeed a retract $T \rightarrow S$ since it is smooth as the composition of smooth maps, and for $p \in S$, $(\exp|_U)^{-1}(p) = (p, 0)$ and thus $r(p) = p$ as desired.

(3) Let us define

$$g: T \rightarrow N \times S, \quad q \mapsto (f(q), r(q)).$$

Then relative to a product chart of $N \times S$, g 's differential is given by

$$dg_q = (df_q, dr_q).$$

So $\ker dg_q = \ker df_q \cap \ker dr_q$. And we know that since S is f 's level set, $\ker df_q = T_q S$ for all $q \in S$. And we have $r|_S = \text{id}_S$, so dr_q acts on $T_q S$ as the identity, thus $\ker df_q \cap \ker dr_q = 0$. So dg_q is injective for all $q \in S$, and thus g has full rank at q .

Furthermore, $\dim S = \dim M - \dim N$ and thus

$$\dim(N \times S) = \dim N + \dim S = \dim M = \dim T,$$

so that dg_q is an isomorphism for $q \in S$. By the inverse function theorem there is an open neighborhood $q \in U_p \subseteq T$ such that g restricts to a diffeomorphism onto its image on U_p . By the same argument as above (when restricting $\exp$ to a diffeomorphism around S_0), g restricts to a diffeomorphism on an open neighborhood $S \subseteq W \subseteq T$.

Now we claim that there is a neighborhood $p \in V$ such that $f^{-1}V \subseteq W$. Suppose not, so focusing on a coordinate chart around p , let $B_{1/n}$ be a coordinate ball (in this coordinate chart) of radius $1/n$ around p . Then $f^{-1}B_{1/n}$ is not a subset of W . So let $x_n \in f^{-1}B_{1/n} - W$ for each n . Since x_n are all contained in the compact $\overline{B}_1$, there is a convergent subsequence $x_{m_n} \rightarrow x$. By continuity, $f(x_{m_n}) \rightarrow f(x)$ and since $f(x_{m_n}) \in B_{1/m_n}$ which is decreasing, we see that $f(x) \in \bigcap_n B_{1/m_n} = \{p\}$ so that $f(x) = p$. Thus $x \in S$, and since W is a neighborhood of S and $x_{m_n} \rightarrow x$, we must have that eventually x_{m_n} is in W , contradicting our choice of x_n to not be in W .

So letting $p \in V$ be a neighborhood such that $f^{-1}V \subseteq W$, we restrict g to $f^{-1}V$. Since g 's restriction to W is a diffeomorphism onto its image, g 's restriction to $f^{-1}V$ is a diffeomorphism as well. Furthermore since $g(f^{-1}V) = V \times S$, we have that $g: f^{-1}V \rightarrow V \times S$ is a diffeomorphism.

And since

$$\text{proj}_V \circ g(q) = \text{proj}_V(f(q), r(q)) = f(q),$$

we have that with $F_p = S = f^{-1}p$ and the diffeomorphism given by g for all $p \in N$, f is a locally trivial fibration as desired.

Let G be a Lie group with Lie algebra $\mathfrak{g}$ equipped with a bi-invariant metric $\langle \bullet, \bullet \rangle$. Show that G 's geodesics starting at the identity are precisely its one-parameter subgroups.

Let $\gamma: \mathbb{R} \rightarrow G$ be the one-parameter subgroup of a left-invariant vector field $Y \in \mathfrak{g}$, so $\gamma'(t) = Y_{\gamma(t)}$. Then its acceleration is

$$D_t \gamma'|_t = \nabla_{\partial/\partial t}^\gamma (Y \circ \gamma)|_t = \nabla_{\gamma'(t)} (Y)|_{\gamma(t)} = \nabla_Y Y|_{\gamma(t)}.$$

So it is sufficient to show that $\nabla_Y Y = 0$ for all left-invariant vector fields $Y \in \mathfrak{g}$.

Koszul's formula gives

$$2\langle X, \nabla_Y Y \rangle = 2Y\langle X, Y \rangle - X\langle Y, Y \rangle + 2\langle Y, [X, Y] \rangle - \langle X, [Y, Y] \rangle.$$

Since $\langle \bullet, \bullet \rangle$ is left-invariant, $\langle X, Y \rangle, \langle Y, Y \rangle$ are constant and thus $Y\langle X, Y \rangle = X\langle Y, Y \rangle = 0$. Since $[Y, Y] = 0$ as well, we have

$$\langle X, \nabla_Y Y \rangle = \langle Y, [X, Y] \rangle.$$

Bi-invariance of the metric implies that for all left-invariant vector fields,

$$\langle [X, Y], Z \rangle = -\langle X, [Z, Y] \rangle.$$

And in particular

$$\langle Y, [X, Y] \rangle = -\langle [Y, Y], X \rangle = 0.$$

So all in all we have that $\langle X, \nabla_Y Y \rangle = 0$ for all left-invariant X . The Levi-Civita connection is left-invariant, so $\nabla_Y Y \in \mathfrak{g}$ and thus it must be zero. So we have shown that its one-parameter subgroup is a geodesic.

If γ is a geodesic starting at e , then $\gamma'(0) \in T_e G$ induces a left-invariant vector field Y . As shown above, Y 's one-parameter subgroup is a geodesic starting at e with initial velocity $Y_e = \gamma'(0)$. By uniqueness of geodesics, γ must be equal to Y 's one-parameter subgroup, as desired.

7.0.4 Problem

Let $f: X \rightarrow Z$ and $g: Y \rightarrow Z$ be smooth maps. We say that f and g are **transverse** to one another, and write $f \pitchfork g$, if for all $x \in X$ and $y \in Y$ such that $f(x) = g(y)$ we have $df_x(T_x X) + dg_y(T_y Y) = T_{f(x)} Z$.

- (1) If $f \pitchfork g$ are transverse to one another, show that they define a categorical pullback $X \times_Z Y$ in the category of smooth manifolds.
- (2) Let $k = \mathbb{R}, \mathbb{C}$, and $\pi_V: V \rightarrow M$ be a smooth vector bundle over k . Show that π_V is a submersion.
- (3) Let $f: N \rightarrow M$ be a smooth map, show that $N \times_M V$ is a smooth manifold.
- (4) Show that there is a smooth map $V \times_M V \rightarrow V$ given by fiberwise addition.
- (5) Let $\underline{k} \rightarrow M$ denote the trivial bundle $k \times M \rightarrow M$. Show that there is a smooth map $\underline{k} \times_M V \rightarrow V$ given by fiberwise scalar multiplication.

- (1) Let $\Delta \subseteq Z \times Z$ denote the diagonal. This is an embedded submanifold as the image of the injective embedding $Z \rightarrow Z \times Z, z \mapsto (z, z)$. This is clearly a topological embedding, and smooth, and it is an immersion since its differential is represented by the matrix $(\text{id}, \text{id})^\top$ in local coordinates (obtained by choosing a coordinate chart in Z , then the product of that coordinate chart with itself is a coordinate chart of its image under the diagonal map, and this matrix is the representation of the differential); which is of full rank.

So Δ is the injective image of a smooth embedding, and is thus an embedded submanifold of $Z \times Z$.

Let $f \times g: X \times Y \rightarrow Z \times Z$ be given coordinate-wise. Notice that

$$(f \times g)^{-1} \Delta = \{(x, y) \in X \times Y \mid (f(x), g(y)) \in \Delta\} = \{(x, y) \mid f(x) = g(y)\} = X \times_Z Y.$$

So to show that $X \times_Z Y$ is an embedded submanifold of $X \times Y$ (and in particular a smooth manifold), it is sufficient to show that $f \times g$ is transverse to Δ .

Indeed, for $fx = gy$ we have, under the identification $T_{x,y}X \times Y \cong T_xX \times T_yY$,

$$d(f \times g)_{x,y}(T_{x,y}X \times Y) \cong df_x(T_xX) \times dg_y(T_yY).$$

Let $z = fx = gy$, and $T_{z,z}\Delta$ is then the diagonal of $T_{z,z}Z \times Z$. This is because if we let $\iota_\Delta: Z \rightarrow Z \times Z$ be the diagonal map, then $T_{z,z}\Delta = d(\iota_\Delta)_z T_z Z$. As explained, ι_Δ 's differential is the matrix $(\text{id}, \text{id})^\top$, which maps $v \in T_z Z$ to (v, v) . Thus $d(\iota_\Delta)_z T_z Z$ is indeed the diagonal of $T_z Z \times T_z Z \cong T_{z,z}Z \times Z$.

In general if we have vector subspaces $U, W \subseteq V$ such that $U + W = V$, then $U \times W + \Delta = V \times V$, where $\Delta \subseteq V \times V$ is the diagonal. For $(x, y) \in V \times V$ we want $u \in U, w \in W, v \in V$ such that $(u + v, w + v) = (x, y)$. This requires $u - w = x - y$; and since $U + W = V$, such u, w exist. Then let $v = x - u$ so $u + v = x$ and $w + v = u + y - x + x - u = y$ as desired.

In our case, $V = T_z Z$, $U = df_x(T_x)$, and $W = dg_y(T_y)$, so indeed we have that $f \times g$ is transverse to Δ and $X \times_Z Y$ is an embedded submanifold of $X \times Y$.

Since it is an embedded submanifold, we can restrict the projection maps $X \times Y \rightarrow X, Y$ to it. Thus the square

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow{p_Y} & Y \\ p_X \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Z \end{array}$$

Indeed commutes since

$$g \circ p_Y(x, y) = g(y) = f(x) = f \circ p_X(x, y).$$

And this indeed is a pullback: if $U \xrightarrow{h} X, U \xrightarrow{k} Y$ commute with f, g , to form $\ell: U \rightarrow X \times_Z Y$ we must have

$$\ell(u) = (h(u), k(u)).$$

(This can be shown directly, also because $X \times_Z Y$ is the pullback in the category of topological spaces.) This is smooth as a map to $X \times Y$, and since $X \times_Z Y$ is an embedded submanifold, we can restrict it to its image, which is $X \times_Z Y$, and it remains smooth. Thus ℓ is unique (since $X \times_Z Y$ is the pullback in the category of topological spaces) and it satisfies the property.

Since the pullback is a universal property; it is unique up to unique diffeomorphism. In particular, if we had another smooth structure on $X \times_Z Y$ which made it the pullback along with p_X, p_Y , then the ℓ defined above would be a diffeomorphism. But in this case

$$\ell(x, y) = \ell(p_X(x, y), p_Y(x, y)) = (x, y) \implies \ell = \text{id}.$$

So id would be a diffeomorphism, meaning they have the same smooth structure.

- (2) Let $\Phi: \pi_V^{-1}U \rightarrow U \times \mathbb{R}^n$ be a local trivialization, where $U \subseteq M$ is open. So it is a diffeomorphism and $\text{proj}_1 \circ \Phi = \pi_V$, where $\text{proj}_1: U \times \mathbb{R}^n \rightarrow U$ is the projection map onto the first factor. Thus for $v \in \pi_V^{-1}U$,

$$d(\pi_V)_v = d(\text{proj}_1)_{\Phi v} \circ d\Phi_v.$$

Since Φ is a diffeomorphism and proj_1 a submersion, $d(\pi_V)_v$ is surjective for all $v \in \pi_V^{-1}U$.

The local trivializations cover V (for $v \in V$, $\pi_V(v)$ is contained in an open subset U over which there is a local trivialization, and $v \in \pi_V^{-1}U$), so $d(\pi_V)_v$ is surjective for all $v \in V$. That means π_V is a submersion.

- (3) Submersions are transverse to every map: $d(\pi_V)_v(T_v V) = T_{\pi_V v} M$ since it is a submersion. In particular $d(\pi_V)_v(T_v V) + df_q(T_q N) = T_{\pi_V v} M$ whenever $f q = \pi_V v$. So $\pi_V \pitchfork f$, and we showed in the first point that this means $N \times_M V$ is a smooth manifold.
- (4) Suppose V has rank n . That is, its local trivializations are maps of the form $\Phi: \pi_V^{-1}U \rightarrow U \times \mathbb{R}^n$ and V has dimension $m + n$ where $m = \dim M$.

We know that $V \times_M V$'s codimension in $V \times V$ is equal to Δ 's (the diagonal's) codimension in $M \times M$ (by the first point). Since $\dim \Delta = \dim M = m$ this tells us

$$\dim(V \times V) - \dim(V \times_M V) = \dim(M \times M) - \dim \Delta \implies \dim(V \times_M V) = 2(m + n) - m = 2n + m.$$

And explicitly,

$$V \times_M V = \{(v, u) \in V \times V \mid \pi_V(v) = \pi_V(u)\} = \bigsqcup_{p \in M} V_p \times V_p,$$

where $V_p = \pi_V^{-1}p$. We want to show that

$$\text{add}: V \times_M V \rightarrow V, \quad (v, u) \mapsto v + u$$

is smooth. It is well-defined since V_p is a vector space by assumption, so we can add vectors in it.

Now, for $(v, u) \in V_p \times V_p$ they both lie in the domain of the same trivialization, say $\Phi: \bar{U} \rightarrow U \times \mathbb{R}^n$ where $\bar{U} = \pi_V^{-1}U$. Let us denote by $\Phi_\ell: \bar{U} \rightarrow \mathbb{R}^n$ Φ 's projection onto the second factor. Since $V \times_M V$ is embedded, $(\bar{U} \times \bar{U}) \cap (V \times_M V)$ is open in $V \times_M V$ (as it is the intersection with an open set). And define

$$\Psi: (\bar{U} \times \bar{U}) \cap (V \times_M V) \rightarrow U \times k^n \times k^n, \quad (v, u) \mapsto (\pi_V(v), \Phi_\ell(v), \Phi_\ell(u)).$$

This is a smooth map, as it is clearly smooth coming from the open set $(\bar{U} \times \bar{U}) \cap (V \times V)$ (as all the coordinates are smooth), and restricting to $V \times_M V$ is restricting to an embedded submanifold, and thus remains smooth.

Ψ is also bijective. It is injective since if $\Psi(v, u) = \Psi(v', u')$ then $\pi_V(v) = \pi_V(u) = \pi_V(v') = \pi_V(u') = p$. Then $\Phi_\ell(v) = \Phi_\ell(v')$, but $\Phi_\ell|_{V_p}$ is a linear isomorphism and thus injective, so $v = v'$. Similar for $u = u'$. And it is surjective: for $(p, a, b) \in U \times k^n \times k^n$, since $\Phi_\ell|_{V_p}$ is a linear isomorphism and thus surjective, there are $v, u \in V_p$ such that $\Phi_\ell(v) = a, \Phi_\ell(u) = b$. Then

$$\Psi(v, u) = (\pi_V(v), \Phi_\ell(v), \Phi_\ell(u)) = (p, a, b)$$

as desired.

Finally, notice that (choosing a product chart)

$$d\Psi = (d\pi_V, d\Phi_\ell, d\Phi_\ell).$$

Since Φ is a diffeomorphism and Φ_ℓ is its projection onto the first factor, it is the composition of submersions (Φ with projection), and thus a submersion. So π_V, Φ_ℓ are all submersions, and thus of ranks m, n respectively. Thus $d\Psi$ has rank $2m + n = \dim U \times k^{2n}$, so Ψ is an immersion and submersion, thus a local diffeomorphism. It is also bijective, and thus a diffeomorphism.

Now consider the composition

$$U \times k^n \times k^n \xrightarrow{\Psi^{-1}} V \times_M V \xrightarrow{\text{add}} V \xrightarrow{\Phi} U \times k^n, \quad (p, a, b) \mapsto (p, a + b).$$

This is clearly smooth (addition in k^n is smooth). Since Φ, Ψ are diffeomorphisms, this means that add is smooth when restricted to $\bar{U} \times \bar{U}$. Since local trivializations cover V , such open sets $\bar{U} \times \bar{U}$ cover $V \times_M V$, and thus add is smooth as desired.

(5) Explicitly we have

$$\underline{k} \times_M V = \{(c, p, v) \in \underline{k} \times V \mid \pi_V(v) = p\}.$$

And we want to show that

$$\text{mult}: \underline{k} \times_M V \rightarrow V, \quad (c, p, v) \mapsto cv$$

is smooth.

Notice that $\underline{k} \times_M V$ is diffeomorphic to $k \times V$. Define

$$\begin{aligned} F: \underline{k} \times_M V &\rightarrow k \times V, & (c, p, v) &\mapsto (c, v), \\ G: k \times V &\rightarrow \underline{k} \times_M V, & (c, v) &\mapsto (c, \pi_V(v), v). \end{aligned}$$

These are both smooth, since F is the restriction of the obvious map $\underline{k} \times V \rightarrow k \times V$ (given by the same formula) which is smooth (as its coordinates are) to an embedded submanifold on its domain. And G is the restriction of the obvious map $k \times V \rightarrow \underline{k} \times V$ (given by the same formula), which is smooth as its coordinates are, to an embedded submanifold on its codomain.

And F, G are inverses:

$$F \circ G(c, v) = F(c, \pi_V(v), v) = (c, v), \quad G \circ F(c, p, v) = G(c, v) = (c, \pi_V(v), v) = (c, p, v).$$

Thus they form diffeomorphisms.

So to show mult is smooth it is sufficient to show $\text{mult} \circ G: k \times V \rightarrow V$ is smooth. We identify mult with $G \circ \text{mult}$, so it is given by

$$\text{mult}(c, v) = cv.$$

Let $\Phi: \bar{U} \rightarrow U \times k^n$ be a local trivialization. Then consider the composition

$$k \times U \times k^n \xrightarrow{\text{id} \times \Phi^{-1}} k \times \bar{U} \xrightarrow{\text{mult}} \bar{U} \xrightarrow{\Phi} U \times k^n.$$

(The target of mult remains $\bar{U}$ since for $v \in \bar{U} = \pi_V^{-1}U$ we have $\pi_V(cv) = \pi_V(v) = p \in U$ so $cv \in \bar{U}$.) This is given by

$$(c, p, a) \mapsto (c, \Phi^{-1}(p, a)) \mapsto c\Phi^{-1}(p, a) \mapsto (p, ca).$$

The final mapping is because Φ is a linear isomorphism on the fibers, so

$$\Phi(c\Phi^{-1}(p, a)) = c(p, a) = (p, ca).$$

Multiplication over k is smooth, so this map is smooth.

Thus mult is smooth over $k \times \bar{U}$, and since local trivializations cover V , it is smooth over all of $k \times V$. And as explained, this means the original mult , defined on $\underline{k} \times_M V$, is smooth.

8 Homework 8

8.0.1 Problem (DoCarmo, Riemannian Geometry, page 83 problem 7)

Let M be a Riemannian manifold, and $p \in M$. Show that there is a neighborhood U of p in M and vector fields $E_1, \dots, E_n \in \mathfrak{X}(U)$ forming an orthonormal local frame such that $\nabla_{E_i} E_j|_p = 0$ for all i, j . Such a frame is called a **local geodesic frame**.

Let B be a geodesic ball around p , so of the form $B = \exp_p(B_\varepsilon(0))$ for an open ball $B_\varepsilon(0) \subseteq T_p M$ contained in the domain of $\exp_p$. Furthermore let $v_1, \dots, v_n \in T_p M$ form an orthonormal basis. Now for every $q \in B$ consider the radial geodesic from p to q , so if $q = \exp_p(w)$ for $w \in B_\varepsilon(0)$ the radial geodesic is given by $\gamma_q: t \mapsto \exp_p(tw)$. Then define $E_i|_q$ to be the parallel transport of $v_i \in T_p M$ along γ_q :

$$E_i|_q = P_{0,1}^{\gamma_q}(v_i).$$

Parallel transport along a curve is a linear isometry. If γ is a curve, $v, u \in T_p M$, and X, Y are their parallel transports along γ , then by definition $P_{0,1}^\gamma(v) = X(1)$ so

$$\langle P_{0,1}^\gamma v, P_{0,1}^\gamma u \rangle = \langle X(1), Y(1) \rangle.$$

On the other hand $\langle X(t), Y(t) \rangle$ is constant along γ , since

$$\frac{d}{dt} \langle X(t), Y(t) \rangle = \langle D_t X, Y \rangle + \langle X, D_t Y \rangle = 0.$$

And so

$$\langle P_{0,1}^\gamma v, P_{0,1}^\gamma u \rangle = \langle X(1), Y(1) \rangle = \langle X(0), Y(0) \rangle = \langle v, u \rangle,$$

as desired. Since $E_i|_q$ is the parallel transport of v_i along γ_q , this means the $E_i|_q$ remain orthonormal.

So we just need to show that the E_i are smooth, and $\nabla_{E_i} E_j|_p = 0$. Consider the definition of E_i : for $q \in B$ we compute $w = \exp_p^{-1}(q)$ and then the parallel transport of v_i along $\gamma = \exp_p(tw)$. We will show that the parallel transport of $v = v_i$ along γ is smooth in w . The parallel transport is given by $X(1)$ where X solves the ODE

$$\frac{\partial X^i}{\partial t}(t) + X^i(t) \dot{\gamma}^j(t) \Gamma_{ij}^k(\gamma(t)) = 0,$$

since γ depends smoothly on w , the solution to this ODE depends smoothly on w . And w depends smoothly on q since $\exp_p^{-1}$ is a diffeomorphism, so in total X depends smoothly on q . And thus $E_i|_q = X(1)$ depends smoothly on q , so E_i is smooth.

To show that $\nabla_{E_i} E_j|_p = 0$, we have that

$$\nabla_{E_i} E_j|_p = \nabla_{E_i(p)} E_j = \nabla_{v_i} E_j.$$

Now let α_i be the geodesic starting at p with initial velocity v_i , so $\alpha_i = \exp_p(tv_i)$. So this is equal to

$$= \nabla_{\alpha_i'(0)} E_j = D_t|_{t=0} (E_j \circ \alpha_i),$$

with the covariant derivative taken along α_i . Now, $E_j(\alpha_i(t))$ is the parallel transport of v_j along $\gamma_{\alpha_i(t)} = \alpha_i$, and thus it is by definition parallel along α_i and its covariant derivative is zero, as desired.

8.0.2 Problem

- (1) Let $O(1, n)$ denote the subgroup of linear transformations of $\mathbb{R}^{n+1}$ which preserve the Lorentz bilinear form. Let $O_+(1, n)$ denote the set of matrices A_j^i in $O(1, n)$ with $A_0^0 > 0$. Show that $O_+(1, n)$ is a subgroup of $O(1, n)$ and its transformations restrict to isometries of H_k^n .
- (2) Determine the geodesics in the hyperboloid model of hyperbolic space as follows. Let P be a two-dimensional linear subspace of $\mathbb{R}^{n+1}$ containing the point $p \in H_k^n$. Show that there is an element of $O_+(1, n)$ which fixes P and acts non-trivially on $\mathbb{R}^{n+1} - P$. Conclude that all geodesics through p can be obtained as intersections of $P \cap H_k^n$ for some P as above.
- (3) Show that the geodesics of point (2) are sent by the isometry $h: H_1^n \rightarrow D^n$ (where D^n is the Klein model of hyperbolic space) to straight lines.
- (4) Show that geodesics in the Poincaré model of hyperbolic space are circles perpendicular to the boundary of the disk.
- (5) Show by example that angles measured with respect to the Klein metric on the disk are not the same as with respect to the Euclidean metric on the disk. On the other hand, show that angles on the Poincaré disk are the same as the Euclidean metric.

- (1) Let us denote by $Q = \text{diag}(-1, 1, \dots, 1)$ the bilinear form, so $(x, y) = x^\top Q y$. If we write $x = (a, u)$ and $y = (b, v)$ then we see $(x, y) = -ab + \langle u, v \rangle$ where $\langle \bullet, \bullet \rangle$ denotes the Euclidean product on $\mathbb{R}^n$.

Then $A \in O(1, n)$ iff

$$(Ax, Ay) = x^\top A^\top Q A y = x^\top Q y = (x, y),$$

i.e. iff $A^\top Q A = Q$. Since Q is its own inverse, clearly $O(1, n)$ consists of invertible matrices. Clearly then $O(1, n)$ is closed under multiplication and contains the identity, and by multiplying $A^\top Q A = Q$ by $(A^{-1})^\top$ and A^{-1} we see it is closed under inversion, so it is a group. Furthermore we see that $A \in O(1, n)$ iff

$$A^{-1} = Q A^\top Q.$$

Then by taking the transpose of both sides we see that this implies

$$(A^\top)^{-1} = (A^{-1})^\top = Q A Q,$$

so that $O(1, n)$ is further closed under transposition.

Now to show that $O_+(1, n)$ is a subgroup, we first note that it clearly contains the identity. And if $A \in O_+(1, n)$ then since $A^{-1} = Q A^\top Q$, we see that $(A^{-1})_0^0 = A_0^0 > 0$ so $O_+(1, n)$ is closed under inversion. And suppose that $A \in O(1, n)$ and let

$$A = \begin{pmatrix} a & u^\top \\ v & * \end{pmatrix}.$$

Then, letting e_i be the standard basis vectors,

$$-1 = (e_0, e_0) = (Ae_0, Ae_0) = ((a, v), (a, v)) = -a^2 + |v|^2 \implies a^2 = |v|^2 + 1,$$

where the norm is the Euclidean norm. In particular we have that $a > |v|$. Since $O(1, n)$ is closed under transposition, we see that $a^2 = |u|^2 + 1$ as well. If $A \in O_+(1, n)$ so that $a > 0$ we see that $a > |u|, |v|$. Now suppose we have

$$A = \begin{pmatrix} a & u^\top \\ v & * \end{pmatrix}, B = \begin{pmatrix} b & z^\top \\ w & * \end{pmatrix} \in O_+(1, n),$$

we want to show that $AB \in O_+(1, n)$. We have

$$AB = \begin{pmatrix} ab + u^\top w & * \\ * & * \end{pmatrix},$$

so we want to show that $ab + u^\top w > 0$. By Cauchy-Schwarz we have

$$ab + u^\top w \geq ab - |u||w| > ab - ab = 0,$$

where the final inequality is because we showed $a > |u|$ and $b > |w|$ above. Thus $O_+(1, n)$ is indeed a subgroup.

By definition $O(1, n)$, and thus $O_+(1, n)$, are isometries of Lorentzian space. Since H_k^n is an embedded submanifold of Lorentzian space, if we can show that maps of $O_+(1, n)$ restrict to it then we will have finished. We want to show that if $A \in O_+(1, n)$ and $x \in H_k^n$ then $Ax \in H_k^n$, so $(Ax)^0 > 0$ and $Q(Ax) = 1/k$. Since A is an isometry, $Q(Ax) = Q(x) = 1/k$ so we need only show that $(Ax)^0 > 0$.

Since $x \in H_k^n$ we have that $x^0 > 0$ and $Q(x) < 0$. Let $x = (x_0, y)$ so $Q(x) = -x_0^2 + |y|^2 < 0$ meaning $x_0 > |y|$. Now let

$$A = \begin{pmatrix} a & u^\top \\ v & * \end{pmatrix},$$

then $(Ax)^0 = ax_0 + u^\top y$ and by Cauchy-Schwarz,

$$(Ax)^0 = ax_0 + u^\top y \geq ax_0 - |u||y| > ax_0 - ax_0 = 0,$$

as desired.

- (2) First we claim that $A \in O(1, n)$ is in $O_+(1, n)$ iff it preserves the sign of the first component of any vector x with $(x) < 0$. If $A \in O_+(1, n)$ then in the previous point we showed that if $x^0 > 0$ and $(x) < 0$ then $(Ax)^0 > 0$ as well. And if $x^0 < 0$ and $(x) < 0$ then $-x^0 > 0$ and $(-x) = (x) < 0$ so $(-Ax)^0 > 0$ so $(Ax)^0 < 0$, so A preserves the sign of the first component all such vectors. And conversely if A preserves the sign of all such vectors, then in particular it does for e_0 , and $(Ae_0)^0 = A_0^0 > 0$ so $A \in O_+(1, n)$.

We also claim that $A \in O_+(1, n)$ iff it preserves the sign of the first component of a *single* x with $(x) < 0$. This is because if $A \notin O_+(1, n)$ then $-A \in O_+(1, n)$ (since $(Ae_0)^0 = -1$ we must have $A_0^0 \neq 0$), so $-A$ must preserve the sign of x . But $(-Ax)^0 = -Ax^0$ which differs from the sign of x since A preserves it, a contradiction.

Let $P \leq \mathbb{R}^{n+1}$ be two-dimensional containing p . Letting x_0 be the normalization of p we can extend x_0 to an orthonormal basis x_0, x_1 of P . This in turn we extend to an orthonormal basis $x_0, \dots, x_n$ of $\mathbb{R}^{n+1}$. By Sylvester's law of inertia, $(x_i, x_j) = \pm \delta_{ij}$ with the negative sign occurring only when $i = j = 0$. Now restricting the bilinear form to $P^\perp = \text{span}\{x_2, \dots, x_n\}$ makes it an inner product since $(x_i, x_j) = \delta_{ij}$ for $i, j \geq 2$. So define A on $P^\perp$ to be a unitary transformation with no fixed points, and to be the identity on P . This preserves the bilinear form, so it is in $O(1, n)$. Furthermore it preserves the sign of x_0 , and since $(x_0) < 0$, by above this means $A \in O_+(1, n)$.

Thus by definition $A \in O_+(1, n)$ fixes P and for every $x \in \mathbb{R}^{n+1} - P$ we can write $x = x' + x''$ for $x' \in P, x'' \in P^\perp$, then $Ax = x' + Ax'' \neq x' + x''$, so every point outside of P is acted on non-trivially.

Now if γ is a geodesic in H_k^n passing through p with velocity v at p . So $\gamma(t_0) = p$ and $\gamma'(t_0) = v \in T_p H_k^n$, which means that $(v, p) = 0$. Let P be spanned by p, v (since $(v, p) = 0$ they are linearly independent so P is two-dimensional). Let $A \in O_+(1, n)$ be as above, so $Ap = p, Av = v$. As A is an isometry, $A \circ \gamma$ is a geodesic of H_k^n , and

$$A \circ \gamma(t_0) = p, \quad (A \circ \gamma)'(t_0) = dA_p \circ \gamma'(t_0) = v,$$

since as a matrix, A 's differential can be identified with A . So by uniqueness, $A \circ \gamma = \gamma$, and since A 's fixed points are P this means that γ 's image lies in $P \cap H_k^n$.

Now, $P \cap H_k^n$ are vectors $\alpha p + \beta v$ such that

$$(\alpha p + \beta v) = (\alpha p) + (\beta v) = -r^2 \alpha^2 + \beta^2 (v) = -r^2,$$

where $r^2 = -1/k$. These are points along a hyperbola, and thus γ 's image is the image of an open interval in this hyperbola. Its image must be the entire hyperbola, since otherwise by parameterizing the hyperbola ($\alpha(t) = \cosh(t)$ and $\beta(t) = r \sinh(t)/\sqrt{(v)}$ since $(v) > 0$) we can take the first t not in γ 's image. Letting $q = \alpha(t)p + \beta(t)v \in H_k^n \cap P$ not be in γ 's image, and suppose γ is defined on (a, b) . But then $\gamma(t) \rightarrow q$ as t approaches the endpoint of γ 's domain, wlog b .

Let W be a uniformly normal neighborhood of q , then eventually $\gamma(t)$ is in W . So eventually $\gamma(t)$ is contained in W , let t_0 be such that $\gamma(t) \in W$ for $t \geq t_0$, and $s = \gamma(t_0)$. Then $\gamma(t) = \exp_s(tw)$ for some $w \in T_s H_k^n$. Since $\gamma(t) \rightarrow q$ this means that $\exp_s(tw) \rightarrow q$, and $q \in W$ so $q = \exp_s(u)$ for some tangent vector u (since W is uniformly normal). Thus $tw \rightarrow u$, and thus $u = bw$, so we can extend γ to its value at its endpoint to be $\gamma(b) = q$ and this remains smooth (since it is given by $\exp_s(tw)$). This is a contradiction to γ 's maximality. Thus γ 's image is the entirety of $P \cap H_k^n$.

- (3) We want to show that h maps $P \cap H_{-1}^n$ to a straight line. Elements of this intersection are vectors $(p_0, p) \in H_{-1}^n$ which are also in P . Since a k -dimensional subspace of n -dimensional space can be written as the kernel of an $(n - k) \times n$ matrix, and thus can be written as solutions to a system of $n - k$ homogeneous linear equations, elements of P are solutions to a system of $n - 1$ linear equations. Now (p_0, p) solves a linear equation $\alpha p_0 + \beta^i p_i = 0$ iff $h(p_0, p) = p/p_0 = q$ solves $\beta^i q_i = -\alpha$. These equations define an affine subspace of $\mathbb{R}^n$, thus $h(P \cap H_{-1}^n) = h(P) \cap D^n$ is the intersection of D^n with an affine subspace of $\mathbb{R}^n$. Since $h(P \cap H_{-1}^n)$ is the image of a geodesic and thus a curve, it is one-dimensional and thus must be the intersection of D^n with a line, as desired.

- (4) First we consider the two-dimensional case, and we consider the stereographic projection $f: H_{-1}^n \rightarrow D^n$ given by

$$f(p_0, p) = \frac{p}{1 + p_0}.$$

This is an isometry, and thus maps geodesics to geodesics. Since geodesics in H_{-1}^n are the solutions to $n - 1$ linear equations, we first restrict to the two-dimensional case, so geodesics are solutions to a single linear equation $-\alpha p_0 + \beta^i p_i = 0$, rewriting it as $\alpha p_0 = \langle \beta, p \rangle$ where the inner product is Euclidean. So given a geodesic parameterized by p , $u = f(p)$ is a geodesic.

If $\alpha = 0$ then $\langle \beta, p \rangle = 0$ iff $\langle \beta, f(p_0, p) \rangle = 0$, so $f(p_0, p)$ forms a straight line orthogonal to β . This is a straight line through the origin and thus orthogonal to the boundary of the disk.

If $\alpha \neq 0$ then by normalizing α and β we can assume $\alpha = 1$ so our equation is $p_0 = \langle \beta, p \rangle$. If we consider f^{-1} given by

$$f^{-1}(u) = \left(\frac{2}{1 - |u|^2} - 1, \frac{2u}{1 - |u|^2} \right),$$

then points on the geodesic are the points $f^{-1}(u)$ where

$$\frac{2}{1 - |u|^2} - 1 = \left\langle \beta, \frac{2u}{1 - |u|^2} \right\rangle \iff |u|^2 - 2\langle \beta, u \rangle + 1 = 0.$$

This can be written as

$$|u - \beta|^2 = |\beta|^2 - 1.$$

If $|\beta|^2 \leq 1$ then this has either no solution or the solution is $u = \beta$ when $|\beta| = 1$ in which case this lies on the boundary of the disk and not within the disk itself.

Otherwise this defines a circle of radius $\sqrt{|\beta|^2 - 1}$ with center β , so the geodesic is the intersection of that circle with D^n . Now consider the intersection of this circle with the boundary of D^n , so the u_0 s where $|u_0 - \beta|^2 = |\beta|^2 - 1$, $|u_0| = 1$. To show that the geodesic is orthogonal to the disk, we show that $0, u_0, \beta$ form a right triangle, that is we want to show $\langle u_0, u_0 - \beta \rangle = 0$. We have that

$$|\beta|^2 - 1 = |u_0|^2 - 2\langle u_0, \beta \rangle + |\beta|^2 \implies \langle u_0, \beta \rangle = 1,$$

and thus

$$\langle u_0, u_0 - \beta \rangle = |u_0|^2 - \langle u_0, \beta \rangle = 1 - \langle u_0, \beta \rangle = 0,$$

as desired.

In the higher-dimension case, a geodesic of H_{-1}^n is determined by a hyperbola on a two-dimensional affine subspace, say $\alpha p + \beta q = v$. Writing our coordinates as $(p_0, p_1, \dots, p_n)$, we can perform an orthogonal transformation on $p_1, \dots, p_n$ (which does not change the metric clearly) to map this affine subspace down to one existing in the coordinates $(p_0, p_1, p_2, 0, \dots, 0)$. Then we are in the same situation as the two-dimensional case.

- (5) To measure the angle between two geodesics γ_1, γ_2 intersecting at a point $p = \gamma_1(0) = \gamma_2(0)$, we compute the angle between $\gamma'_1(0), \gamma'_2(0) \in T_p M$ using the Riemannian metric. In the Klein model, the geodesic connecting any two points is the same as in Euclidean space. Thus to measure the angle between any three points, we are measuring the angle between the same tangent vectors.

If we take an arbitrary point, $p \in D^n$ and two tangent vectors $v, u \in T_p D^n$ then the angle between them is given by

$$\cos \theta = \frac{\langle u, v \rangle}{|u| |v|}.$$

Let us take $p = (1/2, 1/2) \in D^2$ and $u = e_1, v = e_2$ be the standard basis vectors. Then in the Euclidean metric, u and v are orthogonal and thus $\cos \theta = 0$. But in the Klein metric,

$$k_{12} = \frac{p_1 p_2}{(1 - |p|^2)^2} \neq 0,$$

and so e_1, e_2 are not orthogonal, so the angle between them is not 90° , different than in the Euclidean case.

8.0.3 Problem

Let $M \subseteq \mathbb{R}^{n+1}$ be the graph of the smooth function $f: \mathbb{R}^n \rightarrow \mathbb{R}$,

$$M = \{(x_0, \bar{x}) \mid x_0 = f(\bar{x})\}.$$

Give $\mathbb{R}^{n+1}$ either the Euclidean or Lorentzian metric, $G_{ij} = \pm \delta_{ij}$ where the negative sign occurs only potentially when $i = j = 0$. Let g be the pullback of G to M , and let $\mathbf{x}: \mathbb{R}^n \rightarrow M$ be the coordinates

$$\mathbf{x}(x_1, \dots, x_n) = (f(x_1, \dots, x_n), x_1, \dots, x_n).$$

Write $f_i = \partial/\partial x^i f$.

- (1) Show that, with respect to $\mathbf{x}$, $g_{ij} = \delta_{ij} \pm f_i f_j$. Conclude that in a small enough neighborhood U of a critical point of x , g defines a Riemannian metric on $\mathbf{x}(U)$. We implicitly make this restriction for the rest of the problem.
- (2) With respect to $\mathbf{x}$ show that g 's Christoffel symbols are

$$\Gamma_{ij}^k = \pm f_{ij} f_\ell g^{\ell k}.$$

- (3) Show that for a critical point x of f , we have

$$R_{ijk\ell} = \pm(f_{ik} f_{j\ell} - f_{i\ell} f_{jk}).$$

- (1) Let us denote by ∂_i the coordinate frame of Euclidean space, then M 's coordinate frame with respect to $\mathbf{x}$ is given by $\tilde{\partial}_i = d\mathbf{x}(\partial_i)$. As $M \subseteq \mathbb{R}^{n+1}$ is embedded, we can compute this relative to the coordinate frame of $\mathbb{R}^{n+1}$ (also denoted ∂_i),

$$\tilde{\partial}_i = \frac{\partial \mathbf{x}^j}{\partial \partial_i} \partial_j = f_i \partial_0 + \partial_i.$$

Then since g is the pullback of G , we can compute the inner product in Euclidean space, so we have

$$g_{ij} = \langle \tilde{\partial}_i, \tilde{\partial}_j \rangle = \langle f_i \partial_0 + \partial_i, f_j \partial_0 + \partial_j \rangle = f_i f_j G_{00} + G_{ij} = \delta_{ij} \pm f_i f_j,$$

as desired. A critical point x of f is one where $f_i(x) = 0$ for all i , and by smoothness there is a neighborhood of x where $|f_i f_j| < 1$ for all i, j . In such a neighborhood $g_{ij} > 0$ and thus g is Riemannian, as the pullback of G it is already symmetric.

(2) We recall

$$\Gamma_{ij}^k = \frac{1}{2}g^{k\ell}(\tilde{\partial}_i g_{j\ell} + \tilde{\partial}_j g_{i\ell} - \tilde{\partial}_\ell g_{ij}).$$

And $\tilde{\partial}_i = d\mathbf{x}(\partial_i)$ acts on smooth functions by

$$\tilde{\partial}_i f = \partial_i(f \circ \mathbf{x}),$$

thus

$$\tilde{\partial}_i g_{j\ell} = \partial_i(\delta_{ij} \pm f_j f_\ell) = \pm f_{ij} f_\ell \pm f_j f_{i\ell}.$$

Plugging this in we see

$$\Gamma_{ij}^k = \pm \frac{1}{2}g^{k\ell}(f_{ij} f_\ell + f_j f_{i\ell} + f_{ij} f_\ell + f_i f_{j\ell} - f_{i\ell} f_j - f_i f_{j\ell}) = \pm f_{ij} f_\ell g^{k\ell},$$

as desired ($g^{k\ell}$ is symmetric).

(3) If x is a critical point of f , then $f_i = 0$ for all i . Thus $\Gamma_{ij}^k = 0$ and $g_{ij} = \delta_{ij}$ for all i, j, k . We know the components of Riemannian curvature are given by

$$R_{ijkl} = -g_{\ell m}(\tilde{\partial}_i \Gamma_{jk}^m - \tilde{\partial}_j \Gamma_{ik}^m + \Gamma_{jk}^p \Gamma_{ip}^m - \Gamma_{ik}^p \Gamma_{jp}^m) = \tilde{\partial}_j \Gamma_{ik}^\ell - \tilde{\partial}_i \Gamma_{jk}^\ell.$$

And

$$\tilde{\partial}_i \Gamma_{jk}^\ell = \pm \partial_i(f_{jk} f_m g^{m\ell}) = \pm(f_{ijk} f_m g^{m\ell} + f_{jk} f_{im} g^{m\ell} + f_{jk} f_m \partial_i g^{m\ell}) = \pm f_{jk} f_{im} \delta_{m\ell} = \pm f_{jk} f_{i\ell}.$$

Thus plugging in we see

$$R_{ijkl} = \pm(f_{ik} f_{j\ell} - f_{jk} f_{i\ell}).$$

8.0.4 Problem

Let M be a Riemannian manifold, and $x, y \in T_p M$ tangent vectors. Let γ_t be the loop in M based at p obtained by applying γ_t to the parallelogram in $T_p M$ spanned by tX, tY . Parallel transport along a closed loop is called **holonomy**, and thus this defines an operator

$$\text{Hol}_{\gamma_t}: T_p M \rightarrow T_p M.$$

Show that

$$R(X, Y) = \left. \frac{d}{dt} \right|_{t=0} \text{Hol}_{\gamma_{\sqrt{t}}}.$$

Let us define the smooth family of curves $\Gamma: I \times I \rightarrow M$ by $\Gamma(s, t) = \exp(sx + ty)$ for I an open interval containing 0 small enough so that this is defined. We have $\partial_s \Gamma(0, 0) = x$ and $\partial_t \Gamma(0, 0) = y$. The holonomy operator of γ_a is obtained via parallel transport along $\Gamma(s, 0)$ then $\Gamma(a, t)$ then $\Gamma(a - s, a)$ then $\Gamma(0, a - t)$.

Now in general if we have a curve $\gamma: [0, a] \rightarrow M$, then we can define its reverse $-\gamma: [0, a] \rightarrow M$ by $t \mapsto \gamma(a - t)$. Then if X is parallel along γ then X^- is parallel along $-\gamma$ where $X^-(t) = X(a - t)$. This is because for $X \circ \gamma$ extendible $(X \circ \gamma)^- = X \circ (-\gamma)$ and then

$$D_t^{-\gamma}(X \circ (-\gamma))|_t = \nabla_{-\gamma'(a-t)} X = -\nabla_{\gamma'(a-t)} X = -D_t^\gamma(X \circ \gamma)|_t.$$

And so in general $D_t^{-\gamma}(X^-) = -D_t^\gamma(X)$ so clearly X is parallel to γ iff X^- is parallel to $-\gamma$.

Then we have that $P_{a,0}^\gamma = P_{0,a}^{-\gamma}$ since if X is the parallel transport of $v \in T_{\gamma(a)} M = T_{-\gamma(0)} M$ along γ then X^- is its parallel transport along $-\gamma$ and $X(0) = X^-(a)$. Thus we have

$$(P_{a,0}^\gamma)^{-1} = P_{0,a}^\gamma = P_{a,0}^{-\gamma}.$$

So parallel transport along $\Gamma(a - s, a)$ is the inverse of parallel transport along $\Gamma(s, a)$, and parallel transport along $\Gamma(0, a - t)$ is the inverse of parallel transport along $\Gamma(0, t)$.

Now let $v \in T_p M$, and let V be the vector field along Γ obtained by parallel transport of v first along $s \mapsto \Gamma(s, 0)$ and then along $t \mapsto \Gamma(s, t)$. Because parallel transport is the unique solution to a linear ODE and is thus smooth in its input, this is a smooth vector field along Γ . And we have $V(0, 0) = v$, since $s \mapsto V(s, 0)$ is parallel along $\Gamma(s, 0)$ we have that $D_s V(s, 0) = 0$. And $t \mapsto V(s, t)$ is parallel along $t \mapsto \Gamma(s, t)$ so $D_t V(s, t) = 0$ for all s, t . We also know that

$$R(x, y)v = (R(\partial_s \Gamma, \partial_t \Gamma)V)|_{(0,0)} = D_t D_s V(0, 0) - D_s D_t V(0, 0) = D_t D_s V(0, 0).$$

So we just need to compute $D_t D_s V(0, 0)$.

Previously we showed that for a vector field X along a curve γ ,

$$D_t X|_0 = \frac{d}{dt} \Big|_{t=0} (P_{0,t}^\gamma)^{-1}(X(t)) = \lim_{\varepsilon \rightarrow 0} \frac{P_{0,\varepsilon}^{\gamma(\varepsilon-t)}(X(\varepsilon)) - X(0)}{\varepsilon}.$$

And so we have

$$D_s V(0, t) = \lim_{\varepsilon \rightarrow 0} \frac{P_{0,\varepsilon}^{\Gamma(\varepsilon-s,t)}(V(\varepsilon, t)) - V(0, t)}{\varepsilon},$$

and similarly

$$D_t D_s V(0, 0) = \lim_{\delta \rightarrow 0} \frac{P_{0,\delta}^{\Gamma(0,\delta-t)}(D_s V(0, \delta)) - D_s V(0, 0)}{\delta}.$$

Because linear maps are continuous, plugging in the expression for $D_s V(0, \delta)$ in the second expression gives

$$D_t D_s V(0, 0) = \lim_{\varepsilon, \delta \rightarrow 0} \frac{P_{0,\delta}^{\Gamma(0,\delta-t)} \circ P_{0,\varepsilon}^{\Gamma(\varepsilon-s,\delta)}(V(\varepsilon, \delta)) - P_{0,\delta}^{\Gamma(0,\delta-t)}(V(0, \delta)) - P_{0,\varepsilon}^{\Gamma(\varepsilon-s,0)}(V(\varepsilon, 0)) + V(0, 0)}{\delta \varepsilon}.$$

Since by definition $s \mapsto V(s, t)$ and $t \mapsto V(t, 0)$ are parallel we see that

$$\begin{aligned} V(\varepsilon, \delta) &= P_{0,\delta}^{\Gamma(\varepsilon,t)} \circ P_{0,\varepsilon}^{\Gamma(s,0)} v, \\ P_{0,\delta}^{\Gamma(0,\delta-t)} V(0, \delta) &= V(0, 0) = v, \\ P_{0,\varepsilon}^{\Gamma(\varepsilon-s,0)} V(\varepsilon, 0) &= V(0, 0) = v. \end{aligned}$$

The first line is the definition, and the second two lines follow by $P^{-\gamma}$ and P^γ being inverses. Thus we see

$$D_s D_t V(0, 0) = \lim_{\varepsilon, \delta \rightarrow 0} \frac{P_{0,\delta}^{\Gamma(0,\delta-t)} \circ P_{0,\varepsilon}^{\Gamma(\varepsilon-s,\delta)} \circ P_{0,\delta}^{\Gamma(\varepsilon,t)} \circ P_{0,\varepsilon}^{\Gamma(s,0)} v - v}{\delta \varepsilon}.$$

Taking this limit over the path $\varepsilon = \delta$ we see that the parallel transport is just holonomy about γ_ε ,

$$D_s D_t V(0, 0) = \lim_{\varepsilon \rightarrow 0} \frac{\text{Hol}_{\gamma_\varepsilon} v - v}{\varepsilon^2} = \lim_{\varepsilon \rightarrow 0} \frac{\text{Hol}_{\gamma_{\sqrt{\varepsilon}}} v - v}{\varepsilon} = \frac{d}{dt} \Big|_{t=0} \text{Hol}_{\gamma_{\sqrt{t}}}(v).$$

So putting this together we see that for all $v \in T_p M$,

$$R(x, y)v = D_s D_t V(0, 0) = \frac{d}{dt} \Big|_{t=0} \text{Hol}_{\gamma_{\sqrt{t}}}(v)$$

as desired.

8.0.5 Problem

Let $\pi: E \rightarrow N$ be a smooth vector bundle of rank r , and let $f: M \rightarrow N$ be a smooth map. Define

$$f^* E = M \times_N E$$

to be the pullback, and let $\pi_f: f^* E \rightarrow M$ be given by projection onto M . The previous homework showed that $f^* E$ is a smooth manifold.

- (1) Given a local trivialization of E ,

$$\Phi: \pi^{-1}U \rightarrow U \times \mathbb{R}^r,$$

construct a local trivialization of f^*E ,

$$\tilde{\Phi}: \pi_f^{-1}(f^{-1}U) \rightarrow f^{-1}U \times \mathbb{R}^r.$$

- (2) Show that if $g_{\alpha\beta}$ are the transition functions of E , then the transition functions of f^*E are $g_{\alpha\beta} \circ f$. Conclude that with these local trivializations and transition functions, f^*E is a vector bundle of rank r over M with fibers

$$(f^*E)_p \cong E_{fp}.$$

This is the **pullback bundle** of E .

- (3) Prove that there is a natural bijection

$$\Gamma(f^*E) \cong \{X: M \rightarrow E \text{ smooth} \mid \pi \circ X = f\}.$$

In particular there is a natural bijection between sections of f^*TN and vector fields along f .

- (4) Let $f: M \rightarrow N$ be smooth. Show that there is a vector bundle

$$\text{hom}(TM, f^*TN) \rightarrow M$$

whose fibers are given by $\text{hom}(T_pM, T_{fp}N)$. Then prove that $p \mapsto (p, df_p)$ defines a smooth section of $\text{hom}(TM, f^*TN)$.

- (1) We have that

$$\pi_f^{-1}(f^{-1}U) = \{(p, e) \in f^*E \mid fp = \pi e \in U\}.$$

So we can define $\tilde{\Phi}: \pi_f^{-1}(f^{-1}U) \rightarrow f^{-1}U \times \mathbb{R}^r$ by

$$\tilde{\Phi}(p, e) = (p, \text{proj}_2 \circ \Phi(e)).$$

We have that $\text{proj}_1 \circ \tilde{\Phi} = \pi_f$. And the fiber along p of f^*E is

$$(f^*E)_p = \{(p, e) \in f^*E \mid \pi e = fp\},$$

which is isomorphic to E_{fp} via $(p, e) \mapsto e$. For $p \in f^{-1}U$, restricting $\tilde{\Phi}$ to f^*E_p gives us the map $(p, e) \mapsto (p, \text{proj}_2 \circ \Phi(e))$, and since $\text{proj}_2 \circ \Phi$ is a linear isomorphism of E_{fp} , so is $\text{proj}_2 \circ \tilde{\Phi}$.

So we need only show that $\tilde{\Phi}$ is a diffeomorphism. We will explicitly construct an inverse, so a map $\tilde{\Psi}: f^{-1}U \times \mathbb{R}^r \rightarrow \pi_f^{-1}(f^{-1}U) \subseteq f^*E$. To define a map into f^*E it is sufficient to define maps into E and M which commute with π and f . We give these by

$$\begin{aligned} \psi_1: f^{-1}U \times \mathbb{R}^r &\rightarrow M, & (p, v) &\mapsto p, \\ \psi_2: f^{-1}U \times \mathbb{R}^r &\rightarrow E, & (p, v) &\mapsto \Phi^{-1}(fp, v). \end{aligned}$$

Note that ψ_2 is well-defined and smooth as the composition $(p, v) \mapsto (fp, v) \in U \times \mathbb{R}^r \xrightarrow{\Phi^{-1}} \psi_2(p, v) \in E$. And we have

$$\pi \circ \psi_2(p, v) = \pi \circ \Phi^{-1}(fp, v) = \text{proj}_1(fp, v) = fp = f \circ \psi_1(p, v),$$

so by the universal property we have a smooth map $\tilde{\Psi}: f^{-1}U \times \mathbb{R}^r \rightarrow f^*E$ by

$$\tilde{\Psi}(p, v) = (\psi_1(p, v), \psi_2(p, v)) = (p, \Phi^{-1}(fp, v)).$$

Notice that $\tilde{\Psi}$ maps into $\pi_f^{-1}(f^{-1}U)$ since

$$f \circ \pi_f \circ \tilde{\Psi}(p, v) = fp \in U,$$

and as an open submanifold we can restrict $\tilde{\Psi}$'s codomain to it.

Now we have that

$$\tilde{\Psi} \circ \tilde{\Phi}(p, e) = \tilde{\Psi}(p, \text{proj}_2 \circ \Phi(e)) = (p, \Phi^{-1}(fp, \text{proj}_2 \circ \Phi(e))).$$

And we have that since $\text{proj}_1 \circ \Phi = \pi$,

$$\Phi(e) = (\pi(e), \text{proj}_2 \circ \Phi(e)) = (fp, \text{proj}_2 \circ \Phi(e)),$$

we get

$$\tilde{\Psi} \circ \tilde{\Phi}(p, e) = (p, \Phi^{-1} \circ \Phi(e)) = (p, e).$$

And conversely

$$\tilde{\Phi} \circ \tilde{\Psi}(p, v) = \tilde{\Phi}(p, \Phi^{-1}(fp, v)) = (p, \text{proj}_2 \circ \Phi \circ \Phi^{-1}(fp, v)) = (p, v)$$

as desired.

So $\tilde{\Psi} = \tilde{\Phi}^{-1}$, showing that $\tilde{\Phi}$ is indeed a diffeomorphism and thus a local trivialization.

- (2) Let Ψ be another local trivialization of E which gives us a local trivialization $\tilde{\Psi}$ of f^*E . Suppose that τ is the transition function between Φ, Ψ so

$$\Phi \circ \Psi^{-1}(q, v) = (q, \tau(q)v), \quad q \in U, v \in \mathbb{R}^r.$$

Then we get for $p \in f^{-1}U, v \in \mathbb{R}^r$,

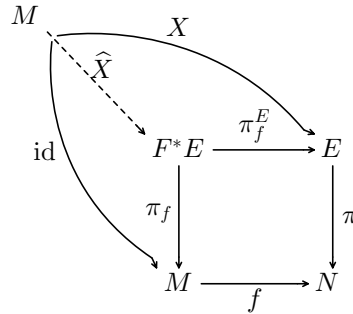
$$\tilde{\Phi} \circ \tilde{\Psi}^{-1}(p, v) = \tilde{\Phi}(p, \Psi^{-1}(fp, v)) = (p, \text{proj}_2 \circ \Phi \circ \Psi^{-1}(fp, v)) = (p, \text{proj}_2(q, \tau(fp)v)) = (p, \tau(fp)v).$$

So we see that the transition function between $\tilde{\Phi}$ and $\tilde{\Psi}$ is $\tilde{\tau} = \tau \circ f$ as desired. Thus f^*E is indeed a vector bundle, and we showed before that its fibers are

$$(f^*E)_p \cong E_{fp}.$$

- (3) By the universal property, a map $\sigma: M \rightarrow f^*E$ is the same as maps $\sigma_1: M \rightarrow E$ and $\sigma_2: M \rightarrow M$ such that $f \circ \sigma_2 = \pi \circ \sigma_1$, where $\sigma_1 = \pi_f^E \circ \sigma$ and $\sigma_2 = \pi_f \circ \sigma$ ($\pi_f^E: f^*E \rightarrow E$ is projection onto E). In particular a section $\sigma: M \rightarrow f^*E$, so $\pi_f \circ \sigma = \text{id}$, corresponds to just giving $\sigma_1 = \pi_f^E \circ \sigma$.

In particular if we have a vector field along f , so $X: M \rightarrow E$ such that $\pi \circ X = f$, then by the universal property this gives rise to a unique map $\hat{X}: M \rightarrow f^*E$ such that $X = \pi_f^E \circ \hat{X}$ and $\pi_f \circ \hat{X} = \text{id}$. So $\hat{X} \in \Gamma(f^*E)$. This is summarized in the following diagram:



Conversely if we have a section $\sigma: M \rightarrow f^*E$, then let $\bar{\sigma} = \pi_f^E \circ \sigma: M \rightarrow E$. This is a vector field along f since

$$\pi \circ \bar{\sigma} = \pi \circ \pi_f^E \circ \sigma = f \circ \pi_f \circ \sigma = f,$$

where the last equality is because σ is a section.

These maps $\widehat{\bullet}$ and $\bar{\bullet}$ are inverses, since for $X: M \rightarrow E$ a vector field along f ,

$$\widehat{\bar{X}} = \pi_f^E \circ \widehat{X} = X,$$

where the last equality was explained above, and is due to the universal property. And if $\sigma: M \rightarrow f^*E$ is a section, then

$$\widehat{\bar{\sigma}} = \pi_f^E \circ \widehat{\sigma}.$$

Since for any X , $\widehat{X}$ is the unique map such that $X = \pi_f^E \circ \widehat{X}$ and $\pi_f \circ \widehat{X} = \text{id}$, and since σ satisfies these, we see that $\widehat{\bar{\sigma}} = \sigma$ as desired.

Thus we have natural bijections

$$\Gamma(f^*E) \xleftrightarrow[\bullet]{\bar{\bullet}} \{X: M \rightarrow E \text{ smooth} \mid \pi \circ X = f\},$$

as desired. In particular a local frame of f^*E corresponds to $X_1, \dots, X_n: M \rightarrow E$ smooth vector fields along f such that $X_i|_p \in E_{fp}$ are linearly independent, by following the isomorphism $f^*E_p \cong E_{fp}$ and $X_i \leftrightarrow \widehat{X}_i$.

- (4) As stated in the recitation, given vector bundles on M $E \rightarrow M, F \rightarrow M$, we can construct dual vector bundles $E^* \rightarrow M$ and tensor product bundles $E \otimes F \rightarrow M$ whose fibers are $(E^*)_p = E_p^*$ and $(E \otimes F)_p = E_p \otimes F_p$ respectively. Putting this together we can construct a vector bundle $E^* \otimes F \rightarrow M$ whose fibers are $E_p^* \otimes F_p$, which is naturally isomorphic to $\text{hom}(E_p, F_p)$. This gives a construction of $\text{hom}(E, F)$ via the natural bijection

$$\coprod_{p \in M} \text{hom}(E_p, F_p) \cong \coprod_{p \in M} E_p^* \otimes F_p = E^* \otimes F.$$

In particular for $E = TM$ and $F = f^*TN$ the fibers are $\text{hom}(E_p, F_p) = \text{hom}(T_pM, T_{fp}N)$.

We will construct the dual bundle and show that if $\sigma_1, \dots, \sigma_n$ is a local frame for $E \rightarrow M$, then $\varepsilon^1, \dots, \varepsilon^n$ is a local frame for $E^* \rightarrow M$ where $\varepsilon^1|_p, \dots, \varepsilon^n|_p$ is the dual basis for $\sigma_1|_p, \dots, \sigma_n|_p$ relative to E_p . Similarly if $(\sigma_i), (\tau_j)$ are local frames for $E \rightarrow M, F \rightarrow M$ respectively on a common open subset, then $(\sigma_i \otimes \tau_j)$ is a local frame for $E \otimes F \rightarrow M$ where $(\sigma_i \otimes \tau_j)|_p = \sigma_i|_p \otimes \tau_j|_p$ in $E_p \otimes F_p$.

Then consider the map $p \mapsto (p, df_p)$. Since f is smooth consider coordinates $(U, (x^i))$ of M and coordinates $(V, (y^j))$ of N such that $f(U) \subseteq V$. Then $\partial/\partial y^j \circ f$ is a vector field $M \rightarrow TN$ along f (as the composition of the vector field $\partial/\partial y^j$ with f it is smooth). This forms a local frame for f^*TN considering dimensions. And let dx^i denote the dual frame for $\partial/\partial x^i$ in U . Then $dx^i \otimes \partial/\partial y^j|_f$ forms a local frame for $T^*M \otimes f^*TN = \text{hom}(TM, f^*TN)$.

Since df_p is linear $T_pM \rightarrow T_{fp}N$, we can represent it relative to this local frame, so

$$df = f_i^j \cdot dx^i \otimes \frac{\partial}{\partial y^j} \Big|_f.$$

Recall that the identification $\text{hom}(V, W) \cong V^* \otimes W$ is given by mapping $T: V \rightarrow W$ to $a_i^j \varphi^i \otimes w_j$ (where φ^i is the dual basis for $(v_i) \in V$ and (w_j) is a basis for W) where $a_i^j w_j = T(v_i)$. Thus for $p \in U$,

$$f_i^j(p) \frac{\partial}{\partial y^j} \Big|_{fp} = df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial f^j}{\partial x^i}(p) \cdot \frac{\partial}{\partial y^j} \Big|_{fp},$$

thus $f_i^j(p) = \frac{\partial f^j}{\partial x^i}(p)$.

This means that df has the local representation

$$df = \frac{\partial f^j}{\partial x^i} \cdot dx^i \otimes \frac{\partial}{\partial y^j} \Big|_f,$$

and since its component functions $\partial f^j / \partial x^i$ are smooth, this means that df is a smooth section as desired.

Now we actually construct E^* . Let $\pi_*: E^* \rightarrow M$ be the projection. Now let $\sigma_1, \dots, \sigma_n$ be a local frame for E , then let $\varepsilon^1, \dots, \varepsilon^n: M \rightarrow E^*$ be defined by $\varepsilon^i|_p$ forming the dual basis for $\sigma_i|_p$. Now if we have another local frame $\tau_1, \dots, \tau_n$ for E and $\delta^1, \dots, \delta^n$ be the dual local frame for E^* , we want to show ε^i, δ^j are compatible. Let us write δ^i in terms of ε^j and σ_i in terms of τ_j ,

$$\delta^i = a_j^i \varepsilon^j, \quad \sigma_i = b_i^j \tau_j.$$

Evaluating at p and composing with $\sigma_j|_p$ on the left and $\delta^j|_p$ on the right shows

$$a_j^i(p) = \delta^i|_p(\sigma_j|_p), \quad b_i^j(p) = \delta^j|_p(\sigma_j|_p).$$

Thus $a_j^i = b_i^j$, and since σ_i, τ_j are compatible this means that b_i^j is invertible and smooth and thus so is a_j^i , so ε^i, τ^j are compatible. Since local frames for E cover M , local dual frames for E^* cover M , so E^* is a smooth vector bundle of the same rank as E , and local frames of E give rise to dual local frames of E^* .

Similarly we construct $E \otimes F$. Let $\pi_\otimes: E \otimes F \rightarrow M$ be the projection. Similar to before we take local frames σ_i, α_j for E, F defined on a common open subset, and construct local frames $\sigma_i \otimes \alpha_j$ for $E \otimes F$ whose value at $p \in M$ is $(\sigma_i \otimes \alpha_j)|_p = \sigma_i|_p \otimes \alpha_j|_p$. These are linearly independent at each point because the tensor of two bases form a basis for the tensor product. If we have other local frames τ_i, β_j for E, F , then we show that $\tau_i \otimes \beta_j$ and $\sigma_i \otimes \alpha_j$ are compatible. Suppose

$$\tau_i = a_i^k \sigma_k, \quad \beta_j = b_j^\ell \alpha_\ell$$

so that a_i^k, b_j^ℓ are the transition maps for the frames of E and F respectively, and are thus smooth. Then by evaluating at each point we see

$$\tau_i \otimes \beta_j = a_i^k b_j^\ell \sigma_k \otimes \alpha_\ell.$$

Thus the transition functions between $\tau_i \otimes \beta_j$ and $\sigma_k \otimes \alpha_\ell$ are given by $a_i^k b_j^\ell$. As the product of smooth real-valued maps, each of these is smooth, and thus the frames are compatible. Thus we have shown that $E \otimes F$ is a vector bundle, and the tensor of local frames of E and F gives a local frame of $E \otimes F$ as desired.

9 Homework 9

9.0.1 Problem

- (1) $f: M \rightarrow N$ be a local isometry, and denote the Riemannian curvature of M, N by Rm_M, Rm_N . For $p \in M$ and $X, Y, Z, W \in T_p M$ show that

$$\text{Rm}_M(X, Y, Z, W) = \text{Rm}_N(df(X), df(Y), df(Z), df(W)).$$

Equivalently, $\text{Rm}_M = f^* \text{Rm}_N$.

- (2) Let M be either the n -sphere of radius r or hyperbolic space H_{-1/r^2}^n . Show that

$$\text{Rm}_M(X, Y, Z, W) = \pm \frac{1}{r^2} (\langle X, Z \rangle \langle Y, W \rangle - \langle X, W \rangle \langle Y, Z \rangle),$$

and deduce that the sphere has positive sectional curvature $1/r^2$ and hyperbolic space has negative sectional curvature $-1/r^2$.

- (3) Show that H_{-1/r^2}^n is geodesically complete.

- (1) Since Riemannian curvature is local we can assume that f is a global isometry. Otherwise, we restrict our attention to open subsets $p \in U \subseteq M$ and $V \subseteq N$ such that f restricts to an isometry $U \rightarrow V$. For an isometry, we know that the pullback of the Levi-Civita connection on N is the Levi-Civita connection on M . So if we let ∇^M, ∇^N be the connections on M, N then

$$\nabla_X^M Y = (f^{-1})_* \nabla_{f_* X}^N f_* Y.$$

So if we let R_M, R_N be the curvature tensors of M, N then for vector fields $X, Y, Z, W \in \mathfrak{X}(M)$,

$$\begin{aligned} R_M(X, Y)Z &= \nabla_Y^M \nabla_X^M Z - \nabla_X^M \nabla_Y^M Z + \nabla_{[X, Y]}^M Z \\ &= (f^{-1})_* (\nabla_{f_* Y}^N \nabla_{f_* X}^N f_* Z - \nabla_{f_* X}^N \nabla_{f_* Y}^N f_* Z + \nabla_{[f_* X, f_* Y]}^N f_* Z) \\ &= (f^{-1})_* R_N(f_* X, f_* Y) f_* Z. \end{aligned}$$

Thus

$$\begin{aligned}\mathrm{Rm}_M(X, Y, Z, W) &= \langle R_M(X, Y)Z, W \rangle = \langle (f^{-1})_* R_N(f_* X, f_* Y) f_* Z, W \rangle \\ &= \langle R_N(f_* X, f_* Y) f_* Z, f_* W \rangle \\ &= \mathrm{Rm}_N(f_* X, f_* Y, f_* Z, f_* W),\end{aligned}$$

and pointwise this means for tangent vectors $X, Y, Z, W \in T_p M$

$$\mathrm{Rm}_M(X, Y, Z, W) = \mathrm{Rm}_N(df(X), df(Y), df(Z), df(W)).$$

- (2) First we claim that it is sufficient to prove this identity at a single point $p \in M$. We will show that isometries act transitively on M , so for any $p, q \in M$ there is an isometry f of M such that $f(p) = q$. Since an isometry is in particular a diffeomorphism df_p is an isomorphism from $T_p M$ to $T_q M$, so for any $X, Y, Z, W \in T_q M$ we have by the first point

$$\mathrm{Rm}_M(X, Y, Z, W) = \mathrm{Rm}_M(df^{-1}X, df^{-1}Y, df^{-1}Z, df^{-1}W),$$

which by assumption is equal to

$$\begin{aligned}\pm \frac{1}{r^2} (\langle df^{-1}X, df^{-1}Z \rangle \langle df^{-1}Y, df^{-1}W \rangle - \langle df^{-1}X, df^{-1}W \rangle \langle df^{-1}Y, df^{-1}Z \rangle) \\ = \pm \frac{1}{r^2} (\langle X, Z \rangle \langle Y, W \rangle - \langle X, W \rangle \langle Y, Z \rangle).\end{aligned}$$

where the final equality is due to f^{-1} being an isometry.

So we need to show two things: that isometries act transitively and that the identity holds at a single point of our choosing. The isometries of M contain either $O(n+1)$ (for the sphere) or $O_+(1, n)$ (for hyperbolic space). $O(n+1)$ clearly acts transitively, as for any two points on the sphere we can normalize them and extend them to orthonormal bases of $\mathbb{R}^{n+1}$ and define an orthogonal transformation from one base to the other. This maps our first vector to the other.

For hyperbolic space, we do essentially the same thing. For $x, y \in H_{-1/r^2}^n$ normalize them to $\hat{x} = x/r$ and $\hat{y} = y/r$ so that $(\hat{x}, \hat{x}) = (\hat{y}, \hat{y}) = -1$. Now extend $\hat{x}, \hat{y}$ to orthonormal bases $\hat{x}, \bar{x}$ and $\hat{y}, \bar{y}$ and define a linear transformation $\hat{x} \mapsto \hat{y}$ and $\bar{x} \mapsto \bar{y}$. Since this is a linear map between orthonormal bases, it preserves the bilinear form and thus is in $O(1, n)$. Furthermore it preserves the sign of the first component of $\hat{x}$, since $\hat{x}$ and $\hat{y}$'s first component is both positive, and as we showed in the previous homework this implies the map is in $O_+(1, n)$. By linearity it maps x to y as desired. So we have shown isometries act transitively in both cases.

Now consider the map $f: \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$f(x) = \sqrt{r^2 \mp |x|^2},$$

where we potentially restrict the domain to the open subset ball of radius r around 0 so that this is smooth. The graph of f is the embedded submanifold of $\mathbb{R}^{n+1}$ given by

$$\mathrm{graph}(f) = \left\{ (\sqrt{r^2 \mp |x|^2}, x) \mid x \in \mathbb{R}^n \right\}.$$

This is contained in M (when M is the sphere we take $-$ in the definition of f and $+$ for hyperbolic space) and is thus an embedded submanifold of M (as both are embedded in $\mathbb{R}^{n+1}$). Let f_i denote the i th partial derivative of f , so that

$$f_i = \mp \frac{x_i}{\sqrt{r^2 \mp |x|^2}}.$$

So we see that $x = 0$ is a critical point of f .

Let G denote either the Euclidean metric (when M is the sphere) or the Lorentzian metric (when M is hyperbolic) of $\mathbb{R}^{n+1}$. The metric on M is the pullback of G along the inclusion, and let us denote by g the pullback of this to $\mathrm{graph}(f)$, which is just the pullback of G . We showed in the previous homework that in a small enough neighborhood of 0, relative to the coordinates $\mathbf{x}(x) = (f(x), x)$,

$$g_{ij} = \delta_{ij} \pm f_i f_j,$$

and when $x = 0$ we have

$$R_{ijk\ell} = \pm(f_{ik}f_{j\ell} - f_{i\ell}f_{jk}).$$

We are going to show that at $x = 0$, so the point $p = (r, 0)$ on M , the Riemannian curvature satisfies the given identity, and as explained this will be sufficient to conclude the identity for all of the points on M . It is further sufficient to show this for a basis of $T_p M$ since Rm is tensorial and thus multilinear, and so is the right-hand side of the identity. So we demonstrate the identity via the local coordinate frame given by $\mathbf{x}$, that is we want to show when $x = 0$,

$$R_{ijk\ell} = \pm \frac{1}{r^2} (g_{ik}g_{j\ell} - g_{i\ell}g_{jk}).$$

Recalling that at $x = 0$ we have $g_{ij} = \delta_{ij}$, to do this it is sufficient to show that

$$f_{ij} = \mp \frac{1}{r} \delta_{ij}.$$

Indeed, for $i \neq j$ we have

$$f_{ij} = \mp \frac{x_i x_j}{(r^2 - |x|^2)^{3/2}},$$

which is zero when $x = 0$. And

$$f_{ii} = \mp \frac{r^2 - |x|^2 + x_i^2}{(r^2 - |x|^2)^{3/2}},$$

which at $x = 0$ is equal to $\mp 1/r$ as desired.

- (3) We claim that for any Riemannian manifold M on which its isometries act transitively, M is geodesically complete. Let $p \in M$, then we know $\exp_p$ is defined at least on some ball $B_\varepsilon(0) \subseteq T_p M$ of radius ε (taken relative to the Riemannian metric). Meaning for $v \in B_\varepsilon(0)$, γ_v is defined on an open interval containing 0 and 1. We claim that γ_v is actually defined on all of $\mathbb{R}$ for all $v \in B_\varepsilon(0)$. Assume not, so there is a geodesic whose domain is bound, we assume without loss of generality it is bound from above. So let

$$a = \inf \{t \in \mathbb{R}_{>0} \mid \text{There is a } v \in B_\varepsilon(0) \text{ such that } \gamma_v \text{ is not defined at } t\},$$

which is finite by assumption and at least 1, so a is positive. So for all $v \in B_\varepsilon(0)$ we have that γ_v is defined on at least $[0, a)$.

For $v \in B_\varepsilon(0)$, let $q = \gamma_v(a/2)$ and $w = \dot{\gamma}_v(a/2)$. Since γ_v is a geodesic,

$$|w|_g = |\dot{\gamma}_v(a/2)|_g = |\dot{\gamma}_v(0)| = |v|_g < \varepsilon.$$

Let f be an isometry of M such that $f(q) = p$, and let $u = df(w)$. Since f is an isometry, $|u| = |w| = |v| < \varepsilon$. Thus γ_u is defined on at least $[0, a)$.

Now consider $f^{-1} \circ \gamma_u$, which is a geodesic since f is an isometry. And

$$f^{-1} \circ \gamma_u(0) = f^{-1}(p) = q, \quad (f^{-1} \circ \gamma_u)'(0) = df_q^{-1}(u) = w.$$

By uniqueness, $f^{-1} \circ \gamma_u$ and γ_w agree on their common domains. Since $w = \dot{\gamma}_v(a/2)$ we have that again by uniqueness, $\gamma_w(t) = \gamma_v(t + a/2)$. Thus

$$f^{-1} \circ \gamma_u(t) = \gamma_v(t + a/2)$$

when both sides are defined, which contains $[0, a/2)$. But then consider

$$\alpha(t) = \begin{cases} \gamma_v(t) & t < a \\ f^{-1} \circ \gamma_u(t - a/2) & a/2 < t < 3a/2 \end{cases}.$$

Since both definitions agree on the intersection of their domains, this is a smooth curve, and as both curves are geodesics it is a geodesic. So α is the geodesic with initial velocity v , and since it is defined up until $3a/2$ we see that all geodesics with initial velocity in $B_\varepsilon(0)$ are defined up until $3a/2$, contradicting the definition of a .

Thus all geodesics in $B_\varepsilon(0)$ are defined on all of $\mathbb{R}$. Now let $v \in T_p M$, then we know that for $c \in \mathbb{R}$

$$\gamma_{cv}(t) = \gamma_v(ct)$$

whenever either side is defined. And in particular if γ_{cv} is defined on all of $\mathbb{R}$, so is γ_v . There is a c such that $cv \in B_\varepsilon(0)$ (e.g. $c = |v|_g + 1$), and we just showed that γ_{cv} is defined on all of $\mathbb{R}$, so that means γ_v is too. Thus M is geodesically complete.

9.0.2 Problem

Let J be a Jacobi field along γ . Show that there is a family of geodesics γ_s such that $\gamma = \gamma_0$ and $J = \frac{\partial \gamma_s}{\partial s} \Big|_{s=0}$.

Any family of geodesics will be of the form

$$\gamma_s(t) = \exp_{\alpha(s)}(tX(s))$$

where α is a smooth curve and X is a vector field along α (so $X(s) \in T_{\alpha(s)}M$ and this is defined). So let us define

$$\Gamma(s, t) = \exp_{\alpha(s)}(tX(s)),$$

and we will search for α, X that satisfy our conditions. First of all we have

$$\Gamma(0, t) = \exp_{\alpha(0)}(tX(0)),$$

so we know that we must have $\alpha(0) = \gamma(0)$ and $X(0) = \dot{\gamma}(0)$.

Since Γ is a smooth family of geodesics by construction, its variation field $V(t) = \partial_s \Gamma(0, t)$ is a Jacobi field along γ . We want to find suitable α, X such that $V = J$. By uniqueness of Jacobi fields it is sufficient to show $J(0) = V(0)$ and $D_t J(0) = D_t V(0)$.

Computing shows

$$\partial_s \Gamma(0, 0) = \frac{\partial}{\partial s} \Big|_{s=0} \Gamma(s, 0) = \frac{\partial}{\partial s} \Big|_{s=0} \exp_{\alpha(s)}(0) = \frac{\partial}{\partial s} \Big|_{s=0} \alpha(s) = \dot{\alpha}(0).$$

So we must have $\dot{\alpha}(0) = J(0)$. Now by symmetry

$$D_t V(0) = D_t \partial_s \Gamma(0, 0) = D_s \partial_t \Gamma(0, 0).$$

And

$$\partial_t \Gamma(s, 0) = \partial_t \exp_{\alpha(s)}(tX(s)) = X(s),$$

so we want

$$D_t J(0) = D_s X(0).$$

So our conditions are

$$\alpha(0) = \gamma(0), \quad X(0) = \dot{\gamma}(0), \quad \dot{\alpha}(0) = J(0), \quad D_s X(0) = D_t J(0).$$

If we take local coordinates around $p = \gamma(0)$ these become

$$\alpha(0) = 0, X^i(0) = \dot{\gamma}^i(0), \quad \dot{\alpha}^i(0) = J^i(0), \quad \dot{X}^i(0) + X^j(0)\dot{\alpha}^k(0)\Gamma_{kj}^i(p) = \dot{J}^i(0) + J^j(0)\dot{\gamma}^k(0)\Gamma_{kj}^i(p).$$

The final condition can be rewritten, by the previous conditions, as

$$\dot{X}^i(0) + \dot{\gamma}^j(0)J^k(0)\Gamma_{kj}^i(p) = \dot{J}^i(0) + J^j(0)\dot{\gamma}^k(0)\Gamma_{kj}^i(p),$$

and by symmetry, $\Gamma_{kj}^i = \Gamma_{jk}^i$ so that this is equivalent to

$$\dot{X}^i(0) = \dot{J}^i(0).$$

So we can define, in these coordinates,

$$\alpha(t) = (J^1(0)x_1, \dots, J^n(0)x_n), \quad X(t) = (J^i(t) + \dot{\gamma}^i(0) - J^i(0))\partial_i|_{\alpha}.$$

And these clearly satisfy the conditions.

9.0.3 Problem (DoCarmo, Riemannian Geometry, page 77 problem 1)

Let (u, v) denote the Cartesian coordinates of $\mathbb{R}^2$. Further let $f, g: (v_0, v_1) \rightarrow \mathbb{R}$ be smooth functions defined on an open interval such that $f'(v)^2 + g'(v)^2 \neq 0$ and $f(v) \neq 0$. Define

$$U = \{(u, v) \in \mathbb{R}^2 \mid u_0 < u < u_1, v_0 < v < v_1\}, \quad \varphi: U \rightarrow \mathbb{R}^3, \quad (u, v) \mapsto (f(v) \cos u, f(v) \sin u, g(v)).$$

- (1) Show that φ is an immersion, and its image $\varphi(U)$ is called a **surface of revolution** S . The image under φ of the constant curves $u = c, v = c$ are called the **meridians** and **parallels** of S respectively.
- (2) Show that the induced metric in the coordinates (u, v) is given by

$$g_{11} = f^2, \quad g_{12} = 0, \quad g_{22} = (f')^2 + (g')^2.$$

- (3) Show that the geodesic equation becomes the system

$$\begin{aligned} \frac{d^2 u}{dt^2} + \frac{2ff'}{f^2} \frac{du}{dt} \frac{dv}{dt} &= 0 \\ \frac{d^2 v}{dt^2} - \frac{ff'}{(f')^2 + (g')^2} \left(\frac{du}{dt} \right)^2 + \frac{f'f'' + g'g''}{(f')^2 + (g')^2} \left(\frac{dv}{dt} \right)^2 &= 0. \end{aligned}$$

- (4) Deduce the following geometric meaning of the geodesic equations above. The second equation means that, except for meridians and parallels, that $|\gamma'(t)|^2$ is constant along $\gamma(t)$. The first equation means that if $\beta(t) < \pi$ is the angle obtained at the intersection of γ with a parallel P intersecting γ at $\gamma(t)$, then $r \cos \beta$ is constant, where r is the radius of the parallel P . The equation $r \cos \beta = \text{const}$ is called **Clairut's relation**.
- (5) Use Clairut's relation to show that a geodesic of the paraboloid, so $f(v) = v, g(v) = v^2$ with $0 < v < \infty$ and $-\varepsilon < u < 2\pi + \varepsilon$, which is not a meridian must intersect itself an infinite number of times.

- (1) We compute the Jacobian of φ :

$$D\varphi = \begin{pmatrix} f'(v) \cos u & -f(v) \sin u \\ f'(v) \sin u & f(v) \cos u \\ g'(v) & 0 \end{pmatrix}.$$

Since $f(v)$ is nonzero, the right column is nonzero, so if $g'(v) \neq 0$ then the two columns are linearly independent. Otherwise $g'(v) = 0$ so $f'(v) \neq 0$ and the determinant of the upper square matrix is

$$f'(v)f(v) \cos^2(u) + f'(v)f(v) \sin^2(u) = f'(v)f(v),$$

which is nonzero. So in all cases, $D\varphi$ has rank 2 and is thus an immersion.

- (2) If we let ∂_i denote the global coordinate frame of $\mathbb{R}^n$, then the coordinate frame of S is given by $\tilde{\partial}_i = d\varphi(\partial_i)$ which in terms of the ∂_j of $\mathbb{R}^3$ is

$$\tilde{\partial}_i = \frac{\partial \varphi^j}{\partial \partial_i} \partial_j,$$

so these are

$$\begin{aligned} \tilde{\partial}_v &= f'(v) \cos(u) \partial_1 + f'(v) \sin(u) \partial_2 + g'(v) \partial_3 \\ \tilde{\partial}_u &= -f(v) \sin(u) \partial_1 + f(v) \cos(u) \partial_2. \end{aligned}$$

Then $\tilde{g}_{ij} = g(\tilde{\partial}_i, \tilde{\partial}_j)$ is given by

$$\begin{aligned} \tilde{g}_{vv} &= f'(v)^2 \cos^2(u) + f'(v)^2 \sin^2(u) + g'(v)^2 = f'(v)^2 + g'(v)^2 \\ \tilde{g}_{uv} &= -f'(v)f(v) \cos(u) \sin(u) + f'(v)f(v) \cos(u) \sin(u) = 0 \\ \tilde{g}_{uu} &= f(v)^2 \sin^2(u) + f(v)^2 \cos^2(u) = f(v)^2, \end{aligned}$$

as desired.

- (3) Let us order the coordinate chart by (u, v) . The Christoffel symbols are given by

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\tilde{\partial}_i g_{j\ell} + \tilde{\partial}_j g_{i\ell} - \tilde{\partial}_\ell g_{ij}).$$

Since g_{ij} is the diagonal matrix

$$g = \begin{pmatrix} f(v)^2 & \\ & f'(v)^2 + g'(v)^2 \end{pmatrix} \Rightarrow g^{-1} = \begin{pmatrix} f(v)^{-2} & \\ & \frac{1}{f'(v)^2 + g'(v)^2} \end{pmatrix},$$

we see that

$$\Gamma_{ij}^k = \frac{1}{2} g^{kk} (\tilde{\partial}_i g_{jk} + \tilde{\partial}_j g_{ik} - \tilde{\partial}_k g_{ij}).$$

Now g_{ij} is either zero, for $i \neq j$, or a function of v . Thus $\partial_1 g_{ij}$ will always be zero, and $\partial_k g_{ij} = 0$ if $i \neq j$. Now computing gives

$$\begin{aligned} \Gamma_{11}^1 &= \Gamma_{22}^1 = \Gamma_{11}^2 = 0 \\ \Gamma_{12}^1 &= \frac{f'}{f} \\ \Gamma_{11}^2 &= -\frac{f'f}{(f')^2 + (g')^2} \\ \Gamma_{22}^2 &= \frac{f'f'' + g'g''}{(f')^2 + (g')^2}. \end{aligned}$$

Plugging these into the geodesic equation $\ddot{x}^k + \dot{x}^i \dot{x}^j \Gamma_{ij}^k = 0$ we get the two equations

$$\begin{aligned} \frac{d^2 u}{dt^2} + 2 \frac{du}{dt} \cdot \frac{dv}{dt} \cdot \frac{f'}{f} &= 0 \\ \frac{d^2 v}{dt^2} - \frac{f'f}{(f')^2 + (g')^2} \left(\frac{du}{dt} \right)^2 + \frac{f'f'' + g'g''}{(f')^2 + (g')^2} \left(\frac{dv}{dt} \right)^2 &= 0, \end{aligned}$$

as desired.

- (4) The energy of a geodesic is given by

$$|\gamma'(t)|^2 = \left| \dot{u} \tilde{\partial}_1 + \dot{v} \tilde{\partial}_2 \right|^2 = f^2 \dot{u}^2 + ((f')^2 + (g')^2) \dot{v}^2.$$

Differentiating this gives (the derivative of $f = f(v(t))$ is $\dot{v}f'$),

$$2(f\dot{v}f'\dot{u}^2 + f^2\dot{u}\ddot{u} + (f'\dot{v}f'' + g'\dot{v}g'')\dot{v}^2 + ((f')^2 + (g')^2)\dot{v}\ddot{v}).$$

We want to show that this is zero. The second geodesic equation gives us, after multiplying by $\dot{v}((f')^2 + (g')^2)$ that the above equation is equal to (after dividing by 2)

$$f f' \dot{v} \dot{u}^2 + f^2 \dot{u} \ddot{u} + f f' \dot{u}^2 \dot{v}.$$

We can divide by f , so we want to show

$$f' \dot{v} \dot{u}^2 + f \dot{u} \ddot{u} + f' \dot{u}^2 \dot{v} = 0,$$

It is sufficient to show that

$$f \ddot{u} + 2f' \dot{v} \dot{u} = 0.$$

This is due to the first equation, and so the derivative of $|\gamma'(t)|^2$ is zero and thus it is constant.

If $P = (\hat{u}, c)$ is a parallel which intersects γ at $\gamma(t)$ then the cosine of the angle between them, $\beta(t)$, is given by

$$\cos \beta(t) = \frac{\langle P', \gamma' \rangle}{|P'| |\gamma'|},$$

and since $P' = \hat{u}' \tilde{\partial}_1$ and $\gamma' = \dot{u} \tilde{\partial}_1 + \dot{v} \tilde{\partial}_2$ we see that this is equal to

$$\cos \beta(t) = \frac{\hat{u}' \dot{u} f^2}{\hat{u}' f |\gamma'|} = \frac{f \dot{u}}{|\gamma'|}.$$

The radius of P is $f(c) = f(v)$, so Clairut's equation says that

$$r \cos \beta(t) = \frac{f^2 \dot{u}}{|\gamma'|}$$

is constant. Since $|\gamma'|$ is constant as γ' is a geodesic, we need to show $f^2 \dot{u}$ is constant. Differentiating gives (recall that the derivative of $f = f(v)$ is $\dot{v}f'$)

$$(f^2 \dot{u})' = 2f' f \dot{v} \dot{u} + f^2 \ddot{u},$$

the first equation means $f^2 \ddot{u} = -2f' f \dot{v} \dot{u}$ so this is zero as desired.

- (5) Let γ be a non-meridian geodesic on the paraboloid, so $r \cos \beta$ is constant by Clairut, and by above this means $f \dot{u}$ is constant. Since u is not constant, that $\dot{u}$ is not identically zero and thus $f \dot{u}$ is not identically zero, and since it is a constant this means that $f \dot{u} \neq 0$ always so $\dot{u} \neq 0$ always. Thus $\dot{u}$ is either always positive or always negative, so by potentially reparameterizing we can assume $\dot{u} > 0$.

Now in our case, $f(v) = v$ and $g(v) = v^2$ so

$$g_{11} = v^2, \quad g_{12} = 0, \quad g_{22} = 1 + 4v^2.$$

And Clairut's relation is

$$v \cos \beta(t) = c,$$

since the radius is $f(v) = v$. Let us reparameterize γ so that it has unit speed: $|\gamma'| = 1$; this does not affect the claim. We know that

$$\cos \beta(t) = v \dot{u},$$

so we get that

$$\dot{u} = \frac{c}{v^2}.$$

Now,

$$1 = |\gamma'|^2 = \dot{u}^2 v^2 + \dot{v}^2 (1 + 4v^2) \geq 4\dot{v}^2 v^2,$$

and so $2\dot{v}v = (v^2)' \leq 1$. Integrating shows that $v^2 \leq t + v(0)^2$, and so

$$\dot{u} \geq \frac{c}{t + v(0)^2},$$

and integrating shows that u approaches ∞ as $t \rightarrow \infty$, since the right-hand side's integral diverges to infinity. Thus u strictly increases to infinity as $t \rightarrow \infty$.

Now we want to show that $v \rightarrow \infty$ as $t \rightarrow \pm\infty$. We have that $v = c/\cos \beta$ and so $v \geq |c|$, and this is an equality iff $\cos \beta = \pm 1$ iff $v \dot{u} = \pm 1$ iff $v^2 \dot{u}^2 = 1$. Since $1 = \dot{u}^2 v^2 + \dot{v}^2 (1 + 4v^2)$, this occurs iff $\dot{v} = 0$.

Now suppose $\dot{v}(t_1) = \dot{v}(t_2) = 0$ with $t_1 < t_2$, then $v(t_1) = v(t_2) = |c|$, and if we consider the interval $[t_1, t_2]$ then v must have zero derivative somewhere in it by Rolle's theorem. But this means that there is a $t_0 \in (t_1, t_2)$ such that $v(t_0) = |c|$ since $\dot{v}(t_0) = 0$. So between every two points where $v(t_1) = v(t_2) = |c|$, there is a third which satisfies this. Since v is smooth, this means that v is identically equal to $|c|$ on $[t_1, t_2]$. But $v = |c|$ implies that $\dot{u} = c/c^2 = 1/c$ is constant and thus $u = t/c$. But this clearly doesn't satisfy the second geodesic equation, since $\dot{u}^2$ is nonzero but all the other terms are.

So $\dot{v}$ is only zero at most at a single point. Suppose this point is t_0 , so that $\dot{v}$ has the same sign on (t_0, ∞) (if t_0 doesn't exist, take $t_0 = -\infty$). Now if $v(t)$ doesn't approach infinity, then since it is monotonic for $t > t_0$ this means it must converge to some value $\lim v(t) = L < \infty$ and $v(t)$ is bound. We know Since $v(t)$ converges to a finite value at infinity, $\lim_t \dot{v}$ must be zero. The second geodesic equation gives

$$\ddot{v} - \frac{v \dot{u}^2}{1 + 4v^2} + \frac{4v \dot{v}^2}{1 + 4v^2} = 0,$$

which means

$$\ddot{v} = \frac{c^2}{v^3(1 + 4v^2)} - \frac{4v \dot{v}^2}{1 + 4v^2},$$

and as $t \rightarrow \infty$ this converges to $c^2/(L^3(1 + 4L^2)) > 0$ so $\ddot{v}$ is eventually greater than a positive constant. Thus $\dot{v}$ must increase to infinity, which is a contradiction to it having limit zero at infinity.

So $\lim v(t) = \infty$ and since $\dot{v}$ has the same sign, for $t > t_0$ this means that it must have positive sign, and thus v increases to infinity as $t \rightarrow \infty$. Thus $v(t) \rightarrow \infty$ as $t \rightarrow \infty$, and by reparameterizing $\gamma(-t)$ we see

that $v(t) \rightarrow \infty$ as $t \rightarrow -\infty$, and this convergence is also monotone. Note that this shows that there must be some t_0 where $\dot{v}(t_0) = 0$, as otherwise we would have to have $v(t)$ always increasing and then we couldn't have that $v(t) \rightarrow \infty$ as $t \rightarrow -\infty$.

Thus v is increasing to infinity on (t_0, ∞) and decreasing from infinity on $(-\infty, t_0)$. In particular v is a diffeomorphism from each interval to $(|c|, \infty)$, and since u also goes to infinity we can write u in terms of v on each interval. That is, $u = u_1(v)$ for $t > t_0$ and $u = u_2(v)$ for $t < t_0$ (or for large/small enough t). Since u is increasing, and v is decreasing on $(-\infty, t_0)$ we see that $u_2(v)$ is a decreasing function of v , and $u_1(v)$ is an increasing function. Thus $u_1(v) - u_2(v)$ is increasing to infinity, and thus there must be infinitely many v such that $u_1(v) - u_2(v)$ is a positive multiple of 2π . For each of those values of v we have that $\cos u_1 = \cos u_2$ and $\sin u_1 = \sin u_2$, so γ intersects itself at (u_1, v) and (u_2, v) , and thus infinitely many times.

9.0.4 Problem

- (1) Let (M, g) be Riemannian, and let $f: M \rightarrow M$ be an isometry. Show that

$$\text{Fix}(f) = \{p \in M \mid f(p) = p\}$$

is a submanifold of M .

- (2) Let $f: M \rightarrow M$ be a diffeomorphism of finite order, so $f^k = \text{id}$ for some k . Show that $\text{Fix}(f)$ is a submanifold.
- (3) Show that the previous point is not true if f doesn't have finite order.

- (1) For $p \in \text{Fix}(f)$ there is an $r > 0$ such that $\exp_p$ is a diffeomorphism from $B_r(0) \subseteq T_p M$. Let $W \subseteq T_p M$ be the subspace fixed by df_p . Then we claim that $\exp_p$ forms a diffeomorphism $W \cap B_r(0) \rightarrow \text{Fix}(f) \cap \exp_p(B_r(0))$. Since $\exp_p$ is a diffeomorphism from $B_r(0)$ we need only show that it maps into $\text{Fix}(f) \cap \exp_p(B_r(0))$ and does so surjectively.

For $v \in W \cap B_r(0)$ we need to show that $\exp_p(v)$ is fixed by f . Since $\exp_p(tv)$ is the unique geodesic starting at p with initial velocity v , $f \circ \exp_p(tv)$ is a geodesic too as f is an isometry. It starts at $f(\exp_p(0)) = f(p) = p$ with initial velocity

$$(f \circ \exp_p(tv))'(0) = df_p(v) = v,$$

so by uniqueness $f \circ \exp_p(tv) = \exp_p(tv)$ and in particular for $t = 1$ we see that $\exp_p(v)$ is fixed by f as desired, meaning $\exp_p$ does indeed map into $\text{Fix}(f) \cap \exp_p(B_r(0))$.

Conversely for $q = \exp_p(v) \in \text{Fix}(f) \cap \exp_p(B_r(0))$ we claim that $v \in B_r(0)$ is fixed by df . As before, $f \circ \exp_p(tv)$ is a geodesic with initial point p and initial velocity $df_p(v)$. Since f is an isometry it preserves the norm, so $df_p(v) \in B_r(0)$. This means that $f \circ \exp_p(tv) = \exp_p(df_p(v))$ by uniqueness of geodesics, and it is defined at $t = 1$ which gives

$$\exp_p(df_p(v)) = f(\exp_p(v)) = \exp_p(v),$$

and since $\exp_p$ is injective on $B_r(0)$ we get $df_p(v) = v$ so $v \in W$. Thus $\exp_p$ is a bijection between Euclidean space $W \cap B_r(0)$ and $\text{Fix}(f) \cap \exp_p(B_r(0))$, we use these as charts for $\text{Fix}(f)$ (since $\exp_p(B_r(0))$ is open in M , $\text{Fix}(f) \cap \exp_p(B_r(0))$ is open in the subspace topology).

These charts are smoothly compatible, since $\exp_p$ is a diffeomorphism onto its image in M . Thus this makes $\text{Fix}(f)$ a submanifold of M .

- (2) Let g_0 be an arbitrary metric on M , define a new metric g by

$$g(X, Y) = \frac{1}{k} \sum_{i=0}^{k-1} g_0(df^i(X), df^i(Y)).$$

Since g_0, df are smooth, so too is g , so this defines a tensor field. It is also positive definite at each point since $g_0(X, X)$ is nonnegative and $g(X, X) = 0$ iff $g_0(df^i(X), df^i(X)) = 0$ for all i , iff $df^i(X) = 0$ iff $X = 0$. And f is an isometry since

$$g(df(X), df(Y)) = \frac{1}{k} \sum_{i=0}^{k-1} g_0(df^{i+1}(X), df^{i+1}(Y)),$$

and by reordering the indices since $df^k = d(f^k) = d \circ id = id$, we see this is equal to $g(X, Y)$. So f is an isometry of (M, g) and by the previous point $\text{Fix}(f)$ is a submanifold of M .

(3) Consider the function

$$h: \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = \exp\left(-\frac{1}{x^2}\right) \sin\left(\frac{1}{x}\right).$$

This is clearly smooth at points other than 0, and computing its first derivative gives

$$h'(x) = \exp\left(-\frac{1}{x^2}\right) \left(\frac{2}{x^3} \sin\left(\frac{1}{x}\right) - \frac{1}{x^2} \cos\left(\frac{1}{x}\right) \right),$$

and inductively we see that

$$h^{(n)}(x) = \exp\left(-\frac{1}{x^2}\right) (P_n(1/x) \sin(1/x) + Q_n(1/x) \cos(1/x)),$$

for polynomials P_n, Q_n . This converges to 0 at $x = 0$, so h is smooth over all of $\mathbb{R}$. The inductive step is just a computation: $h^{(n+1)}(x)$ is equal to

$$\begin{aligned} & 2x^{-3} \exp(-x^{-2}) \cdot (P_n(1/x) \sin(1/x) + Q_n(1/x) \cos(1/x)) \\ & - x^{-1} \exp(-x^{-2}) (P'_n(1/x) \sin(1/x) + P_n(1/x) \cos(1/x) + Q'_n(1/x) \cos(1/x) - Q_n(1/x) \sin(1/x)) \\ & = \exp(-x^{-2}) \cdot (P_{n+1}(1/x) \sin(1/x) + Q_{n+1}(1/x) \cos(1/x)) \end{aligned}$$

for

$$P_{n+1}(x) = 2x^3 P_n(x) - x P'_n(x) + x Q_n(x), \quad Q_{n+1}(x) = 2x^3 Q_n(x) - x P_n(x) + x Q'_n(x).$$

Now consider the vector field given by $X = h(x)dx$, which is the same as ∇H where H is an antiderivative of the smooth function h .

We claim that X is a complete vector field. Notice that $|h(x)|$ is bound, e.g. by 1. So if γ is any maximal integral curve of X , so $\gamma'(t) = h(\gamma(t))$ then

$$\gamma(t) = \gamma(0) + \int_0^t \gamma'(s) ds = \gamma(0) + \int_0^t h(\gamma(s)) ds,$$

which implies that

$$|\gamma(t)| \leq |\gamma(0)| + \int_0^t |h(\gamma(s))| ds \leq |\gamma(0)| + |t|.$$

The escape lemma for integral curves says that if γ is maximal whose domain is bounded by b , then $\gamma[t_0, b)$ is not contained in a compact subset of M , so in $\mathbb{R}$'s case it is unbound. But as we showed, $\gamma[t_0, b)$ is bound, so γ 's domain cannot be bounded, and thus it must be all of $\mathbb{R}$. So X is complete, and so its flow θ is defined on all of $\mathbb{R} \times M \rightarrow M$.

Finally, let Φ be X 's time-one flow, i.e. $\Phi(x) = \theta(1, x)$. As flow is smooth, so is Φ . Furthermore, by the fundamental theorem of flows, Φ is a diffeomorphism $M \rightarrow M$ (since X is complete). But we claim that $\text{Fix}(\Phi) = Z(h)$, the set of h 's zeroes.

Indeed, if $h(x) \neq 0$ then $\theta^{(x)}(t) = \theta(t, x)$ is an immersion and thus has nonzero derivative (integral curves starting at regular points of the field are immersions). Since

$$\dot{\theta}^{(x)}(t) = h(\theta^{(x)}(t)),$$

this means that $h = H'$ is nonzero on the integral curve $\theta^{(x)}(t)$, and thus H is injective on it. If $\Phi(x) = x$ then this means $\theta^{(x)}(1) = \theta^{(x)}(0)$, and thus $H(\theta^{(x)}(1)) = H(\theta^{(x)}(0))$ contradicting h 's injectivity on the integral curve. For $h(x) = 0$, the integral curve starting at x is constant, $\theta^{(x)}(t) = x$, and in particular $\Phi(x) = x$. So

$$\text{Fix}(\Phi) = Z(h).$$

But

$$Z(h) = \left\{ \frac{1}{\pi k} \mid k \in \mathbb{Z} - 0 \right\} \cup \{0\},$$

which is a countable non-discrete subspace of $\mathbb{R}$ (since any neighborhood of 0 contains some other zero of h), and thus cannot be a submanifold (since countable manifolds are discrete).

9.0.5 Problem

Let M be a smooth manifold, and denote its space of differential k -forms by $\Omega^k(M)$, and let $n = \dim M$.

- (1) Let g be a Riemannian metric on M . Define an operator $\operatorname{div}_g: \mathfrak{X}(M) \rightarrow C^\infty(M)$ pointwise by

$$(\operatorname{div}_g X)(p) = \operatorname{tr}(\nabla X|_p),$$

where $\nabla X|_p: T_p M \rightarrow T_p M$ is the linear operator $v \mapsto \nabla_v X|_p$. Show that in a geodesic frame (E_i) at p and $X = f^i E_i$, one has

$$(\operatorname{div}_g X)(p) = (E_i f^i)(p).$$

- (2) Suppose M is orientable. Show that there exists a unique $\nu_g \in \Omega^n(M)$ called the **volume form** induced by g satisfying the equivalent conditions:

- (i) for every oriented local orthonormal frame (E_i) , $\nu_g(E_1, \dots, E_n) = 1$,
- (ii) in every oriented local coordinate coframe,

$$\nu_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n.$$

- (3) Given the volume form $\nu_g \in \Omega^n(M)$, let $\mathcal{L}_X \nu_g$ be its Lie derivative relative to a vector field X . Show that

$$\mathcal{L}_X \nu_g = (\operatorname{div}_g X) \nu_g.$$

- (1) Since $(E_i|_p)$ forms a basis of $T_p M$, we can compute the trace of $\nabla X|_p$ relative to $E_i|_p$. We have that

$$\nabla_{E_i|_p} X|_p = \nabla_{E_i} X|_p = \nabla_{E_i} f^j E_j|_p = (E_i f^j) E_j|_p + f^j \nabla_{E_i} E_j|_p,$$

and since E_i is a geodesic frame at p , this is equal to $(E_i f^j)(p) E_j|_p$. So relative to the basis $(E_i|_p)$, ∇X is represented by the matrix $A_j^i = E_i f^j(p)$, and thus its trace is

$$\operatorname{div}_g X(p) = \sum_i A_i^i = \sum_i E_i f^i(p),$$

as desired.

- (2) First we show the two conditions are equivalent. Let (E_i) be an oriented local orthonormal frame and (x^i) an oriented chart which induces the oriented local coordinate frame (∂_i) . Let A and B be the transition functions between them, so

$$E_i = A_i^j \partial_j, \quad \partial_j = B_j^i E_i$$

meaning A and B are inverse matrices. And we know for any n -form $\omega \in \Omega^n(M)$,

$$\omega(E_1, \dots, E_n) = \det(A_i^j) \omega(\partial_1, \dots, \partial_n), \quad \omega(\partial_1, \dots, \partial_n) = \det(B_j^i) \omega(E_1, \dots, E_n).$$

We will show that $\det(B_j^i) = \sqrt{\det(g_{ij})}$, which shows that if $\nu_g(\bar{E}) = 1$ then $\nu_g(\bar{\partial}) = \sqrt{\det(g_{ij})}$ and vice versa. Since both frames are oriented, their transition maps A, B have positive determinant, so it is sufficient to show $\det(B_j^i)^2 = \det(g_{ij})$. We know that

$$g_{ij} = \langle \partial_i, \partial_j \rangle = \langle B_i^k E_k, B_j^\ell E_\ell \rangle = B_i^k B_j^\ell \delta_{k\ell} = \sum_k B_i^k B_j^k.$$

So we have that

$$g_{ij} = (B^\top \cdot B)_j^i,$$

meaning $g = B^\top B$, so

$$\det(g) = \det(B^\top B) = \det(B^\top) \det(B) = \det(B)^2,$$

as desired. Now, $\nu_g(\partial_1, \dots, \partial_n) = \sqrt{\det(g)}$ is equivalent to $\nu_g = \sqrt{\det(g)} dx^1 \wedge \dots \wedge dx^n$ because the coefficient of $dx^1 \wedge \dots \wedge dx^n$ in the representation of ν_g is given by $\nu_g(\partial_1, \dots, \partial_n)$. So we have shown the equivalence of the two conditions.

Either of the two points uniquely define ν_g , since around any point we have a unique representation of ν_g in terms of a local coordinate coframe, or local orthonormal coframe.

Now we just need to show that ν_g is well-defined. We take the first point as the definition, meaning in terms of any oriented local orthonormal frame (E_i) with dual frame (ε^i) we define

$$\nu_g = \varepsilon^1 \wedge \dots \wedge \varepsilon^n.$$

To show that this is well-defined, we take another oriented local orthonormal frame (F_i) with dual frame (δ^i) and we want to show that

$$\varepsilon^1 \wedge \dots \wedge \varepsilon^n = \delta^1 \wedge \dots \wedge \delta^n.$$

Let A_j^i be the transition matrix, so $F_j = A_j^i E_i$ and thus $\varepsilon^j = A_i^j \delta^i$ so

$$\varepsilon^1 \wedge \dots \wedge \varepsilon^n = \det(A) \delta^1 \wedge \dots \wedge \delta^n.$$

Since A is the transition map between orthonormal bases, it is unitary and thus $|\det(A)| = 1$, since further both E_i, F_i are positively oriented, A has positive determinant and so $\det(A) = 1$ giving us the desired result. So ν_g is well-defined, as desired.

- (3) Let (E_i) be an oriented geodesic frame at p , and write $X = f^i E_i$ locally. Since the Lie derivative of a differential form is itself a differential form, it is sufficient to show that at p we have

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = (\operatorname{div}_g X) \nu_g(E_1, \dots, E_n),$$

then the result also follows for any permutation of $E_1, \dots, E_n$ (by antisymmetry of differential forms) so that we get the result for all vectors $X_1, \dots, X_n \in T_p M$. This is true for all points p so we get the result globally.

For the remainder of this solution, everything is being evaluated at the point p implicitly.

The right-hand side is by definition just $\operatorname{div}_g X$. The left-hand side is, by the formula for the Lie derivative of covariant tensors,

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = X(\nu_g(E_1, \dots, E_n)) - \sum_i \nu_g(E_1, \dots, [X, E_i], \dots, E_n).$$

Since $\nu_g(E_1, \dots, E_n) = 1$ is constant, the first term is zero. By symmetry of the Levi-Civita connection,

$$\nabla_X E_i - \nabla_{E_i} X = [X, E_i],$$

The first term on the left is $f^j \nabla_{E_j} E_i = 0$, so we see

$$[X, E_i] = -\nabla_{E_i} X = -\nabla_{E_i} f^j E_j = -(E_i f^j) E_j.$$

Thus

$$\mathcal{L}_X(\nu_g)(E_1, \dots, E_n) = \sum_i \nu_g(E_1, \dots, (E_i f^j) E_j, \dots, E_n) = \sum_i (E_i f^j) \nu_g(E_1, \dots, E_j, \dots, E_n).$$

Since ν_g is a differential form and thus zero when its arguments are repeated, when $i \neq j$ the term in the sum is zero, so this is equal to

$$= \sum_i (E_i f^i) \nu_g(E_1, \dots, E_n) = \sum_i E_i f^i = \operatorname{div}_g X,$$

as desired.

10 Homework 10

10.0.1 Problem (DoCarmo, Riemannian Geometry, page 103 question 1)

Let G be a Lie group with a bi-invariant metric. Let $X, Y, Z \in \mathfrak{g}$ be left-invariant vector fields.

- (1) Show that $\nabla_X Y = \frac{1}{2}[X, Y]$.
- (2) Conclude that $R(X, Y)Z = \frac{1}{4}[[X, Y], Z]$.
- (3) Show that if X, Y are orthonormal, and σ is the plane generated by X, Y , then the sectional curvature of G at σ is

$$K(\sigma) = \frac{1}{2} \|[X, Y]\|^2.$$

- (1) Consider the map $\mathfrak{g}^2 \rightarrow \mathfrak{g}$ by $(X, Y) \mapsto \nabla_X Y$. This is $\mathbb{R}$ -bilinear, as we know this of connections, and further we know that $\nabla_X X = 0$. This implies that it is antisymmetric; a bilinear map $\alpha: V^2 \rightarrow W$ which is zero when its inputs are equal is antisymmetric:

$$0 = \alpha(v + u, v + u) = \alpha(v, v) + \alpha(v, u) + \alpha(u, v) + \alpha(u, u) = \alpha(v, u) + \alpha(u, v).$$

Thus $\nabla_X Y = -\nabla_Y X$ and by symmetry of the connection,

$$[X, Y] = \nabla_X Y - \nabla_Y X = 2\nabla_X Y$$

as desired.

- (2) By the previous point,

$$R(X, Y)Z = \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X, Y]} Z = \frac{1}{4}[Y, [X, Z]] - \frac{1}{4}[X, [Y, Z]] + \frac{1}{2}[[X, Y], Z].$$

Now by Jacobi,

$$0 = [X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = [X, [Y, Z]] + [Z, [X, Y]] - [Y, [X, Z]]$$

which implies

$$[Y, [X, Z]] - [X, [Y, Z]] = [Z, [X, Y]] = -[[X, Y], Z].$$

Thus we have

$$R(X, Y)Z = \frac{1}{4}[[X, Y], Z]$$

as desired.

- (3) If X, Y are orthonormal then the sectional curvature of σ is just $\text{Rm}(X, Y, X, Y)$ which is

$$K(\sigma) = \langle R(X, Y)X, Y \rangle = \frac{1}{4} \langle [[X, Y], X], Y \rangle = -\frac{1}{4} \langle \nabla_X [X, Y], Y \rangle.$$

Now, because the connection is metric,

$$X \langle [X, Y], Y \rangle = \langle \nabla_X [X, Y], Y \rangle + \langle [X, Y], \nabla_X Y \rangle.$$

And similarly

$$0 = X \langle Y, Y \rangle = 2 \langle \nabla_X Y, Y \rangle,$$

where the left-hand side is zero since $\langle Y, Y \rangle = 1$ is constant. Thus $0 = 2 \langle \nabla_X Y, Y \rangle = \langle [X, Y], Y \rangle$, and so we see

$$0 = \langle \nabla_X [X, Y], Y \rangle + \langle [X, Y], \nabla_X Y \rangle \implies \langle [X, [X, Y]], Y \rangle = -\langle [X, Y], [X, Y] \rangle,$$

and finally

$$K(\sigma) = \frac{1}{4} \langle [X, Y], [X, Y] \rangle$$

as desired.

10.0.2 Problem

- (1) Let ∇ be a connection on a manifold M . For vector fields $X, Y, Z \in \mathfrak{X}(M)$, let $\nabla^2 Z$ be the $(2, 1)$ tensor field obtained by taking the covariant derivative of the $(1, 1)$ tensor field ∇Z . We write $\nabla_{X,Y}^2 Z$ for $\nabla^2 Z(X, Y)$. Show that

$$R(X, Y)Z = \nabla_{Y,X}^2 Z - \nabla_{X,Y}^2 Z.$$

- (2) (DoCarmo, Riemmanian Geometry, page 106 problem 7): prove the second **Binachi identity**:

$$\nabla \text{Rm}(X, Y, Z, W, T) + \nabla \text{Rm}(X, Y, W, T, Z) + \nabla \text{Rm}(X, Y, T, Z, W) = 0.$$

- (1) By the formula for the covariant derivative of a tensor, we see that for a covector field ω ,

$$\begin{aligned} \nabla_{Y,X}^2 Z(\omega) &= \nabla_Y \nabla_X Z(\omega) = Y(\nabla_X Z(\omega)) - \nabla_X Z(\nabla_Y \omega) - \nabla_X Z(\omega, \nabla_Y X) \\ &= Y(\omega(\nabla_X Z)) - \nabla_Y \omega(\nabla_X Z) - \omega(\nabla_{\nabla_Y X} Z). \end{aligned}$$

Now we know that in general,

$$Y(\omega X) = \nabla_Y(\omega X) = \nabla_Y \omega(X) + \omega(\nabla_Y X),$$

and thus

$$Y(\omega(\nabla_X Z)) = \nabla_Y \omega(\nabla_X Z) + \omega(\nabla_Y \nabla_X Z).$$

So

$$\nabla_{Y,X}^2 Z(\omega) = \omega(\nabla_Y \nabla_X Z) - \omega(\nabla_{\nabla_Y X} Z).$$

Since this is true for all ω , we see that

$$\nabla_{Y,X}^2 Z = \nabla_Y \nabla_X Z - \nabla_{\nabla_Y X} Z.$$

Thus

$$\begin{aligned} \nabla_{Y,X}^2 Z - \nabla_{X,Y}^2 Z &= \nabla_Y \nabla_X Z - \nabla_{\nabla_Y X} Z - \nabla_X \nabla_Y Z + \nabla_{\nabla_X Y} Z \\ &= \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{\nabla_X Y - \nabla_Y X} Z \\ &= \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X,Y]} Z \\ &= R(X, Y)Z. \end{aligned}$$

- (2) Everything here is tensorial so we can choose whatever local frame we want, show it for that, and extrapolate the result for all vector fields. Let $p \in M$ and choose a geodesic field (E_i) at p . Then we have,

$$\begin{aligned} \nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) &= \nabla_{E_h} \text{Rm}(E_i, E_j, E_k, E_\ell) = \\ &E_h(\text{Rm}(E_i, E_j, E_k, E_\ell)) - \text{Rm}(\nabla_{E_h} E_i, E_j, E_k, E_\ell) - \cdots - \text{Rm}(E_i, \dots, \nabla_{E_h} E_\ell). \end{aligned}$$

At p we know $\nabla_{E_i} E_j|_p = 0$ for all i, j , so by multilinearity the last terms above are all zero (at p). So we see

$$\begin{aligned} \nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) &= E_h \text{Rm}(E_i, E_j, E_k, E_\ell) = E_h \left\langle \nabla_{E_j, E_i}^2 E_k - \nabla_{E_i, E_j}^2 E_k, E_\ell \right\rangle = \\ &\left\langle \nabla_{E_h} (\nabla_{E_j, E_i}^2 E_k - \nabla_{E_i, E_j}^2 E_k), E_\ell \right\rangle, \end{aligned}$$

the last equality is due to the connection being metric and $\nabla_{E_h} E_\ell = 0$ at p . Now,

$$\left\langle \nabla_{E_h} \nabla_{E_j, E_i}^2 E_k, E_\ell \right\rangle = \left\langle \nabla_{E_h} \nabla_{E_j} \nabla_{E_i} E_k - \nabla_{E_h} \nabla_{\nabla_{E_j} E_i} E_k, E_\ell \right\rangle,$$

and so by this and symmetry of the connection,

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = \left\langle \nabla_{E_h} \nabla_{E_j} \nabla_{E_i} E_k - \nabla_{E_h} \nabla_{E_i} \nabla_{E_j} E_k + \nabla_{E_h} \nabla_{[E_i, E_j]} E_k, E_\ell \right\rangle.$$

Notice that $\text{Rm}(E_i, E_j, E_k, E_\ell) = \text{Rm}(E_k, E_\ell, E_i, E_j)$ so we also can write

$$\nabla \text{Rm}(E_i, E_j, E_k, E_\ell, E_h) = \langle \nabla_{E_h} \nabla_{E_\ell} \nabla_{E_k} E_i - \nabla_{E_h} \nabla_{E_k} \nabla_{E_\ell} E_i + \nabla_{E_h} \nabla_{[E_k, E_\ell]} E_i, E_j \rangle.$$

And so we have that the sum gives the inner product of the following with E_j (writing ∇_i for ∇_{E_i} and $\nabla_{i,j}$ for $\nabla_{[E_i, E_j]}$):

$$\begin{aligned} & \nabla_h \nabla_\ell \nabla_k E_i - \nabla_h \nabla_k \nabla_\ell E_i + \nabla_h \nabla_{k,\ell} E_i \\ & + \nabla_\ell \nabla_k \nabla_h E_i - \nabla_\ell \nabla_h \nabla_k E_i + \nabla_\ell \nabla_{h,k} E_i \\ & + \nabla_k \nabla_h \nabla_\ell E_i - \nabla_k \nabla_\ell \nabla_h E_i + \nabla_k \nabla_{\ell,h} E_i. \end{aligned}$$

Now consider

$$R(E_\ell, E_h) \nabla_k E_i = \nabla_h \nabla_\ell \nabla_k E_i - \nabla_\ell \nabla_h \nabla_k E_i + \nabla_{\ell,h} \nabla_k E_i.$$

These are all zero at p since $\nabla_k E_i = 0$, and the first two terms are the terms from the the above large sum. The final term is zero since $[E_\ell, E_h] = \nabla_\ell E_h - \nabla_h E_\ell = 0$, so the first two terms above cancel out, and so the large sum is just the sum of the terms of the form $\nabla_h \nabla_{k,\ell} E_i$. Also,

$$\nabla_h \nabla_{k,\ell} E_i = \nabla_h \nabla_{k,\ell} E_i - \nabla_{k,\ell} \nabla_h E_i = -R(E_h, [E_k, E_\ell]) E_i + \nabla_{[E_h, [E_k, E_\ell]]} E_i,$$

and since $[E_k, E_\ell] = 0$ at p we see that

$$\nabla_h \nabla_{k,\ell} E_i = \nabla_{[E_h, [E_k, E_\ell]]} E_i.$$

Thus the large sum is the sum of terms of this form, as h, k, ℓ cycle.

Thus the large sum is equal to

$$\sum_{(h,k,\ell)} \nabla_{[E_h, [E_k, E_\ell]]} E_i = \nabla_{\sum [E_h, [E_k, E_\ell]]} E_i,$$

where the sum is over the cycles of (h, k, ℓ) . By the Jacobi identity for the Lie bracket, this is zero. And thus we get that the sum of the covariant derivatives of Rm is the inner product of zero with E_j , which is zero.

10.0.3 Problem

Let $E \rightarrow M$ be a vector bundle with metric h and connection $\bar{\nabla}$. Let $S \subseteq E$ be a subbundle with orthogonal complement $S^\perp \subseteq E$ so that $E = S \oplus S^\perp$. Let $P^\perp: E \rightarrow S^\perp$ denote the orthogonal projection along S . Let $\nabla, \nabla^\perp$ denote the induced connections on $S, S^\perp$ respectively. Finally let $\text{II}: TM \otimes S \rightarrow S^\perp$ denote the second fundamental form of S and $\bar{R}: \Lambda^2 TM \otimes E \rightarrow E$ denote the curvature of $\bar{\nabla}$.

- (1) Show that for $X, Y \in \mathfrak{X}(M)$ and $\xi \in \Gamma(S)$,

$$P^\perp \bar{R}(X, Y) \xi = \nabla_Y^\perp (\text{II}(X) \xi) - \text{II}(X) \nabla_Y \xi - \nabla_X^\perp (\text{II}(Y) \xi) + \text{II}(Y) \nabla_X \xi + \text{II}([X, Y]) \xi.$$

- (2) Let $M \subseteq \bar{M}$ be a submanifold, $\bar{g}$ a Riemannian metric on $\bar{M}$ and g the induced metric on M . Let $E = T\bar{M}|_M$ and $S = TM$ so that $S^\perp = \nu_M$. Let $\bar{\nabla}$ be the connection induced from the Levi-Civita connection of $(\bar{M}, \bar{g})$ so that ∇ is the Levi-Civita connection on (M, g) .

- (i) Define $\nabla \text{II}: \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(\nu_M)$ by

$$\nabla_Y \text{II}(X) Z = \nabla_Y^\perp (\text{II}(X) Z) - \text{II}(\nabla_Y X) Z - \text{II}(X) \nabla_Y Z, \quad X, Y, Z \in \mathfrak{X}(M).$$

Show that ∇II is tensorial, and thus defines a bundle morphism

$$\nabla \text{II}: TM \otimes TM \otimes TM \rightarrow \nu_M.$$

- (ii) Show that

$$P^\perp \bar{R}(X, Y) Z = \nabla_Y \text{II}(X) Z - \nabla_X \text{II}(Y) Z.$$

- (1) The proof here is essentially the same as that of Gauss's theorem's proof from class. By definition we have for $X \in \mathfrak{X}(M)$ and $\xi \in \Gamma(S)$,

$$\bar{\nabla}_X \xi = \nabla_X \xi + \Pi(X)\xi.$$

Since $\nabla_X \xi \in \Gamma(S)$ and $\Pi(X)\xi \in \Gamma(S^\perp)$ repeating this we see

$$\bar{\nabla}_Y \bar{\nabla}_X \xi = \nabla_Y \nabla_X \xi + \Pi(Y)\nabla_X \xi + \nabla_Y^\perp \Pi(X)\xi + \Pi^\perp(Y)\Pi(X)\xi.$$

The first and last terms here lie in $\Gamma(S)$, while the middle two lie in $\Gamma(S^\perp)$. Thus projecting along S gives

$$P^\perp \bar{\nabla}_Y \bar{\nabla}_X \xi = \Pi(Y)\nabla_X \xi + \nabla_Y^\perp \Pi(X)\xi.$$

And for the same reason as before, by definition,

$$\bar{\nabla}_{[X,Y]} \xi = \nabla_{[X,Y]} \xi + \Pi([X,Y])\xi,$$

and here the first term is in $\Gamma(S)$ while the second is in $\Gamma(S^\perp)$ so that

$$P^\perp \bar{\nabla}_{[X,Y]} \xi = \Pi([X,Y])\xi.$$

Putting this all together,

$$\begin{aligned} P^\perp \bar{R}(X, Y)\xi &= P^\perp (\bar{\nabla}_Y \bar{\nabla}_X \xi - \bar{\nabla}_X \bar{\nabla}_Y \xi + \bar{\nabla}_{[X,Y]} \xi) \\ &= \Pi(Y)\nabla_X \xi + \nabla_Y^\perp \Pi(X)\xi - \Pi(X)\nabla_Y \xi - \nabla_X^\perp \Pi(Y)\xi + \Pi([X,Y])\xi, \end{aligned}$$

which is the desired expression.

(2)

- (i) By additivity of $\Pi, \nabla, \nabla^\perp$ in all parameters, $\nabla \Pi$ is clearly additive. And since Π is tensorial and $\nabla, \nabla^\perp$ are tensorial in their lower parameters, $\nabla \Pi$ is tensorial in Y . It remains to show that $\nabla \Pi$ is multiplicative in X, Z :

$$\nabla_Y \Pi(fX)Z = f \nabla_Y \Pi(X)Z, \quad \nabla_Y \Pi(X)(fZ) = f \nabla_Y \Pi(X)Z \quad \text{for } f \in C^\infty(M).$$

For X , we see

$$\begin{aligned} \nabla_Y \Pi(fX)Z &= \nabla_Y^\perp (\Pi(fX)Z) - \Pi(\nabla_Y fX)Z - \Pi(fX)\nabla_Y Z \\ &= \nabla_Y^\perp (f\Pi(X)Z) - \Pi((Yf)X + f\nabla_Y X)Z - f\Pi(X)\nabla_Y Z \\ &= (Yf)\Pi(X)Z + f\nabla_Y^\perp (\Pi(X)Z) - (Yf)\Pi(X)Z - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z \\ &= f(\nabla_Y^\perp (\Pi(X)Z) - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z) \\ &= f\nabla_Y \Pi(X)Z. \end{aligned}$$

And similarly for Z , we see

$$\begin{aligned} \nabla_Y \Pi(X)(fZ) &= \nabla_Y^\perp (\Pi(X)(fZ)) - \Pi(\nabla_Y X)(fZ) - \Pi(X)\nabla_Y (fZ) \\ &= \nabla_Y^\perp (f\Pi(X)Z) - f\Pi(\nabla_Y X)Z - \Pi(X)((Yf)Z + f\nabla_Y Z) \\ &= (Yf)\Pi(X)Z + f\nabla_Y^\perp (\Pi(X)Z) - f\Pi(\nabla_Y X)Z - (Yf)\Pi(X)Z - f\Pi(X)\nabla_Y Z \\ &= f(\nabla_Y^\perp (\Pi(X)Z) - f\Pi(\nabla_Y X)Z - f\Pi(X)\nabla_Y Z) \\ &= f\nabla_Y \Pi(X)Z. \end{aligned}$$

So that $\nabla \Pi$ is tensorial (linear) in all parameters.

- (ii) From our first point, it is sufficient to show that

$$\nabla_Y \Pi(X)Z - \nabla_X \Pi(Y)Z = \nabla_Y^\perp (\Pi(X)Z) - \Pi(X)\nabla_Y Z - \nabla_X^\perp (\Pi(Y)Z) + \Pi(Y)\nabla_X Z + \Pi([X, Y])Z.$$

Direct computation yields

$$\nabla_Y \Pi(X)Z - \nabla_X \Pi(Y)Z = \nabla_Y^\perp (\Pi(X)Z) - \Pi(\nabla_Y X)Z - \Pi(X)\nabla_Y Z - \nabla_X^\perp (\Pi(Y)Z) + \Pi(\nabla_X Y)Z + \Pi(Y)\nabla_X Z.$$

Cancelling out similar terms we see that we need

$$\Pi(\nabla_X Y)Z - \Pi(\nabla_Y X)Z = \Pi([X, Y])Z.$$

And indeed, by tensoriality of Π the left-hand side is equal to

$$\Pi(\nabla_X Y - \nabla_Y X)Z = \Pi([X, Y])Z,$$

where the equality is due to symmetry of the Levi-Civita connection.

10.0.4 Problem (Do Carmo, Differential Geometry of Curves and Surfaces, q7 p169)

Consider the surface of revolution $(\varphi(v) \cos u, \varphi(v) \sin u, \psi(v))$ with constant Gaussian curvature K . Parameterize φ, ψ such that $(\varphi')^2 + (\psi')^2 = 1$.

- (1) Show that φ satisfies $\varphi'' + K\varphi = 0$ and ψ is given by $\psi = \int \sqrt{1 - (\varphi')^2}$.
- (2) Show that all surfaces of revolution with constant curvature $K = 1$ which intersect the xy plane orthogonally are given by

$$\varphi(v) = C \cos v, \quad \psi(v) = \int_0^v \sqrt{1 - C^2 \sin^2 v}.$$

for some constant C .

- (3) Determine all surfaces of revolution with constant curvature $K = -1$, and show that they are all of one of the following forms

$$\varphi(v) = e^v, \quad \varphi(v) = \cosh(v), \quad \varphi(v) = \sinh(v).$$

- (4) Show that the only surface of revolution with $K = 0$ are the right cylinder, the right circular cone, and the plane.

- (1) We know that (Do Carmo page 162)

$$K = -\frac{\varphi''}{\varphi},$$

and thus $\varphi'' + K\varphi = 0$. And we know

$$(\psi')^2 = 1 - (\varphi')^2,$$

we take the square root and integrate, giving us both equations.

- (2) Here we need to solve the ODE $\varphi'' + \varphi = 0$. The characteristic polynomial of this ODE is $x^2 + 1 = 0$, which has roots $\pm i$. Taking the real and imaginary parts, we have solutions $\cos(v), \sin(v)$ and so the general solution is given by $A \cos(v) + B \sin(v)$.

In order for the surface of revolution to intersect xy orthogonally, all meridians must attain critical points when $v = 0$. This is because geometrically, the meridian should be flat when intersecting the xy plane, i.e. have a critical point. This just means $\varphi'(0) = 0$; so $0 = -A \sin(0) + B \cos(0) = B$.

So the surfaces of revolution are all of the form $\varphi(v) = A \cos(v)$ and since $\varphi'(v) = -A \sin(v)$,

$$\varphi(v) = A \cos(v), \quad \psi(v) = \int_0^v \sqrt{1 - A^2 \sin^2(v)}.$$

- (3) We need to solve $\varphi'' - \varphi = 0$, which has characteristic polynomial $x^2 - 1$. The roots are $x = \pm 1$, which means φ has general form $\varphi(v) = Ae^v + Be^{-v}$. If $B = 0$ we are finished, if $A = 0$ then we can reparameterize v as $-v$ to get $\varphi(v) = Be^v$. Otherwise we claim that a shift of v will give us $\varphi(v) = C \cosh(v)$ or $C \sinh(v)$. So reparameterize by $v + c$, then

$$\varphi(v) = Ae^{v+c} + Be^{-v-c} = Ae^c \cdot e^v + Be^{-c} \cdot e^{-v}.$$

So we want to find a c such that

$$|Ae^c| = |Be^{-c}| \iff e^{2c} = \left| \frac{B}{A} \right| \iff c = \frac{1}{2} \log \left| \frac{B}{A} \right|.$$

In such a case, letting $2C = Ae^c$ we see that

$$\varphi(v) = Ce^v \pm Ce^{-v},$$

which is $C \cosh(v)$ or $C \sinh(v)$ (depending on sign), as desired.

The formula for ψ is due to $\cosh'(v) = \sinh(v)$ and $\sinh'(v) = -\cosh(v)$ and $(e^v)' = e^v$.

- (4) We need to solve $\varphi'' = 0$, which has solution $\varphi(v) = Av + B$. Then we have $\psi(v) = Cv$ for $C = \sqrt{1 - A^2}$. When $A = 0$ this gives us

$$(B \cos u, B \sin u, v)$$

which is the cylinder. When $A = 1$ we have

$$((v + B) \cos u, (v + B) \sin u, 0),$$

which is the plane; we can shift v to $v + B$ so this is just polar coordinates of the plane. In general for nonzero A we can shift v so that $B = 0$.

So for all other A , we have

$$(Av \cos u, Av \sin u, Cv),$$

and since $A \neq 1$, $C \neq 0$. This is the cone, as it is the stretching of $(v \cos u, v \sin u, v)$ which is the classic parameterization of the cone.

10.0.5 Problem

- (1) Show that any smooth manifold M admits a Riemannian metric.
 (2) Let ∇ be the Levi-Civita connection on M . Define, for $\omega \in \Omega^k(M)$,

$$(d\omega)(X_1, \dots, X_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} (\nabla_{X_i} \omega)(X_1, \dots, \widehat{X}_i, \dots, X_{k+1}).$$

Show that this defines a linear operator

$$d: \Omega^k(M) \rightarrow \Omega^{k+1}(M).$$

- (3) Show that in any coordinate frame, if

$$\omega = \sum_I \omega_I dx^I \quad \text{summing over increasing multi-indices,}$$

then

$$d\omega = \sum_I d\omega_I \wedge dx^I.$$

- (4) Conclude that d does not depend on the underlying Riemannian metric, and that d coincides with the construction of the exterior derivative from class.
 (5) Prove that

$$\begin{aligned} (d\omega)(X_1, \dots, X_{k+1}) &= \sum_{i=1}^{k+1} (-1)^{i+1} X_i(\omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1})) \\ &\quad + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_{k+1}). \end{aligned}$$

- (1) We can embed M in $\mathbb{R}^n$ for some n (Whitney), and then pull back the Euclidean metric to M . Alternatively, cover M by smooth charts (U_i, φ_i) where $U_i \subseteq M$ are open and $\varphi_i: U_i \rightarrow \mathbb{R}^n$ are diffeomorphisms onto open sets. For each i , pull back the Euclidean metric on $\mathbb{R}^n$ to U_i via φ_i to give $g_i = \varphi_i^* \bar{g}$. This is a metric on U_i as φ_i is a diffeomorphism.

Let (ψ_i) be a partition of unity subordinate to (U_i) , and define on all of M

$$g = \sum_i \psi_i g_i.$$

Explicitly, we extend g_i to be zero outside of U_i , and we define for $p \in M$, $g_p = \sum_i \psi_i(p)g_i|_p$. Around every $p \in M$ there is a neighborhood for which this sum is finite, and thus a finite sum of smooth tensor fields, so g itself is a smooth tensor field. Furthermore it is a metric: g_p is a linear combination of metrics $\psi_i(p)g_i|_p$ and is thus bilinear and symmetric. It is also nonnegative since each g_i and ψ_i are. And $g_p(v, v) = 0$ iff $g_i|_p(v, v) = 0$ for all i where $p \in U_i$ and $\psi_i(p) > 0$. There must be at least one such i , and since g_i is a metric this means $v = 0$, so g is positive definite.

- (2) We show note that $d\omega$ is a tensor. We can write

$$d\omega(X_1, \dots, X_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} \nabla \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}, X_i),$$

so

$$d\omega = \sum \text{sgn} \tau^\tau \nabla \omega,$$

where the sum is over all permutations τ which take a specific index i and move it to the end (meaning i is mapped to $k+1$, and $j > i$ is shifted back one). Since $\nabla \omega$ is a tensor, permuting it is a tensor, and thus $d\omega$ is a tensor.

Now we need to show that $d\omega$ is alternating. While $\nabla \omega$ itself is not alternating, $\nabla_X \omega$ is. This is because for any permutation σ ,

$$\begin{aligned} \nabla_X \omega(Y_{\sigma 1}, \dots, Y_{\sigma k}) &= X(\omega(\bar{Y}_\sigma)) - \sum_i \omega(Y_{\sigma 1}, \dots, \nabla_X Y_{\sigma i}, \dots, Y_{\sigma k}) \\ &= \text{sgn} \sigma X(\omega(\bar{Y})) - \text{sgn} \sigma \sum_i \omega(Y_1, \dots, \nabla_X Y_i, \dots, Y_k) \\ &= \text{sgn} \sigma \nabla_X \omega(\bar{Y}). \end{aligned}$$

Now, if $\bar{X}$ has a repeated value, say $X = X_j = X_{j+1}$,

$$\begin{aligned} d\omega(\bar{X}) &= \sum_i (-1)^{i+1} \nabla_{X_i} \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}) = \\ &= (-1)^{j+1} \nabla_X \omega(\dots, \hat{X}_j, X, \dots) + (-1)^j \nabla_X \omega(\dots, X, \hat{X}_{j+1}, \dots), \end{aligned}$$

since all other terms in the sum have repeated inputs, and $\nabla_{X_i} \omega$ is alternating. Notice that there is an equality $(\dots, \hat{X}_j, X, \dots) = (\dots, X, \hat{X}_{j+1}, \dots)$ and so this sum is zero. A tensor which is zero when it is input an adjacent repeated value is alternating. This is because then swapping adjacent values alters the sign: say α is multilinear and zero when adjacent values are repeated, then

$$\begin{aligned} 0 &= \alpha(\dots, X+Y, X+Y, \dots) = \alpha(\dots, X, X, \dots) + \alpha(\dots, X, Y, \dots) + \alpha(\dots, Y, X, \dots) + \alpha(\dots, Y, Y, \dots) \\ &= \alpha(\dots, X, Y, \dots) + \alpha(\dots, Y, X, \dots). \end{aligned}$$

Since permutations of the form $(i, i+1)$ generate S_n , this means that such an α is then alternating.

So $d\omega$ is alternating, meaning is in $\Omega^{k+1}(M)$ as desired. Clearly by linearity of $\nabla \omega$ in ω , d is linear.

- (3) Let us denote by D the linear map we just defined, and by d the exterior derivative.

First we claim that $\nabla_X(\omega \wedge \eta) = \nabla_X(\omega) \wedge \eta + \omega \wedge \nabla_X \eta$ for all forms ω, η and $X \in \mathfrak{X}(M)$. This is because $\nabla_X^\sigma F = {}^\sigma \nabla_X F$ for all covariant tensors F and permutations σ . This is immediate from

$$\nabla_X F(\bar{Y}) = X(F(\bar{Y})) - \sum_i F(Y_1, \dots, \nabla_X Y_i, \dots, Y_n).$$

Then ∇_X commutes with Alt since it is just a linear combination of σ s. So

$$\begin{aligned} \nabla_X(\omega \wedge \eta) &= \nabla_X \text{Alt}(\omega \otimes \eta) = \text{Alt} \nabla_X(\omega \otimes \eta) = \text{Alt}((\nabla_X \omega) \otimes \eta) + \text{Alt}(\omega \otimes \nabla_X \eta) \\ &= (\nabla_X \omega) \wedge \eta + \omega \wedge \nabla_X \eta. \end{aligned}$$

Now we claim that $D(dx^I)$ for any multi-index I of length k . It is sufficient to show that for any multi-index J of length $k+1$, $D(dx^I)(\partial_J) = 0$, where $\partial_J = (\partial_{j_1}, \dots, \partial_{j_{k+1}})$. For a multi-index $I = (i)$ then we have

$$D(dx^i)(\partial_j, \partial_\ell) = \nabla_{\partial_j} dx^i(\partial_\ell) - \nabla_{\partial_\ell} dx^i(\partial_j).$$

We know that for 1-forms,

$$\nabla_{\partial_j} dx^i(\partial_\ell) = \partial_\ell(dx^i(\partial_j)) - dx^i(\nabla_{\partial_j} \partial_\ell) = -dx^i(\nabla_{\partial_j} \partial_\ell),$$

since $dx^i \partial_j$ is constant. So this is equal to

$$D(dx^i)(\partial_j, \partial_\ell) = dx^i(\nabla_{\partial_\ell} \partial_j - \nabla_{\partial_j} \partial_\ell) = dx^i[\partial_\ell, \partial_j] = 0.$$

So we have our base case.

Now consider the multi-index (I, ℓ) , where I has length k . Writing $\widehat{J}_i$ for the multi-index obtained by removing the index i from J , we have

$$D(dx^I \wedge dx^\ell)(\partial_J) = \sum_i (-1)^{i+1} \nabla_{\partial_{j_i}} dx^I \wedge dx^\ell(\partial_{\widehat{J}_i}) + \sum_i (-1)^{i+1} dx^I \wedge \nabla_{\partial_{j_i}} dx^\ell(\partial_{\widehat{J}_i}).$$

We claim that both sums here are zero.

For the first sum, if $\ell \notin J$ then it is trivially zero. Otherwise, we can assume that it is the last index in J (because we can otherwise permute J and this will affect the sum only in sign), so let $J = (J', \ell)$. Then the first sum is

$$\sum_{i \leq k+1} (-1)^{i+1} \nabla_{\partial_{j_i}} dx^I(\partial_{\widehat{J}_i}) = D(dx^I)(\partial_{J'}) = 0$$

inductively.

For the second sum, if I is not a sub-multi-index of J then this is trivially zero again. Otherwise we can assume I occurs as the first k indices of J , $J = (I, \partial_j, \partial_m)$ (otherwise we permute). Then in the sum, we can take i to be greater than k as otherwise we remove an index from I in J and the resulting wedge will be zero. So the sum is then equal to

$$\pm(\nabla_{\partial_j} dx^\ell(\partial_m) - \nabla_{\partial_m} dx^\ell(\partial_j)) = \pm D(dx^\ell)(\partial_j, \partial_m) = 0$$

from our base case. So both sums are zero and thus inductively we have $D(dx^I) = 0$.

By linearity of D it is sufficient to show that $D(u dx^I) = du \wedge dx^I$ for $u \in C^\infty(M)$. Indeed, from what we just showed:

$$\begin{aligned} D(u dx^I)(\partial_J) &= \sum_i (-1)^{i+1} \nabla_{\partial_{j_i}} u dx^I(\partial_{\widehat{J}_i}) \\ &= \sum_i (-1)^{i+1} (\partial_{j_i} u) dx^I(\partial_{\widehat{J}_i}) + u \sum_i (-1)^{i+1} \partial_{j_i} dx^I(\partial_{\widehat{J}_i}) \\ &= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^I + u D(dx^I)(\partial_J) \\ &= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^I. \end{aligned}$$

On the other hand,

$$du \wedge dx^I(\partial_J) = \sum_i \frac{\partial u}{\partial x_i} dx^i \wedge dx^I(\partial_J),$$

and we can sum over i occurring in J as otherwise the term will be zero,

$$= \sum_i \frac{\partial u}{\partial x_{j_i}} dx^{j_i} \wedge dx^I(\partial_J) = \sum_i \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^{j_i I}.$$

Since j_i occurs at the i th index of J , a permutation of J bring it to $j_i I$ must map i to 1 and gives a permutation of $\widehat{J}_i$ mapping to I . The sign of this permutation is $(-1)^{i+1}$ times the sign of the sub-permutation mapping $\widehat{J}_i$ to I , as we must offset all the indices before i by 1. So this is equal to

$$= \sum_i (-1)^{i+1} \frac{\partial u}{\partial x_{j_i}} \delta_{j_i}^I = D(u dx^I),$$

as desired.

It is sufficient to show this equality for ∂_J since they are both alternating and ∂_J forms local frames.

(4) The formula

$$d\omega = \sum_I d\omega_I \wedge dx^I$$

is independent on choice of metric and connection, and it coincides with the construction from class.

(5) We know that

$$\nabla_Y \omega(\bar{X}) = Y(\omega(\bar{X})) - \sum_i \omega(X_1, \dots, \nabla_Y X_i, \dots, X_k).$$

Thus

$$\begin{aligned} d\omega(X_1, \dots, X_{k+1}) &= \sum_i (-1)^{i+1} \nabla_{X_i} \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}) \\ &= \sum_i (-1)^{i+1} X_i(\omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1})) - \sum_i (-1)^{i+1} \sum_{j \neq i} \omega(\dots, \hat{X}_i, \dots, \nabla_{X_i} X_j, \dots). \end{aligned}$$

In the final sum, $\hat{X}_i$ may come after $\nabla_{X_i} X_j$. This final sum can thus be written as

$$\sum_{i < j} (-1)^{j+1} \omega(\dots, \nabla_{X_j} X_i, \dots, \hat{X}_j, \dots) + \sum_{i < j} (-1)^{i+1} \omega(\dots, \hat{X}_i, \dots, \nabla_{X_i} X_j, \dots).$$

Permuting the inputs so that the covariant derivative appears first has a sign of $(-1)^{i+1}$ in the first sum and $(-1)^j$ in the second ($i-1$ and j transpositions respectively). So this is equal to

$$= \sum_{i < j} (-1)^{i+j} (\omega(\nabla_{X_j} X_i, \dots) - \omega(\nabla_{X_i} X_j, \dots)),$$

and by symmetry of the connection, this is equal to

$$= \sum_{i < j} (-1)^{i+j} \omega([X_j, X_i], \dots) = - \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], \dots).$$

We subtract this from the first sum, so we see that

$$d\omega(X_1, \dots, X_{k+1}) = \sum_i (-1)^{i+1} X_i(\omega(\dots \hat{X}_i \dots)) + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], \dots)$$

as desired.

11 Homework 11

11.0.1 Problem (DoCarmo, Riemannian Geometry, page 46 question 7)

Let G be a compact connected Lie group of dimension n .

- (1) Show that if ω is a left-invariant n -form, then it is right-invariant.
- (2) Show that there exists a non-vanishing left-invariant n -form ω on G .
- (3) Let $\langle \bullet, \bullet \rangle$ be a left invariant metric on G , and ω a positive n -form on G which is left-invariant. Define a new Riemannian metric $\langle\langle \bullet, \bullet \rangle\rangle$ on G by

$$\langle\langle u, v \rangle\rangle_y = \int_G \langle (dR_x)_y u, (dR_x)_y v \rangle_{yx} \omega_x, \quad x, y \in G, \quad u, v \in T_y G.$$

Show that this is a bi-invariant metric.

- (1) Let $a \in G$, then $R_a^*\omega$ is left-invariant as for $x \in G$ we have

$$L_x^* R_a^* \omega = R_a^* L_x^* \omega = R_a^* \omega.$$

All left-invariant forms are determined by their values at the identity, and since $\Lambda^n T_e^* G$ is one-dimensional, this means that all left-invariant n -forms are scalar multiples of one another. Thus $R_a^* \omega = f(a)\omega$ for some $f(a) \in \mathbb{R} - 0$ (since ω is non-vanishing, so too is $R_a^* \omega$ since R_a is a diffeomorphism).

Now,

$$R_{ba}^* \omega = R_a^* (R_b^* \omega) = R_a^* (f(b)\omega) = f(b) R_a^* \omega = f(b)f(a)\omega,$$

which means that $f(ba) = f(b)f(a)$ so f is a group morphism from G to the multiplicative group of reals $\mathbb{R} - 0$. Furthermore it is continuous, as we can compute $f(a)$ by taking a basis $v_1, \dots, v_n \in T_e G$ then

$$f(a) = \frac{R_a^* \omega(v_1, \dots, v_n)}{\omega(v_1, \dots, v_n)},$$

and $a \mapsto R_a^* \omega$ is smooth, so this is just scaling a smooth map.

Since G is compact and connected, $f(G) \subseteq \mathbb{R} - 0$ is then a compact connected subgroup. It is thus a closed interval $[c, d]$, which means $c^2 \geq c, d^2 \leq d$ which implies $d \leq 1 \leq c$ so $f(G) = \{1\}$. Thus $R_a^* \omega = \omega$ for all $a \in G$, and ω is right-invariant.

- (2) Let (E_i) be a left-invariant global frame of TG , i.e. left-invariant vector fields obtained from a basis of $T_e G$. Take (ε^j) its dual global coframe, then define

$$\omega = \varepsilon^1 \wedge \dots \wedge \varepsilon^n.$$

Then for $a \in G$ we have

$$L_a^* \omega = (L_a^* \varepsilon^1) \wedge \dots \wedge (L_a^* \varepsilon^n).$$

Now we have

$$L_a^* \varepsilon^j(E_i) = \varepsilon^j(L_a^* E_i) = \varepsilon^j(E_i) = \delta_i^j,$$

which shows that $L_a^* \varepsilon^j = \varepsilon^j$, meaning $L_a^* \omega = \omega$, so ω is left-invariant and non-vanishing.

- (3) Since all non-vanishing n -forms on a connected manifold are either positive or negative. This is because if we take a positive n -form ω , all n -forms are multiples of this by a smooth function $u\omega$. If they are nonvanishing, then u is never zero. If the manifold is connected, this means u is either strictly positive or strictly negative.

So let ω be a nonvanishing left-invariant n -form. By potentially multiplying by -1 , we can assume it is positive. Then we define $\langle\langle \bullet, \bullet \rangle\rangle$ as in the question.

By linearity of the differential, it is immediate that $\langle\langle \bullet, \bullet \rangle\rangle$ is bilinear. It is also symmetric by symmetry of $\langle \bullet, \bullet \rangle$. Furthermore, $\langle\langle (dR_x)_y u, (dR_x)_y v \rangle\rangle_{yx}$ is a positive n -form when $u \neq 0$ (since the inner product is positive), and thus its integral is positive. Thus for $u \neq 0$, $\langle\langle u, u \rangle\rangle > 0$. For $u = 0$, the integral is zero, so $\langle\langle u, u \rangle\rangle = 0$.

Now, we have

$$\begin{aligned} \langle\langle d(L_a)_y u, d(L_a)_y v \rangle\rangle_{ay} &= \int_G \langle\langle (dR_x)_{ay} d(L_a)_y u, (dR_x)_{ay} d(L_a)_y v \rangle\rangle_{ayx} \omega \\ &= \int_G \langle\langle d(R_x \circ L_a)_y u, d(R_x \circ L_a)_y v \rangle\rangle_{ayx} \omega \\ &= \int_G \langle\langle d(L_a \circ R_x)_y u, d(L_a \circ R_x)_y v \rangle\rangle_{ayx} \omega \\ &= \int_G \langle\langle d(L_a)_{yx} d(R_x)_y u, d(L_a)_{yx} d(R_x)_y v \rangle\rangle_{ayx} \omega \\ &= \int_G \langle\langle d(R_x)_y u, d(R_x)_y v \rangle\rangle_{yx} \omega \\ &= \langle\langle u, v \rangle\rangle_y, \end{aligned}$$

where the final equality is due to the left-invariance of $\langle \bullet, \bullet \rangle$. So we see that $\langle\langle \bullet, \bullet \rangle\rangle$ remains left-invariant.

Now we claim right-invariance. Let us write $f_{u,v}: G \rightarrow \mathbb{R}$ for the map $f_{u,v}(x) = \langle\langle (dR_x)_y u, (dR_x)_y v \rangle\rangle$. Thus

$$\langle\langle u, v \rangle\rangle_y = \int_G f_{u,v} \omega.$$

And we have

$$L_a^*(f_{u,v}\omega) = (f_{u,v} \circ L_a)(L_a^*\omega) = (f_{u,v} \circ L_a)\omega$$

since ω is left-invariant. Furthermore, for all $a \in G$, L_a is orientation-preserving. This is because $L_a^*\omega = \omega$, and ω is the orientation form.

Thus we have

$$\begin{aligned} \langle\langle d(R_a)_y u, d(R_a)_y v \rangle\rangle_{ya} &= \int_G \langle\langle d(R_x)_{ya} d(R_a)_y u, d(R_x)_{ya} d(R_a)_y v \rangle\rangle_{yax} \omega \\ &= \int_G \langle\langle d(R_x \circ R_a)_y u, d(R_x \circ R_a)_y v \rangle\rangle_{yax} \omega \\ &= \int_G \langle\langle d(R_{ax})_y u, d(R_{ax})_y v \rangle\rangle_{yax} \omega \\ &= \int_G (f_{u,v} \circ L_a) \omega \\ &= \int_G L_a^*(f_{u,v}\omega) \\ &= \int_G f_{u,v}\omega \\ &= \langle\langle u, v \rangle\rangle_y. \end{aligned}$$

Showing right-invariance, as desired. G being compact is necessary for these integrals to be defined, as otherwise ω may not be compactly supported.

Finally smoothness follows from left-invariance. This is because for any vector fields X, Y , by left invariance

$$\langle\langle X, Y \rangle\rangle_a = \langle\langle X_a, Y_a \rangle\rangle_a = \langle\langle d(L_{a^{-1}})X_a, d(L_{a^{-1}})Y_a \rangle\rangle_e,$$

and $a \mapsto d(L_{a^{-1}})X_a$ is smooth as a composition of smooth functions. Then inner product is taken in the identity which is smooth. Thus we have constructed a bi-invariant Riemannian metric.

11.0.2 Problem

Let (M, g) be Riemannian. For a smooth real-valued function $f: M \rightarrow \mathbb{R}$ we define its **gradient** to be the unique vector field ∇f which satisfies $g(\nabla f, X) = df(X) = Xf$ for all vector fields X . The **Hessian** of f is the covariant 2-tensor defined by

$$\text{Hess}f(X, Y) = g(\nabla_X \nabla f, Y).$$

For $p \in M$ define $f_p: M \rightarrow \mathbb{R}$ by $f_p(q) = \frac{1}{2}d(p, q)^2$.

- (1) Show that for $f: M \rightarrow \mathbb{R}$, $\text{Hess}f$ is symmetric.
- (2) Suppose there is a constant $c \geq 0$ such that $\text{Hess}f(v, v) \geq cg(v, v)$ for all tangent vectors v . Let γ be a unit speed geodesic. Show that

$$\frac{d^2}{dt^2}(f \circ \gamma) \geq c.$$

In particular $f \circ \gamma$ is convex, and strictly convex for $c > 0$.

- (3) Let B be a normal ball around p . Show that f_p is smooth on B . Let $\gamma: [0, 1] \rightarrow B$ be a geodesic such that $\gamma(0) = p$ and $\gamma(1) = q$. Show that $\nabla f_p(q) = \dot{\gamma}(q)$.
- (4) Continuing with the case in the previous point, let $v, w \in T_p M$ be such that $\exp_p(v) = \gamma(1)$. For sufficiently small $\varepsilon > 0$ and $(s, t) \in (-\varepsilon, \varepsilon) \times [0, 1]$, let

$$\bar{\gamma}(s, t) = \bar{\gamma}_s(t) = \exp_p(t(v + sw)).$$

Then define

$$J = \left. \frac{d}{ds} \right|_{s=0} \bar{\gamma}$$

to be the Jacobi field along $\gamma = \bar{\gamma}_0$ such that $J(0) = 0$ and $J'(0) = w$. Show that

$$\text{Hess}f_p(J(1), J(1)) = g(J'(1), J(1)).$$

- (5) Again we continue with the previous point. Assume the sectional curvature satisfies $K \leq 0$. For $t > 0$, let

$$\lambda(t) = \frac{g(J'(t), J(t))}{|J(t)|^2}.$$

Prove that $\lambda(t)$ is well-defined and strictly positive for $t > 0$. Prove that $\dot{\lambda}(t) \geq -\lambda^2$, and show this is strict when $K < 0$ and v, w are linearly independent.

- (6) Again we continue with the previous point. Show that $\lambda(t) \rightarrow \infty$ as $t \rightarrow 0$. Using the previous point, show that $\frac{\dot{\lambda}}{\lambda^2} + 1$ is well-defined and non-negative for $t > 0$. Integrate over $t \in [0, 1]$ to obtain $\lambda(1) \geq 1$.
- (7) Let M have nonpositive sectional curvature and $B \subseteq M$ be a normal ball centered at p . Conclude from the previous point that for all $q \in B$ and $v \in T_q M$, $\text{Hess}f_p(v, v) \geq g(v, v)$. Show that if $K < 0$ and $v, \nabla f_p(q)$ are linearly independent, then this inequality is strict.
- (8) Let M be complete and simply connected, with $K \leq 0$. Show that f_p is smooth on all of M .
- (9) Continuing the previous point, let $\varphi: M \rightarrow M$ be an isometry of finite order k . Show that φ has a fixed point as follows: for $p \in M$ let

$$h = \sum_{i=0}^{k-1} f_{\varphi^i(p)}.$$

Show that $h \circ \varphi = h$, and then from point (7) conclude $\text{Hess}h > 0$. Using Hopf-Rinow and part (2), show that h has a unique minimum, which must be a fixed point of φ .

- (10) Continue with point (8). Let $p \in M$ and $\gamma: [0, t] \rightarrow M$ be a geodesic. Combining points (2) and (7), show that

$$\frac{d^2}{dt^2}(f_p \circ \gamma) \geq 1,$$

and integrating twice we obtain

$$d(p, \gamma(t))^2 \geq d(p, \gamma(0))^2 + 2g(\nabla f_p(\gamma(0)), \dot{\gamma}(0))t + t^2.$$

Using part (3), interpret this as a generalized law of cosines. In particular for any geodesic triangle in M with side lengths a, b, c , and angle α opposite side a , we have

$$a^2 \geq b^2 + c^2 - 2bc \cos \alpha.$$

Show that this inequality is strict for $K < 0$ and a non-degenerate triangle.

- (11) Continue with part (10). Show that the sum of angles in a geodesic triangle in M is less than or equal to π . Furthermore this inequality is strict for $K < 0$ and a nondegenerate triangle.

- (1) We have by metric-compatibility

$$\text{Hess}f(X, Y) = g(\nabla_X \nabla f, Y) = X(g(\nabla f, Y)) - g(\nabla f, \nabla_X Y) = X(Yf) - \nabla_X Y(f).$$

And thus

$$\text{Hess}f(Y, X) = Y(Xf) - \nabla_Y X(f).$$

Subtracting both sides gives

$$\text{Hess}f(X, Y) - \text{Hess}f(Y, X) = X(Yf) - Y(Xf) - (\nabla_X Y(f) - \nabla_Y X(f)),$$

and both terms are $[X, Y]f$: the first by definition and the second by symmetry of the connection. Thus $\text{Hess}f(X, Y) = \text{Hess}f(Y, X)$ as desired.

- (2) For t_0 in γ 's domain, let $p = \gamma(t_0)$ and consider a geodesic frame centered at p , (E_i) . Then (E_i) are orthonormal, and so the components of ∇f with respect to this local frame are given by

$$g(\nabla f, E_i) = df(E_i) = E_i(f).$$

Thus

$$\nabla f = \sum_i (E_i f) E_i.$$

Let us also represent $\dot{\gamma}$ in terms of this geodesic frame: $\dot{\gamma} = a^i(t) E_i(\gamma(t))$. Then we have

$$\begin{aligned} \text{Hess}f(\dot{\gamma}(t), \dot{\gamma}(t)) &= g\left(\nabla_{a^i E_i(\gamma)} \sum_j (E_j f) E_j, a^k E_k(\gamma)\right) \\ &= a^i a^k g\left(\nabla_{E_i} \sum_j (E_j f) E_j, E_k\right)_{\gamma} \\ &= \sum_j a^i a^k g(E_i(E_j f) E_j + (E_j f) \nabla_{E_i} E_j, E_k). \end{aligned}$$

Evaluating at t_0 , this becomes

$$= \sum_j a^i a^k g(E_i(E_j f) E_j, E_k) = a^i(t_0) a^j(t_0) E_i(E_j f)|_p.$$

On the other hand,

$$\frac{d}{dt}(f \circ \gamma) = df(a^i(t) E_i(\gamma(t))) = a^i(t) (E_i f) \circ \gamma.$$

Differentiating again gives

$$\frac{d^2}{dt^2}(f \circ \gamma) = \dot{a}^i(t) (E_i f) \circ \gamma + a^i(t) d(E_i f)(a^j E_j|_{\gamma}) = \dot{a}^i(t) (E_i f)|_{\gamma} + a^i a^j E_j(E_i f)|_{\gamma}.$$

Notice that the first term is just $D_t \dot{\gamma} = 0$, so we see that

$$\frac{d^2}{dt^2}(f \circ \gamma) = a^i a^j E_j(E_i f)|_{\gamma},$$

which is equal to $\text{Hess}f(\dot{\gamma}, \dot{\gamma})$ at t_0 .

This is true for all t_0 , so we see that

$$\frac{d^2}{dt^2}(f \circ \gamma) = \text{Hess}f(\dot{\gamma}, \dot{\gamma}) \geq cg(\dot{\gamma}, \dot{\gamma}) = c|\dot{\gamma}|^2 = c$$

as desired.

- (3) Let $\tilde{B} \subseteq T_p M$ be the tangent ball mapped diffeomorphically to B by $\exp_p$. Radial geodesics are minimizing, so for $v \in \tilde{B}$ we have $d(p, \exp_p(v)) = |v|$. Thus

$$f_p(\exp_p(v)) = \frac{1}{2} |v|^2.$$

Thus $f_p \circ \exp_p$ is smooth, and since $\exp_p$ is a diffeomorphism $\tilde{B} \rightarrow B$, we see that f_p is smooth on B .

Let $\tilde{g}$ denote the Euclidean metric on $T_p M$, so $\exp_p: \tilde{B} \rightarrow B$ is an isometry. In general if we have an isometry $\varphi: (N, \tilde{g}) \rightarrow (M, g)$ and $f: M \rightarrow \mathbb{R}$, then

$$\tilde{g}(\nabla(f \circ \varphi), X) = d(f \circ \varphi)(X) = df(\varphi_* X) = g(\nabla f, \varphi_* X).$$

Since φ_* is an isometry, this is equal to

$$= \tilde{g}(\varphi_*^{-1} \nabla f, X).$$

This is true for all X , so

$$\nabla(f \circ \varphi) = \varphi_*^{-1} \nabla f \implies \nabla f = \varphi_* \nabla(f \circ \varphi).$$

In our case,

$$\nabla f_p = (\exp_p)_* \nabla(f \circ \exp_p),$$

and $f \circ \exp_p(v) = |v|^2/2$ whose gradient is then just v (gradients in the Euclidean case are computed classically). The geodesic connecting p and q has the form $\gamma(t) = \exp_p(tv)$ for $q = \exp_p v$, then

$$\nabla f_p(q) = \nabla f_p(\exp_p v) = d(\exp_p)v = \dot{\gamma}(1).$$

(4) Notice that

$$J'(1) = \partial_t \partial_s \bar{\gamma}(0, 1) = \partial_s \partial_t \bar{\gamma}(0, 1) = \partial_s \dot{\bar{\gamma}}_s(1) \Big|_{s=0}$$

where the final equality is due to symmetry. By the previous point

$$\nabla f_p(\bar{\gamma}_s(1)) = \dot{\bar{\gamma}}_s(1).$$

And by definition we have

$$\text{Hess} f_p(J(1), J(1)) = g(\nabla_{J(1)} \nabla f_p, J(1)).$$

Consider the first term:

$$\nabla_{J(1)} \nabla f_p = \nabla_{\partial_s \bar{\gamma}(0, 1)} \nabla f_p,$$

which is equal to the covariant derivative of $\nabla f_p(\bar{\gamma}_s(1))$ with respect to s taken along $\bar{\gamma}_s(1)$, evaluated at 0. That is, it is equal to

$$D_s \nabla f_p(\bar{\gamma}_s(1)) \Big|_{s=0} = D_s \dot{\bar{\gamma}}_s(1) \Big|_{s=0} = J'(1),$$

where the final equality is due to the first equation we showed. Thus we have

$$\text{Hess} f_p(J(1), J(1)) = g(J'(1), J(1))$$

as desired.

(5) Since M has nonpositive sectional curvature, no two points are conjugate along any geodesic. Since $J(0) = 0$, this means that $J(t) \neq 0$ for all $t \neq 0$. Thus λ is well-defined.

Computing the derivative of λ , we get

$$\dot{\lambda} = \frac{(g(J'', J) + g(J', J')) |J|^2 - 2g(J', J)^2}{g(J, J)^2},$$

and

$$\lambda^2 = \frac{g(J', J)^2}{g(J, J)^2}.$$

So $\dot{\lambda} \geq -\lambda^2$ iff

$$(g(J'', J) + |J'|^2) |J|^2 \geq g(J', J)^2.$$

Cauchy-Schwarz gives us

$$g(J', J) \leq |J'| |J|$$

and so if we can show that $g(J'', J)$ is nonnegative we will have the inequality.

By the Jacobi equation

$$g(J'', J) = g(-R(\dot{\gamma}, J)\dot{\gamma}, J) = -\text{Rm}(\dot{\gamma}, J, \dot{\gamma}, J).$$

If $\dot{\gamma}, J$ are linearly dependent this is zero and we are finished. Otherwise this is just a negative multiple of the sectional curvature at $\dot{\gamma}, J$. Since the sectional curvature is nonpositive, this is nonnegative and we are finished.

Notice that this also implies $\lambda(t) > 0$. This is because as we showed,

$$g(J', J)' = |J'|^2 + g(J'', J) \geq 0.$$

Thus $g(J'(t), J(t))$ is nondecreasing, and at t it is zero. Thus it is nonnegative for $t > 0$, and since at $t = 0$ we have $|J'|^2 = |w|^2 > 0$ it is strictly increasing in a neighborhood of 0, so $g(J', J) > 0$ for $t > 0$. Thus $\lambda(t) > 0$ for $t > 0$ as desired.

Now if $K < 0$ and v, w are linearly independent, we want to show this inequality is strict. In lieu of our previous analysis, it is sufficient to show that $\dot{\gamma}(t), J(t)$ is linearly independent. We know that, since $J(0) = 0$, $J'(0) = w$, and $\dot{\gamma}(0) = v$,

$$g(J(t), \dot{\gamma}(t)) = g(J'(0), \dot{\gamma}(0))t + g(J(0), \dot{\gamma}(0)) = g(w, v)t.$$

If J and $\dot{\gamma}$ were dependent, this would mean

$$J = \frac{g(J, \dot{\gamma})}{g(\dot{\gamma}, \dot{\gamma})} \dot{\gamma} = \frac{g(w, v)t}{g(v, v)} \dot{\gamma}.$$

Let us denote the right-hand side by $\hat{J}$. This is a Jacobi field because fields of the form $at\dot{\gamma}(t)$ are Jacobi fields: their first derivative is $a\dot{\gamma}$, and so their second is zero. Then $R(at\dot{\gamma}, \dot{\gamma})\dot{\gamma} = 0$, so they are Jacobi fields.

Since Jacobi fields form a vector space, this means $J - \hat{J}$ is a Jacobi field. But $J(0) = \hat{J}(0) = 0$ and they also agree at another point t . But since M has nonpositive sectional curvature, no two points are conjugate along any geodesic. This means that $J - \hat{J}$ cannot be a nontrivial Jacobi field, as otherwise $\gamma(0), \gamma(t)$ would be conjugate. Thus $J = \hat{J}$, but since v, w are linearly independent

$$\hat{J}'(0) = \frac{g(w, v)}{g(v, v)} v \neq w = J'(0).$$

- (6) We note that as $t \rightarrow 0$, both the numerator and denominator approach 0 since $J(t)$ is smooth and $J(0) = 0$. Thus by Lhopital, the limit is equal to

$$\lim_{t \rightarrow 0} \lambda(t) = \lim_{t \rightarrow 0} \frac{g(J'', J) + g(J', J')}{2g(J', J)}.$$

The numerator of the limit approaches $g(J'(0), J'(0)) > 0$, while the denominator approaches 0. Thus the limit is infinite (everything in the limit is positive, so this converges to $+\infty$), as desired.

Since $\lambda(t) > 0$ for $t > 0$, $\dot{\lambda}/\lambda^2 + 1$ is well-defined for $t > 0$. Since $\dot{\lambda} \geq -\lambda^2$, we see that $\dot{\lambda}/\lambda^2 \geq -1$ so this is nonnegative for $t > 0$. Integrating yields

$$\int_0^1 \frac{\dot{\lambda}}{\lambda^2} + 1 dt = -\frac{1}{\lambda} \Big|_0^1 + 1 = -\frac{1}{\lambda(1)} + 1,$$

as we showed the limit of $\lambda(t)$ as $t \rightarrow 0$ is infinite, and thus $1/\lambda \rightarrow 0$. Since the integrand is nonnegative, we see

$$-\frac{1}{\lambda(1)} + 1 \geq 0 \implies \frac{1}{\lambda(1)} \leq 1 \implies \lambda(1) \geq 1$$

as desired.

Note that if $K < 0$ and v, w are linearly independent then the inequality $\dot{\lambda} > -\lambda^2$ is strict. Thus the integrand is strictly positive and so we get that

$$-\frac{1}{\lambda(1)} + 1 > 0 \implies \lambda(1) > 1.$$

- (7) From (4) we have that

$$\text{Hess}f_p(J(1), J(1)) = g(J'(1), J(1)) = \lambda(1)g(J(1), J(1)) \geq g(J(1), J(1)).$$

For $q \in B$ and $v \in T_q M$, we want a Jacobi field J such that $J(1) = v$.

Let us write $q = \exp_p(u)$, so we want $x, y \in T_p M$ such that $J = \partial_s \exp_p(t(x + sy))|_{s=0}$ maps $J(1) = v$. Since

$$J(1) = \frac{\partial}{\partial s} \Big|_{s=0} \exp_p(x + sy) = d(\exp_p)_x(y),$$

we need $\exp_p(x) = q$, meaning we need $x = u$. Then since $\exp_p$ is a diffeomorphism to B , there will exist a y which satisfies this. So this gives us a Jacobi field J where $J(1) = v$, and thus

$$\text{Hess} f_p(v, v) \geq g(v, v).$$

This inequality is strict when x, y are linearly independent, as then $\lambda(1) > 1$. Since $x = u$ and $y = d(\exp_p^{-1})_u(v)$, this is iff $v = d\exp_p(y)$ and $d\exp_p(u)$ are linearly independent. Now,

$$\nabla f_p(q) = \dot{\gamma}(1),$$

where γ is the geodesic connecting p and q , so $\gamma(t) = \exp_p(tu)$. Then

$$\nabla f_p(q) = d\exp_p(u),$$

so x, y are linearly independent iff $v, \nabla f_p(q)$ are linearly independent as desired.

- (8) By Hadamard, $\exp_p: T_p M \rightarrow M$ is a diffeomorphism. This means that M is itself a normal ball, and as we showed f_p is smooth on normal balls. Thus f_p is smooth, as desired.
- (9) Notice that since φ is an isometry,

$$f_{\varphi^i(p)} \circ \varphi(q) = \frac{1}{2} d(\varphi^i(p), \varphi(q))^2 = \frac{1}{2} d(\varphi^{i-1}(p), q) = f_{\varphi^{i-1}(p)}(q).$$

Thus $f_{\varphi^i(p)} \circ \varphi = f_{\varphi^{i-1}(p)}$ and so

$$h \circ \varphi = \sum_{i=0}^{k-1} f_{\varphi^i(p)} \circ \varphi = \sum_{i=0}^{k-1} f_{\varphi^{i-1}(p)} = h.$$

Now notice that given two smooth real-valued functions f, f' , we have $\nabla(f + f') = \nabla f + \nabla f'$. This is because $d(f + f') = df + df'$, and the musical isomorphisms being linear. Thus

$$\begin{aligned} \text{Hess}(f + f')(X, Y) &= g(\nabla_X \nabla(f + f'), Y) = g(\nabla_X (\nabla f + \nabla f'), Y) = \\ &= g(\nabla_X \nabla f, Y) + g(\nabla_X \nabla f', Y) = \text{Hess} f(X, Y) + \text{Hess} f'(X, Y). \end{aligned}$$

Thus $\text{Hess}(f + f') = \text{Hess} f + \text{Hess} f'$.

And thus

$$\text{Hess} h = \sum_{i=0}^{k-1} \text{Hess}(f_{\varphi^i(p)}).$$

Since $\text{Hess} f_p(v, v) \geq g(v, v) > 0$ for all p , we see that $\text{Hess} h \geq kg(v, v) > 0$ as desired.

Now suppose h had two minima, $p \neq q$. Since M is complete, there is a minimizing unit geodesic γ from p to q , so $p = \gamma(0)$ and $q = \gamma(a)$ say. Then from (2) we have

$$\frac{d^2}{dt^2}(h \circ \gamma) \geq k > 0,$$

which means that $h \circ \gamma$ is strictly convex, and thus can only have a single minimum, contradicting both $h \circ \gamma(0)$ and $h \circ \gamma(a)$ both being minima.

Now since $h(p) \geq 0$, h 's image is bounded from below. Let $c = \inf h(M)$, and let $x_n \in M$ be a sequence such that $h(x_n) \rightarrow c$. For sufficiently large n , we then have $h(x_n) < c + 1$, and since $d(p, x)^2 \leq h(x)$ we have $d(p, x_n) \leq \sqrt{c + 1}$. Thus x_n eventually lies within the closed metric ball around p of radius $\sqrt{c + 1}$. This is bound and closed, so by Hopf-Rinow, compact. Thus x_n has a convergent subsequence x_{m_n} , say $x_{m_n} \rightarrow x_0$.

Since $h(x_n) \rightarrow c$, we have that $h(x_{m_n}) \rightarrow c$ and by continuity $h(x_{m_n}) \rightarrow h(x_0)$. Thus $h(x_0) = c$, and since c is the infimum of the image of h , x_0 is a minimum of h . As we showed the minimum is unique, it is the only minimum.

Now, $h \circ \varphi = h$ means that $h(\varphi(x_0)) = h(x_0)$ is minimal. Since h has a unique minimum, this means $\varphi(x_0) = x_0$ so x_0 is a fixed point of φ .

- (10) From (7) we have $\text{Hess}f_p(v, v) \geq g(v, v)$ which means by (2) that

$$\frac{d^2}{dt^2}(f_p \circ \gamma) \geq 1$$

for all unit velocity geodesics γ . Integrating once we see that

$$\frac{d}{dt}f_p \circ \gamma(t) = \int_0^t \frac{d^2}{dt^2}(f_p \circ \gamma) + \frac{d}{dt}f_p \circ \gamma(0) \geq t + \frac{d}{dt}f_p \circ \gamma(0).$$

Integrating again gives

$$f_p \circ \gamma(t) \geq \frac{t^2}{2} + t \frac{d}{dt}f_p \circ \gamma(0) + f_p \circ \gamma(0).$$

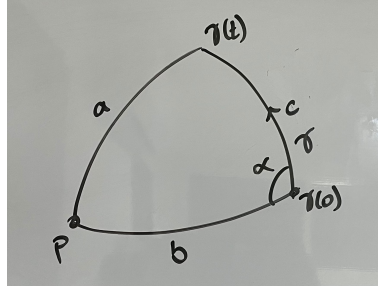
Recall that

$$g(\nabla f_p(\gamma(0)), \dot{\gamma}(0)) = df_p(\dot{\gamma}(0)) = \frac{d}{dt}f_p \circ \gamma(0).$$

Thus multiplying the above inequality by two, we get the desired

$$d(p, \gamma(t))^2 \geq d(p, \gamma(0))^2 + 2g(\nabla f_p(\gamma(0)), \dot{\gamma}(0))t + t^2.$$

Consider the following geodesic triangle, where γ is a minimizing unit-velocity geodesic



Then $a^2 = d(p, \gamma(t))^2$, $b = d(p, \gamma(0))^2$, and $c = L(\gamma)^2 = t^2$. If we let β be the geodesic from $\beta(0) = \gamma(0)$ to $\beta(1) = p$, then we know that $\nabla f_p(\gamma(0)) = -\dot{\beta}(0)$ (since $\beta(1-t)$ is the geodesic in the reverse direction). Thus

$$g(\nabla f_p(\gamma(0)), \dot{\gamma}(0)) = -g(\dot{\beta}(0), \dot{\gamma}(0)).$$

The angle α is given by the angle between $\dot{\beta}(0)$ and $\dot{\gamma}(0)$, so

$$\cos \alpha = \frac{g(\dot{\beta}(0), \dot{\gamma}(0))}{|\dot{\beta}|}.$$

And since β is a geodesic, $b = L(\beta) = |\dot{\beta}|$. Thus

$$2g(\nabla f_p(\gamma(0)), \dot{\gamma}(0))t = -2bt \cos \alpha.$$

Since $c = t$ we see that this is equal to $-2bc \cos \alpha$.

Thus we have

$$a^2 \geq b^2 + c^2 - 2bc \cos \alpha$$

as desired.

This inequality is strict when $K < 0$ and the triangle is non-degenerate because then the inequality $\frac{d^2}{dt^2}(f_p \circ \gamma) > 1$ is strict. Then integrating preserves this strict inequality.

- (11) Let T be the triangle in M , with side lengths a, b, c and associated angles α, β, γ . By the triangle inequality, $c \leq a + b$ so there exists a Euclidean triangle T' with the same side lengths. Let the associated angles be α', β', γ' . Then we have

$$a^2 = b^2 + c^2 - 2bc \cos \alpha' \geq b^2 + c^2 - 2bc \cos \alpha.$$

This implies that $\cos \alpha \geq \cos \alpha'$. Since in $[0, \pi]$ cosine is decreasing, this means that $\alpha' \geq \alpha$. The same holds for β, γ .

Thus we have

$$\alpha + \beta + \gamma \leq \alpha' + \beta' + \gamma' = \pi$$

as desired. When $K < 0$ and the triangle is non-degenerate, then the inequality above is strict: $\cos \alpha > \cos \alpha'$. Thus $\alpha < \alpha'$ and the inequality of the sum is strict.

12 Homework 12

12.0.1 Problem

Let G be a Lie group equipped with a bi-invariant metric. We show in this exercise that a 1-parameter subgroups and geodesics coincide. Let $\gamma: \mathbb{R} \rightarrow G$ be a geodesic starting at e .

- (1) Let B be a normal ball around e of radius $\varepsilon > 0$. Let s be small enough so that $\gamma|_{[0,s]} \subseteq B$. Let $\varphi = L_{\gamma(s)}$ be the isometry given by left-multiplication by $\gamma(s)$. Show that

$$\left. \frac{d}{dt} \right|_{t=0} (\varphi \circ \gamma) = \dot{\gamma}(s).$$

- (2) Conclude that $\varphi \circ \gamma(t) = \gamma(s+t)$ for all t . Thus for s sufficiently small and all t , $\gamma(s)\gamma(t) = \gamma(s+t)$.
- (3) Conclude $\gamma(s)\gamma(t) = \gamma(s+t)$ for all s, t .

- (1) Since $\gamma|_{[0,s]}$ is contained in a normal ball, it is a radial geodesic and thus minimizing. Let $s' \in (0, s)$, then consider the curve α obtained by concatenating $\gamma|_{[s',s]}$ and $\varphi \circ \gamma|_{[0,s']}$. This is a curve $\gamma(s')$ to $\varphi \circ \gamma(s')$. Furthermore, since $\gamma|_{[s',s]}$ is minimizing and φ is an isometry, so is $\varphi \circ \gamma|_{[0,s']}$. Thus
- $$L(\alpha) = L(\gamma|_{[s',s]}) + L(\varphi \circ \gamma|_{[0,s']}) = d(\gamma(s'), \gamma(s)) + d(\varphi(e), \varphi \circ \gamma(s')) = d(\gamma(s'), \gamma(s)) + d(e, \gamma(s')).$$
- Since all these points, $e = \gamma(0), \gamma(s'), \gamma(s)$, lie on the minimizing γ this is equal to $d(e, \gamma(s))$. Since right-multiplication by $\gamma(s')$ is also an isometry,

$$L(\alpha) = d(e, \gamma(s)) = d(e \cdot \gamma(s'), \gamma(s) \cdot \gamma(s')) = d(\gamma(s'), \varphi \circ \gamma(s')).$$

Thus α is minimizing and thus a geodesic. Since $\alpha(t) = \gamma(t+s')$ for $t < s-s'$, by the uniqueness of geodesics we must have $\alpha(t) = \gamma(t+s')$ for all t . In particular

$$\gamma(s+t) = \alpha(s-s'+t) = \varphi \circ \gamma(t),$$

and thus differentiating both sides and evaluating at $t=0$ gives

$$\left. \frac{d}{dt} \right|_{t=0} \varphi \circ \gamma(t) = \dot{\gamma}(s)$$

as desired.

- (2) As φ is an isometry, $\varphi \circ \gamma$ is a geodesic starting at $\varphi(e) = \gamma(s)$ with initial velocity $\dot{\gamma}(s)$. By uniqueness of geodesics, $\varphi \circ \gamma(t) = \gamma(t+s)$ for all t , since $\gamma(t+s)$ is another geodesic with these same initial conditions. By definition of φ this means $\gamma(s)\gamma(t) = \gamma(t+s)$ for all t .
- (3) Let s be arbitrary, then there is some natural n such that s/n is small enough to apply the above point (i.e. $\gamma|_{[0,s/n]}$ is contained in B). Then for all t we have $\gamma(s/n+t) = \gamma(s/n)\gamma(t)$. In particular,

$$\gamma(s+t) = \gamma\left(n \frac{s}{n} + t\right) = \gamma\left(\frac{s}{n}\right) \gamma\left((n-1) \frac{s}{n} + t\right),$$

and continuing this inductively,

$$\gamma(s+t) = \gamma\left(\frac{s}{n}\right)^n \gamma(t).$$

In particular for $t=0$ we see that $\gamma(s) = \gamma(s/n)^n$, and so for arbitrary t we have

$$\gamma(s+t) = \gamma\left(\frac{s}{n}\right)^n \gamma(t) = \gamma(s)\gamma(t)$$

as desired.

12.0.2 Problem

Let M be an orientable Riemannian n -manifold with positive curvature and even dimension. Let γ be a closed geodesic in M , i.e. an immersion $S^1 \rightarrow M$ which is a geodesic at all points. Prove that γ is homotopic to a closed curve whose length is strictly less than γ 's.

Let $w = \dot{\gamma}(0)$, where 0 denotes any distinguished point on S^1 , we will also view γ as a smooth geodesic $[0, 1] \rightarrow M$. Let $e_1, \dots, e_{n-1}$ be an orthonormal basis for the orthogonal complement $w^\perp$. Now consider parallel transport P^γ ; we claim there is a tangent vector $v \in w^\perp$ such that $P^\gamma v = v$.

Firstly, P^γ is an isometry because the connection is metric (since $\langle V, W \rangle$ is constant when V, W are parallel along γ). Thus $P^\gamma(e_i)$ forms an orthonormal basis, so there is an orthogonal matrix A such that $P^\gamma(e_i) = A_i^j e_j$. Furthermore we know that $\dot{\gamma}$ is parallel along γ as it is a geodesic, so $P^\gamma w = \dot{\gamma}(1) = \dot{\gamma}(0) = w$.

Let ω be an orientation form for M . Let V_i be the parallel transports of e_i , and $W = \dot{\gamma}$ is w 's; then $\omega(V_1(t), \dots, V_{n-1}(t), W(t)) \neq 0$ for all t as $V_1(t), \dots, V_{n-1}(t), W(t)$ forms a basis for $T_{\gamma(t)}M$ as parallel transport is a linear isomorphism. So $\omega(\bar{V}(t), W(t))$ is a continuous non-vanishing map, since its domain is connected this means $\omega(\bar{V}(1), W(1)) = \omega(\bar{V}(1), w)$ and $\omega(\bar{V}(0), W(0)) = \omega(e_1, \dots, e_{n-1}, w)$ have the same sign.

Since

$$\omega(\bar{V}(1), w) = \omega(A\bar{e}, w) = \det(A \oplus (1))\omega(\bar{e}, w) = \det A \omega(\bar{e}, w),$$

this means that $\det A > 0$. Thus A is an orthogonal matrix with positive determinant of odd dimension $(n-1)$; we showed in class that this means that it has a nonzero fixed point $(a^1, \dots, a^{n-1})$. Let $v = a^i e_i$, so that

$$P^\gamma(v) = a^i P^\gamma(e_i) = A_i^j a^j e_j = a^i e_i = v.$$

Let $V(t)$ be v 's parallel transport along γ . Since $V(1) = v = V(0)$, this means we can view V as a smooth vector field $S^1 \rightarrow M$ along γ . Let Γ be a proper variation of γ with variation field V , and let $E(s) = E(\Gamma_s)$. Since γ is a geodesic $E'(0) = 0$. Furthermore since V is smooth and proper, and Γ is smooth and proper as well (or at least can be taken to be so),

$$\frac{1}{2}E''(0) = - \int_0^1 \langle V(t), V''(t) + R(\dot{\gamma}(t), V(t))\dot{\gamma}(t) \rangle dt.$$

Since V is parallel along γ , $V' = 0$ and thus $V'' = 0$. So this is equal to

$$\frac{1}{2}E''(0) = - \int_0^1 \langle V(t), R(\dot{\gamma}(t), V(t))\dot{\gamma}(t) \rangle dt = - \int_0^1 \text{Rm}(\dot{\gamma}(t), V(t), \dot{\gamma}(t), V(t)) dt.$$

The integrand is a positive multiple of the sectional curvature (since $v = V(0)$ and $w = \dot{\gamma}(0)$ are linearly independent, $V(t), \dot{\gamma}(t)$ remain linearly independent), so this integral is positive. Thus $E''(0)$ is negative, and since $E'(0) = 0$, this means 0 is a local maximum of $E(s)$.

Let $s > 0$ be such that $E(s) < E(0)$. Then $\gamma_s(t) = \Gamma(s, t)$ is a smooth curve where

$$L(\gamma_s) \leq E(\gamma_s)^2 = E(s)^2 < E(0)^2 = E(\gamma)^2 = L(\gamma),$$

where the last equality is because γ is a geodesic defined on $[0, 1]$. And γ, γ_s are homotopic, as Γ forms a homotopy between them (since by definition $\gamma(t) = \Gamma(0, t)$ and $\gamma_s(t) = \Gamma(s, t)$).

12.0.3 Problem

Let $f: \bar{M} \rightarrow M$ be a Riemannian submersion, meaning it is a surjective submersion and it preserves the metric between vectors orthogonal to its fibers.

A vector $\bar{x} \in T_{\bar{p}}\bar{M}$ is **horizontal** if it is orthogonal to the fiber $f^{-1}p$ ($p = f\bar{p}$). Similarly it is **vertical** if it is tangent to the fiber. Thus there is a decomposition,

$$T_{\bar{p}}\bar{M} = (T_{\bar{p}}\bar{M})^h \oplus (T_{\bar{p}}\bar{M})^v, \quad (T_{\bar{p}}\bar{M})^v = T_{\bar{p}}f^{-1}p, \quad (T_{\bar{p}}\bar{M})^h = (T_{\bar{p}}\bar{M})^v{}^\perp.$$

For $X \in \mathfrak{X}(M)$, the **horizontal lift** of $\bar{X}$ of X is the horizontal vector field defined by $df_{\bar{p}}(\bar{X}|_{\bar{p}}) = X|_p$.

- (1) Show that $\bar{X}$ is differentiable.
- (2) Let $\nabla, \bar{\nabla}$ be the Riemannian connections of $M, \bar{M}$ respectively. Show that

$$\bar{\nabla}_{\bar{X}}\bar{Y} = \bar{\nabla}_X Y + \frac{1}{2}[\bar{X}, \bar{Y}]^v, \quad X, Y \in \mathfrak{X}(M),$$

where Z^v is the vertical component of $Z \in \mathfrak{X}(\bar{M})$.

- (3) Show that $[\bar{X}, \bar{Y}]_{\bar{p}}$ depends only on $\bar{X}|_{\bar{p}}$ and $\bar{Y}|_{\bar{p}}$.

- (1) Notice that $df_{\bar{p}}$ is an isomorphism $T_{\bar{p}}\bar{M}^h \rightarrow T_p M$, as f is constant on its fibers and thus df_p is zero on $T_{\bar{p}}\bar{M}^v$. Since df_p is surjective as f is a submersion, and the dimensions of the horizontal part and $T_p M$ are equal, this means that this must be an isomorphism. Thus $\bar{X}$ is uniquely determined; we need only show it is smooth.

Since f is a submersion, there are local charts $\bar{p} \in (U, (\bar{x}, \bar{y}))$ and $p \in (V, (\tilde{x}))$ such that f has the local representation $(\bar{x}, \bar{y}) \mapsto \tilde{x}$. Thus $df(\partial/\partial \bar{x}^i) = \partial/\partial \tilde{x}^i \circ f$ and $df(\partial/\partial \bar{y}^j) = 0$. This means that the $\partial/\partial \bar{y}^j$ span $T_{\bar{p}}\bar{M}^v$. Applying Gram-Schmidt to $\partial/\partial \bar{y}^j, \partial/\partial \bar{x}^i$ (in this order) gives a smooth local orthonormal frame $\bar{F}, \bar{E}$. The span of $\bar{F}$ and $\partial/\partial \bar{y}^j$ are the same, so $\bar{F}$ is an orthonormal frame of $T_{\bar{p}}\bar{M}^v$ and thus $\bar{E}$ is an orthonormal frame of $T_{\bar{p}}\bar{M}^h$.

Now, there are smooth $a^i \in C^\infty(V)$ such that locally $X = a^i \partial/\partial \tilde{x}^i$. Define $\bar{Y} = (a^i \circ f) \partial/\partial \bar{x}^i$, which is a local vector field of $\bar{M}$ and

$$df\bar{Y} = (a^i \circ f) \frac{\partial}{\partial \tilde{x}^i} \circ f = X \circ f.$$

We can project $\bar{Y}$ onto its $E_1, \dots, E_n$ components, which is smooth since F_j, E_i form a local frame. Since the other components F_j are in df 's kernel, this does not affect the image. That is to say, $\bar{Y}^h$ is smooth and $df\bar{Y}^h = df\bar{Y} = X \circ f$.

By uniqueness, $\bar{Y}^h = \bar{X}$ in this neighborhood. Since $\bar{Y}^h$ is smooth, so too must $\bar{X}$ be in this neighborhood. This neighborhood was chosen around an arbitrary point, so $\bar{X}$ is smooth.

- (2) Let $X, Y, Z \in \mathfrak{X}(M)$ and $T \in \mathfrak{X}(\bar{M})$ be a vertical field. Notice that every vector field of $\mathfrak{X}(\bar{M})$ has a horizontal and vertical component. By the same argument as the previous point, the horizontal and vertical components of a vector field are smooth. This is because we can find a local orthonormal frame $\bar{F}, \bar{E}$ for the vertical and horizontal tangent bundles and then the components of a vector field in $\mathfrak{X}(\bar{M})$ are just the projections onto $\bar{F}$ and $\bar{E}$ respectively, which is smooth.

Anyway, since $\bar{X}$ is horizontal and T is vertical, $\langle \bar{X}, T \rangle = 0$ (at every point they are by definition orthogonal). Since $\bar{Y}, \bar{Z}$ are horizontal, and f is Riemannian and thus preserves the metric between them,

$$\bar{X} \langle \bar{Y}, \bar{Z} \rangle = \bar{X} \langle df\bar{Y}, df\bar{Z} \rangle = \bar{X} (\langle Y, Z \rangle \circ f) = df(\bar{X}) \langle Y, Z \rangle = X \langle Y, Z \rangle.$$

The second-to-last equality is because $df(v)(u) = v(u \circ f)$ for all smooth real-valued u and derivations v . Similarly,

$$T \langle \bar{X}, \bar{Y} \rangle = T(\langle X, Y \rangle \circ f) = df(T) \langle X, Y \rangle = 0$$

for the same reason.

We also have $df[\bar{X}, T] = [df\bar{X}, df(T)] = [X, 0] = 0$. And $df[\bar{X}, \bar{Y}] = [df\bar{X}, df\bar{Y}] = [X, Y]$.

Notice that for $V \in \mathfrak{X}(\bar{M})$ arbitrary, since $\bar{X}$ is horizontal

$$\langle V, \bar{X} \rangle = \langle V^h, \bar{X} \rangle = \langle df(V^h), df\bar{X} \rangle = \langle df(V), X \rangle.$$

This is because projecting onto the horizontal component is an orthogonal projection, and $\bar{X}$ is already horizontal; and $df(V) = df(V^h) + df(V^v) = df(V^h)$ since df is zero on vertical vectors. Thus we have

$$\langle [\bar{X}, \bar{Y}], \bar{Z} \rangle = \langle df[\bar{X}, \bar{Y}], df\bar{Z} \rangle = \langle [X, Y], Z \rangle,$$

and

$$\langle [\bar{X}, T], \bar{Y} \rangle = \langle df[\bar{X}, T], df\bar{Y} \rangle = 0.$$

By Koszul's formula,

$$\langle \nabla_{\bar{X}} \bar{Y}, \bar{Z} \rangle = \frac{1}{2} (\bar{X} \langle \bar{Y}, \bar{Z} \rangle + \bar{Y} \langle \bar{Z}, \bar{X} \rangle - \bar{Z} \langle \bar{X}, \bar{Y} \rangle - \langle \bar{Y}, [\bar{X}, \bar{Z}] \rangle - \langle \bar{Z}, [\bar{Y}, \bar{X}] \rangle + \langle \bar{X}, [\bar{Z}, \bar{Y}] \rangle).$$

Our computations showed that this is equal to the same expression but with X, Y, Z instead of $\bar{X}, \bar{Y}, \bar{Z}$. By Koszul's formula, this is just $\langle \nabla_X Y, Z \rangle$ so

$$\langle \nabla_{\bar{X}} \bar{Y}, \bar{Z} \rangle = \langle \nabla_X Y, Z \rangle.$$

Again by Koszul,

$$\langle \nabla_{\bar{X}} \bar{Y}, T \rangle = \frac{1}{2} (\bar{X} \langle \bar{Y}, T \rangle + \bar{Y} \langle T, \bar{X} \rangle - T \langle \bar{X}, \bar{Y} \rangle - \langle \bar{Y}, [\bar{X}, T] \rangle - \langle T, [\bar{Y}, \bar{X}] \rangle + \langle \bar{X}, [T, \bar{Y}] \rangle).$$

As we showed, all of these terms are zero except for $-\langle T, [\bar{Y}, \bar{X}] \rangle = \langle [\bar{X}, \bar{Y}], T \rangle$. Since T is vertical, this is equal to $\langle [\bar{X}, \bar{Y}]^v, T \rangle$. Thus

$$\langle \nabla_{\bar{X}} \bar{Y}, T \rangle = \frac{1}{2} \langle [\bar{X}, \bar{Y}]^v, T \rangle.$$

And so

$$\langle \nabla_{\bar{X}} \bar{Y}, \bar{Z} + T \rangle = \langle \nabla_X Y, Z \rangle + \left\langle \frac{1}{2} [\bar{X}, \bar{Y}]^v, T \right\rangle.$$

Since $\langle X, Z \rangle = \langle \bar{X}, \bar{Z} \rangle$, we see (also because $\bar{Z}$ is horizontal and thus orthogonal to vertical vectors, and T is orthogonal to horizontal vectors),

$$= \left\langle \nabla_X Y + \frac{1}{2} [\bar{X}, \bar{Y}]^v, \bar{Z} + T \right\rangle.$$

If we evaluate at a specific point $\bar{p}$ where all tangent vectors can be decomposed to their vertical and horizontal components, and the horizontal component is the value of a horizontal lift of some vector field of M , we get that at $\bar{p}$ for all $v \in T_{\bar{p}} \bar{M}$

$$\langle \nabla_{\bar{X}} \bar{Y}|_{\bar{p}}, v \rangle = \left\langle \nabla_X Y|_{\bar{p}} + \frac{1}{2} [\bar{X}, \bar{Y}]_{\bar{p}}^v, v \right\rangle,$$

and thus we have equality

$$\nabla_{\bar{X}} \bar{Y} = \nabla_X Y + \frac{1}{2} [\bar{X}, \bar{Y}]^v$$

at $\bar{p}$, and thus over all of $\bar{M}$ as $\bar{p}$ is arbitrary.

More explicitly, for tangent v , write $v = v^v + v^h$ and then let $Z \in \mathfrak{X}(M)$ be such that $Z_p = df_{\bar{p}}(v^h)$ so that $\bar{Z}_{\bar{p}} = v^h$. And extend v^v to a vertical vector field T . Then we get the desired result.

- (3) By additivity, it is sufficient to show that if $\bar{X}(\bar{p}) = 0$ then $[\bar{X}, \bar{Y}]_{\bar{p}}^v = 0$ (and similarly for $\bar{Y}$, but this is symmetrical). To show that $[\bar{X}, \bar{Y}]^v = 0$ it is sufficient to show that $\langle [\bar{X}, \bar{Y}]^v, V \rangle = 0$ for arbitrary $V \in \mathfrak{X}(\bar{M})$. For horizontal V this is clearly true, so it is sufficient to show this for $V = T$ vertical in which case

$$\langle [\bar{X}, \bar{Y}]^v, T \rangle = \langle [\bar{X}, \bar{Y}], T \rangle = \langle \nabla_{\bar{X}} \bar{Y} - \nabla_{\bar{Y}} \bar{X}, T \rangle.$$

(Everything here is being evaluated at $\bar{p}$).

Since $\nabla \bar{Y}$ is tensorial, since $\bar{X}_{\bar{p}} = 0$ we have $\nabla_{\bar{X}} \bar{Y} = 0$ at $\bar{p}$. So we just need to show that $\nabla_{\bar{Y}} \bar{X}$ is horizontal. Let (E_i) be a local horizontal frame around $\bar{p}$, so that $\bar{X} = u^i E_i$ for smooth real-valued u^i (since $\bar{X}$ is horizontal it has a representation by E_i). Since $X|_{\bar{p}} = 0$ this means $u^i(\bar{p}) = 0$ for all i . Then

$$\nabla_{\bar{Y}} \bar{X} = \nabla_{\bar{Y}} u^i E_i = u^i \nabla_{\bar{Y}} E_i + (\bar{Y} u^i) E_i.$$

Evaluating at $\bar{p}$, the first term vanishes as $u^i(\bar{p}) = 0$. The second term is horizontal as the E_i are. Thus at $\bar{p}$, $\nabla_{\bar{Y}} \bar{X}$ is horizontal and thus orthogonal to T . So $[\bar{X}, \bar{Y}]^v$ is orthogonal to all vertical T , and thus is zero.

Thus we are done, but I spell out the end. Firstly, clearly $\overline{X' - X} = \bar{X} - \bar{X}'$ as both are horizontal lifts of $X - X'$ by linearity of df ; also projection $\bullet^v$ is linear. If $\bar{X}'_{\bar{p}} = \bar{X}_{\bar{p}}$ then at $\bar{p}$, $[\bar{X}, \bar{Y}]^v - [\bar{X}', \bar{Y}]^v = [\bar{X} - \bar{X}', \bar{Y}]^v = 0$ as we just showed. Thus $[\bar{X}, \bar{Y}]^v = [\bar{X}', \bar{Y}]^v$ at $\bar{p}$.

For $\bar{Y}$, if $\bar{Y}_{\bar{p}} = 0$ then $[\bar{Y}, \bar{X}]^v = 0$ at $\bar{p}$ and thus $[\bar{X}, \bar{Y}]^v = -[\bar{Y}, \bar{X}]^v = 0$. Thus we obtain the result for $\bar{Y}$. So $[\bar{X}, \bar{Y}]^v$'s value at $\bar{p}$ is determined only by $\bar{X}$ and $\bar{Y}$'s.

12.0.4 Problem

For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, let $E \rightarrow M$ be a smooth $\mathbb{K}$ -vector bundle of rank r over a smooth manifold M .

- (1) Consider the r -fold fiber product $E^r = E \times_M \cdots \times_M E \rightarrow M$, which is a smooth manifold. By construction, its fibers over $p \in M$ are $E_p \oplus \cdots \oplus E_p$. Let $\text{Fr}(E) \subseteq E^r$ be the subset given by tuples $v_1, \dots, v_r \in E_p$ of bases, as p ranges. Show that $\text{Fr}(E)$ is open and thus a submanifold.
- (2) Show that $\text{GL}_r(\mathbb{K})$ acts properly, smoothly, and freely on $\text{Fr}(E)$ by $(a_i^j) \cdot (p, (v_i)) = (p, (a_i^j v_j))$. In the case that $\mathbb{K} = \mathbb{R}$, show that this action restricts to a proper, smooth, free action by the subgroup $\text{GL}_r^+(\mathbb{R})$ of matrices of positive determinant.
- (3) Show that $\text{Fr}(E)/\text{GL}_r(\mathbb{K})$ is diffeomorphic to M . When $\mathbb{K} = \mathbb{R}$, define the quotient $\text{Or}(E) = \text{Fr}(E)/\text{GL}_r^+(\mathbb{R})$. Show that $\text{Or}(E) \rightarrow M$ is a two-fold cover of M .
We say that E is **orientable** if there is a section of the covering $\text{Or}(E) \rightarrow M$. An **orientation** is a choice of a section.
- (4) Show that M is orientable (as a smooth manifold) iff TM is (as a vector bundle).
- (5) The Möbius band M is the quotient of $S^1 \times \mathbb{R}$ by the action of $\mathbb{Z}/2$ where the action by the non-trivial element is given by $(\theta, x) \mapsto (\theta + \pi, -x)$. Show that M is not orientable by showing that TM is not.
- (6) Let $\pi: \text{Or}(E) \rightarrow M$ be the two-fold cover of M from point (3). Show that the pullback bundle $\pi^*E \rightarrow \text{Or}(E)$ has a natural orientation. Deduce that $\text{Or}(TM)$ is orientable as a manifold.

- (1) Let $\pi_E: E \rightarrow M$ be the projection map. Let $\Phi: \pi_E^{-1}U \rightarrow U \times \mathbb{K}^r$ be a local trivialization, and define its r -fold product $\Phi^{\times r}: (\pi_E^{-1}U)^{\times r} \rightarrow U^{\times r} \times \mathbb{K}^{r \times r}$. Since E^r is an embedded submanifold of $E^{\times r}$ (as shown during the construction of the fiber product), restricting $\Phi^{\times r}$ to E^r remains smooth; call this restriction $\tilde{\Phi}^r$. In fact as the fiber product of vector bundles, $\tilde{\Phi}^r$ is a local trivialization of E^r . So we have in particular $(\pi_E^{-1}U)^{\times r} = \pi_{E^r}^{-1}U$.

Now consider the composition

$$\tilde{\Phi}^r: \pi_{E^r}^{-1}U \xrightarrow{\Phi^r} U^{\times r} \times \mathbb{K}^{r \times r} \xrightarrow{\text{proj}} \mathbb{K}^{r \times r} \xrightarrow{\det} \mathbb{K}.$$

This maps a tuple $(v_1, \dots, v_r)$ in E^r to their representation by the local frame associated to Φ , and then to the determinant of this matrix. This tuple is linearly independent iff the representation by the local frame is, which is iff the representation matrix is invertible. Thus

$$\text{Fr}(E) \cap (\pi_{E^r}^{-1}U = (\tilde{\Phi}^r)^{-1}(\mathbb{K} - 0),$$

which is open as $\tilde{\Phi}^r$ is continuous. Since the trivializations cover E^r , this means that $\text{Fr}(E)$ is open as desired.

- (2) This is clearly a group action. We consider the representation of the action in each local trivialization of E^r , under which $\text{Fr}(E)$ has the image of $\text{GL}_r(\mathbb{K})$. In each local trivialization, this group action corresponds to multiplication $\text{GL}_r(\mathbb{K}) \times \text{GL}_r(\mathbb{K}) \rightarrow \text{GL}_r(\mathbb{K})$ (this follows from the linearity of the trivializations on the fibers). This is smooth in each local trivialization, and thus smooth on all of $\text{Fr}(E)$. The action of $\text{GL}_r(\mathbb{K})$ on itself is free and proper as well, and thus so is this action.
Since the action of $\text{GL}_r(\mathbb{R})$ is free and smooth, so is $\text{GL}_r^+(\mathbb{R})$'s as an open subgroup. And it is proper since $\text{GL}_r^+(\mathbb{R})$ is a connected component of $\text{GL}_r(\mathbb{R})$ and thus closed in $\text{GL}_r(\mathbb{R})$. In general if $H \subseteq G$ is a closed subgroup and G acts properly on M , so does H : for $K \subseteq M$ compact, $H_K = G_K \cap H$. Since G_K is compact and H is closed, H_K is a closed subset of G_K and thus compact.
- (3) The projection $\bar{\pi}: \text{Fr}(E)/\text{GL}_r(\mathbb{K}) \rightarrow M$ is bijective. This is because we can construct an inverse by mapping $p \in M$ to $(p, [e_1, \dots, e_r])$ where $e_1, \dots, e_r$ is any basis of E_p . This is independent on choice of basis, as they can be mapped to one another via a transition matrix.

So we have the commutative diagram

$$\begin{array}{ccc}
 \text{Fr}(E) & & \\
 \pi_{\text{GL}} \downarrow & \searrow \pi_{\text{Fr}} & \\
 \text{Fr}(E)/\text{GL}_r(\mathbb{K}) & \xrightarrow{\bar{\pi}} & M
 \end{array}$$

The maps π_{GL} and π_{Fr} are submersions (the first because $E^r \rightarrow M$ is a vector bundle and $\text{Fr}(E) \subseteq E^r$, the second because quotient maps are). So

$$\bar{\pi} \circ \pi_{\text{GL}} = \pi_{\text{Fr}},$$

and thus $\bar{\pi}$ is smooth (because π_{GL} is a submersion, $\bar{\pi}$ is smooth iff $\bar{\pi} \circ \pi_{E^r}$ is). And similarly

$$\bar{\pi}^{-1} \circ \pi_{\text{Fr}} = \pi_{\text{GL}}$$

means that $\bar{\pi}^{-1}$ is smooth as well. Thus $\bar{\pi}$ is a diffeomorphism $\text{Fr}(E)/\text{GL}_r(\mathbb{K}) \cong M$.

Now consider projection $\pi: \text{Or}(E) \rightarrow M$, so we have the commutative diagram

$$\begin{array}{ccc}
 \text{Fr}(E) & & \\
 \pi_{\text{Or}} \downarrow & \searrow \pi_{\text{Fr}} & \\
 \text{Or}(E) & \xrightarrow{\pi} & M
 \end{array}$$

For the same reason as before, π is then smooth.

If we take a local trivialization Φ^r of $U \subseteq M$, then $\pi^{-1}U = \pi_{\text{Or}}(\pi_{\text{Fr}}^{-1}(U))$. Under the trivialization Φ^r , $\pi_{\text{Fr}}^{-1}U$ corresponds to $U \times \text{GL}_r(\mathbb{R}) = (U \times \text{GL}_r^+(\mathbb{R})) \sqcup (U \times \text{GL}_r^-(\mathbb{R}))$. And thus

$$\pi^{-1}U = \pi_{\text{Or}}(U \times \text{GL}_r^+(\mathbb{R})) \sqcup \pi_{\text{Or}}(U \times \text{GL}_r^-(\mathbb{R})),$$

the union is disjoint because $\text{GL}_r^+(\mathbb{R})$'s action preserves the sign of the determinant.

π is bijective on both of these components: its inverse for the first is given by mapping $p \in U$ to $(p, [e_1, \dots, e_r])$ where $e_1, \dots, e_r$ is the local frame associated with Φ^r . The inverse for the second is given by mapping $p \mapsto (p, [-e_1, e_2, \dots, e_r])$.

Both of these are smooth for the same reason as above: calling them π_1^{-1}, π_2^{-1} we have $\pi_{\text{Or}} = \pi_i^{-1} \circ \pi_{\text{Fr}}$, and since $\pi_{\text{Or}}, \pi_{\text{Fr}}$ are submersions this implies smoothness. Thus π is indeed a two-fold cover of M as desired.

- (4) If M is orientable, let $(U_i, (x_i^j))$ be a consistently oriented atlas for M . Define $\psi_i: U_i \rightarrow \text{Fr}(TM)$ by $\psi_i(p) = \left(\partial/\partial x_i^1|_p, \dots, \partial/\partial x_i^n|_p \right)$. This is smooth as $\text{Fr}(TM)$ is a submanifold of $TM^{\times r}$, and this is a local section of TM^r (if we take the local trivialization Φ associated with $\partial/\partial x_i^1|_p$, this map under Φ^r corresponds to $p \mapsto (p, \text{id})$ which is smooth). Thus this defines a map to the quotient $\bar{\psi}_i: U_i \rightarrow \text{Or}(TM)$ which is a local section.

Since the atlases are consistently oriented, $\psi_j(p) = \left(\partial/\partial x_j^1|_p, \dots, \partial/\partial x_j^n|_p \right)$'s transition matrix with $\psi_i(p)$ has positive determinant. Thus $\pi_{\text{Or}}(\psi_j(p)) = \pi_{\text{Or}}(\psi_i(p))$, which means $\bar{\psi}_i = \bar{\psi}_j$ along their intersections. So we can glue these local sections together to get a global section, so $\text{Or}(TM)$ is oriented as desired.

Conversely let $\sigma: M \rightarrow \text{Or}(TM)$ be a global section of $\text{Or}(TM) \rightarrow M$. Then for each $p \in M$, given $T_p M$ the orientation determined by $\sigma(p)$. That is, $e_1, \dots, e_n \in T_p M$ is positively oriented iff $[e_1, \dots, e_n] \in \sigma(p)$.

We claim that this is a continuous orientation. For $p \in M$ take a connected local chart $p \in (U, (x^i))$. If $[\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p] \notin \sigma(p)$ then since the fibers of $\text{Or}(TM)$ have cardinality 2 this means that $[-\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p] \in \sigma(p)$, so we can swap x^1 with $-x^1$ to obtain a chart whose coordinates are in $\sigma(p)$. Then define $\hat{\sigma}: U \rightarrow \text{Fr}(E)$ by $\hat{\sigma}(q) = (\partial/\partial x^1|_q, \dots, \partial/\partial x^n|_q)$, so we have that $\pi_{\text{Or}} \circ \hat{\sigma}(p) = \sigma(p)$; we claim this is true for all $q \in U$.

Now consider $\text{sgn}: \text{Fr}(TM) \rightarrow \{\pm 1\}$ which maps $(e_1, \dots, e_n) \in E_q$ to $+1$ if $(e_1, \dots, e_n)$ and $\hat{\sigma}(q)$ are consistently oriented, and -1 otherwise. This just maps $(e_1, \dots, e_n)$ to the sign of the determinant of its transition matrix with $\hat{\sigma}(q)$. This is thus continuous, as taking the determinant and sign are.

Since sgn is invariant under $\text{GL}_r^+(\mathbb{R})$'s action, it quotients to give a continuous map which we also denote by $\text{sgn}: \text{Or}(TM) \rightarrow \{\pm 1\}$. This is just the indicator for whether or not $\hat{\sigma}$ is an orientation: $\text{sgn}(\mathcal{O}_q) = +1$ if $\hat{\sigma}(q) \in \mathcal{O}_q$ and -1 otherwise. Now consider $\text{sgn} \circ \sigma: U \rightarrow \{\pm 1\}$: it maps q to $+1$ if $\hat{\sigma}(q) \in \sigma(q)$ and -1 otherwise. As a composition of continuous maps, it is continuous; since $\{\pm 1\}$ is discrete it is locally constant. Since U is connected it is constant, and since $\hat{\sigma}(p) \in \sigma(p)$ it is identically 1.

Thus $\partial/\partial x^1, \dots, \partial/\partial x^n$ are in σ for all points in U , meaning it is an oriented local coordinate frame of M . So $(U, (x^i))$ is an oriented local coordinate chart. Thus every point lies in an oriented local coordinate chart, and thus the orientation is continuous as desired.

- (5) Let $q: S^1 \times \mathbb{R} \rightarrow M$ be the quotient map, and let $\varphi \in \mathbb{Z}/2$ be the non-trivial element. Since q is a submersion between manifolds of the same dimension (since $\mathbb{Z}/2$ is discrete), dq_x is an isomorphism of tangent spaces for all x .

Suppose there is a global section $\sigma: M \rightarrow \text{Or}(TM)$. Since dq_x is an isomorphism, for each $(\theta, x) \in S^1 \times \mathbb{R}$ let $\hat{\sigma}(\theta, x) = dq_{\theta, x}^{-1}(\sigma[\theta, x])$. This maps a basis in $\sigma[\theta, x]$ to a basis of $T_{\theta, x}S^1 \times \mathbb{R}$, all the bases are consistently oriented. This is because essentially we are mapping a basis B to AB (where $A = dq^{-1}$); if C is another basis with X the transition matrix, so $XB = C$, then $Y = AXA^{-1}$ is the transition matrix from AC to AB , and $\det Y = \det X$.

Now $q \circ \varphi = q$ which means that

$$\hat{\sigma} = dq^{-1} \circ \sigma = d(q \circ \varphi)^{-1} \circ \sigma = d\varphi^{-1} \circ dq^{-1} \circ \sigma = d\varphi \circ dq^{-1} \circ \sigma.$$

But $d\varphi$ has a representation $\text{diag}(1, -1)$ which has determinant -1 . Thus $d\varphi$ reverses orientation, so $d\varphi^{-1} \circ dq^{-1} \circ \sigma$ would have orientation opposite to $dq^{-1} \circ \sigma$, a contradiction.

- (6) Let us write

$$\text{Or}(E) = \{(p, \mathcal{O}_p) \mid p \in M, \mathcal{O}_p \text{ is an orientation of } E_p\}.$$

We have that

$$\pi^*E = \{(e, \mathcal{O}_p) \mid e \in E_p, \mathcal{O}_p \in \text{Or}(E)_p\},$$

The fibers of π^*E over $\text{Or}(E)$ are isomorphic to E_p . Thus

$$\text{Or}(\pi^*E) = \{(e, \mathcal{O}_p, \mathcal{O}'_p) \mid e \in E_p, \mathcal{O}_p \text{ and } \mathcal{O}'_p \text{ are orientations of } E_p\},$$

since $\pi^*E_{(e, \mathcal{O}_p)} \cong E_p$ so orientations of the fibers of π^*E are orientations of the fibers of E .

Now define $\sigma: \text{Or}(E) \rightarrow \text{Or}(\pi^*E)$ which maps $\mathcal{O}_p \in \text{Or}(E)_p$ (an orientation of E_p) to $(0_p, \mathcal{O}_p, \mathcal{O}_p)$, where $0_p \in E_p$ is the zero vector. This is a smooth section, meaning $\pi^*E \rightarrow \text{Or}(E)$ is orientable as desired.

Now we claim that π^*TM and $T\text{Or}(TM)$ are isomorphic bundles. Since π^*TM is orientable, this would mean $T\text{Or}(TM)$ is an orientable bundle and thus by (4), $\text{Or}(TM)$ is an orientable manifold. We have

$$\begin{aligned} \pi^*TM &= \{(v, \mathcal{O}_p) \mid v \in T_pM, \mathcal{O}_p \in \text{Or}(TM)_p\}, \\ T\text{Or}(TM) &= \{(\mathcal{O}_p, v) \mid \mathcal{O}_p \in \text{Or}(TM), v \in T_{\mathcal{O}_p}\text{Or}(TM)\}. \end{aligned}$$

Now, since $\pi: \text{Or}(TM) \rightarrow M$ is a two-fold cover, it is in particular a local diffeomorphism. Thus $d\pi_{\mathcal{O}_p}: T_{\mathcal{O}_p}\text{Or}(TM) \rightarrow T_pM$ is an isomorphism.

Consider the map

$$\theta: T\text{Or}(TM) \rightarrow TM \times \text{Or}(TM), \quad \theta(\mathcal{O}_p, v) = (d\pi_{\mathcal{O}_p}v, \mathcal{O}_p).$$

This is smooth since its first component is just the differential $d\pi$, and the second is just a projection. Furthermore it lands in $\pi^*TM = TM \times_M \text{Or}(TM)$ since both coordinates map to p by π_{TM} and π . Since $\pi^*TM \subseteq TM \times \text{Or}(TM)$ is embedded, we can restrict θ to a smooth map

$$\theta: T\text{Or}(TM) \rightarrow \pi^*TM.$$

It is bijective: if $\theta(\mathcal{O}_p, v) = \theta(\mathcal{O}_q, u)$ then $\mathcal{O}_p = \mathcal{O}_q$ and since $d\pi_{\mathcal{O}_p}$ is an isomorphism, $v = u$. And for $(w, \mathcal{O}_p) \in \pi^*TM$, $\theta(\mathcal{O}_p, d\pi_{\mathcal{O}_p}^{-1}w) = (w, \mathcal{O}_p)$.

It also clearly commutes with the projections $T\text{Or}(TM) \rightarrow \text{Or}(TM)$ and $\pi^*TM \rightarrow \text{Or}(TM)$, and on the fibers it is linear: for $\mathcal{O}_p \in \text{Or}(E)$ θ restricts to $d\pi_{\mathcal{O}_p}$ on the fiber of $\mathcal{O}_p$. Thus θ is a bundle morphism, and as it is bijective, it is a bundle isomorphism.

So $\pi^*TM \cong T\text{Or}(TM)$ and as the former is orientable, so too is the latter. Then by (4) $\text{Or}(TM)$ is orientable as desired.

12.0.5 Problem

Let M be a smooth manifold. Denote interior multiplication by $X \in \mathfrak{X}(M)$ by ι_X .

- (1) Prove **Cartan's formula**: for every $\omega \in \Omega^k(M)$ and $X \in \mathfrak{X}(M)$,

$$\mathcal{L}_X \omega = d\iota_X \omega + \iota_X d\omega.$$

- (2) Suppose M is oriented with Riemannian metric g . Give ∂M its induced orientation, and let $\nu_{\partial M} \rightarrow \partial M$ be the normal bundle of $\partial M \subseteq M$. Prove that there exists a unique section $N \in \Gamma(\nu_{\partial M})$ with norm 1 such that for every $p \in \partial M$, if $(e_1, \dots, e_{n-1})$ is an oriented basis of $T_p \partial M$, then $(N_p, e_1, \dots, e_{n-1})$ is an oriented basis for $T_p M$.

- (3) Let $\nu_g \in \Omega^n(M)$ be M 's volume form. Let

$$\nu_{i^*g} \in \Omega^{n-1}(\partial M)$$

be ∂M 's volume form with respect to the metric i^*g induced from the inclusion $i: \partial M \hookrightarrow M$. Prove that

$$\nu_{i^*g} = i^*(\iota_N \nu_g).$$

- (4) Since $\nu_g \in \Omega^n(M)$, for every $f \in C^\infty(M)$ the integral $\int_M f \nu_g$ is defined. Prove the **divergence theorem**: if $X \in \mathfrak{X}(M)$ is compactly supported then

$$\int_M \operatorname{div} X \nu_g = \int_{\partial M} \langle X, N \rangle_g \nu_{i^*g}.$$

- (1) For $Y_1, \dots, Y_k \in \mathfrak{X}(M)$ we know

$$\mathcal{L}_X \omega(Y_1, \dots, Y_n) = X(\omega(\bar{Y})) - \sum_i \omega(Y_1, \dots, [X, Y_i], \dots, Y_k).$$

Now we compute $d\iota_X \omega(\bar{Y})$ and $\iota_X d\omega(\bar{Y})$. For an increasing multi-index I , let us denote by $\bar{Y}_{-I}$ the tuple obtained from removing the indices I from $\bar{Y}$:

$$\bar{Y}_{-i_1, \dots, i_\ell} = (Y_1, \dots, \widehat{Y}_{i_1}, \dots, \widehat{Y}_{i_\ell}, \dots, Y_k).$$

From the previous homework,

$$\begin{aligned} d\iota_X \omega(\bar{Y}) &= \sum_i (-1)^{i-1} Y_i (\iota_X \omega(\bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \iota_X \omega([Y_i, Y_j], \bar{Y}_{-i,j}) \\ &= \sum_i (-1)^{i-1} Y_i (\omega(X, \bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \omega(X, [Y_i, Y_j], \bar{Y}_{-i,j}) \\ &= \sum_i (-1)^{i-1} Y_i (\omega(X, \bar{Y}_{-i})) - \sum_{i < j} (-1)^{i+j} \omega([Y_i, Y_j], X, \bar{Y}_{-i,j}). \end{aligned}$$

The last equality is due to the antisymmetry of ω .

For $\iota_X d\omega(\bar{Y}) = d\omega(X, \bar{Y})$, we consider the removal of the first index and all other indices independently. Notice that in the second sum in the exterior derivative (removing two indices), we consider two cases: when $i = 1$ and $1 < j \leq k+1$ (removing X and Y_{j-1} and multiplying by $(-1)^{1+j}$), and when $1 < i < j$ (removing Y_{i-1} and Y_{j-1} and multiplying by $(-1)^{i+j}$). By permuting the summation index, in the first case we sum over all $j \leq k$, and multiply the term by $(-1)^j$. In the second case, we sum over all $i < j \leq k$, and multiply the term by $(-1)^{(i+1)+(j+1)} = (-1)^{i+j}$. Thus,

$$d\omega(X, \bar{Y}) = X(\omega(\bar{Y})) + \sum_i (-1)^i Y_i (\omega(X, \bar{Y}_{-i})) + \sum_{i < j} (-1)^{i+j} \omega([Y_i, Y_j], X, \bar{Y}_{-i,j}) + \sum_j (-1)^j \omega([X, Y_j], \bar{Y}_{-j}).$$

By our previous computation, we see that this is equal to

$$d\omega(X, \bar{Y}) = X(\omega(\bar{Y})) + \sum_j (-1)^j \omega([X, Y_j], \bar{Y}_{-j}) - d\iota_X \omega(\bar{Y}).$$

Furthermore, since $\omega([X, Y_j], \bar{Y}_{-j}) = (-1)^{j-1} \omega(Y_1, \dots, [X, Y_j], \dots, Y_k)$ since we apply $j-1$ transpositions to move the first index to the j th, this is equal to

$$\iota_X d\omega(\bar{Y}) = X(\omega(\bar{Y})) - \sum_j \omega(Y_1, \dots, [X, Y_j], \dots, Y_k) - d\iota_X \omega(\bar{Y}) = \mathcal{L}_X \omega(\bar{Y}) - d\iota_X \omega(\bar{Y})$$

as desired.

- (2) Let $n = \dim M$. Then since $\dim \partial M = n-1$, the dimension of the fibers of $\nu_{\partial M}$ are all 1. So there are only two vectors in each fiber with norm 1, each with opposite sign. Since $v, e_1, \dots, e_{n-1}$ and $-v, e_1, \dots, e_{n-1}$ have opposite orientations, for each $p \in \partial M$ there is a unique choice for N_p .

Let $(U, (x^i))$ be a boundary chart for M . The induced orientation of ∂M is such that if $e_1, \dots, e_{n-1}$ is oriented for ∂M then $-\partial_n, e_1, \dots, e_{n-1}$ is oriented. Here we use the convention that Euclidean half-space is nonnegative in its last coordinate. For $p \in U \cap \partial M$, let v be the orthogonal projection of $-\partial_n|_p$ to $\nu_{\partial M}|_p$.

Since $\nu_{\partial M}$ is orthogonal to $e_1, \dots, e_{n-1}$, $v, e_1, \dots, e_{n-1}$ remains a basis. Furthermore, since this is a projection orthogonal to the e_i we have that $-\partial_n = v + b^i e_i$ for some coefficients b^i . Thus the transition matrix between $-\partial_n, e_1, \dots, e_{n-1}$ and $v, e_1, \dots, e_{n-1}$ is of the form

$$\begin{pmatrix} 1 & & & \\ | & 1 & & \\ * & & \ddots & \\ | & & & 1 \end{pmatrix},$$

which has determinant 1. Thus $-\partial_n, e_1, \dots, e_{n-1}$ and $v, e_1, \dots, e_{n-1}$ are consistently oriented, so the list $v, e_1, \dots, e_{n-1}$ is oriented for M . Thus N_p must be the normalization of v , i.e. $N_p = v/|v|_g$, because N_p is unique and this vector has the desired property (since v does, and multiplying by a positive constant does not affect orientation). So locally in U , N is the projection of $-\partial_n$ onto $\nu_{\partial M}$ and then normalization. Both of these preserve smoothness (since $\nu_{\partial M}$ is a vector bundle), so N is smooth in U .

By uniqueness, if V is any other boundary chart, the N we defined for U and the N we defined for V must coincide by uniqueness of N_p at all points p . Since N is smooth over all boundary charts, it is smooth over all ∂M . It has the desired property in each U , and thus over all of ∂M .

- (3) Let $e_1, \dots, e_{n-1}$ be an oriented orthonormal basis in $T_p \partial M$. Then

$$\iota^*(\iota_N \nu_g)(e_1, \dots, e_{n-1}) = \nu_g(N_p, e_1, \dots, e_{n-1}).$$

By the previous point, $N_p, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis of $T_p M$. The characterizing property of ν_g is that it maps oriented orthonormal bases to 1, so $\iota^*(\iota_N \nu_g)(e_1, \dots, e_{n-1}) = 1$.

This is true for all oriented orthonormal bases, and thus by the characterizing property of the Riemannian volume form, $\iota^*(\iota_N \nu_g) = \nu_{\iota^* g}$ as desired.

- (4) Since X is compactly supported, so too is $\operatorname{div} X$. This is because if $X = f^i E_i$ in a geodesic frame centered at p then $\operatorname{div} X(p) = (E_i f^i)(p)$. So if $X(p) = 0$ then $f^i = 0$ and thus $\operatorname{div} X(p) = 0$, meaning $\operatorname{supp} \operatorname{div} X \subseteq \operatorname{supp} X$ which is compact. So integrating $\operatorname{div} X \nu_g$ makes sense.

Previously we showed that $\mathcal{L}_X \nu_g = (\operatorname{div} X) \nu_g$. By Cartan's formula,

$$(\operatorname{div} X) \nu_g = \mathcal{L}_X \nu_g = d\iota_X \nu_g + \iota_X d\nu_g.$$

Since ν_g is a top-form, its exterior derivative is zero (all $(n+1)$ -forms are zero) and thus we can simplify

$$(\operatorname{div} X) \nu_g = d\iota_X \nu_g.$$

By Stokes' theorem,

$$\int_M (\operatorname{div} X) \nu_g = \int_M d\iota_X \nu_g = \int_{\partial M} \iota^*(\iota_X \nu_g).$$

Now we claim that $\iota^*(\iota_X \nu_g) = \langle X, N \rangle \nu_{\iota^*g}$ on ∂M .

To show the equality of two top-forms, it is sufficient to show that at any point and for a single oriented basis they agree. This is because top-forms are determined by their value at any oriented basis (by multi-linearity and antisymmetry). So for $p \in \partial M$ let $e_1, \dots, e_{n-1}$ be an oriented orthonormal basis of $T_p \partial M$. Then $N_p, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis of $T_p M$ and so $X = \langle X, N \rangle N + \sum_i \langle X, e_i \rangle e_i$ and thus

$$\begin{aligned} \iota^*(\iota_X \nu_g)(e_1, \dots, e_{n-1}) &= \nu_g(X, e_1, \dots, e_{n-1}) = \\ &= \langle X, N \rangle \nu_g(N, e_1, \dots, e_{n-1}) + \sum_i \langle X, e_i \rangle \nu_g(e_i, e_1, \dots, e_{n-1}) = \langle X, N \rangle \end{aligned}$$

where the last equality is because $N, e_1, \dots, e_{n-1}$ is an oriented orthonormal basis and thus, by the characterization of the Riemannian volume form, $\nu_g(N, e_1, \dots, e_{n-1}) = 1$; the other terms are all zero by antisymmetry.

On the other hand, since $\nu_{\iota^*g}(e_1, \dots, e_{n-1}) = 1$ we see that

$$\langle X, N \rangle \nu_{\iota^*g}(e_1, \dots, e_{n-1}) = \langle X, N \rangle = \iota^*(\iota_X \nu_g)(e_1, \dots, e_{n-1})$$

as desired.

All in all, we have shown $\iota^*(\iota_X \nu_g) = \langle X, N \rangle \nu_{\iota^*g}$ and thus

$$\int_M (\operatorname{div} X) \nu_g = \int_{\partial M} \iota^*(\iota_X \nu_g) = \int_{\partial M} \langle X, N \rangle \nu_{\iota^*g}$$

as desired.

13 Homework 13

13.0.1 Problem (DoCarmo, Riemannian Geometry, page 106 question 8)

Let M be a connected Riemannian manifold of dimension $n \geq 3$. Suppose that M is **isotropic**: its sectional curvature at each point is independent on choice of subspace $\sigma \subseteq T_p M$. Prove that M has constant sectional curvature; that is its sectional curvature is further independent on choice of point.

Let us define the 4-tensor

$$\operatorname{Rm}'(W, Z, X, Y) = \langle W, X \rangle \langle Z, Y \rangle - \langle Z, X \rangle \langle W, Y \rangle.$$

Because sectional curvature $K(p, \sigma) = K(p)$ is independent of σ , we know that $\operatorname{Rm} = K \operatorname{Rm}'$. Notice that for $U \in \mathfrak{X}(M)$,

$$\nabla_U \operatorname{Rm}'(W, Z, X, Y) = U(\operatorname{Rm}'(W, Z, X, Y)) - \langle \nabla_U W, X \rangle \langle Z, Y \rangle + \langle Z, X \rangle \langle \nabla_U W, Y \rangle - \dots$$

Because the connection is metric, further expansions of the terms show that this all cancels out. This is lengthy so I will not write everything out, but for example expanding once gives

$$U(\operatorname{Rm}'(W, Z, X, Y)) = U\langle W, X \rangle \langle Z, Y \rangle - U\langle Z, X \rangle \langle W, Y \rangle = \langle \nabla_U W, Z \rangle \langle Z, Y \rangle - \langle Z, X \rangle \langle \nabla_U W, Y \rangle + \dots$$

which cancels out with the first term above.

Thus

$$\nabla_U \operatorname{Rm}' = 0$$

and so

$$\nabla \operatorname{Rm}(\bullet, \bullet, \bullet, \bullet, U) = \nabla_U \operatorname{Rm} = \nabla_U (K \operatorname{Rm}') = (U K) \operatorname{Rm}' + K \nabla_U \operatorname{Rm}' = (U K) \operatorname{Rm}'.$$

The second Binachi identity gives

$$\nabla \operatorname{Rm}(W, Z, X, Y, U) + \nabla \operatorname{Rm}(W, Z, Y, U, X) + \nabla \operatorname{Rm}(W, Z, U, X, Y) = 0.$$

Thus

$$(U K) \operatorname{Rm}'(W, Z, X, Y) + (X K) \operatorname{Rm}'(W, Z, Y, U) + (Y K) \operatorname{Rm}'(W, Z, U, X) = 0.$$

Fix a $p \in M$ and let $X \in T_p M$ be nonzero. Because $n \geq 3$ we can find $Y, Z \in T_p M$ such that $\langle X, Y \rangle = \langle Y, Z \rangle = \langle Z, X \rangle = 0$ and $\langle Z, Z \rangle = 1$ (extend X to an orthogonal basis). Letting $U = Z$, and letting W be arbitrary we get

$$(ZK)\text{Rm}'(W, Z, X, Y) + (XK)\text{Rm}'(W, Z, Y, Z) + (YK)\text{Rm}'(W, Z, Z, X) = 0.$$

Expanding definitions,

$$\begin{aligned}\text{Rm}'(W, Z, X, Y) &= \langle W, X \rangle \langle Z, Y \rangle - \langle Z, X \rangle \langle W, Y \rangle = 0, \\ \text{Rm}'(W, Z, Y, Z) &= \langle W, Y \rangle \langle Z, Z \rangle - \langle Z, Y \rangle \langle W, Z \rangle = \langle W, Y \rangle, \\ \text{Rm}'(W, Z, Z, X) &= \langle W, Z \rangle \langle Z, X \rangle - \langle Z, Z \rangle \langle W, X \rangle = -\langle W, X \rangle.\end{aligned}$$

Thus we see that

$$(XK)\langle W, Y \rangle - (YK)\langle W, X \rangle = \langle W, (XK)Y - (YK)X \rangle = 0.$$

Since this is true for arbitrary W , this means that $(XK)Y - (YK)X = 0$. Because X, Y are linearly independent this implies $XK = 0$.

So for all $X \in T_p M$ we have $XK = 0$ (for $X = 0$ this is trivial). Thus K is locally constant, and since M is connected this means K is constant as desired.

13.0.2 Problem (DoCarmo, Riemannian Geometry, page 187 question 10)

Let $f: \bar{M} \rightarrow M$ be a Riemannian submersion; $X, Y, Z, W \in \mathfrak{X}(M)$ and $\bar{X}, \bar{Y}, \bar{Z}, \bar{W}$ be their horizontal lifts. Let $R, \bar{R}$ be the curvature tensors of $M, \bar{M}$ respectively. Prove the following.

(1)

$$\langle \bar{R}(\bar{X}, \bar{Y})\bar{Z}, \bar{W} \rangle = \langle R(X, Y)Z, W \rangle - \frac{1}{4} \langle [\bar{X}, \bar{Z}]^v, [\bar{Y}, \bar{W}]^v \rangle + \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle - \frac{1}{2} \langle [\bar{Z}, \bar{W}]^v, [\bar{X}, \bar{Y}]^v \rangle.$$

(2) Let σ be the plane generated by orthonormal $X, Y \in \mathfrak{X}(M)$, and $\bar{\sigma}$ the plane generated by $\bar{X}, \bar{Y}$. Then

$$K(\sigma) = \bar{K}(\bar{\sigma}) + \frac{3}{4} |[\bar{X}, \bar{Y}]^v|^2 \geq \bar{K}(\bar{\sigma}).$$

(1) Notice that, as per a previous exercise,

$$\bar{X} \langle \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{W} \rangle = \bar{X} \left\langle \bar{\nabla}_{\bar{X}} \bar{Y} + \frac{1}{2} [\bar{X}, \bar{Y}]^v, \bar{W} \right\rangle = \bar{X} \langle \bar{\nabla}_{\bar{X}} \bar{Y}, \bar{W} \rangle = X \langle \nabla_X Y, W \rangle$$

where the second equality is because $\bar{W}$ is horizontal and thus orthogonal to vertical vectors.

Because the connection is metric,

$$\langle \bar{\nabla}_{\bar{X}} \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{W} \rangle = \bar{X} \langle \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{W} \rangle - \langle \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{\nabla}_{\bar{X}} \bar{W} \rangle.$$

The first term is equal to $X \langle \bar{\nabla}_Y Z, W \rangle$. The second term is equal to

$$\begin{aligned}\langle \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{\nabla}_{\bar{X}} \bar{W} \rangle &= \left\langle \bar{\nabla}_Y \bar{Z} + \frac{1}{2} [\bar{Y}, \bar{Z}]^v, \bar{\nabla}_X \bar{W} + \frac{1}{2} [\bar{X}, \bar{W}]^v \right\rangle = \langle \bar{\nabla}_Y \bar{Z}, \bar{\nabla}_X \bar{W} \rangle + \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle \\ &= \langle \nabla_Y Z, \nabla_X W \rangle + \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle.\end{aligned}$$

Thus

$$\begin{aligned}\langle \bar{\nabla}_{\bar{X}} \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{W} \rangle &= X \langle \nabla_Y Z, W \rangle - \langle \nabla_Y Z, \nabla_X W \rangle - \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle \\ &= \langle \nabla_X \nabla_Y Z, W \rangle - \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle.\end{aligned}$$

If $T \in \mathfrak{X}(\bar{M})$ is vertical, then

$$\langle \bar{\nabla}_T \bar{X}, \bar{Y} \rangle = \langle \bar{\nabla}_{\bar{X}} T, \bar{Y} \rangle + \langle [T, \bar{X}], \bar{Y} \rangle.$$

The second term is zero, as per the previous exercise. Thus this is equal to (since $\langle T, \bar{Y} \rangle = 0$)

$$= \bar{X} \langle T, \bar{Y} \rangle - \langle T, \bar{\nabla}_{\bar{X}} \bar{Y} \rangle = -\langle T, \bar{\nabla}_{\bar{X}} \bar{Y} \rangle = -\frac{1}{2} \langle T, [\bar{X}, \bar{Y}]^v \rangle.$$

Now we have

$$\langle \bar{\nabla}_{[\bar{X}, \bar{Y}]} \bar{Z}, \bar{W} \rangle = \langle \bar{\nabla}_{[\bar{X}, \bar{Y}]^h} \bar{Z}, \bar{W} \rangle + \langle \bar{\nabla}_{[\bar{X}, \bar{Y}]^v} \bar{Z}, \bar{W} \rangle.$$

As we just showed, the second term is equal to

$$\langle \bar{\nabla}_{[\bar{X}, \bar{Y}]^v} \bar{Z}, \bar{W} \rangle = -\frac{1}{2} \langle [\bar{X}, \bar{Y}]^v, [\bar{Z}, \bar{W}]^v \rangle.$$

As per the previous homework $\langle [\bar{X}, \bar{Y}]^h, \bar{Z} \rangle = \langle [\bar{X}, \bar{Y}], \bar{Z} \rangle = \langle [X, Y], Z \rangle$. Thus

$$\langle [\bar{X}, \bar{Y}]^h, \bar{Z} \rangle = \langle [\bar{X}, \bar{Y}], \bar{Z} \rangle = \langle [X, Y], Z \rangle$$

and since all horizontal vectors are the lift of some vector field this means that $[\bar{X}, \bar{Y}]^h = \overline{[X, Y]}$. Thus

$$\langle \bar{\nabla}_{[\bar{X}, \bar{Y}]} \bar{Z}, \bar{W} \rangle = \langle \nabla_{[X, Y]} Z, W \rangle - \frac{1}{2} \langle [\bar{X}, \bar{Y}]^v, [\bar{Z}, \bar{W}]^v \rangle.$$

Putting this together,

$$\begin{aligned} \langle \bar{R}(\bar{X}, \bar{Y}) \bar{Z}, \bar{W} \rangle &= \langle \bar{\nabla}_{\bar{Y}} \bar{\nabla}_{\bar{X}} \bar{Z}, \bar{W} \rangle - \langle \bar{\nabla}_{\bar{X}} \bar{\nabla}_{\bar{Y}} \bar{Z}, \bar{W} \rangle + \langle \bar{\nabla}_{[\bar{X}, \bar{Y}]} \bar{Z}, \bar{W} \rangle \\ &= \langle R(X, Y) Z, W \rangle - \frac{1}{4} \langle [\bar{X}, \bar{Z}]^v, [\bar{Y}, \bar{W}]^v \rangle \\ &\quad + \frac{1}{4} \langle [\bar{Y}, \bar{Z}]^v, [\bar{X}, \bar{W}]^v \rangle - \frac{1}{2} \langle [\bar{Z}, \bar{W}]^v, [\bar{X}, \bar{Y}]^v \rangle. \end{aligned}$$

(2) Because df preserves the metric of horizontal vectors, $\bar{X}, \bar{Y}$ remain orthonormal. Thus

$$\bar{K}(\bar{\sigma}) = \overline{\text{Rm}}(\bar{X}, \bar{Y}, \bar{X}, \bar{Y}) = \text{Rm}(X, Y, X, Y) - \frac{1}{4} \langle [\bar{X}, \bar{X}]^v, [\bar{Y}, \bar{Y}]^v \rangle + \frac{1}{4} \langle [\bar{Y}, \bar{X}]^v, [\bar{X}, \bar{Y}]^v \rangle - \frac{1}{2} \langle [\bar{X}, \bar{Y}]^v, [\bar{X}, \bar{Y}]^v \rangle,$$

and because the Lie bracket is antisymmetric we get that the first inner product is zero, the second is equal to $-|[\bar{X}, \bar{Y}]^v|^2/4$ and so

$$K(\sigma) = \text{Rm}(X, Y, X, Y) = \bar{K}(\bar{\sigma}) + \frac{3}{4} |[\bar{X}, \bar{Y}]^v|^2.$$

13.0.3 Problem (DoCarmo, Riemannian Geometry, page 188 question 12)

Define a Riemannian metric on $\mathbb{C}^{n+1} - 0$ as follows. For $x \in \mathbb{C}^{n+1} - 0$ and $V, W \in T_x(\mathbb{C}^{n+1} - 0) \cong \mathbb{C}^{n+1}$ define

$$\langle V, W \rangle_x = \frac{\text{Re}(V, W)}{(Z, Z)}.$$

Notice that the metric $\langle \bullet, \bullet \rangle$ restricted to $S^{2n+1} \subseteq \mathbb{C}^{n+1} - 0$ coincides with the one induced from $\mathbb{R}^{2n+2}$.

- (1) Show that for all $0 \leq \theta \leq 2\pi$, $e^{i\theta}: S^{2n+1} \rightarrow S^{2n+1}$ is an isometry. Thus it is possible to define a Riemannian metric on $\mathbb{CP}^n$ such that $f: S^{2n+1} \rightarrow \mathbb{CP}^n$ is a Riemannian submersion.
- (2) Show that in this metric, the sectional curvature of $\mathbb{CP}^n$ is

$$K(\sigma) = 1 + 3 \cos^2 \varphi$$

where σ is generated by the orthonormal pair X, Y and $\cos \varphi = \langle \bar{X}, i\bar{Y} \rangle$ where $\bar{X}$ is the horizontal lift of X .

- (1) Multiplication by $z \in \mathbb{C}$ has differential $dz: T_v \mathbb{C}^n \rightarrow T_{zv} \mathbb{C}^n$, after identifying both naturally with $\mathbb{C}^n$, given by multiplication by z . Thus multiplication by a $\mu = e^{i\theta}$ satisfies for $V, W \in T_z S^{2n+1}$,

$$\langle d\mu(V), d\mu(W) \rangle = \langle \mu V, \mu W \rangle = \operatorname{Re}(\mu V, \mu W) = \operatorname{Re}|\mu|^2(V, W) = \operatorname{Re}(V, W) = \langle V, W \rangle.$$

Thus μ is an isometry as desired.

In order for f to be a Riemannian submersion, the isomorphism

$$df_p: (T_p S^{2n+1})^h \xrightarrow{\sim} T_{fp} \mathbb{CP}^n$$

must preserve the metric. This uniquely defines the metric on $T_{fp} \mathbb{CP}^n$: define $\langle v, u \rangle_{\mathbb{CP}^n} = \langle \bar{v}, \bar{u} \rangle_{S^{2n+1}}$ where $\bar{v}, \bar{u}$ are the unique horizontal lifts of v, u .

Now if $fp = fq$ then $q = \mu p$ for some $\mu = e^{i\theta}$ and because $f = f \circ \mu$,

$$df_q = df_p \circ d\mu_p$$

so that if $df_q(v') = df_p(v)$ then as μ is an isometry, $v' = d\mu_p(v)$ and since μ is an isometry this means the metric on $\mathbb{CP}^n$ is well-defined.

This requires that μ be an isometry between the horizontal subspaces, so that $d\mu_p$ maps horizontal vectors to horizontal vectors. Because horizontal vectors are those in the kernel of df , and since $f = f \circ \mu$, if $df(v) = 0$ then $df \circ d\mu(v) = df(v) = 0$ as desired.

Note that the horizontal lift of a vector field $X \in \mathfrak{X}(M)$ exists independently of whether or not M 's metric is smooth; as long as f is a surjective submersion which maps horizontal vectors isometrically. And the horizontal lift is smooth (when $\bar{M}$'s metric is); the proof given in the previous homework doesn't have an assumption on M 's metric's smoothness. But then $g(X, Y) \circ f = \bar{g}(\bar{X}, \bar{Y})$ is smooth (since $\bar{g}, \bar{X}, \bar{Y}$ are) and since f is a surjective submersion, $g(X, Y)$ is smooth for all $X, Y \in \mathfrak{X}(M)$ so g is. That is to say, if there is a surjective submersion $f: \bar{M} \rightarrow M$ between manifolds equipped with metrics, where $\bar{M}$'s is smooth, such that f is an isometry on $\bar{M}$'s horizontal vectors (relative to f), then M 's metric is smooth. This is satisfied in our case, and so the metric defined on $\mathbb{CP}^n$ is smooth. Thus this defines a smooth metric on $\mathbb{CP}^n$ as desired.

- (2) Let Z be the position vector describing S^{2n+1} , i.e. $Z_v = v$ for $v \in S^{2n+1}$. Because its extension to $\mathbb{C}^n$ is smooth, so $Z \in \Gamma(T\mathbb{C}^n|_{S^{2n+1}})$.

We know that S^n is the level set of 1 under $N: \mathbb{R}^{n+1} - 0 \rightarrow \mathbb{R}, N(x) = |x|^2$ which is a submersion. Thus $T_p S^n$ consists of vectors $v \in T_p \mathbb{R}^{n+1}$ where $dN(v) = 0$. And the Jacobian of N at p is $2p$, so $dN(v) = 2(p, v)$, i.e. $T_p S^n = p^\perp \subseteq \mathbb{R}^{n+1}$.

In particular, $ip \in T_p S^{2n+1}$ since

$$\langle p, ip \rangle_p = \frac{\operatorname{Re}(p, ip)}{(p, p)} = \frac{-\operatorname{Re} i(p, p)}{(p, p)} = 0.$$

Thus iZ (which maps $p \mapsto ip$) is tangent to S^{2n+1} , so it defines a smooth vector field $iZ \in \mathfrak{X}(S^{2n+1})$.

Let $X, Y \in \mathfrak{X}(\mathbb{CP}^n)$, and let $\alpha: (-\varepsilon, \varepsilon) \rightarrow S^{2n+1}$ be a smooth curve with $\alpha(0) = p$ and $\alpha'(0) = \bar{X}(p)$. Let $\bar{\nabla}$ be the Levi-Civita connection of S^{2n+1} , then

$$(\bar{\nabla}_{\bar{X}} iZ)_p = \left. \frac{d}{dt} \right|_{t=0} iZ \circ \alpha(t) = \left. \frac{d}{dt} \right|_{t=0} i\alpha(t) = i\alpha'(0) = i\bar{X}(p).$$

That is to say,

$$\bar{\nabla}_{\bar{X}} iZ = i\bar{X}.$$

Now,

$$\langle \bar{\nabla}_{\bar{X}} \bar{Y}, iZ \rangle = \bar{X} \langle \bar{Y}, iZ \rangle - \langle \bar{Y}, \bar{\nabla}_{\bar{X}} iZ \rangle = -\langle \bar{Y}, i\bar{X} \rangle,$$

where the last equality is due to $\bar{Y}$ being horizontal and iZ being vertical, as well as what we just showed. So we have

$$\langle [\bar{X}, \bar{Y}], iZ \rangle = \langle \bar{\nabla}_{\bar{X}} \bar{Y} - \bar{\nabla}_{\bar{Y}} \bar{X}, iZ \rangle = -\langle \bar{Y}, i\bar{X} \rangle + \langle \bar{X}, i\bar{Y} \rangle.$$

By definition we have

$$\begin{aligned} \langle v, iu \rangle_p &= \frac{\operatorname{Re}(v, iu)}{(p, p)} = \frac{-\operatorname{Re} i(v, u)}{(p, p)} = \frac{\operatorname{Im}(v, u)}{(p, p)}, \\ \langle iv, u \rangle_p &= -\frac{\operatorname{Im}(v, u)}{(p, p)}, \end{aligned}$$

so that $\langle iv, u \rangle_p = -\langle v, iu \rangle_p$. Thus

$$\langle [\bar{X}, \bar{Y}], iZ \rangle = 2\langle \bar{X}, i\bar{Y} \rangle = 2 \cos \varphi.$$

By the previous question,

$$K(\sigma) = \bar{K}(\bar{\sigma}) + \frac{3}{4} |[\bar{X}, \bar{Y}]^v|^2.$$

Because the vertical tangent space has dimension 1, it is spanned by iZ , which has unit norm. Thus

$$[\bar{X}, \bar{Y}]^v = \langle [\bar{X}, \bar{Y}], iZ \rangle iZ = (2 \cos \varphi) iZ$$

which has norm squared of $4 \cos^2 \varphi$. Because the unit sphere has sectional curvature of 1, we get that

$$K(\sigma) = 1 + 3 \cos^2 \varphi$$

as desired.

13.0.4 Problem

Let $U \subseteq \mathbb{R}^n$ be open and $f: U \rightarrow \mathbb{R}$ be smooth. Define

$$M = \{(\bar{x}, y) \mid \bar{x} \in U, y = f(\bar{x})\} \subseteq \mathbb{R}^{n+1}$$

and consider the coordinate chart $\mathbf{x}: U \rightarrow M$ by $\mathbf{x}(\bar{x}) = (\bar{x}, f(\bar{x}))$. Prove the following.

- (1) A unit normal field to M is given by

$$\nu \circ \mathbf{x} = \frac{(-\nabla f, 1)}{\sqrt{1 + |\nabla f|^2}}.$$

- (2) The second fundamental form of M is given by

$$\langle \Pi(\partial_i, \partial_j), \nu \rangle \circ \mathbf{x} = \frac{\text{Hess} f(\partial_i, \partial_j)}{\sqrt{1 + |\nabla f|^2}}.$$

- (3) The mean curvature of M is given by

$$\langle H, \nu \rangle \circ \mathbf{x} = \text{div} \left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}} \right).$$

- (4) If $U = \mathbb{R}^n$ and f is linear, then M is a minimal surface.

- (5) If $U = \{u \in \mathbb{R}^2 \mid |u| > a\}$ and $f = a \cosh^{-1}(|u|/a)$ then M is a minimal surface.

- (1) Define $F: U \times \mathbb{R} \rightarrow \mathbb{R}$ by $F(\bar{x}, y) = y - f(\bar{x})$. Then $\nabla F = (-\nabla f, 1)$ so that F is a submersion and thus $M = F^{-1}0$ is embedded. And thus for $p \in M$, $T_p M = \ker \nabla F$ which are all the vectors orthogonal to $(-\nabla f, 1)$. Normalizing this, we see that a normal vector field is

$$\frac{(-\nabla f, 1)}{\sqrt{1 + |\nabla f|^2}}$$

as desired.

- (2) Let us denote by $\tilde{\partial}_i$ the coordinate frame induced by $(U, \mathbf{x})$, and ∂_i the Euclidean coordinate frame. We want to show

$$\langle \Pi(\tilde{\partial}_i, \tilde{\partial}_j), \nu \rangle \circ \mathbf{x} = \frac{\text{Hess}f(\partial_i, \partial_j)}{\sqrt{1 + |\nabla f|^2}}.$$

Because $\Pi(X, Y) = \bar{\nabla}_X Y - \nabla_X Y$ where $\bar{\nabla}$ is the Euclidean connection and ∇ is M 's, since $\nabla_X Y \in \mathfrak{X}(M)$ is orthogonal to ν ,

$$\langle \Pi(X, Y), \nu \rangle = \langle \bar{\nabla}_X Y, \nu \rangle.$$

From exercise 8 we know

$$\tilde{\partial}_i = f_i \partial_{n+1} + \partial_i, \quad f_i = \frac{\partial f}{\partial x_i}.$$

And thus

$$\langle \Pi(\tilde{\partial}_i, \tilde{\partial}_j), \nu \rangle = \langle \bar{\nabla}_{\tilde{\partial}_i} \tilde{\partial}_j, \nu \rangle = \langle \bar{\nabla}_{\partial_i + f_i \partial_{n+1}} (\partial_j + f_j \partial_{n+1}), \nu \rangle.$$

Because $\bar{\nabla}_{\partial_i} \partial_j = 0$ for all i, j , we see that this is equal to

$$= \langle \bar{\nabla}_{\partial_i} (f_j \partial_{n+1}) + f_i \bar{\nabla}_{\partial_{n+1}} (f_j \partial_{n+1}), \nu \rangle = \langle f_{ij} \partial_{n+1}, \nu \rangle.$$

Note the last equality is due to f not being a function of x_{n+1} so that $\partial_{n+1} f_j = 0$.

From the previous point, this is equal to

$$\langle \Pi(\tilde{\partial}_i, \tilde{\partial}_j), \nu \rangle = \frac{f_{ij}}{\sqrt{1 + |\nabla f|^2}}.$$

Now we compute $\text{Hess}f(\partial_i, \partial_j)$ from definition:

$$\text{Hess}f(\partial_i, \partial_j) = \langle \bar{\nabla}_{\partial_i} \nabla f, \partial_j \rangle = \left\langle \bar{\nabla}_{\partial_i} \sum_{\ell} f_{\ell} \partial_{\ell}, \partial_j \right\rangle = \sum_{\ell} \langle f_{i\ell} \partial_{\ell}, \partial_j \rangle = f_{ij}$$

which gives us the desired result.

- (3) Firstly, mean curvature can be computed in the coordinate frame $\tilde{\partial}_i$ by

$$H = g^{ij} \Pi(\tilde{\partial}_i, \tilde{\partial}_j),$$

and so by the previous point

$$\langle H, \nu \rangle = g^{ij} \langle \Pi(\tilde{\partial}_i, \tilde{\partial}_j), \nu \rangle = g^{ij} \frac{f_{ij}}{\sqrt{1 + |\nabla f|^2}}.$$

Recall that

$$g_{ij} = f_i f_j + \delta_{ij}$$

so that

$$g = I + (\nabla f)(\nabla f)^{\top}.$$

The ShermanMorrison formula then gives us g^{-1} :

$$g^{-1} = I - \frac{(\nabla f)(\nabla f)^{\top}}{1 + |\nabla f|^2} \Rightarrow g^{ij} = \delta_{ij} - \frac{f_i f_j}{1 + |\nabla f|^2}.$$

Thus

$$\langle H, \nu \rangle = \left(\delta^{ij} - \frac{f_i f_j}{1 + |\nabla f|^2} \right) \frac{f_{ij}}{\sqrt{1 + |\nabla f|^2}} = \frac{f_{ii}}{\sqrt{1 + |\nabla f|^2}} - \frac{f_i f_j f_{ij}}{(1 + |\nabla f|^2)^{3/2}}$$

where the free indices are summed over.

On the other hand, because ∂_i forms a geodesic frame we know that

$$\text{div} \left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}} \right) = \partial_i \frac{f_i}{\sqrt{1 + |\nabla f|^2}} = \frac{f_{ii}}{\sqrt{1 + |\nabla f|^2}} - \frac{f_i \partial_i (1 + |\nabla f|^2)}{2(1 + |\nabla f|^2)^{3/2}}.$$

A quick computation shows

$$\partial_i(1 + |\nabla f|^2) = 2f_j f_{ij}$$

and so

$$\operatorname{div} \left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}} \right) = \frac{f_{ii}}{\sqrt{1 + |\nabla f|^2}} - \frac{f_i f_j f_{ij}}{(1 + |\nabla f|^2)^{3/2}} = \langle H, \nu \rangle$$

as desired.

- (4) In this point and the next, let

$$X = \frac{\nabla f}{\sqrt{1 + |\nabla f|^2}}.$$

If f is linear, then ∇f is constant and so X is constant. Thus ∇X is zero, so its trace is zero, and thus $\operatorname{div} X = 0$ meaning $\langle H, \nu \rangle = 0$. Since H is orthogonal to M , this means $H = 0$ so M is minimal.

- (5) Let us write (x, y) instead of (u_1, u_2) and let $r = \sqrt{x^2 + y^2}$. Then our function is $f(r) = a \cosh^{-1}(r/a)$ whose derivative is

$$f'(r) = \frac{1}{\sqrt{r^2/a^2 - 1}} = \frac{a}{\sqrt{r^2 - a^2}}$$

Because $\nabla r = (x/r, y/r)$ we get that

$$\nabla f = \frac{a}{\sqrt{r^2 - a^2}} \left(\frac{x}{r}, \frac{y}{r} \right) \Rightarrow |\nabla f|^2 = \frac{a^2}{r^2 - a^2}.$$

And so

$$\sqrt{1 + |\nabla f|^2} = \frac{r}{\sqrt{r^2 - a^2}},$$

thus

$$X = \frac{a}{r} \left(\frac{x}{r}, \frac{y}{r} \right) = \frac{a}{r^2} (x, y).$$

Its divergence is then

$$\frac{1}{a} \cdot \operatorname{div} X = \operatorname{div} \left(\frac{x}{r^2}, \frac{y}{r^2} \right) = \frac{\partial}{\partial x} \frac{x}{r^2} + \frac{\partial}{\partial y} \frac{y}{r^2}.$$

Computing shows that

$$\frac{\partial}{\partial x} \frac{x}{r^2} = \frac{\partial}{\partial x} \frac{x}{x^2 + y^2} = \frac{r^2 - 2x^2}{r^4}$$

so that

$$\frac{1}{a} \cdot \operatorname{div} X = \frac{2}{r^2} - 2 \frac{x^2 + y^2}{r^4} = 0,$$

meaning $\langle H, \nu \rangle = \operatorname{div} X = 0$ and thus $H = 0$ so M is minimal.